

The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties

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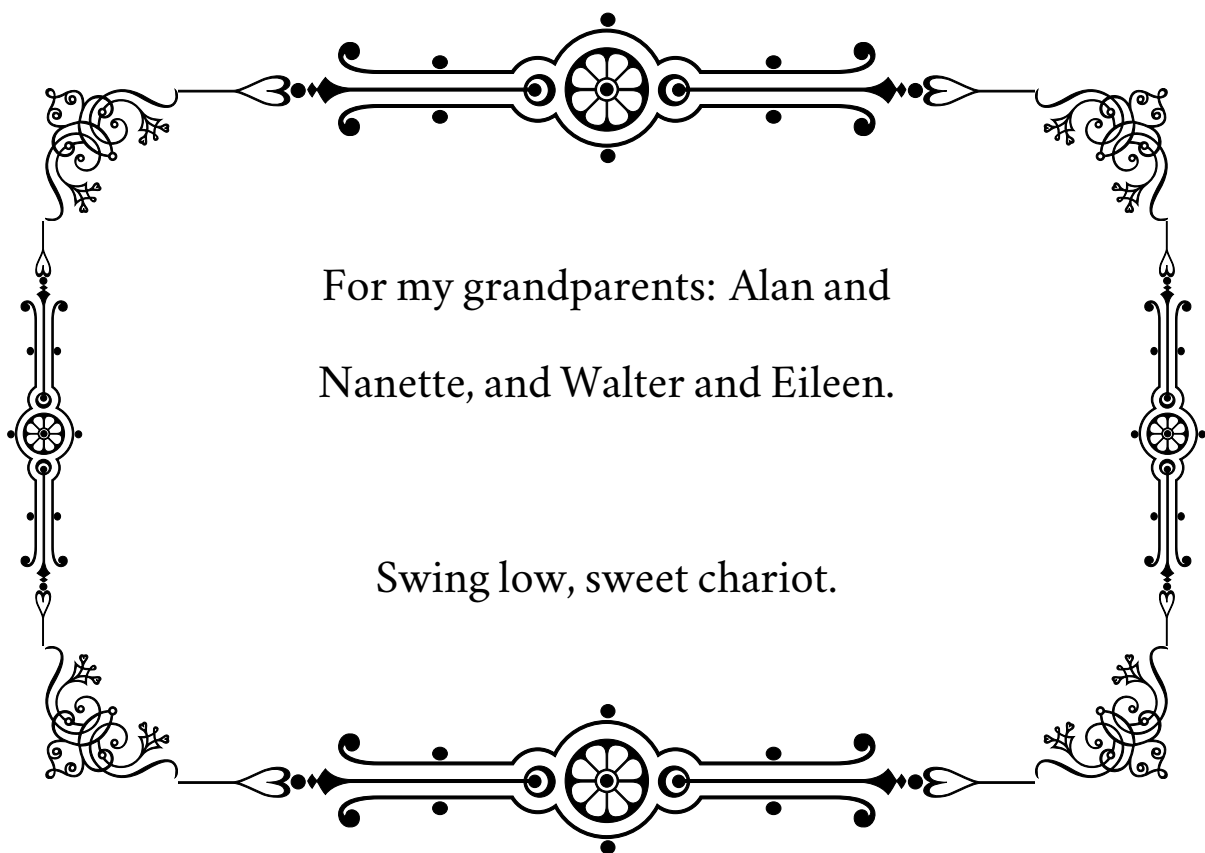
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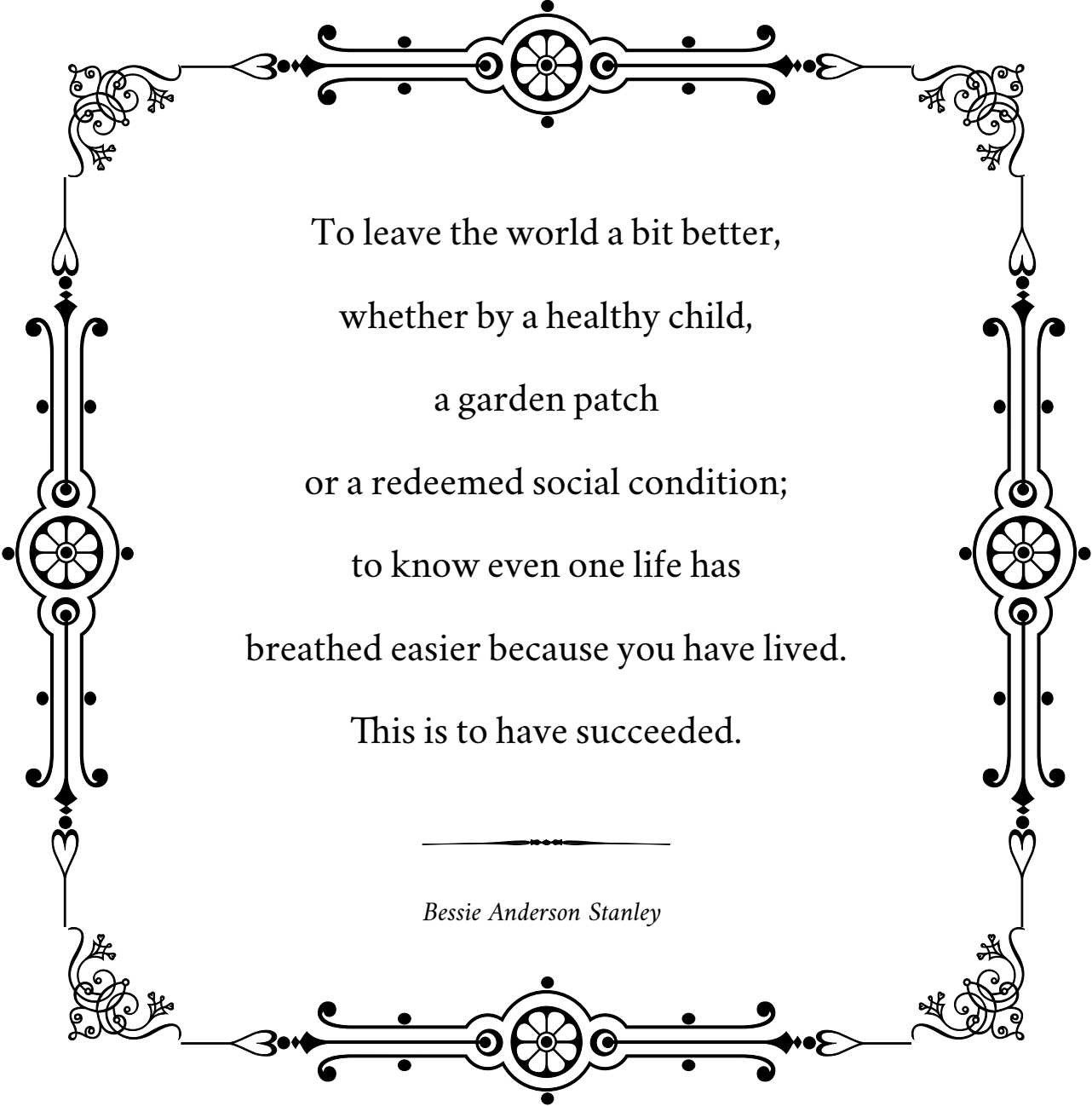
Abstract

As a field, stellar modelling is now seven decades old. Yet, there remain challenges associated with the numerical method; chief among them the treatment of secondaries, and modelling the core-collapse supernova and its aftermath. In this thesis, we present self-consistent calculations for three major phases of stellar evolution. We display a numerical example of a system in which helium-rich material is transferred between the constituent stars, and compare this to one where the composition of the accreted mass is fixed to the companion's surface composition. We find that this results in accretors which are hotter and at times more luminous than when material is assumed to reflect the surface composition of the secondary. We present a method for determining the goodness-of-fit of a natal supernova kick, and display this method for a kick linear in the ejecta-to-remnant mass ratio Q : $v_{\text{kick}} = \alpha Q + \beta$, with free parameters α and β . We find that the parameters $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}$ and $\beta = 15^{+10}_{-15} \text{ km s}^{-1}$ are able to replicate the merger rate of observed galactic double neutron stars, period eccentricity distributions, and isolated pulsar velocities, and produce ultra-stripped supernovae. We present a small grid of overcontact binary stars, where both stars have been evolved in detail. We use the BPASS v2.2 models in order to constrain the initial zero-age main sequence parameters of six observed O+O overcontact systems. We use these parameters as inputs to a variant of the Cambridge STARS code, and produce matches to one of the binaries, with candidate matches for two more. We discuss avenues for future research, including modifications to the mass-loss prescriptions already in use within the code, as well as population studies of overcontact binaries.

Cover Image: The Milky Way over Uluru. Image by Eddie Yip, released under a CC-BY-SA license:

<https://www.flickr.com/photos/eddiextcteam/28311312421/>. Image taken on Arrernte Country.





To leave the world a bit better,
whether by a healthy child,
a garden patch
or a redeemed social condition;
to know even one life has
breathed easier because you have lived.
This is to have succeeded.

Bessie Anderson Stanley

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*Nei rā te kaupapa o te reo māori e / He taonga tuku iho nō nga tipuna
he parekarakawa, he mea hirahira e / Toku reo, toku ohoohe mapihi maurea whakakaimarihi*

— NEI RĀ TE KAUPAPA, BY TE OKEROA HURIWAKA AND TAWHIRO MAXWELL

Waipapa Taumata Rau (The University of Auckland) sits within the rohe of several Māori iwi/hapū, including Ngāti Whātua, Ngāti Whātua o Ōrākei, Te Kawerau a Maki, Ngāti Tamaoho, Te Ākitai Waiohū, Ngāti Maru, Te Patukirikiri, Ngāti Pāoa, Ngāti Tamatera, Ngāi Tai ki Tāmaki, Ngāti Te Ata, and Ngāti Whanaunga, and Waikato-Tainui. Ipipiri (the Bay of Islands) sits within the rohe of Ngāpuhi and Ngāti Rēhia. I acknowledge these iwi and hapū as the tāngata whenua of Tāmaki Makaurau, Ipipiri, and Aotearoa, I acknowledge that sovereignty was never ceded under Te Tiriti o Waitangi, and I pay my respects to kuia and kaumātua of these iwi and hapū, past, present, and future. Tūturu whakamaui, kia tina! Tihei mauri ora!

And I came from the Dreaming / From the dusty red-soil plain

I am the ancient heart / The keeper of the flame.

I stood upon that rocky shore / I watched those tall ships come

when for eighty thousand years I've been / The first Australian

— I AM AUSTRALIAN, AS PERFORMED BY THE BUSHWACKERS

Throughout my travels for this PhD, I was privileged to visit several lands I never dreamed I could as a child. In Australia, I worked on lands traditionally occupied by peoples of the Kulin Nations (predominantly the Wurundjeri people), and the Jagera, Yuggera and Ugarapul peoples. I acknowledge their sovereignty and traditional ownership and custodianship of the lands and waterways on which Swinburne University of Technology and the University of Southern Queensland (Springfield) are located. I acknowledge their Elders, past, present, future and emerging, and I acknowledge that sovereignty was never ceded. I thank the kind hospitality of the Polish, Belgian, Singaporean, German, Dutch, and Australian peoples, all of whose lands I visited throughout this thesis. I have learned so much, sitting on the shores of the Eternal Sea that is the

Milky Way.

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This work made use of v2.2 of the Binary Population and Spectral Synthesis (BPASS) models as described in [Eldridge et al. \(2017\)](#) and [Stanway & Eldridge \(2018\)](#).

*Raise a glass to all of us
Tomorrow there’ll be more of us
Telling the story of tonight.*

— FOUND/TONIGHT (BEN PLATT AND LIN-MANUEL MIRANDA)

I am the proud grandchild of immigrants, and by my count, the first person who identifies as Sāmoan to graduate with a PhD from the Department of Physics at the University of Auckland in (at least) a quarter century. I’m also one of the few gender non-conforming students in my Department. To all my fellow BIPOC students, all my queer whānau, and all my other underrepresented students in the sciences, I see you, I recognise you, I feel you. Kia kaha, kia māia, kia manawanui.

It would be impossible for me to list my entire whakapapa. It is known as far back as nine generations in parts, tracing back to around the mid-1700s – including to the first Lord Mayor of the City of Sheffield, a Prime Minister of Samoa, several Ministers in Samoa – including a former Minister of Finance – and Chair of Samoa’s National Bank, and some of the major clans in Samoa. We suspect there may be an American President in there too. At last count we know of about 500 relatives – my Great-Great-Aunt’s father-in-law had thirteen children in the mid-1800s. But I want to especially mihi to my mother Sue, my late aunt Leslie, my late grandfather Alan, my grandmother Nanette and her late mother Mary “Sugalu” Scott, my aunt Alison and uncle Pete, my great-uncle Ben and great-aunt Lulu, my cousins Jeremy, Jaime, Charlotte, Candice, Ngareta, Wairaka, Marihi, and Tama. To the rest of my ‘aiga: *e lautele le vateatea, ae le o le avanoa iinei: Ou te alofa ia te outou uma.*

Life before death. Strength before weakness. Journey before destination.

— FIRST IDEAL OF THE KNIGHTS RADIANT

For Jaffe, Bailey, Riegel, Ray, Mercer, Johnson, Willingham, and O’Brien: Vox Machina, the Mighty Nein, Bells Hells, Gilmore, Allura, Delilah, Pumat, Xandis, and Ryn. For Iyengar, Carazo, Ray, Riegel, Willingham, Wilson, and Mulligan: the Ring of Brass, Vespina, Maya, and Kir. For Lewis: Peter, Edmund, Susan, Lucy, Aslan and Jadis. For Sanderson: Kaladin, Syl, Shallan, Jasnah, Vin, Elend, Kelsier, Rysn, and Hoid. For Vincke, Wilde, English, Newbon, Downie, Solomon, Béart, Jones, Wiles, Mercer, Gregory, Simmons, Isaacs,

and Robertson: Lae'zel, Shadowheart, Astarion, Gale, Wyll, Karlach, Halsin, Jaheira, Minsc, Minthara, Ketheric, Gortash, and Orin the Red. For those who imagine the strange and beautiful and complex and weird and mystical worlds of fantasy, whether betwixt the pages of a book, on the silver screen, or upon the roll of a d20, I pay my enormous regards to you.

I argue that we choose to be good because of our bonds with other people and our innate desire to treat them with dignity. Simply put, we are not in this alone.

— CHIDI ANAGONYE

It occurs to me that there are precious few moments in one's life to acknowledge and thank the people who contributed to where one is now. I hold the belief that a person is defined by the sum of their actions and interactions, so I would like to acknowledge a few of the people that have made me the person I am today.

I'd like at this point to acknowledge friends: Connor, Frazer, Girish, Hunter, Pierce, Ryan, Scott, and Sean, with whom I shared some of my first physics and mathematics courses. Emma, the self-appointed creative director of this thesis. Josh, Tessa, Hamish, Jasmine, Ryan: my first experience with D&D was with Eldrin. Andrew, George, Hamish, Connor, Callum, and Mark: Katerina was my longest player experience; playing her taught me so much about myself (and Andrew). Josh, Liam, Arthur, Connor, Andrew, Oliver: Ayo and the rest of the Blue Originators (you never learned their party's name) owe you a debt of gratitude. Even good old Necromancer Bob says hi. Connor, Kali, Jackson, Andrew: I can't wait for you to see what other war crimes Matilda has in store.

I'm going to thank Jasmine again. It is a well kept secret in our friend group that Jasmine got top in course in Linguistics 203 (Applied English Grammar) during undergrad. She has helped me time and time again, and especially during the writeup of the paper version of Chapter 3. Ngā mihi e hoa.

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Once a king or queen in Narnia, always a king or queen.

— ASLAN

I've lost touch with some of the people I did high school with but the formative experiences there still sit inscribed in indelible ink in my mind. Caitlin, who has been one of my best friends for the better part of a quarter-century (and who would *not* let me live it down were I to forget her). Callum, Susannah, Kyle, Grayson, Tomek "if you're going to try to pronounce my last name, put some **EMPHASIS** on it" Pietkiewicz and the entire business and commerce crew. Sam, Louise, Crooks, the IT Ninjas. Isabella, Chyna, Evan,

Parth, Jay, Matt, all the people at NZSDC 2015, even if I had to argue in favour of alternative medicine and exclusive schools for LGBTQIA youth, those moments will be with me forever. Finn and Susannah: I know they said they never ranked us in maths classes, but they ranked us in maths classes, and I know we always vied for the top three places. Finn is an architect now and Susannah is admitted to the High Court – does that mean I finally win the ranking?

I'd also like to thank Di Eastwood, Maddy Haydock, Melissa Johansson, and Lauren Mulcahy (our debating coach, and my Year 13, 12, and 11 English teachers). Not all of you are teaching anymore, but I still remember vivid parts of your lessons. Miss Mulcahy commending my essay structure (for not opening an essay with "The question I am answering is..." but actually motivating why I enjoyed *To Kill a Mockingbird* and *Romeo+Juliet*.). Miss Johansson, for showing faith in the process of my creative writing portfolio ("Sean, I want to send you to counselling for this story. Change nothing, this is an E."). Mrs Haydock, firstly for Mufasa but also for believing in my weird and experimentalist poetry (I was going through a bit of a fourth-wall breaking phase with poetry), for letting me rant for 30 minutes on why *The Matrix* is actually a Marxist allegory and the (concerningly frequent) themes from Freudian psychoanalysis we can pull from it. Moreover, I remember in English class, you congratulating me for using "moreover" correctly in a sentence. Mrs Eastwood for always challenging me to do better in debating. I've said it before and I will say it again: I credit a mix of you four for my erudition and eloquence.

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Right, I'm almost done here, I swear.

Physics is a collaborative endeavor, and I would be remiss to not acknowledge my research compatriots. Thank you to the BPASS team, both in Auckland and abroad: John,² Heloise, Max, Sohan, Wouter, Petra, Erin, Adib, Conor, Gareth, Ashley, and Ankur. We've also been joined by a few students who have since left the team (because they graduated!): Gleb, Emma, Louis, Ethan, Siris, Celina, and Amanda. Thank you to our PIs, Dr Elizabeth Stanway and Prof. Jan Eldridge.

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¹This is not a joke or a quip, my Year 13 physics teacher played as a number eight in the 1999 Rugby World Cup for the USA. They lost 53-8 to Ireland, 27-25 to Romania, and 55-19 to Australia. They also lost 106-8 to England, outside of the RWC.

²John is responsible for me adding exactly one of the quotes in this thesis. I'll leave the reader to guess which one.

to fill out my first year, as there were only three required papers for my (at that time) Computer Science degree. I took PHYSICS 120 and 150 because I was good at it in High School (see Mr Lumkong, Mr Dyson, and Mr Whitaker above), and PHYSICS 107/G because it sounded fun. I left that course a physics double major, and no joke, restructured my entire degree around taking PHYSICS 356. Jan, Nick, and Richard are a large part of why I am a scientist today. I cannot thank them enough for that. From the bottom of my heart, thank you, ngā mihi, vinaka vakalevu, fa’afetai lava.

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*That you are here - that life exists, and identity;
that the powerful play goes on and you may contribute a verse.*

— O ME! O LIFE! BY WALT WHITMAN

When I started this PhD, I had two alive grandparents. I now have only one, my grandmother Nanette. For Alan, Walter, and Eileen, I carry your memories with me always. In the words of Liam O’Brien: *I was richer for having known you, friend.*

I could not be more keen, for some plain old boring precedented times.

— CR. RICHARD HILLS

I’ve noticed that the more theses I write, the more people I thank. My PGDipSci dissertation thanked five people. My Masters thesis thanked some fifty-five, a Government, and several coffee companies. Did I beat it this time?

Run, you clever [girl], and be a Doctor.
No, thank YOU. One thing about trains... it doesn't matter where they're going. What matters is deciding to get on.

— CLARA OSWIN OSWALD, DOCTOR WHO

— THE CONDUCTOR, THE POLAR EXPRESS

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Glossary



Notation	Description
<i>pp</i> -chain	proton-proton chain.
<i>pp</i> 1-chain	path 1 proton-proton chain.
<i>pp</i> 2-chain	path 2 proton-proton chain.
<i>pp</i> 3-chain	path 3 proton-proton chain.
AGB	Asymptotic Giant Branch.
BBH	Binary Black Hole.
BH	Black Hole.
BH-WD	Black Hole - White Dwarf Binary.
BNS	Binary Neutron Star.
BPASS	Binary Population and Spectral Synthesis.
CCSN	core-collapse supernovae.
CDF	Cumulative Distribution Function.
CEE	Common Envelope Evolution.
CO	Compact Object.
CO core	Carbon-Oxygen core.
CPTA	Chinese Pulsar Timing Array.
DTD	Delay-Time Distribution.
EOS	Equation of State.

Notation	Description
EPTA	European Pulsar Timing Array.
GR	General Relativity.
HRD	Hertzsprung-Russell Diagram.
IMF	Initial Mass Function.
ISCO	innermost stable circular orbit.
ISM	Interstellar Medium.
JWST	James Webb Space Telescope.
KS	Kolmogorov-Smirnov.
LIGO	Laser Interferometer Gravitational Wave Observatory.
LISA	Laser Interferometer Space Antenna.
LMC	Large Magellanic Cloud.
LVK Collaboration	LIGO-Virgo-KAGRA Collaboration.
MLT	Mixing Length Theory.
MS	Main Sequence.
MWEG	Milky-Way Equivalent Galaxy.
NANOGrav	North American Nanohertz Observatory for Gravitational Waves.
NS	Neutron Star.
NS-BH	Neutron Star - Black Hole Binary.
NS-WD	Neutron Star - White Dwarf Binary.
O3	Operating Run 3.
O4	Operating Run 4.
ONe core	Oxygen-Neon core.

Notation	Description
PDF	Probability Density Function.
PPTA	Parkes Pulsar Timing Array.
RGB	Red Giant Branch.
RLOF	Roche Lobe overflow.
SFH	Stellar Formation History.
SFRD	Star Formation Rate Density.
SMC	Small Magellanic Cloud.
SPH	Smoothed-particle hydrodynamics.
TŻO	Thorne-Żytkow Object.
TAMS	Terminal-Age Main Sequence.
USSN	ultra-stripped supernova.
WD	White Dwarf.
WR	Wolf Rayet.
ZAMS	Zero-Age Main Sequence.

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Parts of Chapter 3 are based on “The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material”, submitted to MNRAS and available on the NASA ADS here:

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Nature of contribution by PhD candidate	Performed data analysis, wrote majority of text	
Extent of contribution by PhD candidate (%)	70	

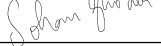
CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Wrote part of text, criticism, commentary, and review
Max Briel	Criticism, commentary, and review
Sohan Ghodla	Criticism, commentary, and review

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The undersigned hereby certify that:

- ☐ the above statement correctly reflects the nature and extent of the PhD candidate’s contribution to this work, and the nature of the contribution of each of the co-authors; and
- ☐ that the candidate wrote all or the majority of the text.

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Nature of contribution by PhD candidate	Wrote majority of the paper, developed most parts of the testing pipeline, later data generation, performed the statistical tests.	
Extent of contribution by PhD candidate (%)	90	

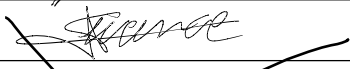
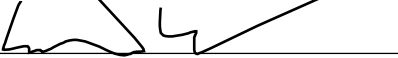
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Name	Nature of Contribution
Jan Eldridge	Original data generation, guidance on project, wrote clarification on the difference between BPASS primary and secondary models, criticism on manuscript
Max Briel	Developing ideas, code commentary, criticism on manuscript
Héloïse Stevance	Developing the test for Ultra-Stripped Supernovae, guidance on project, criticism on manuscript
Reinhold Willcox	Providing single-star pulsar dataset from Willcox+ 2021, criticism on manuscript

Certification by Co-Authors

The undersigned hereby certify that:

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- ☐ that the candidate wrote all or the majority of the text.

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Reinhold Willcox		21/11/2024



CHAPTER 1

Introduction



The only way to learn is by playing. The only way to win is by learning. And the only way to begin is by beginning.

— *Sam Reich* —





ARGUABLY THE GREATEST ADVANCEMENT in modern astronomy was the direct detection of gravitational waves – the rippling of space-time as a result of the motion of massive celestial bodies – on September 14, 2015. The Laser Interferometer Gravitational Wave Observatory (LIGO) collaboration detected the chirp of merging binary black holes, with masses $m_1 = 36^{+5}_{-4} M_\odot$ and $m_2 = 29^{+4}_{-4} M_\odot$ (Abbott et al., 2016). This detection sparked the so-called “gravitational wave era”, with detections in Operating Run 4 (O4) reading a cadence of hundreds per year.¹ These detections afford researchers probes of various physical phenomena. For instance, these merger events are some of the most energetic events in the Universe – GW150914 peaked at a gravitational-wave luminosity of $3.6 \times 10^{56} \text{ erg s}^{-1}$ (Abbott et al., 2016), compared with the typical energy release of a core-collapse supernovae (CCSN) of $\sim 10^{53} \text{ erg}$ (Hartmann, 1999) – and the merger of binary neutron stars are the predominant formation channels for most of the elements heavier than iron (see, among others, Eichler et al., 1989; Arcavi et al., 2017; Pian et al., 2017; Smartt et al., 2017).

In addition, the stochastic background of gravitational waves is a direct probe on early-Universe cosmology (Christensen, 2018), though the frequency band that these gravitational waves span is between 10^{-9} and 10^{14} Hz (Christensen, 1992, 2018), making detection of the stochastic background difficult. Indeed, it took eight years after the detection of gravitational waves for the stochastic background to be detected by the collaborative effort of North American Nanohertz Observatory for Gravitational Waves (NANOGrav) (Agazie et al., 2023), the European Pulsar Timing Array (EPTA) (Antoniadis et al., 2023), the Parkes Pulsar Timing Array (PPTA) (Reardon et al., 2023), and the Chinese Pulsar Timing Array (CPTA) (Xu et al., 2023).

Beyond the stochastic background, the vast majority of gravitational wave sources involve stars at some point in their lives – whether that be the merger of Compact Objects, the detonation of CCSN, or the rotation of an isolated massive source such as a Neutron Star (NS) or Black Hole (BH). In these cases stars are either the immediate sources of gravitational waves, or the progenitors of the objects that are the sources. Only in extreme cases, such as the merger of galaxies or the motion of supermassive black holes, are stars not directly involved in the generation of gravitational waves (and even then, this is debatable).

Ergo, any attempt at understanding gravitational waves must be founded in a strong theoretical understanding of stellar structure and evolution. Without theory, any observation is no more than a curiosity; it is with theoretical underpinning and models that insight can be gained. As most astrophysical sources of gravitational waves (i.e., not the stochastic background) are formed by the merger of two Compact Objects (COs) – black holes, white dwarfs, and neutron stars – one must understand their formation and dynamics, and thus the evolution of progenitor stars into these remnants, as well.

Furthermore, a sound understanding of the evolution of stars has impacts beyond gravitational wave astronomy. As the interiors of stars are complex nuclear furnaces, they are responsible for producing most of the elements in the Universe heavier than Hydrogen (Burbidge et al., 1957). In fact, of the elements lighter than Uranium, the only elements not substantially produced in stellar processes (such as through kilonovae or in supernovae) are technetium, promethium (which both lack stable isotopes),² boron, beryllium (which

¹From the start of O4 in May 2023 to the end of July 2025, approximately 219 unretracted significant (false alarm rate of $< 1 \text{ month}^{-1}$) detections, averaging to a cadence of $\sim 97 \text{ yr}^{-1}$.

²There are technetium stars, and promethium has been observed in some atmospheres (Cowley et al., 2004), however these processes do not generally pollute their environment.

both are produced via cosmic ray fission), and some formation channels of lithium.

Starlight is also a critical requirement for probing the observable Universe. Photons are the means by which many astrophysical observations are made. Thus, a complete picture of the births, lives, and deaths of stars – and the compact remnants that they may leave after the supernova – is essential to understand spectra, nucleosynthetic yields, chemical abundances, and supernovae. These models also allow us to probe remnant mass distributions, characteristics of supernovae progenitors, and the binary interactions that define the majority of the stars in the Universe. This picture is incomplete without stellar structure and evolution models.

From the perspective of Universal structure and habitability, stars are also the focal point of planetary systems, and provide the theoretical right conditions for the formation of life. The protoplanetary discs that form around stars depend on the host star’s mass and luminosity. Without a complete picture of stellar evolution, one is unable to describe the likelihood of extraterrestrial life within the Universe.

1.1 | Observational astronomy

If the stars should appear one night in a thousand years, how would men believe and adore; and preserve for many generations the remembrance of the city of God which had been shown!

— “NATURE”, BY RALPH WALDO EMERSON

There are approximately 9000 stars visible in the night sky to the naked eye (with magnitudes brighter than 6.5), across both hemispheres (Hoffleit, 1964), which are displayed in Figure 1.1. The dark-adapted eye can see approximately as far as η Carinae,³ which is only approximately 2300 pc away.

2300 pc is an order of magnitude less than the size of Galaxy, which is approximately 26.8 kpc in diameter (Goodwin et al., 1998), which in turn pales in comparison to the Local Group at ~ 3 Mpc (Redd, 2018), and the observable Universe at 28.5 Gpc (Bars & Terning, 2010). There are, in addition, $\sim 1.7 \times 10^{22}$ stars in the observable Universe, assuming 1.7×10^{11} galaxies, with approximately 10^{11} stars in each (Marov, 2015). Thus, the birth, evolution, and death of stars is of fundamental importance to understanding the local structure and evolution of astrophysical objects on any scale.

Given the number of stars visible in the vast expanse of night, and their relatively stable positions in the sky, it is no surprise that they are a fixture in many religions and cultures around the world. Ancient Greece saw a difference in movement of some celestial objects, and called these $\pi\lambda\alpha\nu\eta\tau\eta\varsigma$, which morphed into *planētes* and eventually the English word “planets”. Babylonia knew of Halley’s comet and tracked observations of it as late as 87 BCE (Stephenson et al., 1985). Records of eclipses in Chinese culture exist as far back as 1600

³This is a debated topic; we use the definition that a “star” includes binaries and that the star is in the catalogue of Hoffleit (1964), which rules out V762 Cas, the other commonly held “farthest star”.

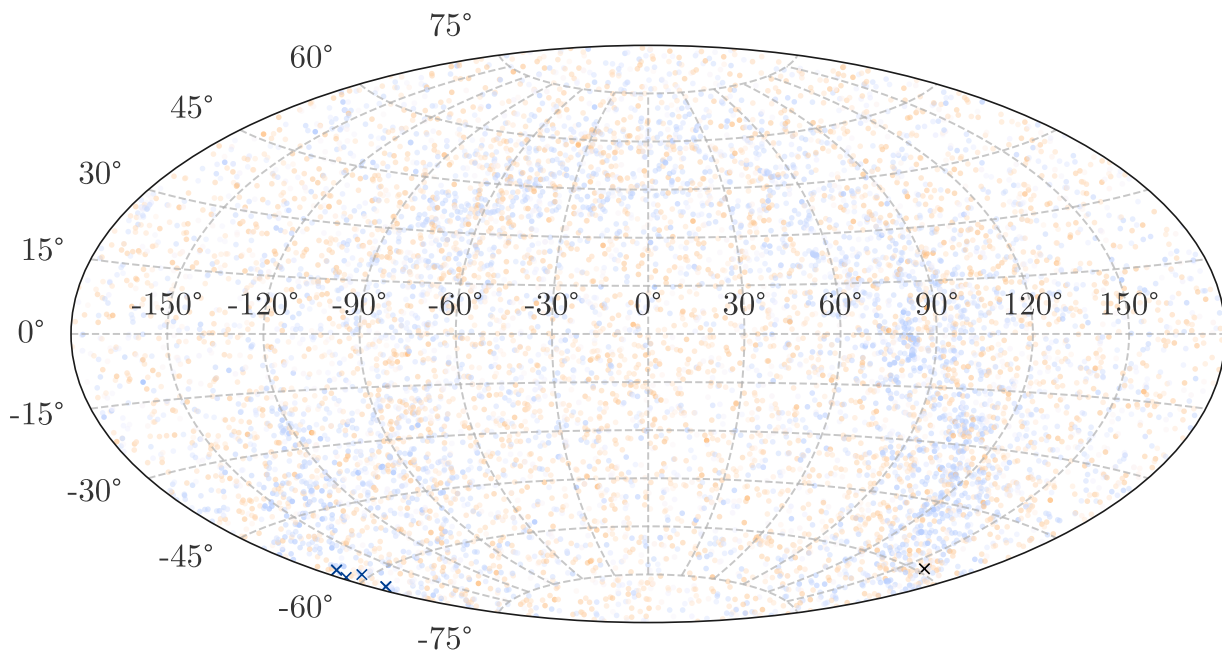


Figure 1.1: The locations of the ~ 9000 stars in the night sky visible to the naked eye, as reported in Hoffleit (1964). The colors are computed by converting the B-V color indices to temperatures using the fits in Ballesteros (2012), and then converted to RGB via the algorithm in this StackOverflow post: <https://stackoverflow.com/a/45497817>. The Southern Cross (Crucis) is shown fully opaque in blue towards the bottom left of the plot as a point of reference, and η Carinae is shown fully opaque in black in the bottom right.

BCE (Fang, 2015). The Nazca lines, geoglyphs carved into the Nazca desert in Peru at some point between 500 BCE and 500 CE, may be early astronomical calendars (Kosok & Reiche, 1949), though this theory was later placed into doubt (Hawkins, 1973). Knowledge of the movements of stars guided Polynesian wayfarers as they crossed the South Pacific (Buck, 1938; Hamacher, 2022), were embedded in the cultural myths of First Nations peoples in Australia (Noon & De Napoli, 2022), allowed them to forecast weather patterns (Hamacher et al., 2019), and were plausibly recorded into oral histories (Fuller & Hamacher, 2017). Knowledge of the lunar perigee was even written into *Othello* by Shakespeare:

It is the very error of the moon;
She comes more near the earth than she was wont,
And makes men mad.

– Shakespeare (ca. 1603)

Knowledge of oceanic tides existed in the time of Ancient Greece, and was described by Ptolemy, who wrote in *Tetrabiblos* that

ἀμπώτεις καὶ αἱ παλίρροιαι, καὶ αἱ τῶν πνευμάτων δὲ τροπαὶ μάλιστα περὶ τὰς τοιαύτας τῶν φωτῶν κεντρώσεις ἀποτελοῦνται πρὸς οὓς ἂν τῶν ἀνέμων ἐπὶ τὰ αὐτὰ τὸ πλάτος τῆς σελήνης τὰς προσνεύσεις ποιούμενον καταλαμβάνηται.	For the hour by hour intensifications and relaxations of the weather vary in response to such positions of the stars as these, in the same way that the ebb and flow of the tide respond to the phases of the moon, and the changes in the air-currents are brought about especially at such appearances of the luminaries at the angles, in the direction of those winds towards which the latitude of the moon is found to be inclining.
Ptolemy & Proclus (ca. 168)	Robbins (1940)

This knowledge flowed through to Dante, who wrote in the third part of the Divine Comedy, *Paradiso*:

E come 'l volger del ciel de la luna cuopre e discuopre i liti sanza posa, così fa di Fiorenza la Fortuna:	And, as the turning of the lunar sphere covers and endlessly uncovers the edges of the shore, thus does fortune deal with Florence.
Alighieri (ca. 1317)	Corbett & Webb (2016)

Despite this deep cultural connection to the stars, what we would now call Astronomy is a somewhat recent invention. Arguably conventional (and in some sense, “correct”) astronomy arose in the 16th century CE with Copernicus proposing the heliocentric model of the solar system, which was confirmed by Galileo in the 17th century CE ([Carroll & Ostlie, 2017](#)) – although Galileo’s refusal to recant his claims did result in his being convicted of heresy for 300 years. The 17th century brought about the establishment of orbital mechanics with the work of Kepler (based upon the observations of Tycho Brahe), and the discovery of gravity by Newton.

The early 20th century saw a veritable renaissance in both the observational and theoretical spheres. On the one hand, developments such as the theories of Special Relativity in 1905 ([Einstein, 1905](#)), General Relativity (GR) in 1917 ([Einstein, 1917](#)), and the predictions of gravitational waves ([Poincaré, 1906](#); [Einstein, 1916](#)) led to theoretical revolutions. Around this time, Arthur Eddington began working on determining the internal structure and evolution of stars ([Eddington, 1916, 1920](#)), and [Gamow \(1928\)](#) put forth the idea of quantum tunneling allowing elements to overcomb the Coloumb barrier. On the other hand, the innovations of [Payne \(1925\)](#) in showing that stars are composed of mostly Hydrogen and Helium, [Baade & Zwicky \(1934\)](#) in distinguishing “super-novae” from “common novae”, [Chandrasekhar \(1935\)](#) in establishing the idea of the “Chandrasekhar mass” (the limiting mass of a white dwarf), [Hoyle \(1946\)](#) in developing the foundation of the

nuclear reaction of elements heavier than Hydrogen in stars, [Salpeter \(1955\)](#) in developing the Initial Mass Function (IMF), [Burbidge et al. \(1957\)](#) in truly developing the theory of stellar nucleosynthesis, [Heney et al. \(1955\)](#) and [Hayashi \(1961\)](#) in exploring the pre-main sequence evolution of stars were key to kickstarting entire branches of astronomy such as spectroscopy and computational astrophysics.

The pace of discovery of new and exotic astronomical phenomena has not slowed into the 21st century. Thorne-Żytkow Objects (TŻOs) – giant stars with NS cores ([Thorne & Żytkow, 1977](#)) – once only existed as theoretical curiosities, yet now have credible observational candidates in [Levesque et al. \(2014\)](#). The aforementioned gravitational waves were detected in 2015 ([LIGO Scientific Collaboration and Virgo Collaboration, 2016](#)), a century after they were demonstrated by [Einstein \(1916\)](#) to be a central prediction of GR. More recently, [Lau et al. \(2022\)](#) directly imaged 17 carbon-rich dust rings around WR140 with James Webb Space Telescope (JWST), enriching the Interstellar Medium (ISM) with carbon-rich compounds. In the coming years, the Vera C. Rubin Observatory ([Ivezić et al., 2019](#)) and Laser Interferometer Space Antenna (LISA) are expected to vastly increase the number of short-period eclipsing double white dwarfs ([Burdge et al., 2020](#)), with the latter able to detect gravitational waves with frequencies between 0.1 mHz and 1 Hz ([Amaro-Seoane et al., 2012](#)), compared to the $\sim 10 \text{ Hz} - 1000 \text{ Hz}$ that LIGO-Virgo-KAGRA Collaboration (LVK Collaboration) can detect ([Schmitz, 2021](#)). The stochastic gravitational-wave background was detected by [Agazie et al. \(2023\)](#); [Antoniadis et al. \(2023\)](#); [Reardon et al. \(2023\)](#); [Xu et al. \(2023\)](#), via a search of pulsar-timing arrays.

Astronomy as a field has blossomed since the naked-eye era – entering the era of the telescope, the era of the observatory, and now the gravitational wave era. The next era of astronomy is the big data era. With observations reaching sub-day cadences, a large array of theoretical models, and perhaps the development of “on-demand” detailed models is required.

1.2 | Stellar Modelling

Modelling the evolution of a star is a nontrivial endeavour. Stars reach internal temperatures, pressures, and densities unsustainable on Earth. For instance, the core of a star is capable of nuclear fusion, which has not been sustainably recreated on Earth. Moreover, the density of Sirius B – a White Dwarf (WD) in the constellation of Canis Major – is $2.14 \times 10^{10} \text{ kg m}^{-3}$ ([Barstow et al., 2005](#)) and the mean density of the Sun is $\sim 1410 \text{ kg m}^{-3}$ ([Eddy & Ise, 1979](#)). On Earth, the highest density measured in a laboratory is $\sim 27\,000 \text{ kg m}^{-3}$, requiring pressures on the order of 2.3 TPa ([Fratanduono et al., 2020](#)), and observed for only nanoseconds.

It is nigh impossible to achieve and sustain these conditions to experimentally test any physical phenomena. Nevertheless, missions such as the Parker Solar Probe have made flyby missions into the atmosphere of the Sun, though broadening the sample size to two stars – using Proxima Centauri, the nearest star to Earth aside from the Sun – would require travelling at approximately $0.1c$ for 40 years to reach. If instead we were to send the Parker Solar Probe travelling at a constant velocity of its top velocity of $\sim 176\,471 \text{ m s}^{-1}$, it would take approximately 7200 years to reach Proxima Centauri. Needless to say, physical experimentation with a

star is difficult.

1.2.1 | Stellar Modelling and Population Synthesis

Because of these factors, one often turns to computer simulations to model the interiors of stars. The question turns from physical experimentation to mathematical ones, and the idea of a *stellar model* becomes useful. A stellar model is a calculation performed such that the internal structure (the present state of physical quantities) and evolution (how these quantities change over time) are determined. Computing these parameters is nontrivial. As will be developed in Chapter 2, the simplest model of a star is a series of four coupled differential equations describing the internal pressure, temperature, luminosity, and mass distributions. However, including “extra” physics greatly increases the complexity of the problem; [Henyey et al. \(1964\)](#) developed methods of solving the equations of structure and evolution that turn the problem into matrix inversion, at the cost of the central “Henyey matrix” being thousands of elements in size.

The diversity and complexity of stellar profiles, interiors, and characteristics discussed earlier demands a careful and varied approach. One cannot abstract away much of the important physics. Although it is common to use approximations such as spherical symmetry to reduce the dimensionality of the problem, and to only evaluate the equations of stellar structure at a fixed number of locations in this one remaining dimension, chosen sufficiently intelligently, a complex computational effort must be undertaken.

This problem is then magnified when attempting to extrapolate out to describe an entire population of stars, such as from a burst of stellar formation activity, or in a shared environment such as a globular cluster. It is inefficient to, for example, model all approximately 10^{11} stars in the Milky Way ([Marov, 2015](#)) as independent entities. One must develop a way to take a library of models and determine the large-scale behaviour this way.

Broadly speaking, there are three approaches generating a population of stars, all of which have both merits and drawbacks. These three methods are:

1. Rapid Population Synthesis,
2. Detailed Population Synthesis, and
3. 3D Simulations.

Each of these approaches informs the other two, and all play important roles in stellar modelling. Although we focus on detailed methods, we will briefly outline all three approaches here.

Population Synthesis with Detailed Binary Models

A detailed population synthesis code uses detailed model sets, where the interior state of the stars is computed, and coupled to the orbital evolution equations. In this regime, observables such as mass transfer are explicitly calculated and solved for from the exact equations of state, structure, and evolution of the binary. By doing this, one follows the binary evolution in a reasonable amount of detail, obfuscating fine detail and (generally) asymmetry in the star in favour of recovering the larger-scale structure.

An example of such a code is the BPASS code (Eldridge et al., 2017), which uses a custom version of the Cambridge STARS code for the stellar modelling and a bespoke proprietary code named TUI for the population synthesis. The key approach in the present version of BPASS is to model the primary star in detail until it undergoes a supernova, whilst approximating the evolution of the secondary. After the primary supernova, the secondary is then evolved in detail with the primary treated as a point mass. After the secondary supernova, the parameters of the remnants – if any – can thus be fed into a postprocessing code to have their orbital equations solved for further.

Rapid Binary Population Synthesis

Rapid binary population synthesis models are instead either compilations of single-star evolutionary tracks that have been solved for using a code such as MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023), fit to using approximations such as those of Hurley et al. (2000) or Hurley et al. (2002), or interpolated between. These models are then coupled to the orbital evolution *a posteriori* using a binary evolution framework, effectively approximating the computationally expensive part of binary evolution models to varying degrees of detail. This allows them to generate binary models on-the-fly, and optimise for this interaction, at the potential cost of missing more precise facets of binary evolution.

3D Simulations

3D simulations are the most detailed method of stellar modelling. The base principle is that the star is evaluated as a network of particles in a continuous media, and the evolution of the star is followed in theoretically atomic detail. For example, PHANTOM (Price et al., 2018) is a Smoothed-particle hydrodynamics (SPH) code which essentially models the star as a fluid, where the Lagrangian hydrodynamical equations are solved for in two-dimensional cells (termed “particles”). Although these present the most detailed models, they are also the most time-intensive to solve. For a SPH code, the famous Navier-Stokes equation is invoked, which has no analytical solution but is computed in excruciating detail. As a result, most SPH models take weeks to run and simulate a star for kiloseconds of its lifetime, rather than for the millions (if not billions) of years detailed models are able to compute. Full population synthesis using 3D simulations is thus unlikely to be tenable without significant advances in computational techniques.

Though it may seem that they serve the same niche and are therefore competing to be the “better” method of stellar modelling, detailed 1D models and 3D models exist harmoniously with each other. The advancements in 3D modelling inform the evolution of 1D models in a process known as “3-to-1”, and the computational flexibility of 1D models offers insights into where 3D models wish to target to evolve, in a lock-step advancement method.

1.3 | This work

We now describe the placement of this work within the field, and outline the key aims this thesis sets out to answer. This thesis can be summarised as an endeavour to model the interior physics of interacting binary stars over certain near-extreme phenomena. We also aim to examine the post-supernova kinematics of binary systems. Thus, the area of interest of this thesis is effectively the entire lifetime of a binary system post-birth, with the critical exceptions of the moment of supernovae, and the interior physics of compact objects.

1.3.1 | Research themes

There are essentially two major themes guiding this research.

- I. Can we study contact binary systems, and refine our treatment of Roche Lobe overflow (RLOF) to overcome numerical issues inherent in these models?
- II. Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?

1.3.2 | Thesis outline

Six impossible things. Count them, Alice.

— ALICE IN WONDERLAND

This thesis is, broadly, a tour of the structure and evolution of binary stars from Zero-Age Main Sequence (ZAMS) to their eventual merger, and has been structured to approximately follow this life cycle. The thesis is structured as follows:

1. In Chapter 2, we provide the necessary theoretical background for future chapters, critically examine the pieces of software and numerical methods employed in this work, and discuss the logistics of running this single-thread grid-based code as a multiprocessed code on a distributed cluster.
2. In Chapter 3, we explore the regime in which mass transfer is assumed to become helium-rich, and the effects on the future evolution of these binaries. (Theme I)
3. In Chapter 4, we examine the kinematics of the post-supernova inspiral and the mechanics of the natal supernova kick. (Theme II)
4. In Chapter 5, we present the results of our stellar evolution codes in the interesting case when both stars almost (or in some instances, do) touch. (Theme I)
5. In Chapter 6, we review the thesis and conclude. (Themes I, II)
6. In Appendix A, we present additional derivations for key theoretical expressions from Chapter 2.

7. In Appendix B, we list the codes employed and their versions, as well as the default parameter choices we make for our stellar evolution models (Theme I).
8. In Appendix C, we present period-eccentricity distributions for the LVK Collaboration O3 contour (Theme II).
9. In Appendix D, we present the bespoke contact binary models that we discuss in Chapter 5 (Theme I).
10. In Appendix E, we enumerate the predicted remnants produced by the overcontact binaries in Chapter 5.
11. In Appendix F, we provide the KAITIAKI code necessary to generate the models for certain stellar evolution plots throughout the thesis. (Theme I)

We now move into the more detailed background, and outlay the theory that underpins all of our models.



CHAPTER 2

Theoretical Background



There is a theory which states that if ever anyone discovers exactly what the Universe is for and why it is here, it will instantly disappear and be replaced by something even more bizarre and inexplicable.

There is another theory which states that this has already happened.

— *Douglas Adams* —



IN THIS CHAPTER, WE PROVIDE A brief account of the evolution of a star from a theoretical perspective, from its formation to post-supernova. We will discuss both the case of a single isolated star, and an isolated binary star system. It is pertinent to note that our binaries will be assumed to have formed together at birth – thus having the same age – rather than by, for example, the dynamical capture of a runaway star in a globular cluster, which introduces extra effects and considerations.

In this chapter, unless context otherwise requires, we use the following conventions surrounding the standard model and particle physics. γ refers to a photon, e^+ refers to a positron, e^- refers to an electron, ν refers to a neutrino, ν_e refers to an electron neutrino specifically, and A_ZX refers to atom X with atomic number Z and mass number A .

2.1 | Stellar Formation

*Nature and nature's laws lay hid in night:
God said, "Let Newton be!" and all was light.*

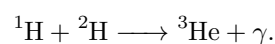
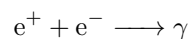
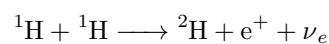
— ALEXANDER POPE

From the point of view of stellar evolution, which starts in earnest at the ZAMS, the formation of stars is interesting but ultimately has little to no impact on the nuclear evolution of the star besides setting the initial mass. We provide a short recap of stellar formation here as scene-setting, but our stellar evolution models assume that a gas cloud has already collapsed and we have a star at its ZAMS.

Stars are born in clouds of gas and dust. Due to inhomogeneities in the dust formation, some regions become more dense than others. When a region of this gas with adiabatic constant γ and mean molecular weight μ reaches the so-called Jeans Mass,

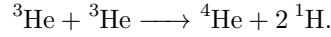
$$M_J = \left(\frac{10k_B T}{3(\gamma - 1)\mu m_H G} \right)^{\frac{3}{2}} \left(\frac{3}{4\pi\rho} \right)^{\frac{1}{2}} \sim \left(\frac{10k_B T}{G} R \right)^{\frac{3}{2}} \left(\frac{1}{M} \right)^{\frac{1}{2}}, \quad (2.1)$$

where the similarity is taking $\gamma = \frac{5}{3}, \mu = \frac{1}{2}$ for an adiabatic gas composed of H(II), then gravity becomes the dominant force in the clumps of gas. This causes a protostar to form. Generally speaking, this protostar slowly gains more mass from its surroundings (McKee & Ostriker, 2007) and, as it does so, increases in temperature. When the core temperature reaches 10^7 K, the proton-proton chain (pp -chain) begins. Bare protons are combined to form deuterium, positrons, and neutrinos. The positrons annihilate with electrons, producing photons, and the deuterium fuses with another proton to produce helium-3:

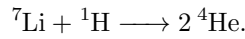
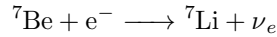
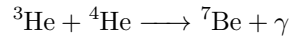


The three constituent parts of the *pp*-chain are the path 1 proton-proton chain (*pp1*-chain), the path 2 proton-proton chain (*pp2*-chain), and the path 3 proton-proton chain (*pp3*-chain). Which path the star takes is dependent on the temperature:

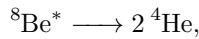
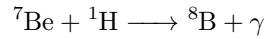
- The *pp1*-chain dominates at low temperatures, and is given by the reaction



- The *pp2*-chain dominates at temperatures $> 1.4 \times 10^7$ K, and is given by the reaction chain



- The *pp3*-chain dominates at temperatures $> 2.3 \times 10^7$ K, and is given by the reaction chain



where a superscript asterisk denotes an unstable excited state.

All three paths in the *pp*-chain ultimately produce ${}^4\text{He}$, with the key output difference being the total energy per ${}^4\text{He}$ atom; the energy yield is highest for the *pp1*-chain, and smallest for the *pp3*-chain.

The burning of ${}^2\text{H}$ – deuterium – occurs for a short fraction of the lifetime of the star, before core hydrogen burning begins in earnest. This energy release will sustain the star throughout its lifetime, providing outwards pressure resisting gravity pulling the star inwards, in a balance termed *hydrostatic equilibrium*. Once hydrogen fusion has begun, we say that the star has reached ZAMS, and is now on the Main Sequence (MS), where it will spend approximately 90% of its lifetime.

2.2 | Stellar Evolution

*It did not last: the Devil howling “Ho!
Let Einstein be!” restored the status quo.*

— “ANSWER TO POPE’S

EPITAPH FOR SIR ISAAC NEWTON”, BY J. C. SQUIRE

Now that our protostar has evolved onto the MS, it has become a star in its own right. Over the next few million years, it will continue to burn hydrogen into helium, forming a helium core. It is at this point that the evolution encounters a number of bifurcation points, depending on the mass and chemical composition of the star. In general, while the star lives, the four equations of stellar structure hold,

$$\left. \begin{aligned} \frac{dm}{dr} &= 4\pi r^2 \rho && \text{(mass continuity),} \\ \frac{dp}{dr} &= -\frac{GM\rho}{r^2} && \text{(hydrostatic equilibrium),} \\ \frac{dT}{dr} &= -\frac{3\kappa\rho L}{64\pi r^2 \sigma T^3} && \text{(energy transport), and} \\ \frac{dL}{dr} &= 4\pi r^2 \rho \epsilon && \text{(thermal equilibrium),} \end{aligned} \right\} \quad (2.2)$$

where ϵ is the rate of energy eneration per unit mass, κ is the opacity of the material, σ is the Stefann-Boltzmann constant, m is the mass internal to radius r , p is pressure, ρ is density, M is total mass of the star, and L is the luminosity.

In addition to Eqn(2.2), an Equation of State (EOS) demanding $\rho = \rho(P, T)$ is desired. We take our EOS from [Pols et al. \(1995\)](#), which built upon that in [Eggleton et al. \(1973\)](#). Critically, Pols’ EOS includes the dissociation of molecular hydrogen, Coulomb interactions, and pressure ionization, which [Eggleton et al. \(1973\)](#) lacked. The EOS is an approximation to Fermi-Dirac statistics, and originates from the free energy

$$F(V, T, N_e, N_e) = -\frac{1}{3}aT^4V + kT \sum_i N_i \left(\ln \frac{N_i h^3}{\omega_i V (2\pi m_i kT)^{\frac{3}{2}}} - 1 + \frac{\chi_i}{kT} \right) + F_e(N_e, V, T) + \Delta F_e(N_e, V, T). \quad (2.3)$$

These equations represent conservation of mass, hydrostatic equilibrium, energy transport, and thermal equilibrium, respectively. Knowing constants ϵ, κ , we have four equations with four unknowns. It is usually convenient to use the *Rosseland mean* opacity ([Rosseland, 1924](#)),

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu, \quad (2.4)$$

where a is the radiation density constant ($a = 7.57 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$) and B is the black-body intensity at frequency ν and temperature T . The function κ_ν is obtainable from atom-electron-photon interaction

(Eldridge & Tout, 2004a): bound-bound, bound-free, free-free (bremsstrahlung) transitions, and electron scattering. κ_ν is thus related to the scattering cross-section $\sigma_\nu = \int_0^{2\pi} \int_0^\pi \frac{d\sigma}{d\Omega} \sin\theta d\theta d\phi$. The Rosseland mean becomes

$$\kappa = \frac{T^5 a c^3 k_B}{\pi h^2 \int_0^\infty \frac{\nu^4}{\left(\cosh\left(\frac{h\nu}{T k_B}\right) - 1\right) \kappa_\nu} d\nu}, \quad (2.5)$$

though the integral cannot be analytically solved further. On the other hand, ϵ is obtainable from knowing the nuclear reactions occurring in the star's core, namely

$$\epsilon = \sum_i^N \text{energy output of reaction } i. \quad (2.6)$$

The reaction rates and energy outputs are typically inferred from empirical data; we take the interpolation in temperature from Caughlan & Fowler (1988), and correct the rates to approximate the cross-sections of Bahcall & Pinsonneault (1992); Bahcall et al. (1995).

All told, in the simplest sense, the internal *structure* of the star can be modelled by hand by solving Eqn(2.2) from $m = 0$ to $m = M$.

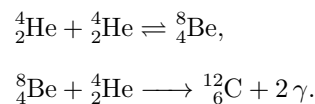
2.2.1 | The chemical composition of stars

There's a potion that can make you shrink.

— FIRST OF THE SIX IMPOSSIBLE THINGS

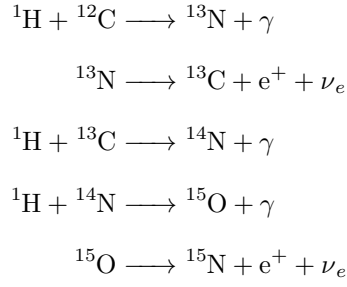
A star contains at its core a complex, multi-layered, nuclear furnace. As one atom approaches another, it must overcome the natural electrostatic repulsion so that the strong nuclear force ensures fusion occurs. This requires the bombarding atom to tunnel through the Coulomb barrier Gamow (1928), instigating fusion.

We have already discussed the *pp*-chain, fusing hydrogen into helium and up to beryllium. The two remaining parts of the story are the triple- α process and the CNO cycle. The triple- α process generates carbon, and is given by the reactions (Carroll & Ostlie, 2017)¹

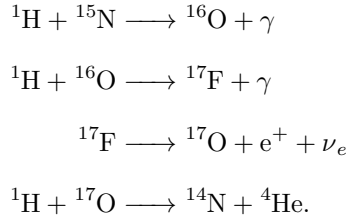


The CNO cycle generates elements up to fluorine via a branched cyclic reaction. The reactions are presented here, and are summarised in a flow chart in Figure 2.1. The start of the cycle, pre-branch, is given by

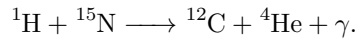
¹The reversible reaction is due to Beryllium-8 being unstable; its half-life is on the order of 1×10^{-16} s, and it decays back to $2 {}^4_2\text{He}$ if it does not react further.



Here, the reaction branches when ${}^{15}\text{N}$ undergoes a proton capture. In 1 in every 2500 captures, the reaction branches to produce (at its heaviest) fluorine, in the *NO cycle*:



In the remaining cases, the reaction instead produces carbon-12, in the conclusion of the *CN cycle*:



Further reactions essentially proceed similarly, and the core can produce up to iron before the reaction becomes endothermic. However, these elements do not dominate the star, with the exception of forming a Carbon-Oxygen core (CO core) and Oxygen-Neon core (ONe core) later in the evolution of the star. We thus group the elements into hydrogen, helium, and metals (which we define as anything with atomic number ≥ 3 , i.e., lithium and heavier elements). It is convenient to speak of the mass fraction of the element, i.e., the fraction by mass of a star that is a certain element. For our purposes, this is expressible as

$$X + Y + \sum_{i=3}^{118} \sum_j {}^i_j\text{X} = 1, \quad (2.7)$$

where X is the mass fraction of hydrogen, Y of helium, and ${}^i_j\text{X}$ as the metal² with atomic number i and mass number j . When speaking about the chemical composition of a star it is also convenient to speak of the *metallicity*, the fraction at ZAMS of the star that is metal. For this, we define the metallicity as

²It is custom in astronomy to refer to anything with an atomic number greater than 2 as a metal, rather than the conventional definition of a material conductive at absolute zero, including most of the d-block, s-block, f-block, and about half of the p-block on the periodic table.

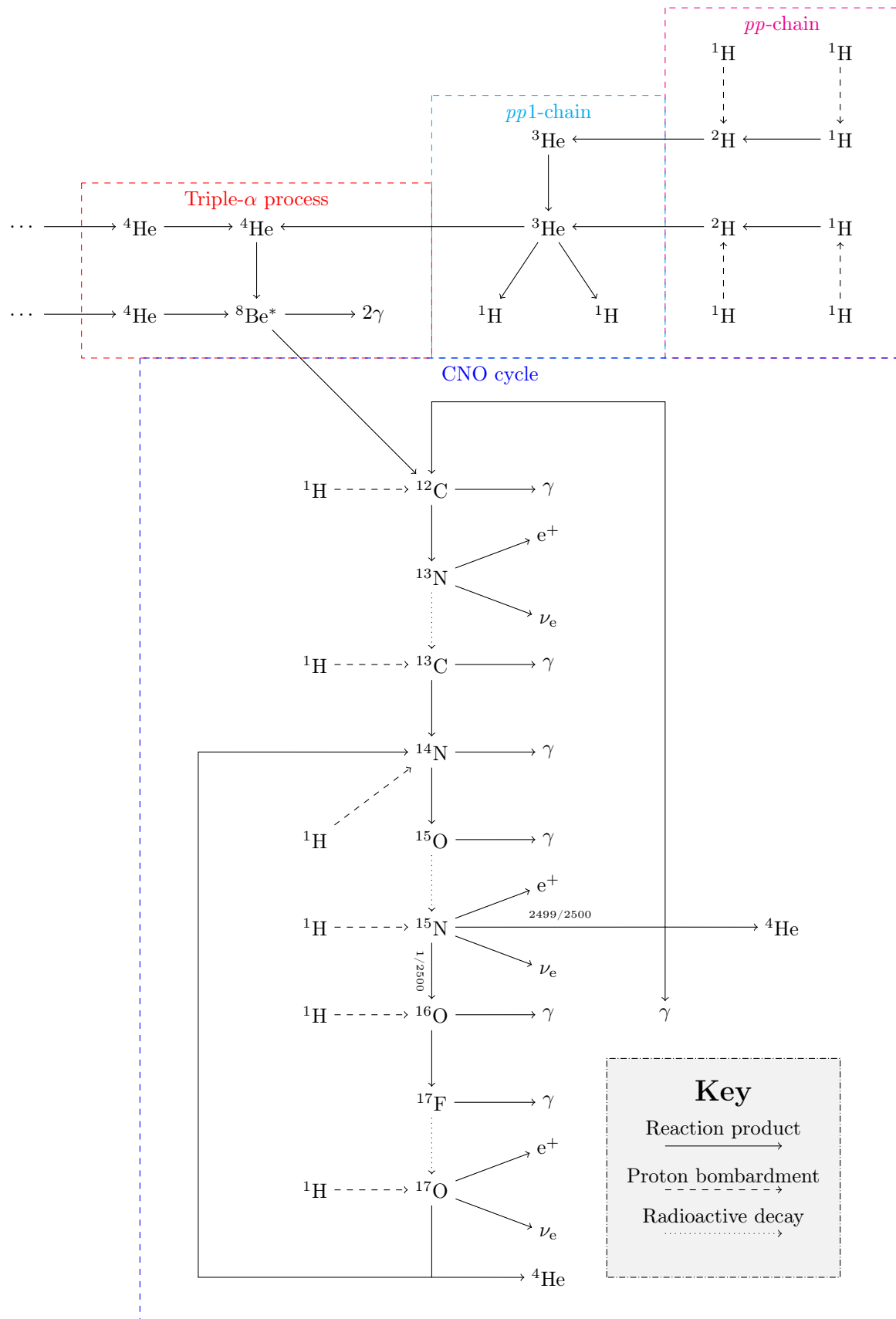


Figure 2.1: A schematic of the reactions leading up to the CNO cycle. The *pp1*-chain chain is replaced by the *pp2*-chain/*pp3*-chain chain, depending on the temperature of the star. The *pp*-chains generating the 4He atoms on the leftmost side of the chain are omitted for simplicity. Once the 15N atom has been produced, the reaction cycle is essentially self-sustaining, provided there are protons available to bombard the reactants.

$$Z \equiv \sum_{i=3}^{118} \sum_j^i X, \quad (2.8)$$

such that Eqn(2.7) becomes

$$X + Y + Z = 1. \quad (2.9)$$

Of note, abundances are not always in the literature referred to in this fashion. In some works, such as [Asplund et al. \(2021\)](#), abundances are reported in the form $\mathcal{X}_i = 12 + \log_{10}(X_i/H)$. To convert between the two, the relationship

$$\mathcal{X}_i = 12 + \log_{10} \left(\frac{Z \times \frac{X_{i,\text{init}}}{56}}{0.75 - 2.5Z} \right) \quad (2.10)$$

is employed, where $X_{i,\text{init}}$ is the initial mass fraction of a given element (which we take from [Grevesse & Noels 1993](#), summarised in Table 2.1).

Element	Initial abundance	Element	Initial abundance
Hydrogen	0.7	Neon	0.099
Carbon	0.173	Magnesium	0.038
Nitrogen	0.053	Silicon	0.080
Oxygen	0.482	Iron	0.072

Table 2.1: The unweighted initial abundances by mass of our models, unless otherwise stated. Metals are then weighted by metallicity, such that, for example, the carbon abundance is $0.173 \times 0.020 = 0.00346$. The balance is assumed to be in Helium.

2.2.2 | Evolutionary Pathways

In this section, we examine the three major regimes of stellar evolution for single-stars: low-mass, intermediate-mass, and massive stars. Stars above $M_{\text{ZAMS}} \approx 80 M_{\odot}$ are out of the scope of this thesis – although they exist, and the BPASS IMF reaches $300 M_{\odot}$ ([Eldridge et al., 2017](#)), the systems of interest in future chapters are at most $\sim 80 M_{\odot}$, and thus very massive stars are out of scope for this work.

Low-Mass Stars: $0.08 M_{\odot} < M < 2 M_{\odot}$

Stars below $\approx 0.5 M_{\odot}$ do not ignite helium, and become helium WDs. Above this, they ignite helium degenerately in their cores, leading to thermal runaway. The star can peak at a luminosity close to $10^{11} L_{\odot}$ (approximately 100 times the peak luminosity of SN 1987A, per Figure 5 of [Arnett et al. \(1989\)](#)). However, this peak in luminosity is attributable to nuclear fusion in the core, and the energy does not reach the surface, let alone escape into the ISM – rather, it is absorbed into the outer layers of the star. During this extremely brief thermal runaway, the luminosity begins to settle and the star is able to continue evolving. Models of low

mass stars fail at this stage and care is required to evolve them. This phenomena is called the *helium flash* and is discussed further in Section 2.6.1.

After the ignition of helium, these stars cross the so-called Hertzsprung Gap – the region where very few stars are observed empirically. This region is bounded (on a colour-magnitude diagram, which is a transform of the Hertzsprung-Russell Diagram (HRD)) by the points $(M_v, B - V) = \{(2, 0.5), (-7, 0.1), (-6, 1.6)\}$ (Hoyle, 1960), where M_v is the V-band magnitude (magnitude in the visual band), and $B - V$ is the colour index of the star. The phenomenon of a gap existing is attributable to the fact that stars cross this gap exceedingly quickly: whereas it spends $\sim 90\%$ of its life on the MS, it spends $\sim 2\%$ of its life crossing the Hertzsprung gap (Hoyle, 1960).

Intermediate-Mass Stars: $2 M_\odot < M < 8 M_\odot$

Intermediate mass stars exist above the threshold for the helium flash to occur. Helium ignition is not degenerate and the development of a helium core is not impeded. As the star ascends the Asymptotic Giant Branch (AGB), the core becomes helium-exhausted and it becomes a red giant, with a radius many hundreds or even thousands of times greater than the Sun. Its mass, sustained prior to the AGB via nuclear fusion, begins to decrease. An example of intermediate-mass evolution is shown in Figure 2.2.

Massive Stars: $8 M_\odot < M < 80 M_\odot$

Massive stars are the key focal objects of this thesis, and in many ways offer some of the most exciting phenomena we can probe. In Figure 2.3 we show the variety of evolution that this regime has, alongside the ZAMS track from $8 M_\odot$ to $90 M_\odot$. The key features are as follows. When a massive star exhausts its supply of hydrogen in its core and evolves off the MS to begin burning helium, it generally bypasses the Hertzsprung gap and evolves up the Red Giant Branch (RGB). From here, stars below $\approx 15 M_\odot$ generally engage in a “blue loop”, where it remains at a roughly constant radius whilst increasing in temperature, before cooling at a slightly higher luminosity. This is due to excess helium being burned in shells above the core (Walmswell et al., 2015). We note that our models include blue loops, as shown in Figure 2.3 and Figure 2.4.

2.2.3 | Stellar Atmospheres

As a final note in the evolution of a single star, we must briefly address what occurs about the photosphere of the star. A star is not a planet – there is no firm, opaque boundary demarcating the inner of a star from the outer. The common definition of the “surface” of a star is the surface of last scattering (equivalently, the photosphere), which is the point radially in which the mean free path ℓ_γ of a photon emitted from the core of the star becomes meaningfully long (internally, $\ell_\gamma \sim 2$ cm, externally, $\ell_\gamma \sim 4.4$ pc).³ However, outflows from the star as well as the potential interaction of a partner in a common-envelope event mean that this surface is not necessarily firm: there is no sharp turn from stellar material to empty space. Thus, the simulation of stellar atmospheres is of interest to the stellar astrophysicist.

³The mean free path is a function of the particle under consideration: for a neutrino, internally $\ell_\nu \sim 32$ pc and externally $\ell_\nu \sim 81$ Rpc $\sim 3 \times 10^{18}$ times the diameter of the observable Universe.

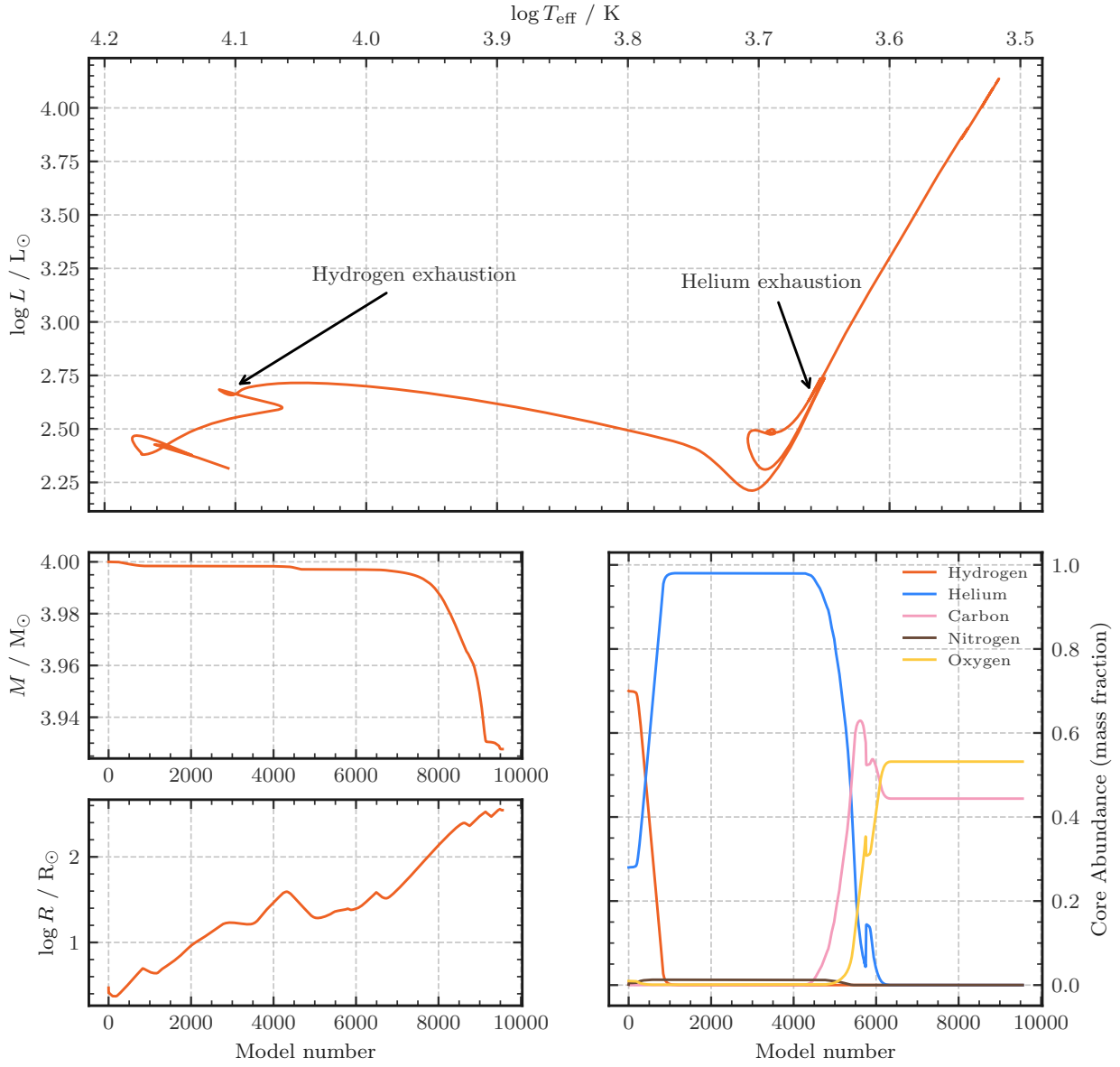


Figure 2.2: The evolution of an $4 M_{\odot}$ star from ZAMS until CO core development. The final core abundances are $Z_{\text{C}} \approx 0.44$ and $Z_{\text{O}} \approx 0.53$.

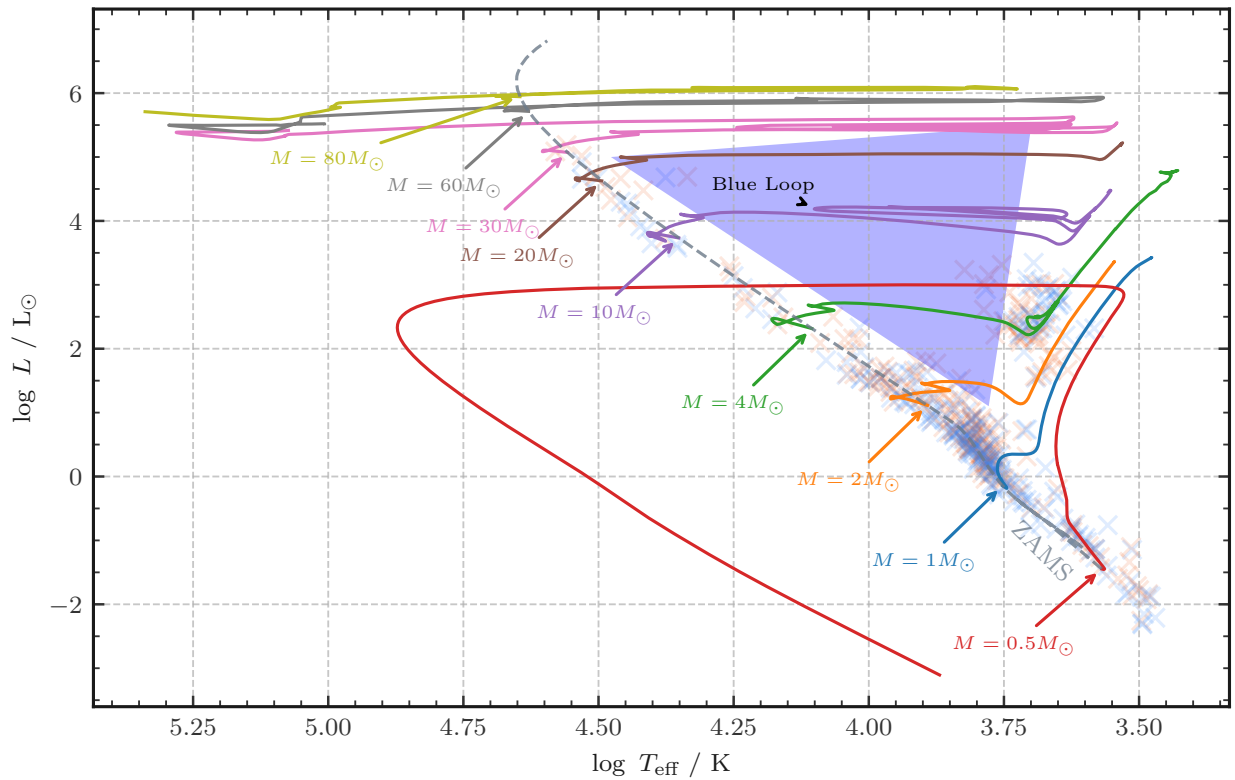


Figure 2.3: The evolution of stars in the range $0.5 - 80 M_{\odot}$. The dashed black line represents the Zero-Age Main Sequence (ZAMS), which is the origin point of all of our models, and (in this plot) extends from $0.5 M_{\odot}$ to $300 M_{\odot}$. The blue triangle is the (highly approximate) bounds of the Hertzsprung gap. Crosses indicate binaries in the Double Eclipsing Binaries catalogue (DEBCAT; [Southworth, 2015](#)), and illustrate the apparent “zone of avoidance” that defines the Hertzsprung gap. Blue crosses are primaries, and orange crosses are secondaries.

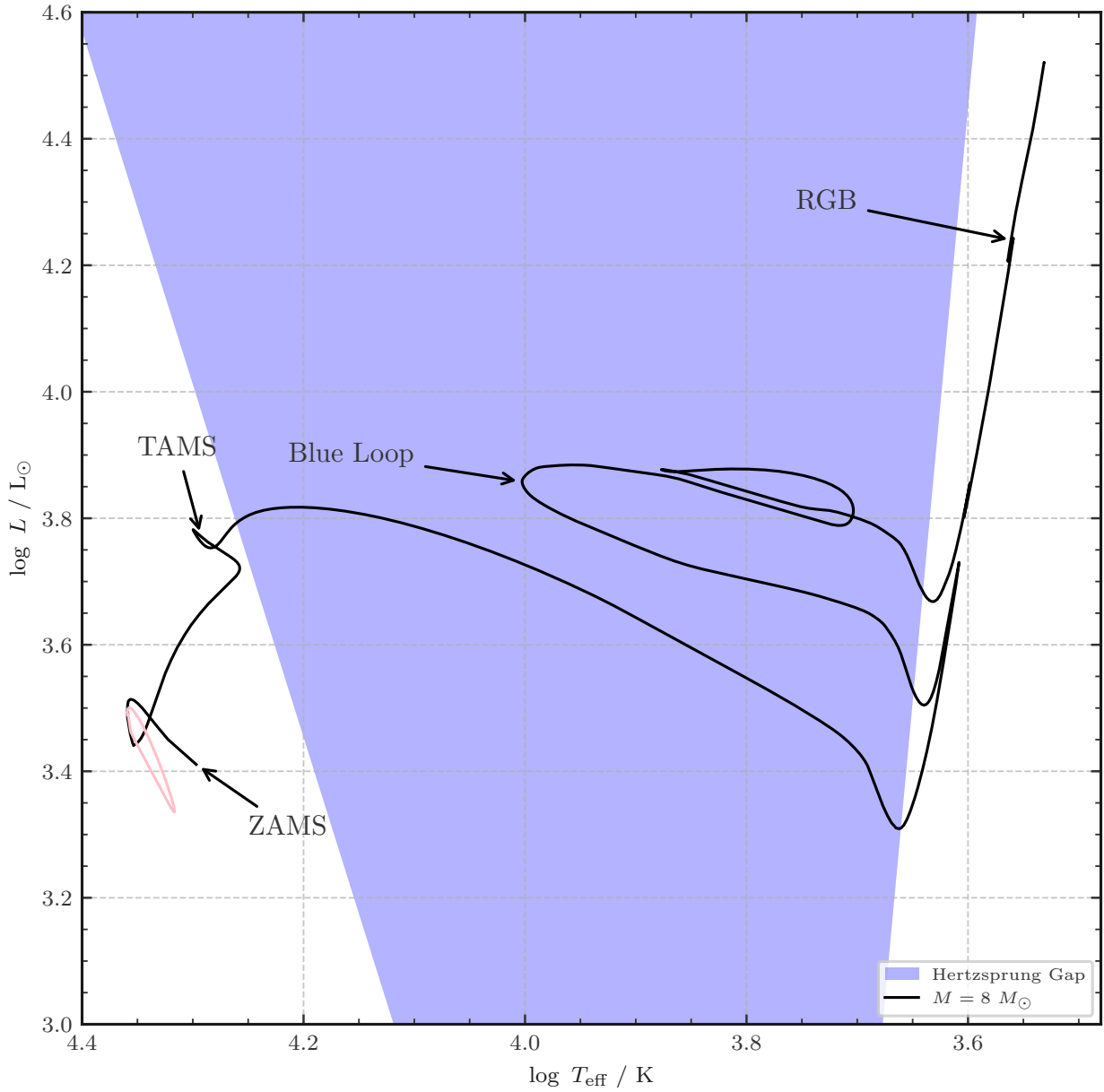


Figure 2.4: The evolution of an $8 M_{\odot}$ star. Highlighted is the star's Zero-Age Main Sequence (ZAMS), Terminal-Age Main Sequence (TAMS), the blue loop, and the Asymptotic Giant Branch (AGB). The Hertzsprung gap is also highlighted. In this particular model, the star achieves CNO equilibrium in the leftmost (hottest) region of the plot, highlighted in pink.

However, this introduces another rather large layer of complexity. The synthesis of stellar spectra is not a trivial task, even without considering the complexity of evolving the star simultaneously. [Groh et al. \(2014\)](#) is one of the very few instances of coupled spectroscopic calculations with evolutionary calculations, which only evolved a single star. Moreover, this thesis is concerned with detailed modelling of the stellar *interior*, and thus the atmosphere is an interesting sidenote, but one we shall not pursue.

2.3 | Supernovae

<i>And on the pedestal these words appear:</i>	<i>Pallida mors æquo pulsat pede pauperum</i>
<i>“My name is Ozymandias, king of kings:</i>	<i>tabernas regumque turres.</i>
<i>Look on my works, ye Mighty, and despair!”</i>	<i>Pale death knocks with the same tempo upon the</i>
<i>Nothing beside remains.</i>	<i>huts of the poor and the towers of kings.</i>
— PERCY BYSSHE SHELLEY, <i>OZYMANDIAS</i>	— HORACE

A star’s lifetime is not infinite. The key assumption underpinning stellar evolution is that of hydrostatic equilibrium: the counterplay of gravity against outwards pressure. In short, a dynamically stable, non-rotating star composed of an ideal gas with adiabatic constant γ and gravitational potential energy Ω radiates energy via

$$L = -\dot{E} = -\frac{3\gamma - 4}{3(\gamma - 1)} \frac{d\Omega}{dt} - \frac{d\gamma}{dt} \left(\frac{1}{(\gamma - 1)^2} \right), \quad (2.11)$$

If the chemical gradient is largely unchanged, $\frac{d\gamma}{dt} \sim 0$ so the luminosity reduces to

$$L = -\frac{3\gamma - 4}{3(\gamma - 1)} \frac{d\Omega}{dt} \quad \text{where} \quad U = -\frac{1}{3(\gamma - 1)} \Omega, \quad (2.12)$$

where U is the total internal energy of the star. Since $\dot{E} < 0$ for a star radiating energy, and $\Omega < 0$ by definition (since assembling a star is energetically favourable), we have that $\dot{\Omega} < 0$ and $\dot{U} > 0$, i.e., that as the star radiates energy it attempts to shrink. This shrinking is arrested by the nuclear fusion in the core providing outwards pressure, meaning the star is stable and resists explosion. As the star continues to burn through its fuel supply and eventually exhaust it, gravity will win as the dominant force, and result in the star imploding in a CCSN. From a thermodynamic perspective, as the star evolves, it shrinks as energy is bled off into the interstellar medium. This shrinkage compresses and heats the core, meaning the star’s specific heat capacity is negative. This system is thermodynamically unstable, and thus every star either collapses into a remnant (cf. BH, NS) or is sustained in some form by electron degeneracy pressure (cf. WD).

In any case, the breaking of hydrostatic equilibrium results generally results in⁴ a supernova, which can be classified into subtypes, as shown in Figure 2.5. In this event, gravity pulls the stellar core inwards, before

⁴The extent of the veracity of this statement is an area of active research: [Reynolds et al. \(2015\)](#) presents a short summary.

“bouncing” and ejecting the envelope into the interstellar medium. The physics of this approximately second-long event is highly uncertain, and is in general unable to be modelled by detailed stellar evolution codes. All models we generate – assuming they have not been terminated earlier – will terminate at the moment of supernova.

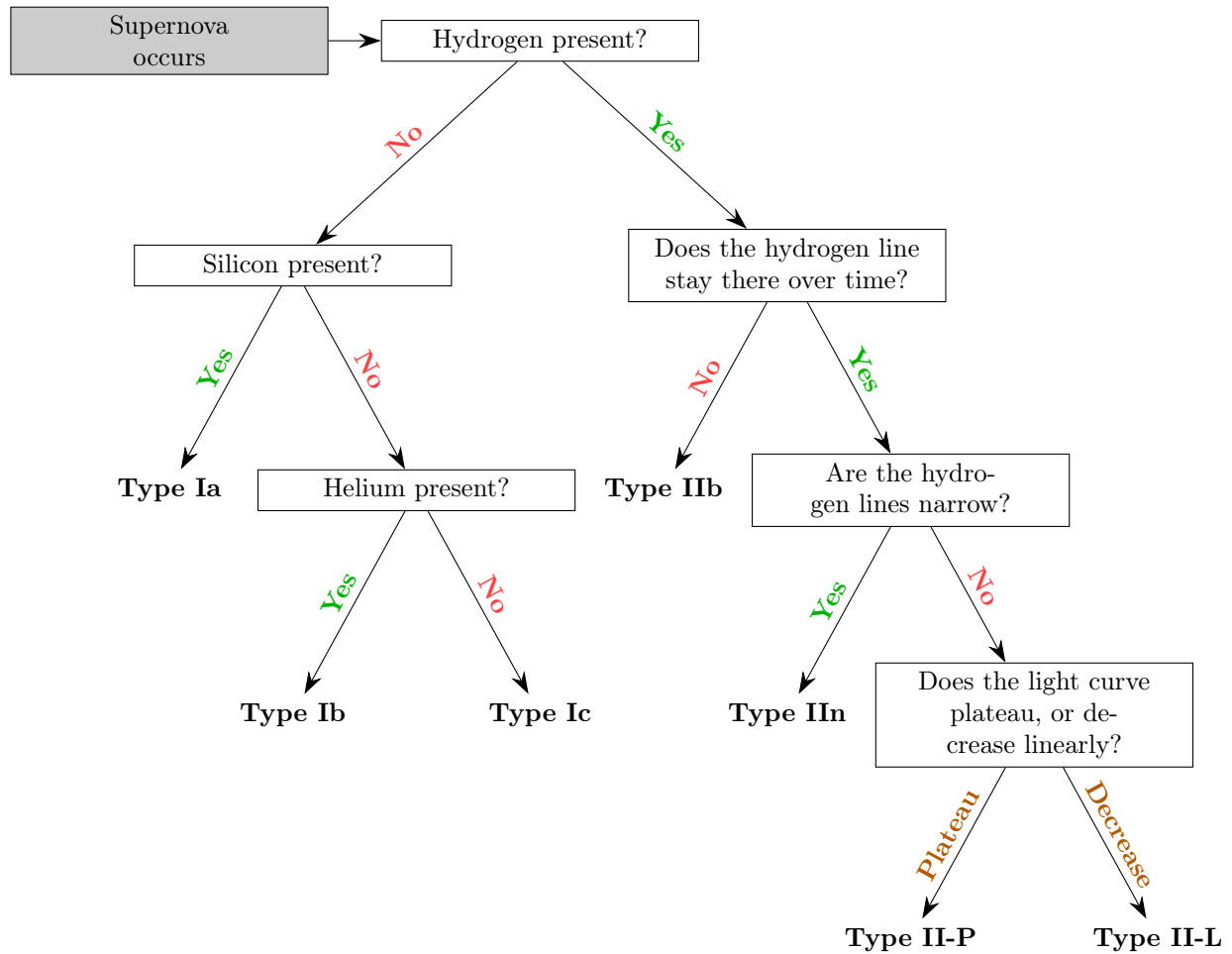


Figure 2.5: A rough outline of supernova classification. One can see that the presence of hydrogen lines distinguishes a Type I from Type II supernova. Type II supernova subtypes are determined by the behaviour of their light curves – how the apparent magnitude changes against time – and thus are determined by the luminosity of the supernova, something detailed models are not able to model. On the other hand, the presence of hydrogen/silicon/helium in the supernova is something that detailed codes *are* able to model. There are further delineations into Type Ia/b/c-pec and Type IIpec, denoting peculiar supernovae, however these are rare and are in some sense exception categories, where the behaviour does not fit nicely into a taxonomy tree.

Importantly for the nucleosynthesis of iron-group and heavier elements (which cannot be formed via hydrostatic nuclear fusion,⁵ as shown in Figure 2.6), the supernova shockwave propagating through the core causes a new phase of nucleosynthesis, albeit briefly.

2.3.1 | Type Ia supernovae

There is another class of supernovae we consider in this thesis, namely type Ia supernovae. These are formed when a WD star exceeds the maximum mass by which electron degeneracy pressure can sustain the star. This mass is termed the Chandrasekhar mass, $M_{\text{ch}} \approx 1.44 M_{\odot}$. When material accretes onto a white dwarf, its

⁵Strictly speaking, fusion can occur for elements heavier than Iron-56, but the reaction is endothermic; it requires more energy than is released, meaning it is unsustainable.

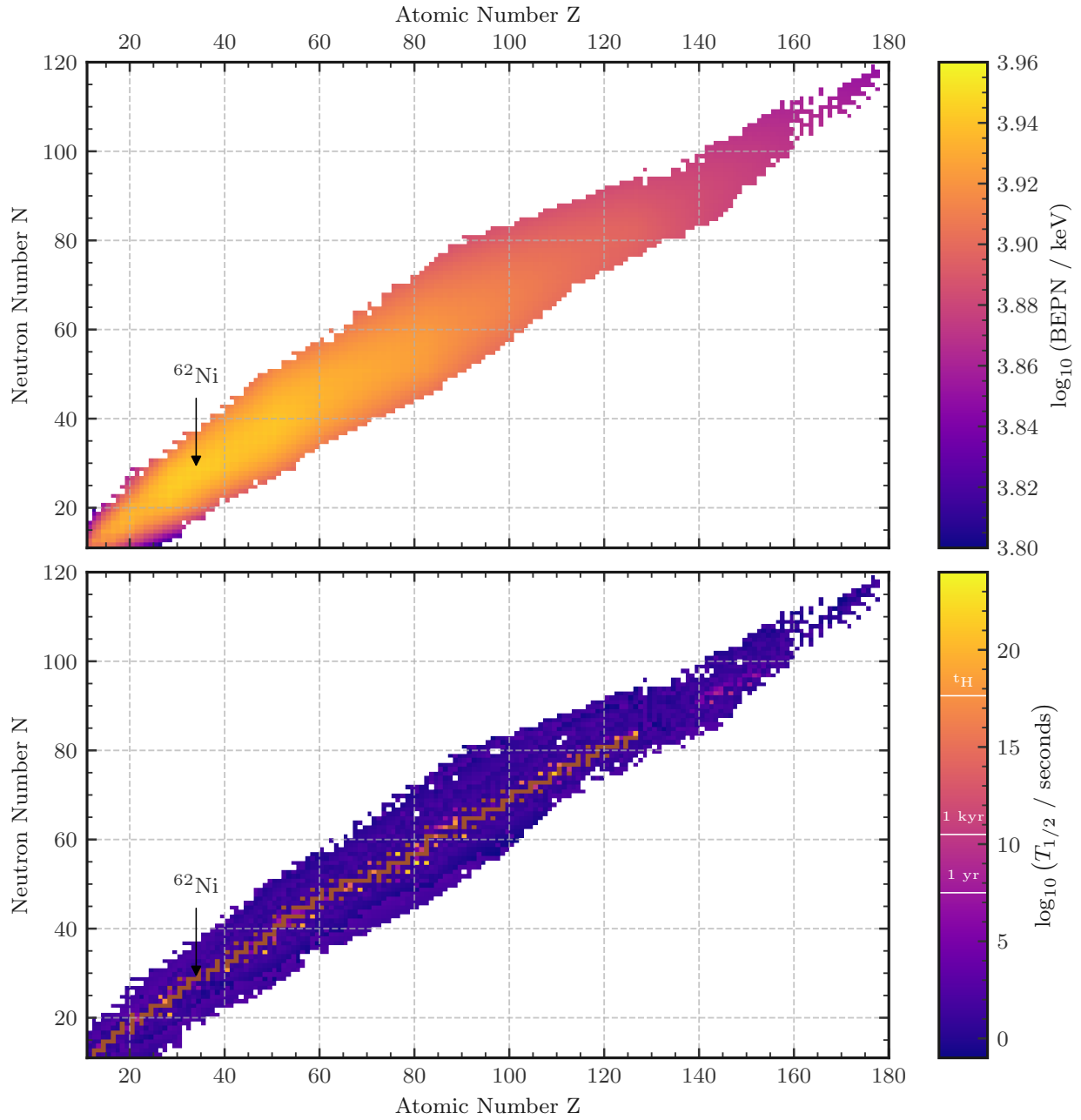


Figure 2.6: *Top:* The binding energy per nucleon and half-lives for elements up to Oganesson. The most stable element of fusion here is Nickel-62. Elements lighter than Nickel-62 will fuse towards it, and elements heavier will undergo fission to reach it. *Bottom:* The half-life in seconds across the same isotopes. Marked on the colorbar axis are 1 yr, 1000 yr, and $t_H = 13.8 \text{ Gyr}$. Stable elements are marked in brown. Underlying data for both plots are from the International Atomic Energy Agency.

mass will increase, and if the accretion does not stop, the mass will exceed the Chandrasekhar mass. This means electron degeneracy pressure is no longer able to sustain the star, and the core collapses further into a Type Ia (or thermonuclear) supernova.

2.4 | Two to tango: Binary Evolution

Introducing a second star into the mix complicates the evolution. Whereas before a star losing mass (through, for example, stellar winds) was assumed to eject it into the ISM, it may now accrete that material directly onto its companion. In addition, either star may in principle increase in radius enough to overlap with its companion, or even touch surfaces – introducing new dynamics into the mix.

The most marked change for a binary compared to a single star is that its trajectory on the HRD is altered. The MS is changed somewhat, but (for longer than ~ 5 d periods) the interaction takes place post-MS and is shown in Figure 2.7.

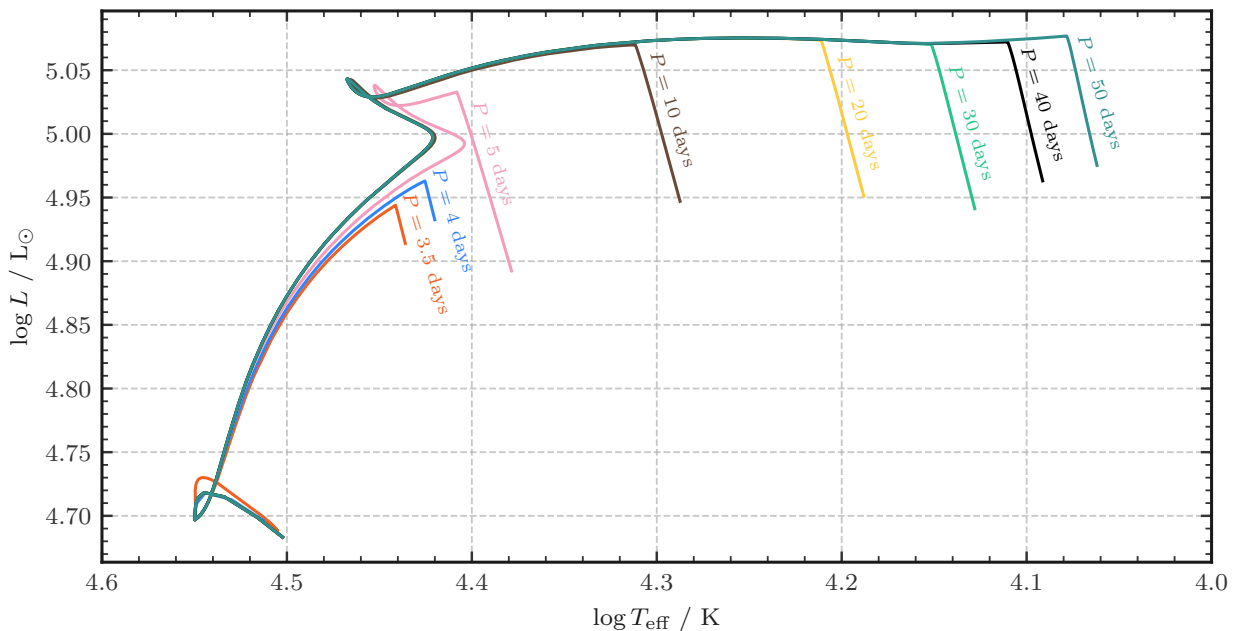


Figure 2.7: The HRD for an $21 M_{\odot}$ star with an (unshown) $10 M_{\odot}$ companion with varying period. In this figure, only the initial binary period was free to change between runs. Short periods interact closer to the MS. The evolution was terminated 1,000 models after the primary star achieved RLOF for the first time.

2.4.1 | Binary star kinematics

There are several formation channels for a binary star system. Binaries may be formed from birth (as discussed in Section 2.1, with the caveat that in this scenario two clumps of gas form near each other), via fragmentation of a protostar’s disc, dynamical capture of one star by another (comparable to one of the suggested origins of Phobos and Deimos, per [Burns \(1978\)](#)), or via a higher multiplicity system ejecting one or more of its constituent stars, perhaps via a natal supernova kick.

In any case, once formed and the orbit is stabilised, binary systems obey Kepler’s three laws. In the

generalised binary regime, however, the assumption that $M_1 \gg M_2$ (which underpins most statements of Kepler's laws) is not necessarily true – for instance, the system LSS 3074 has two stellar components with ZAMS masses of approximately $30 M_\odot$ and $10 M_\odot$ (Raucq et al., 2017). Moreover, the fact that there are two stars in orbit rather than one object orbiting another means that Kepler's laws must be stated more precisely. Formally, they can be expressed as (see Appendix A for derivations):

1. All orbits are ellipses, with both stars orbiting a common barycentre at the focus of the ellipse,
2. A line segment joining the two stars sweeps out equal areas in equal times,
3. The square of the period of the system is proportional to the cube of the semimajor axis of the ellipse (the binary separation), with the proportionality term dependent on the sum of the masses.

The third law can be stated more precisely as

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.13)$$

where T is the period of the binary, M_1 and M_2 the constituent masses, $G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the universal gravitational constant, and a is the semimajor axis of the binary. A schematic of the orbit of a binary system is given in Figure 2.8.

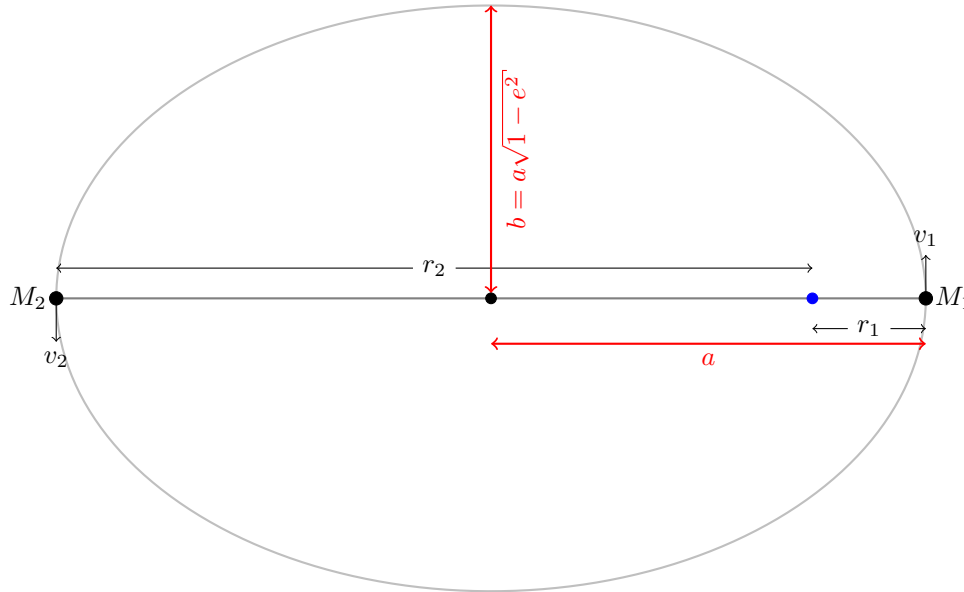


Figure 2.8: A schematic of a simple binary star system with eccentricity e , semimajor axis a , and semiminor axis b . The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance r_1 to the barycentre by Kepler's first and Newton's second laws. The elliptical orbital trajectory is shown in grey. Figure adapted from Richards (2021).

2.4.2 | Roche Lobes

A binary system is complicated by the presence of its companions in several different ways. As alluded to in the previous section, the presence of a companion warps space-time in such a way that material can be quite distorted from a sphere or oblate spheroid but still remain gravitationally bound to the system.

In the extremal case, a star can grow to fill or even exceed its Roche Lobe, which is the tear-drop shaped equipotential surface within which material remains gravitationally bound to its star. In Figure 2.9 we display these equipotential surfaces for a range of different mass ratios. It is pertinent to note that the shape of all of the Roche Lobes change rapidly at low mass ratios, however at higher mass ratios the shapes change less rapidly. In an equal-mass ratio binary, the binary configuration has rotational symmetry about the $\mathcal{L}_1$ point, and reflectional symmetry about the x -axis and the line $y = \mathcal{L}_1$.

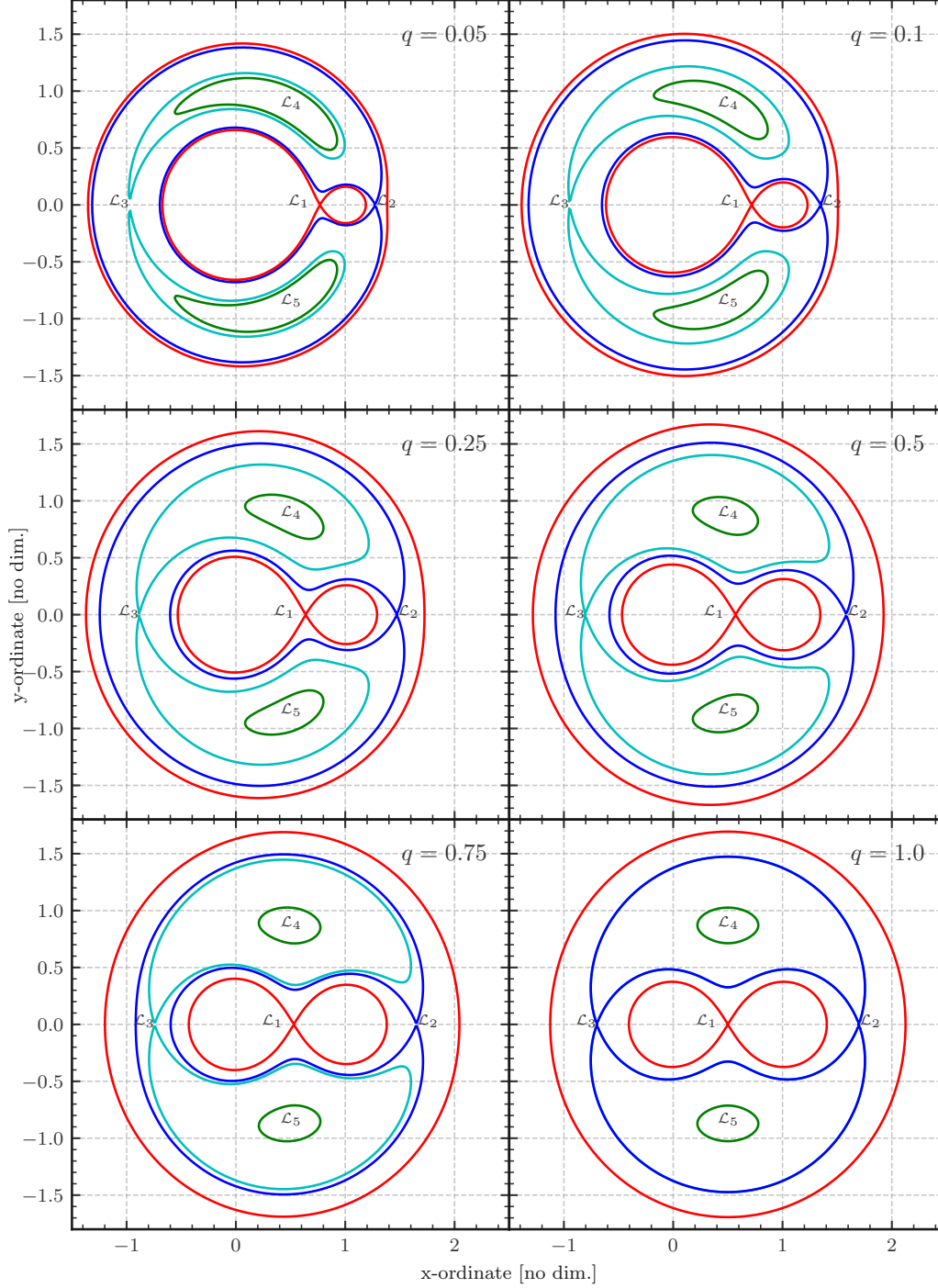


Figure 2.9: The dimensionless Roche potential as a function of mass ratio q . Note that the shape of the Roche potential changes markedly at low mass ratio, and changes more slowly as the mass ratio approaches unity. Distances have been normalised such that the primary star is at $(x, y, z) = (0, 0, 0)$ and the secondary star is at $(x, y, z) = (1, 0, 0)$, i.e., we measure distances in units of separation and that the coordinate system is centered on the primary star.

There are two primary Roche Lobe points we consider: the $\mathcal{L}_1$ point and the $\mathcal{L}_2$ point. The $\mathcal{L}_1$ point (also known as the inner Lagrangian point) exists on the line connecting the centres of the two stars, the $\mathcal{L}_2$ point (the outer Lagrangian point) is located on the other side of the secondary to the inner point. The dimensionless x -ordinate of the $\mathcal{L}_1$ point, ξ , measured in terms of the separation a , is given as the solution to

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \quad (2.14)$$

where $\xi_1 = \frac{M_2}{M_1+M_2}$, and $\xi_2 = \frac{M_1}{M_1+M_2}$. This expression is derived in Appendix A

This is a quintic expression in ξ , and solving this results in the real ξ -ordinates of five extrema. The extrema are labelled $\mathcal{L}_1$ through $\mathcal{L}_5$. $\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3$ are saddle points of the potential, and $\mathcal{L}_4$ and $\mathcal{L}_5$ are maxima. No analytical function for ξ exists unless either or both of ξ_1, ξ_2 are zero (which would only be the case for a single star, and thus the Roche lobe would not exist in the first instance), thus we must make approximations.

The $\mathcal{L}_1$ point

The inner Lagrangian point has the approximation from [Eggleton \(1983\)](#), which is

$$\frac{R_L}{a} \approx \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} + \ln\left(1 + q^{\frac{1}{3}}\right)}. \quad (2.15)$$

We show the circle with this radius R_L in Figure 2.10 plotted against the exact solution for $\eta = \mathcal{L}_1$. Eggleton's formula is a volume-equivalent one; although his radius $R_L < \xi$, the volume of an Eggleton-Roche sphere $\frac{4}{3}\pi R_L^3$ is approximately equal (to within 1 per cent) of the numerically integrated volume, as is shown in Figure 2.11.

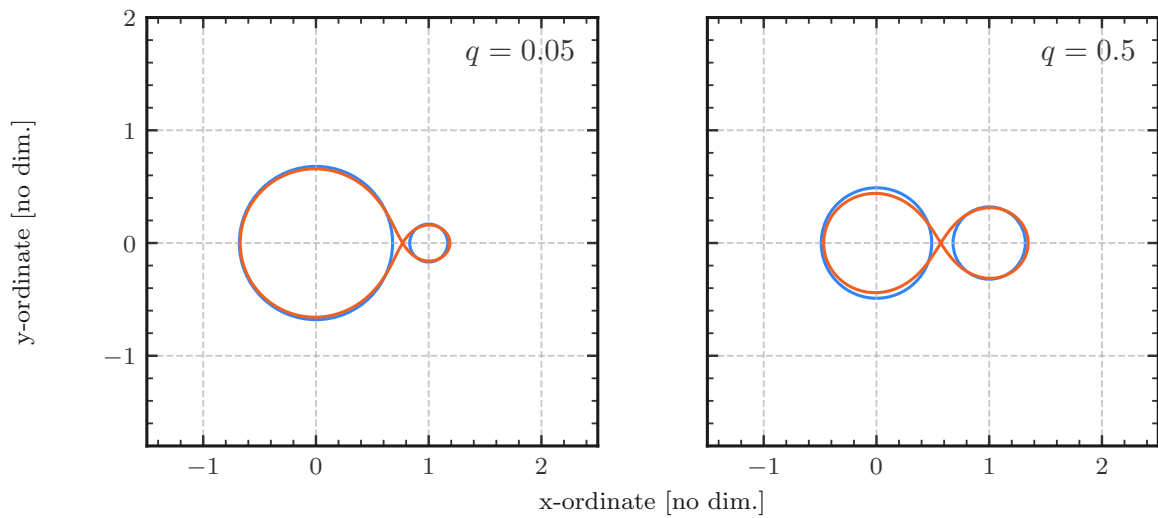


Figure 2.10: A schematic of the Roche lobe formula Eqn(2.15). Here, the exact solution to the Roche lobe formula ξ in Eqn(2.14) (orange) vs the fit provided by [Eggleton \(1983\)](#) (blue). Although precise details of the Roche lobe, such as the apex of the teardrop at the point ξ , are lost, the volume is preserved to within 1 per cent accuracy over all mass ratios.

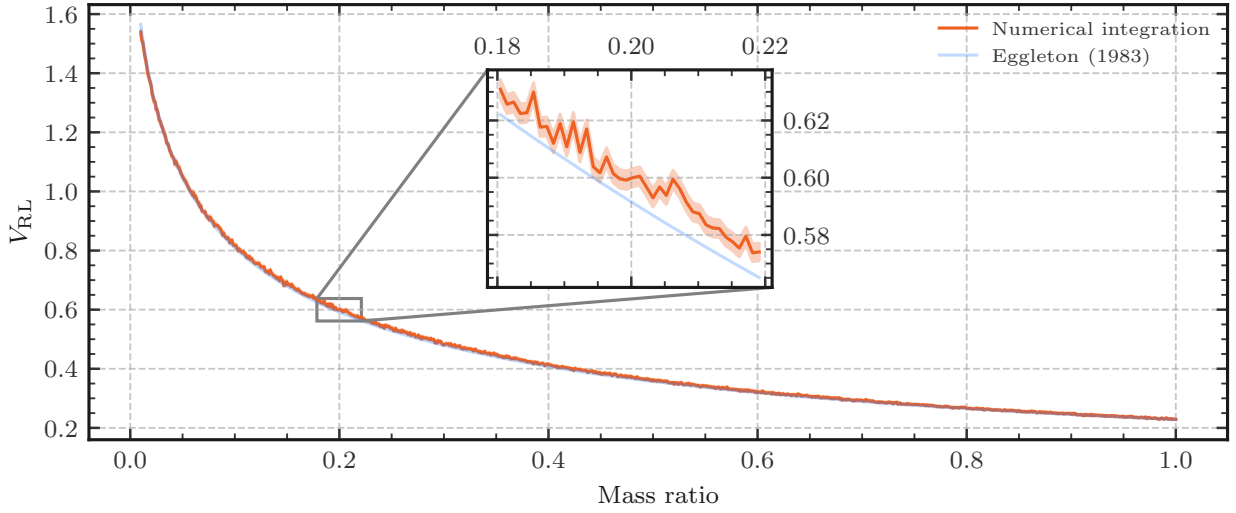


Figure 2.11: The volume of the Roche Lobe computed numerically through Monte-Carlo integration (orange) vs the volume assuming Eqn(2.15) (blue). The Eggleton approximation fits well to the volume, with around 1 per cent error across the entire mass ratio range (truncated here to $q \leq 1$).

The $\mathcal{L}_2$ point

The outer Lagrangian point, $\mathcal{L}_2$, is given by a different zero of the potential function. The approximation we use is that of [Marchant et al. \(2016\)](#), where

$$R_{L2} = R_L(q) \left(1 + \frac{0.299 \arctan(1.84q^{-0.397})}{q^{0.520}} \right), \quad (2.16)$$

and $R_L(q)$ is given by Eqn(2.15).

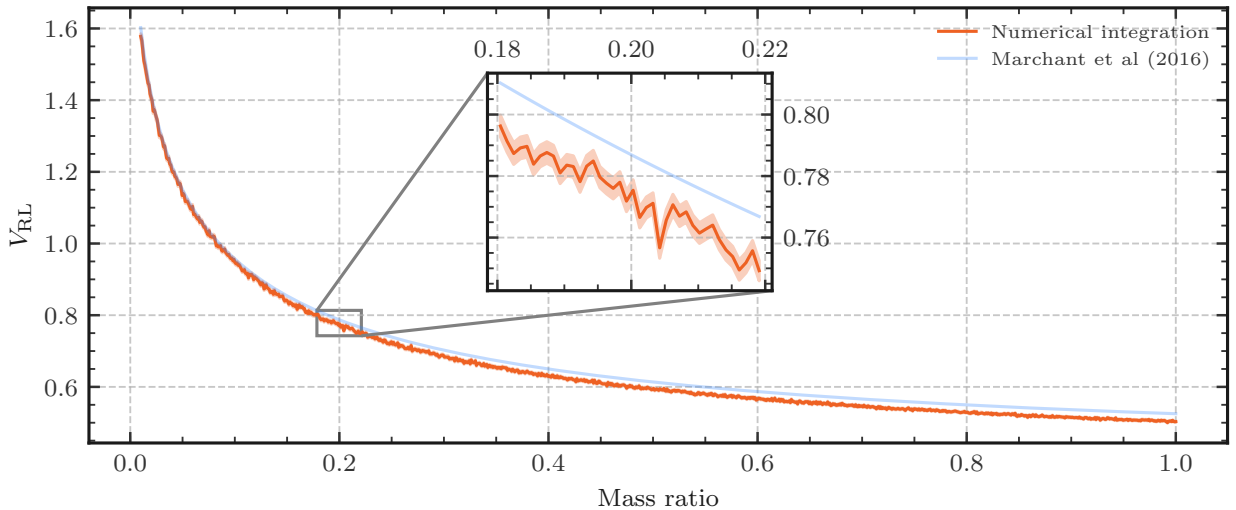


Figure 2.12: As for Figure 2.11, but for the $\mathcal{L}_2$ point with formula given by Eqn(2.16) ([Marchant et al., 2016](#)).

We note that [Marchant et al. \(2016\)](#) uses an inverse mass ratio definition compared to [Eggleton \(1983\)](#), that is, [Marchant et al. \(2016\)](#) takes $q_{\text{Marchant}} = \frac{M_1}{M_2}$ whereas [Eggleton \(1983\)](#) takes $q_{\text{Eggleton}} = \frac{M_2}{M_1} = \frac{1}{q_{\text{Marchant}}}$. This is accounted for in Eqn(2.16) such that the mass ratios are consistent (and we take $q = q_{\text{Eggleton}}$), but

other texts do not necessarily differentiate or describe the difference.

2.4.3 | Mass Transfer

And a cake that can make you grow.

— SECOND OF THE SIX IMPOSSIBLE THINGS

When two stars are sufficiently close in proximity, surface and near-surface events can cause accretion of material from one star to another due to the primary’s stellar winds (whereas in the single-star case, this would be matter lost to the ISM). The nature of mass transfer is a complicated affair; most binary star population synthesis codes typically inherit their mass loss prescriptions from papers such as [Bondi & Hoyle \(1944\)](#); [Hurley et al. \(2002\)](#), and this approach is what we detail in this section.

Mass transfer can broadly be sectioned into four main categories:

1. $\dot{M}_{\text{wind}}$, where matter is lost through stellar winds,
2. $\dot{M}_{\text{RLOF}}$, where matter is transferred through one (or more) of the Lagrange points in a binary (cf. Section 2.4.2),
3. $\dot{M}_{\text{acc}}$, where matter is transferred through accretion (or interaction with a disk), and
4. $\dot{M}_{\text{Bondi}}$, where matter proceeds through a [Bondi & Hoyle \(1944\)](#) type mechanism.

Our mass loss rate for each star is constructed from these four terms as

$$\dot{M} = \dot{M}_{\text{wind}} - \dot{M}_{\text{Bondi}} + \dot{M}_{\text{RLOF}} - \dot{M}_{\text{acc}}. \quad (2.17)$$

Wind mass-loss

Wind mass loss computations are (generally) performed using a mix of theoretical and empirical mass-loss rate prescriptions of [de Jager et al. \(1988\)](#) and [Vink et al. \(2001\)](#). The precise scheme is detailed in [Eldridge et al. \(2017\)](#).⁶ There are two critical modifications we make to their scheme.

The first modification surrounds the transition into the Wolf Rayet (WR) regime. Traditionally this has been a sharp transition when the surface hydrogen mass fraction becomes less than 0.4 and the surface temperature is above 10^4 K. This is a reasonable approximation, though if a star reaches this phase and is *not* yet a WR star, the code will fail. To account for this we gradually turn on these enhanced WR rates when this condition approaches, smoothing the transition.

The second modification we have made is that if a star has a hydrogen exhausted core and begins approaching the WR temperature range, we assume that WR mass loss must begin and thus turn on the slowly-turning on WR regime.

⁶In other codes such as MESA, this is referred to as the “Dutch” mass loss prescription owing to the nationality of the authors of [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#).

Mathematically, the computation for $\dot{M}_{\text{wind}}$ is more involved, though follows the same essential structure as Vink et al. (2001). Details are provided in Section A.4.

Roche Lobe overflow (RLOF)

Matter extruded from the Lagrangian points is assumed to transfer with a mass loss rate similar to Hurley et al. (2002). We take

$$\dot{M}_{\text{RLOF},i} = k \cdot \max\left(\ln\left(\frac{R_i}{R_{\text{RL},i}}\right), 0\right)^3 \left(\frac{M_i}{M_{\odot}}\right)^2 M_{\odot} \text{ yr}^{-1}. \quad (2.18)$$

Here, k (referred to in our code as RMT) parameterises effectively how much of the star we permit to exist outside of its Roche lobe before it is accreted – i.e., the voracity of RLOF. The choice of this can be somewhat arbitrary. Hurley et al. (2002) for example took $k = 3 \times 10^{-6} \min\{M_1, 5\}^2$, where the coefficient was “chosen by experiment to ensure that the mass transfer is steady”, and warned that for degenerate stars a more appropriate choice would be to increase this by a factor of 10^3 , corroborated by Claeys et al. (2014). We choose to adopt $k = 3 \times 10^{-6} \times 10^3 = 3 \times 10^{-3}$, with no dependence on the mass. This permits the star to overfill its Roche lobe even in tight binaries, but still allows efficient mass transfer.

Matter exceeding $\mathcal{L}_2$ (the outer Lagrange point) does not currently receive special treatment. The implications of $\mathcal{L}_2$ overflow are significant. This ejection will lead to unstable mass transfer, shrinking the orbit and likely inducing a Common Envelope Evolution (CEE) event (Fragos et al., 2023). There is debate about the appropriate treatment here. Linial & Sari (2017) indexes the $\mathcal{L}_2$ rate to the $\mathcal{L}_1$ rate via an efficiency parameter. Li & Li (2013) considers the case analogous to $\mathcal{L}_1$ outflow, where material outflowing from $\mathcal{L}_2$ acts to retard the angular momentum. In their notation $\frac{dJ}{dt} = -\epsilon \dot{M}_1 a_{L2}^2 \omega$ (compare $\frac{dJ}{dt} = -\beta \dot{M}_1 a_{L1}^2 \omega$ for RLOF through $\mathcal{L}_1$) for controllable parameters β and ϵ . The ambiguity of treatment leads to some codes to terminate at the onset of $\mathcal{L}_2$ outflow, such as POSYDON (Fragos et al., 2023). $\mathcal{L}_2$ outflow requires stars to be in reasonably tight orbits (~ 2 d), which we will approach, but none of our stars will enter this regime, and thus we do not afford it special treatment.

Accretion

Our wind accretion and RLOF accretion rates are taken from Bondi & Hoyle (1944); Hurley et al. (2002), and limits the accretion rate by the thermal timescale of the star. The prescription is that

$$\dot{M}_{\text{acc}} = \min\left(\dot{M}_{\text{RLOF}}, \frac{M}{T_{\text{KH}}} \max\left(1 - \frac{\omega}{0.8\omega_{\text{crit}}}, 0\right)\right), \quad (2.19)$$

where ω and ω_{crit} are the spin rate and critical spin rate of the star. We generally assume that the star is composed of individually rotating shells of variable thickness.⁷ If N is the number of meshpoints that the star is composed of (for our cases, normally $N = 199$), then the moment of inertia is given by

⁷Also known as shellular rotation.

$$I = \frac{2}{5} M_c R_c^2 + \sum_{k=1}^N \frac{2}{3} (m_k - m_{k+1}) r_k^2, \quad (2.20)$$

where m_k is the mass internal to meshpoint k (similar for r_k), and a subscript of c refers to the central meshpoint. It is assumed that the central meshpoint sits atop a solid sphere, hence $\frac{2}{5}$ as its prefactor.

Bondi-Hoyle wind accretion

Bondi-Hoyle accretion for a circular orbit is given by

$$\dot{M}_{\text{Bondi}} = \min \left[- \left(\frac{GM_2}{v_W^2} \right)^2 \frac{\alpha_W}{2a^2} \frac{1}{(1+v^2)^{3/2}}, 0.8 \right] \dot{M}_{\text{wind}}, \quad (2.21)$$

where we have assumed that the orbital velocity is given by

$$v^2 = \frac{G(M_1 + M_2)}{av_W^2}, \quad (2.22)$$

and that the wind velocity is given by

$$v_W = 2\beta_W \frac{GM_1}{R_1}. \quad (2.23)$$

Generalising this to an eccentric orbit causes the first argument in Eqn(2.21) to pick up a coefficient of $\frac{1}{\sqrt{1-e^2}}$. There are two free parameters to select: α_W and β_W . These ostensibly depend on stellar types, though we select for simplicity $\alpha_W = \frac{3}{2}$ (Boffin & Jorissen, 1988) and $\beta_W = 7$ (Lamers et al., 1995) for O-type stars. This parameterisation is likely incomplete; Hurley et al. (2002) argues that as per Lamers et al. (1995), β_W should vary between 7 for O-type stars to 0.5 for A and F-type stars, perhaps descending as low as $\beta_W \approx \frac{1}{8}$ for cool supergiants. We leave this implementation for future work under the assumption that our mass-loss rate is correct to within an order of magnitude.

2.5 | Post-Supernova

Wit nodded.

“Oh, is this the place where I talk?” Kaladin said.

“If you wish.”

“I don’t wish. Get on with the story.”

— “RHYTHM OF WAR”, BY BRANDON SANDERSON

After a supernova, the star leaves behind its core as a compact remnant. Generally, these are either black holes or neutron stars. The interior of a neutron star is reasonably complex, with superfluid neutrons within the crust (Lattimer & Prakash, 2001) and a latticed core-crust interface (Rueda et al., 2014), with “pasta

phases” at the interface (Zdunik et al., 2017). The interior of a black hole is impossible to observationally probe without violating causality.

The supernova ejects some fraction of material M_{ej} , leaving behind a remnant of mass M_{rem} . Eldridge & Tout (2004b) gives a relationship between ejecta, remnant, and stellar masses by assuming a supernova releases 10^{46} J ($=10^{53}$ erg) of energy, of which 1 per cent is transferred into the envelope. The full relation between the binding energy U , the stellar mass M_* , and the ejecta and remnant masses is thus

$$\int_{M_{\text{rem}}}^{M_*} \left(U - \frac{GM}{r(M)} \right) dM = 10^{44} \text{ J}, \quad (2.24)$$

where $r(M)$ is the radius of the sphere containing mass M of the star. The post-supernova geometry is given by Tauris & Takens (1998).

The orbital motions of a binary system post-secondary supernova are now deterministic from four initial parameters: m_1, m_2, a , and e . Given their high masses and small radii, we may approximate them as point masses, and thus their motion is now wholly determined by the emission of gravitational radiation through (detectable) gravitational waves. The pertinent equations of motion are given by Peters (1964), namely,

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right), \quad (2.25)$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right), \quad (2.26)$$

with the constant β defined as

$$\beta \equiv \frac{64}{5} \frac{G^3 m M (m + M)}{c^5}. \quad (2.27)$$

Now if one defines the constant c_0 through

$$c_0 = a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304} e_0^2 \right)^{-\frac{870}{2299}}, \quad (2.28)$$

then the time from the second supernova to the conclusion of the inspiral, termed the *coalescence time*, will be given by

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de. \quad (2.29)$$

The derivations for Eqns(2.25), (2.26) and (2.29) are given in Appendix A. The integral has no closed-form solution, and a numerical integrator is typically used to compute the coalescence time. However, some approximations such as Mandel (2021) do exist and approximate the coalescence time rather well over the entire range of mass ratios.

2.5.1 | Merger rates of compact binaries

The merger rate of a population of compact binaries at time t after formation can be formalised via the heuristic

$$m(t) = \text{number of binaries for which } T_c + T_{\text{rej}} + T_{\text{evo}} < t, \quad (2.30)$$

where T_c is the coalescence time, T_{rej} is the rejuvenation age (the age at which the primary star last accreted mass), and T_{evo} is the evolution time-scale (the length of time the primary spends on the MS). $m(t)$ is called the Delay-Time Distribution (DTD) of a population.

However, the underlying assumption in this is that the population was formed after an instantaneous burst of star-forming activity at time $t = 0$, which is clearly unphysical. We must account for the so-called Star Formation Rate Density (SFRD) $\psi(z)$, assumed to be taken from [Madau & Dickinson \(2014\)](#):

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \text{ M}_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (2.31)$$

where z is the redshift. For the purposes of converting between lookback time t and redshift z , we assume a standard Λ CDM cosmology with $\Omega_M = 0.3, \Omega_\Lambda = 0.7, \Omega_k = 0, H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Of course, this ignores the effect of mixed-metallicity models, which we account for using the prescription of [Langer & Norman \(2006\)](#), whereupon the SFRD is multiplied by a factor χ :

$$\chi(z, Z) = \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)}, \quad (2.32)$$

where $\hat{\Gamma}$ and Γ are the upper incomplete and complete Gamma functions respectively. This means our Stellar Formation History (SFH) – which takes into account both redshift and metallicity – denoted $\Psi(z, Z)$, is given by

$$\Psi(z, Z) = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)} dz, \quad (2.33)$$

where z_1 and z_2 are the redshifts of our bin edges. The event rate as a function of time, $\mathcal{R}(t)$, is therefore given by the convolution

$$\mathcal{R}(t) = \Psi(z_t) * m(t), \quad (2.34)$$

where z_t is the redshift corresponding to lookback time t , given by the solution to the equation

$$t = t_H \int_0^{z_t} \frac{dz'}{(1+z') \sqrt{\Omega_M (1+z')^3 + \Omega_k (1+z')^2 + \Omega_\Lambda}}, \quad (2.35)$$

with t_H the Hubble time, and $\Omega_M, \Omega_k, \Omega_\Lambda$ the matter density, curvature parameter, and cosmological

constant density parameter of the Universe. In general this integral does not have a closed-form solution, however in the case of a flat Universe with $\Omega_k = 0$, an exact solution for redshift⁸ is

$$z = \left(\frac{a}{\sinh\left(\operatorname{asinh}(a) + \frac{t}{b}\right)} \right)^{\frac{2}{3}} - 1, \quad (2.36)$$

where $a = \sqrt{\frac{\Omega_\Lambda}{\Omega_M}}$, and $b = \frac{1}{H_0} \frac{2}{3\sqrt{\Omega_M}}$. Detail on how the convolution in Eqn(2.34) is computed may be found in [Richards \(2021\)](#).

2.6 | Numerical implementation

I think there is a world market for maybe five computers.

— THOMAS WATSON, PRESIDENT OF IBM, 1943

Having now discussed the evolution of stars from a theoretical perspective, it becomes imperative to discuss the methods of modelling them. In this section, we provide an overview of the four codes we either use extensively or have developed to carry out the work of this thesis. These four codes are:

1. The Cambridge STARS code (Section 2.6.1), originally developed by Peter Eggleton, which we use to solve the equations of stellar structure and evolution,
2. The TUI code (Section 2.6.2), developed by Jan Eldridge, which performs our binary population synthesis of gravitational wave sources,
3. The KAITIAKI code (Section 2.6.3), developed for this thesis, which is our grid-manager code and allows modification of the input parameters to the Cambridge STARS code programmatically, and
4. The TAKAHE code (Section 4.1.3), developed by Sean Richards for [Richards \(2021\)](#), which evolves the orbits of our binary compact objects.

⁸Integral was computed via SYMPY.

2.6.1 | The Cambridge STARS code

Brief History

*And through the night, behind the
wheel, the mileage clicking west
I think upon Mackenzie, David Thompson and the rest
Who cracked the mountain ram-
parts and did show a path for me
To race the roaring Fraser to the sea.*

— “NORTHWEST PASSAGE”, BY STAN ROGERS

In 1959, Louis George Henyey published his method for automatic stellar evolution ([Henyey et al., 1959](#)). Prior to this – and the widespread availability of the computer – a model was treated as a mathematical process and thus was computed, often labouriously, by hand. For instance, in [Eggleton \(1967\)](#), Eggleton writes that

Since we have seen in Section 3 that the structure of the flame zone can be obtained independently (almost) of the rest of the star, it is reasonable to expect that [thermal] instabilities of the type considered by Schwarzschild & Härm can also be examined in this way, so that the considerable labour of computing the structure of an entire star can be avoided.

– [Eggleton \(1967\)](#)

The Cambridge STARS code was one of the earliest detailed stellar evolution codes to be written, though initially it only modelled one star in detail. To model a binary system, tweaks were made whereupon the primary would be modelled in detail whilst approximating the secondary with the equations of [Hurley et al. \(2002\)](#) until the primary supernova, at which point the secondary would be evolved in detail. This had its merits; notably computation time was saved, and the results were generally accurate anyway.

However, [Stancliffe & Eldridge \(2009\)](#) introduced detailed modelling of *both* stars in a binary system into the STARS code – that is, the interiors of both stars were computed in detail, alongside the simultaneous computation of the orbital parameters. Since then, a wide array of improvements have been made to the numerical stability of the code, as well as introducing or refining new physics.

There are four variants of this code:

1. The Cambridge STARS code, which is the version originally developed by Peter Eggleton and originally described in [Eggleton \(1971\)](#). This version was originally only capable of solving the equations for a single star.

2. The ELDRIDGE STARS code described in [Eldridge & Tout \(2004b,a\)](#); [Eldridge et al. \(2008, 2017\)](#), which is the version used to generate the fiducial BPASS v2 grids. This version models the primary in detail while approximating the secondary using the equations of [Hurley et al. \(2002\)](#).
3. The BSTARS code, which is the unmodified version lately available from Cambridge, described in [Stancliffe & Eldridge \(2009\)](#). This version models both stars in detail throughout their lives.
4. The AOTEAROA STARS code, which is the version of the code incorporating the changes we have made.

Unless context otherwise requires, when we refer to “the STARS code” with no further qualification, we are referring specifically to the AOTEAROA STARS code.

Matrix Inversion

The heart of STARS is a matrix inversion algorithm. Although computing the inverse of a matrix is a polynomial time process, it can still take considerable time to invert. STARS implements a Henyey matrix ([Henyey et al., 1964](#)), which is of size $n = (4 + n_{\text{comp}})N_{\text{mesh}}$, with n_{comp} the number of equations to be solved and N_{mesh} the number of meshpoints in the model. An individual element of the Henyey matrix is given by

$$\mathbf{H}_{ij} = \frac{\partial (\text{equation } i)}{\partial (\text{variable } j)}. \quad (2.37)$$

For a “typical” binary computation, with $n_{\text{comp}} = 30$ equations and $N_{\text{mesh}} = 199$ meshpoints, the matrix is already 6766×6766 , for a binary which evolves up the AGB (the theoretically most complex calculation we perform) with $N_{\text{mesh}} = 999$, the matrix is 33966×33966 . Introducing k new equations into the code causes the runtime of the matrix inversion calculation to scale like $\mathcal{O}(n^3 + k^3 N_{\text{mesh}}^3)$.

The matrix inversion is simple Gauss-Jordan inversion, which is of complexity class (for an $n \times n$ matrix) $\mathcal{O}(n^3)$. [Coppersmith & Winograd \(1987\)](#) showed that this could be optimised to $\mathcal{O}(n^{2.376})$ and conjectured reducibility to $\mathcal{O}(n^2)$.⁹ Since our code is embarrassingly parallelisable, and can be executed in parallel simply by increasing the number of cores available, the payoff for decreasing the runtime by one polynomial order is inconsequential, and thus we use the straightforward implementation.

Initial Condition Selection

In this section we provide an account for the selection of our general initial conditions. Unless otherwise noted in this thesis, it is assumed that the values will be using the parameters described here and in Appendix B.

In our implementation of Mixing Length Theory (MLT) ([Vitense, 1953](#); [Böhm-Vitense, 1958](#); [Cox & Giuli, 1968](#)), we use an MLT parameter of $\alpha_{\text{MLT}} = 1.985$, selected via calibration against a model of the Sun. This is effectively a dissipativity parameter, and represents the depth at which a parcel of matter penetrates into a lower layer before becoming fully dissipated. The value was selected by modelling a $1 M_{\odot}$ star and varying only the mixing length parameter until it had a luminosity of $1 L_{\odot}$ and a radius of $1 R_{\odot}$, at the present age of the Sun.

⁹The current “best” optimisation is $\mathcal{O}(n^{2.3727})$ ([Williams, 2012](#)). DeepMind’s AlphaTensor has claimed a faster result in [Fawzi et al. \(2022\)](#), though they do not state a complexity class. This is because “...this does not yield practical algorithms.”

Mesh spacing

A natural question that arises when modelling a star is where internally to model it – there is a counterbalance between the level of detail and the time cost of the matrix inversion. The obvious answer would be to distribute a mesh equally spaced within radius of the star, however the nonhomogeneity of a star, plus the potential for very thin shells of physically important material, necessitate a more intelligent approach.

Instead of being equally spaced *radially*, these mesh points are taken to be equally spaced within a function $\mathcal{Q}$, which is taken from [Stancliffe et al. \(2004\)](#) and given for a binary that does not have AGB-specific treatment enabled by

$$\begin{aligned} \mathcal{Q}(P, T, M, R) = & c_4 \ln P + c_5 \ln (P + c_H) + c_2 \ln (P + 3P_{\text{He}}) + c_2 \ln (P + 0.3P_{\text{He}}) + c_7 \ln \left(\frac{T}{T + 2 \times 10^4} \right) \\ & + \ln \left(\frac{c_6 M^{\frac{2}{3}}}{M + M^{\frac{2}{3}}} \right) - c_3 \ln \left(\frac{R^2}{c_8} + 1 \right), \end{aligned} \quad (2.38)$$

where P, T, M, R are the total pressures, temperatures, masses, and radii of each star, the constants c_i are user controllable and given in Appendix B, P_{He} is the pressure in the helium-burning shell, and

$$c_H = \begin{cases} 0.1P_H & X_H \geq 0.15 \\ 0.1\mathcal{P} & \text{otherwise} \end{cases} \quad (2.39)$$

where P_H is the pressures in the hydrogen-burning shell. Here, $\mathcal{P}$ represents the pressure in the hydrogen-burning shell when $X_H = 0.15$, and its inclusion obviates numerical issues if the envelope is removed and there is no hydrogen burning occurring. The choice of mesh spacing coefficients $c_1 \dots c_{10}$ leads to different areas of the star being prioritised. Coordinates c_1, c_9 , and c_{10} are not used except in AGB-specific evolution, which is not required for most cases (and requires a change of parameterisation in the pressure-shell terms). During binary evolution, c_5 is hardcoded to be 0.15, and c_2 to be 0.0, to focus meshpoints into the hydrogen-burning shell.

Placement of the mesh is controlled by the derivative $\frac{\partial \mathcal{Q}}{\partial k}$, where k is the meshpoint number ([Stancliffe, 2016](#)). Specifically, meshpoints are placed at locations where

$$\left. \frac{\partial \mathcal{Q}}{\partial k} \right|_k - \left. \frac{\partial \mathcal{Q}}{\partial k} \right|_{k+1} = 0. \quad (2.40)$$

The mesh is thus adaptive (insofar as it changes to place meshpoints where they are needed for evolution), non-Lagrangian (insofar as the derivative is not at constant mass coordinate), and inviscid (insofar as the gradient of $\mathcal{Q}$ is constant through meshpoint number). This combination of requirements is sometimes referred to as an Eggletonian mesh. If one desires to make the mesh viscous, one can follow [Stancliffe et al. \(2004\)](#) and write, for $0 \leq \gamma \leq 1$,

$$\left(\frac{\partial \mathcal{Q}}{\partial k} \Big|_k - \frac{\partial \mathcal{Q}}{\partial k} \Big|_{k+1} \right) (1 - \gamma) + \gamma \left(\frac{\partial m}{\partial t} \Big|_k \right) = 0, \quad (2.41)$$

where now the parameter γ is a weighting coefficient such that if $\gamma = 1$, then the mesh spacing function is irrelevant and the parameters are spaced equally in mass, and if $\gamma = 0$, the original mesh spacing function is used. In between, if $\gamma = \gamma(k, dt)$, then parts of the mesh are allowed to be Lagrangian and parts non-Lagrangian. We typically keep the mesh non-viscous and non-Lagrangian, as a viscous mesh is most useful for AGB evolution and evolving through thermal pulses.

More detail is provided in [Eggleton \(1971\)](#) for the original mesh spacing function, and [Stancliffe et al. \(2004\)](#) for the binary mesh spacing function.

Timestep control

The STARS code utilises an adaptive timestep control mechanism, whereby the timestep is allowed to shrink to model rapid changes or grow if the star is broadly inert and unchanging. The core computation is a parameter Δ , given by

$$\Delta = \frac{d_{\text{opt}}}{\sum_i \sum_j |\delta x_{ij}|}, \quad (2.42)$$

where d_{opt} is a pre-defined target timestep (typically taken to be ~ 1 , unless otherwise specified; see Appendix B for our selections), and δx_{ij} is the change in variable x_i at meshpoint j . In this calculation, luminosity is explicitly excluded as this would cause the timestep to explode during certain points of the star's lifetime. A new timestep, d' , is selected from the previous timestep d by

$$d' = \begin{cases} \min[d_1, \Delta] d & \Delta < 1 \\ d & \Delta = 1 \\ \max[d_2, \Delta] d & \Delta > 1 \end{cases}, \quad (2.43)$$

where d_1 and d_2 are pre-determined control parameters. Most commonly we take $d_1 = 0.1$ and $d_2 = 1.2$ unless otherwise specified.

Numerical instabilities

Stellar modelling is a nontrivial endeavour and there are many cases where the code can fail seemingly arbitrarily. The largest cause of numerical instability is the helium flash, which occurs in low-mass stars between about $0.8 M_{\odot} \lesssim M_{\text{ZAMS}} \lesssim 2.5 M_{\odot}$. In this band, the triple- α process occurs within a highly electron-degenerate helium core. As a result, the temperature of the star increases sharply during this process, leading to thermal runaway. This temperature increase in turn results in an increase in the luminosity (which peaks at $\sim 10^{10} L_{\odot}$). The flash itself – being the period of thermal runaway – occurs on the order of a few minutes, though the spike in luminosity lasts approximately 7 Myr (for the $2.51 M_{\odot}$ star shown in Figure 2.13).

A method termed “pseudoevolution” can be used to overcome this source of error. Briefly, a complete model (termed the *reference model*) of a $3 M_{\odot}$ star is evolved with a full model description saved at every timestep. When the helium flash occurs in a model (heuristically, $0.1 \leq M_{\text{He}}/M_{\odot} \leq 0.55$ and $\log T_c/\text{K} > 7.9$), we query the reference model for a model with the same helium core mass as the failed model. Using a custom subroutine in STARS, we turn off all nuclear and thermal burning and remove mass from the reference model to match the mass of the failed model. This model is then allowed to evolve as per normal, having overcome the helium flash. We note that this is not a *perfect* method, as this does induce discontinuities in the plots, as shown in Figure 2.14, though these are essentially insignificant.

The other stage where low-mass stars are prone to fail is during a thermal pulse (Schwarzschild & Härm, 1965, 1967). The mass range for thermal pulses is broader, from $\sim 0.8 M_{\odot}$ to $\sim 8 M_{\odot}$. A thermal pulse occurs as a star ascends the AGB; as a CO core or ONe core has developed, shells of burning helium and hydrogen (separated by a region of almost pure helium) sit atop the core (Pols et al., 2001). As the intershell region dissipates due to dredge up or helium burning, hydrogen burning reignites in the hydrogen shell. As both shells burn (at the same rate), the helium-burning shell becomes unstable and eventually a process of thermal runaway begins (similar to the helium flash, except off-core). This causes a spike in the luminosity, termed a thermal pulse. This can persist for many pulses – Stancliffe (2006) observes pulses before model termination, while Stancliffe (2005) observes 17 pulses for a model similar to those in Figure 2.13.

The method to overcome this numerical instability is involved and not fully implemented in STARS. Briefly, the number of meshpoints must be drastically increases near the onset of thermal pulses; up to 999 instead of the typical 199/499. The timestep must also be forcibly decreased; thermal pulses typically last for around a year (Pols et al., 2001) so the timestep must be reduced to not miss them (and to not have the modulus of the change in parameters – $\sum_i \sum_j |\delta x_{ij}|$ in Eqn(2.42) – and thus the error, increase too much). This method was implemented to generate Figure 2.13, though only two pulses were achievable.

Critical modifications

The major modifications we have made to the STARS code are as follows.

Firstly, the floating-point precision has been increased to double precision for all variables. This has minor effects on the evolved models, insofar as they are generally able to evolve to slightly further stages. This step has previously been undertaken, however not all variables were correctly converted. This did not invalidate any models previous versions of the STARS code has generated; rather it means they terminated slightly earlier than the point to which they may have been able to evolve.

As part of this, we also undertook an extensive refactor of the codebase. Whilst some of the code remains as originally written, it is now more compliant with modern programming practices and is in some sense easier to alter with new routines.

We have also improved mass loss interpolation in the MASSLOSS.F routine. Whilst not changing the underlying assumption that mass transfer is governed by de Jager et al. (1988); Vink et al. (2001) (Section 2.4.3), we have improved the transition between typical O-type mass loss and WR mass loss rates by slowing turning

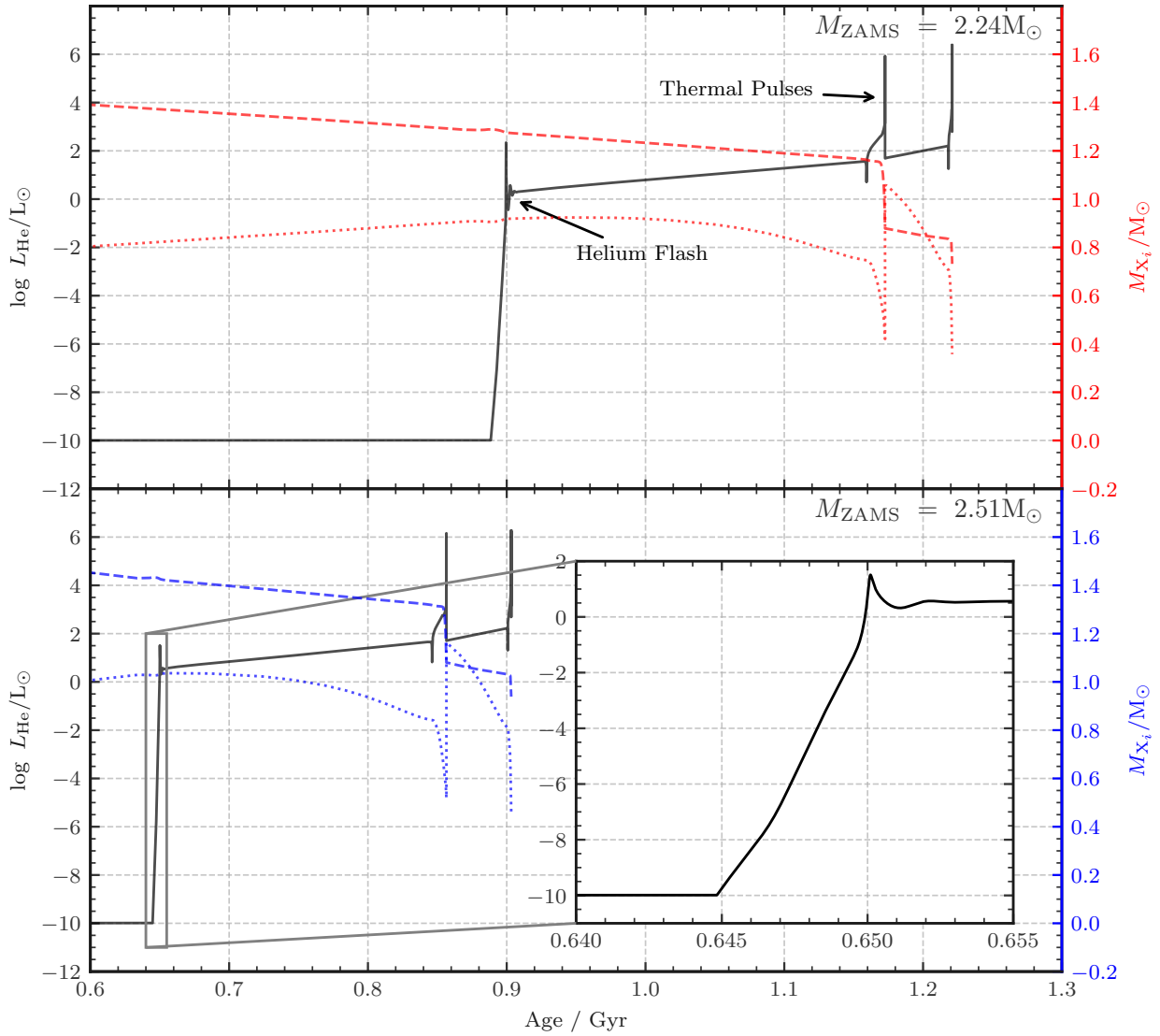


Figure 2.13: Two stars of a similar mass that undergo a helium flash and two thermal pulses. Shown in the inset is a zoomed-in helium flash that lasts ~ 7 Myr. Both stars undergo two thermal pulses before the code crashes due to the relative error in the calculation increasing too rapidly. The two stars could, in principle, undergo many more thermal pulses before their deaths.

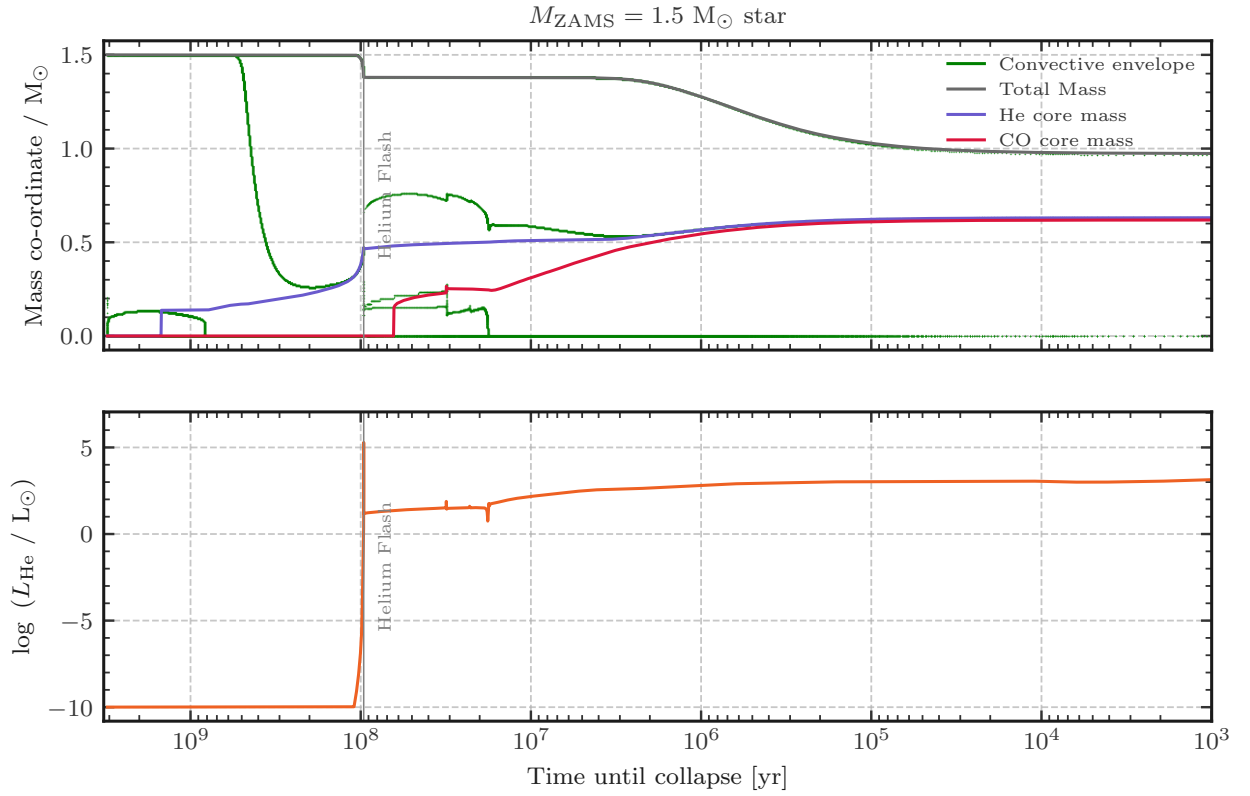


Figure 2.14: The helium flash (gray line) shown on a Kippenhahn diagram. The total mass is shown as the solid gray line, and the CO core mass is shown in red. The green lines show the boundaries of the convective regions of the star.

on the WR mass loss rate as the surface hydrogen abundance drops below 0.4. The transition was previously abrupt and would cause the code to fail due to numerical instability if a non-WR star entered this regime.

Similarly to this, the code now manually reduces the timestep in the run-up to, during, and after the contact binary phase. Evolution during this phase breaks thermal equilibrium (i.e., the star evolves on a longer timescale than its Kelvin-Helmholtz time-scale) and hence the evolution can change rapidly as the star attempts to adjust back to thermal equilibrium. The contact phase, for example, typically last for between 100-200 kyr in our models.

We have also begun implementing an $\alpha - \lambda$ treatment of CEE, though we stress that this has not been rigorously tested yet, and is thus disabled for the majority of our models.

2.6.2 | The TUI code

TUI is the core population synthesis code within the BPASS project. TUI takes as input the output from the STARS code. TUI was first described in [Ghodla et al. \(2023\)](#), and was further described in [Stevance et al. \(2023\)](#), though we note that their version is technically a refactor of the version we use. The predominant function we use TUI for is the matching of pre- and post- supernova models for Chapter 4, and in this section we provide a brief account of that functionality of TUI (though the control of the population synthesis models is beyond the scope of this thesis).

TUI has precompiled “input_” files which report a STARS plot file, an IMF weight, and a model type

(merger/normal primary/normal secondary/single model/quasi-homogeneous evolution binary/single star). After loading in a model, TUI applies a given kick and computes the new orbital parameters from [Tauris & Takens \(1998\)](#). Crucially, a core assumption of BPASS is that the orbit remains circular after the first supernova and is allowed to be eccentric after the second. Of course, nature cares little for our assumptions, and the supernova may cause the binary to become eccentric after the first supernova; TUI accounts for this by setting the separation after the first supernova to be the semi-latus rectum of the eccentric binary. TUI repeats this process after the second supernova.

For the purposes of Chapter 4, if the binary remains bound (has $0 \leq e < 1$) after the second supernova, we record the following parameters:

1. The mass of the primary,
2. The mass of the secondary,
3. The orbital separation,
4. The eccentricity,
5. A weight term proportional to the IMF weighting,
6. The rejuvenation age, which is the time since the companion last accreted material, if it accreted more than five per cent of its initial mass (see [Richards et al., 2023](#)), and
7. The evolution age, which is the length of time from the secondary's ZAMS until its death.

This provides us with the post-secondary supernova orbital parameters which determine the properties of the merger, such as the event rate $\mathcal{R}(z)$ and its path through period-eccentricity space.

2.6.3 | The KAITIAKI grid-watcher code

KAITIAKI is a grid-watcher code we have developed that sits atop the STARS code and manages its execution. It was developed primarily to manipulate the fixed-width text files that STARS uses, in contrast to the “typical” manipulation done through the command line. However development proceeded in such a way that it now provides an interface to all of the (relevant to this thesis) output and input files. Further, it also now allows the user to control jobs through a PYTHON interface, greatly simplifying grid calculations.

The core of KAITIAKI is the `STARSController()` interface. The most naïve run would be to initialise a model of, say, $8 M_{\odot}$ and evolve it until it evolves up the AGB. A sample of code for this is shown in Code Sample 2.1 which evolves a $10 M_{\odot}$ star.

Code Sample 2.1: A KAITIAKI script to evolve a $10 M_{\odot}$ star

```
1 import kaitiaki
2
3 # First, instantiate the STARS code. run_bs is the location of the run_bs
4 # bash script, which sets up the symlinks and actually runs the code.
5 STARS = kaitiaki.STARS.STARSController(run_bs='..')
6
```

```

7  # Loads the default data file and solar metallicity opacity table
8  STARS.blit('z020')
9
10 # Loads a 1 Msun model
11 STARS.load_default_modin()
12
13 # Sets the code up to target a ZAMS mass of 10 Msun.
14 STARS.setup_zams_inflation(10)
15
16 # Executes the code, such that the final modout will be a 10 Msun model.
17 STARS.run()
18
19 # Equivalent to "tail -399 modout > modin" -- transfers the last 2*199+1
20 # lines from modout (the last converged model) to modin
21 STARS.modout_to_modin()
22
23 # Again, to reload data
24 STARS.blit()
25
26 # Set up the key parameters for our run
27 STARS.configure_parameters({
28     'DT1': 0.1, # Fraction of timestep to allow for lower variation
29                 # (if the timestep is cut, it can be cut to 10% of its
30                 # present value)
31     'DDD': 2,   # Modulus of total desired timestep
32     'IML1': 5,  # WR mass-loss rates
33     'RML': 0,   # This was set to 10 Msun during inflation -- should be
34                 # zero so no mass is added except via the usual ML channels
35                 # (this parameter, formally, sets a constant ML rate so
36                 # that in the absence of other mechanisms, the mass would
37                 # always decrease)
38     'IMODE': 1, # Single star mode
39     'IX': 1,   # Hydrogen burning
40     'IY': 1,   # Helium burning
41     'IZ': 1,   # Metal burning
42     'ITH': 1,  # Thermal energy generation
43     'ISTART': 1 # Reset age, NMOD, and the timestep (to 10% of T_KH)
44 })
45
46 # Execute the model, run until the AGB.
47 STARS.run()

```

2.7 | Chapter Summary

*If in the first act you have hung a pistol
on the wall, then in the following one it
should be fired. Otherwise don't put it there.*

— ANTON CHEKHOV

In this Chapter, we have explored the evolution of stars from pre-MS evolution to supernova, accounting for the effects of binary interaction. This represents the state of the field as at the commencement of this work. In future chapters we will detail the relevant changes we have made to these prescriptions, motivating the work we have performed.



CHAPTER 3

Helium Rich Mass Transfer



All hands, this is the captain. Begin prepping all stations for first contact.

— *Captain Ed Mercer* —



SINCE THE FORCE BETWEEN TWO objects due to gravity scales like $F \sim \frac{1}{r_{\text{sep}}}$, another question naturally arises during the consideration of contact binaries. For sufficiently close binaries, one can imagine that mass transfer through RLOF or tidal effects could become reasonably strong, stripping the hydrogen envelope of a main sequence (or shortly post-main sequence) star and exposing the helium-enriched core underneath. If this stripping does not cause the star to retreat within its Roche lobe, then this core material would continue to be accreted, creating helium-enriched matter streams accreting onto its companion.

As discussed in Section 2.4, the presence of a companion measurably alters the evolution of a star. Typically this manifests as allowing matter to be exchanged between the components or the gravity of one star warping the photosphere of the other. In this chapter we focus on the first instance, where mass transfer occurs between two stars. In particular, we consider a binary system with ZAMS masses of $30 + 10 M_{\odot}$, and an initial period of $10^{0.6} \approx 3.98$ d. We specifically explore this case as a case study on the effect of helium enriched mass transfer. Future work will be needed to examine the prevalence of the effect of this mass transfer on a grid of close binaries.

3.1 | Motivation

The recent discovery of intermediate-mass helium stars in the “gap” between subdwarf helium stars and Wolf–Rayet (WR) stars (Drout et al., 2023; Götzberg et al., 2023) is a significant shift in the understanding of the importance of interacting binary stars for stellar populations. These stars have $\sim 8\text{--}25 M_{\odot}$ progenitors that have had their hydrogen envelope stripped via binary interaction. Although many groups have investigated the importance of these objects for stellar populations, such as sources of ionising photons (e.g. Stanway et al., 2016; Götzberg et al., 2017; Xiao et al., 2018), hydrogen-poor core-collapse supernovae (e.g. De Donder & Vanbeveren, 1998; Smith et al., 2011; Eldridge et al., 2013; Eldridge & Maund, 2016; Laplace et al., 2021) and gravitational wave transients and/or γ -ray bursts (e.g. Eldridge et al., 2011; Qin et al., 2018; Bavera et al., 2020; Fuller & Lu, 2022; Stevance et al., 2023), until direct confirmation of their existence, uncertainty remained about their validity. Importantly, now with the potential detection of these objects, we can begin to test and verify our models of these stars.

The evolution of a star in a binary is complicated in many ways due to the presence of its companion. Notably, the two stars may exchange mass, altering the orbit dependent on the mass of the donor and accretor stars. Much study has been devoted to the nature of binary mass transfer (among others, Delgado & Thomas, 1981; Soberman et al., 1997; Davis et al., 2013; Landri et al., 2024). In some studies, such as Dutta & Klencki (2023), which concentrate on the evolution of the donor star, the accretor is assumed to be a point mass. However, studies where the companion is followed in detail show that the evolution of the accretor can be exotic compared to that of a star from single stellar evolutionary pathways, (e.g. Wellstein et al., 2001; Renzo & Götzberg, 2021; Renzo et al., 2023; Menon et al., 2024; Schneider et al., 2024; Wagg et al., 2024).

Most previous work has focused on the structure of the donor as it becomes a helium star. Epochs of mass

transfer alter the structure of the constituent stars in a binary (Law-Smith et al., 2020; Klencki et al., 2021; Renzo & Götzberg, 2021), but the accretor has only been investigated superficially (e.g., Packet, 1981; Cantiello et al., 2007; de Mink et al., 2013; Renzo & Götzberg, 2021). These studies indicate that the accretor requires further investigation. The accretors might, as a group, provide the most unusual and exotic stellar objects we observe in the Universe, such as chemically homogeneous stars (de Mink et al., 2013; Ghodla et al., 2023) and certain types of chemically peculiar stars (e.g., Hunter et al., 2008; Langer, 2012). Thus, the accretors in a binary merit further investigation.

There are open questions in the evolution of massive binary stars that a further study of accretors may provide insights into. These include, for example, the discrepancy between the spectroscopic mass and the evolutionary mass (see, for example, Herrero et al., 1992; Massey et al., 2009; Bouret et al., 2013; Massey et al., 2013; McEvoy et al., 2015; Markova et al., 2018; Ramachandran et al., 2018; Bestenlehner et al., 2020). Mass transfer alone seems insufficient to explain this, given that most codes use an implementation or variant of the so-called “Dutch” mass loss scheme of de Jager et al. (1988); Vink et al. (2001). Rather, this discrepancy has commonly been attributed to deficiencies in the physics of model atmospheres (Herrero et al., 1992) or, more recently, missing or uncertain physics in the stellar evolution models (Bestenlehner et al., 2020). It has been observed through multiple modelling efforts such as BONNSAI (work in Bestenlehner et al. 2020, BONNSAI in Schneider et al. 2014), the work of Maeder (1990) (Herrero et al., 1992), the Geneva tracks (work by Markova et al. 2018, Geneva code by Ekström et al. 2012) and the PoWR code (work by Ramachandran et al. 2018, PoWR code in Gräfener et al. 2002; Hamann & Gräfener 2003; Sander et al. 2015).

Previous studies of close binaries suggest that they experience Case A mass transfer, i.e., mass transfer during the main sequence, and produce Algol systems in which the less massive component fills its Roche lobe (e.g. Leung, 1989; Pols, 1994; Vanbeveren et al., 1998; Wellstein et al., 2001; de Mink et al., 2007; Sen et al., 2022). In tight binaries, this can lead to a contact phase, with both stars filling their Roche lobes. Contact binaries were studied as early as the 1970s (Robertson & Eggleton, 1977), though it is only in the last few decades that progress has moved from the lower mass W Ursae Majoris binaries to massive contact binaries – Menon et al. (2021a) was the first, providing a study of the LMC and SMC. Most work focuses on the population properties of these contact binaries, with few in-depth analyses of individual systems. As in Sen et al. (2022), if the mass transfer rate exceeds the slow Case A regime (where mass transfer is on the nuclear time-scale) and enters the fast Case A regime (where it is on the thermal time-scale), the hydrogen-rich envelope of the star may become depleted; thus, the helium core is exposed and the material being accreted may become helium rich. An alternative scenario is a main sequence merger event if the orbit hardens sufficiently; this generally requires filling of the second Lagrange point as well (Tylanda et al., 2011; Nandez et al., 2014).

The effect of helium-enriched mass transfer has previously been studied (e.g., Delgado & Thomas, 1981). Most authors, however, treat the problem of Case BB helium mass transfer as having originated from a helium star on the helium main sequence, rather than as emerging from previously hydrogen-dominant mass transfer. In the latter case, it is specifically binaries formed as a pair from *birth* that experience this mass transfer,

with helium-rich material originating from core hydrogen burning in the donor star (e.g., [Renzo & Götberg, 2021](#)).

Mass transfer of particularly helium-rich material would result in a binary in which thermohaline mixing becomes important to include. This is the process by which an inverted mean molecular weight profile is close to its stability limit due to the temperature gradient (e.g. [Ulrich, 1972](#); [Kippenhahn et al., 1980](#); [Charbonnel & Zahn, 2007](#); [Angelou et al., 2011](#); [Fraser et al., 2022](#)). Thermohaline mixing was quantitatively introduced by [Ulrich \(1972\)](#) by analogy to a saline solution such as ocean water ([Stern, 1960](#)). The inclusion of the thermohaline mixing mechanism allows material with a greater mean molecular weight to settle towards the core of the star, which one expects to become dominant during the accretion of helium-rich material. It is commonly included in most binary evolution models (e.g., [Fragos et al., 2023](#); [Sen et al., 2023](#)). However, it is difficult to directly constrain the strength of thermohaline mixing by direct measurement.

In this Chapter, we investigate both the impact of helium-rich material on an accretor star, and the thermohaline mixing that this induces, with an example binary stellar evolution model. Our aim is to demonstrate that it is important to account for the varying composition of material transferred between the donor and accretor in a binary. We also aim to determine the effect of the helium-rich material on the accretor and investigate the impact of thermohaline mixing on the resultant composition profiles created in the accretor.

3.2 | Numerical Method

Our general method remains the same as outlined in Chapter 2, and we use the AOTEAROA STARS code outlined in Section 2.6.1. We outline here the key changes made for this work.

During accretion, the infalling material may have a composition different from that of the accretor’s surface layer. We allow for *variable composition accretion* (VCA) in the following way. If one star is filling its Roche lobe, we set the abundance of the accreted material to the surface abundance of this star. If both stars are filling their Roche lobes, we set the abundance of the accreted material to the *average* of the two surface abundances of both stars. The primary reason for this modification is that if both stars are filling their Roche lobes, not averaging their abundances would mean that we would be artificially altering the composition of the accretor’s surface.

To evaluate the importance of this VCA, we also create a model where we do not link the surface composition of the accretor to that of the donor, with this being our *non-variable composition accretion* (NVCA) model. In these models, we set the abundance of the accreted material to that of the surface of the accretor, which is in line with the default behaviour of MESA ([Paxton et al., 2015](#)).

This variable accretion will result in an inverted mean molecular weight profile, resulting in thermohaline mixing. To model this, we use the prescription of [Kippenhahn et al. \(1980\)](#) and [Stancliffe et al. \(2007\)](#). Thermohaline mixing in the accretor is treated as a diffusive process, where the diffusion coefficient D_{th} is given by

$$D_{\text{th}} = \alpha_{\text{th}} \frac{16aT^3 H_{\text{P}}}{(\nabla_{\text{ad}} - \nabla) c_{\text{P}} \rho \kappa} \left| \frac{d\mu}{dr} \right| \frac{1}{\mu}, \quad (3.1)$$

where H_{P} is the pressure scale height, a is the radiation pressure constant, ∇_{ad} is the adiabatic temperature gradient, ∇ is the temperature gradient, μ is the mean molecular weight, c_{P} is the specific heat at constant pressure, κ is the opacity, and ρ is the density. We also introduce α_{th} , a thermohaline mixing efficiency parameter. $\alpha_{\text{th}} = 0$ represents no thermohaline mixing, and $\alpha_{\text{th}} = 1$ represents efficient thermohaline mixing. We note that this definition differs slightly from the definition in other works, most notably MESA (Paxton et al., 2013). Our efficiency parameter can be converted to theirs using the relationship

$$\alpha_{\text{th}} = \alpha_{\text{th}}^{\text{MESA}} \cdot \underbrace{\sum_k \frac{1}{16} \frac{g_k}{P_k} (\mu_{k+1} + \mu_k) \left(\frac{\varphi}{\delta} \nabla_{\mu} - \nabla_{\text{ad}} \right) \left| \frac{\partial r}{\partial \mu} \right|_k}_{\equiv \mathcal{C}}, \quad (3.2)$$

where g is the local gravity, $\varphi = \frac{\partial \ln \rho}{\partial \ln \mu}$, $\delta = \frac{\partial \ln \rho}{\partial \ln T}$, r is the radius, and subscript k refers to a quantity internal to the meshpoint k . As this is meshpoint-dependent, and thus structure-dependent, in general the meshpoint sum will vary throughout the evolution of the star. At the onset of thermohaline mixing, the meshpoint sum in the presented model evaluates to $\approx 1.62 \times 10^{-4}$. In general throughout the VCA model the constant is between 1×10^{-6} and 1, and this is shown in Figure 3.1.

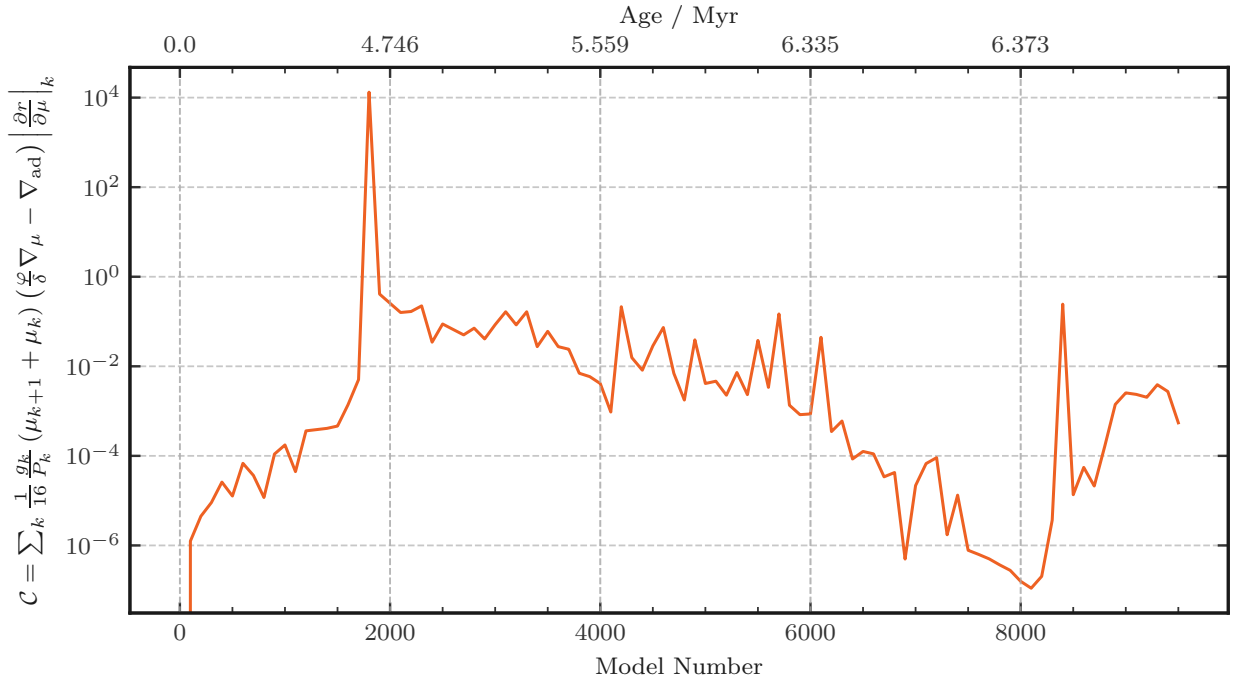


Figure 3.1: The value of the meshpoint sum $\mathcal{C}$ in Eqn(3.2) throughout VCA.

We note that our mass loss prescription is given by Eqn(2.18), with the choice of $k = 3 \times 10^{-3}$. This scaling factor is a combination of the efficiency parameters 3×10^{-6} from Hurley et al. (2002, equation 59) and 1000 from Claeys et al. (2014, equation 11). The first choice was made by Hurley et al. (2002) as a free parameter by experiment to ensure numerical stability. In the second, Claeys et al. (2014) included the additional factor to ensure that thermal time-scale mass transfer occurs with a higher mass-transfer rate while also retaining

numerical stability. We note that this is an incomplete implementation of the [Claeys et al. \(2014\)](#) prescription of RLOF, and we do not interpolate between subunity and superunity mass ratios. We have run a model with the full implementation, and find the only difference in evolution is towards the end of evolution of the donor, where the evolution becomes numerically unstable. There is also an inconsistency because in contact systems the value of k given in [Claeys et al. \(2014\)](#) depends on the mass ratio – which means that it is different for the donor and accretor, even if they are both main sequence stars. To remove this inconsistency, we leave the value of k constant as it provides numerical stability in our models. We note that this should be replaced in the future by a more physically motivated prescription such as that given by [Kolb & Ritter \(1990\)](#).

3.3 | Evolution overview

Stage	Letter	Section described
Model begins	A	Section 3.3.1
RLOF begins	B	Section 3.3.2
Contact begins	C	Section 3.3.3
Contact ends	D	Section 3.3.4
RLOF ends	E	Section 3.3.5

Table 3.1: The five major stages of the evolution of our binary.

The binary’s evolution can be broken up into five main stages, explored in the following section. Each stage is assigned a label, which is delineated in Table 3.1 – in brief, the lexicographic ordering reflects temporal evolution of the binary. That is to say, (A) represents the ZAMS, (B) the start of RLOF on the donor, (C) the start of RLOF for the accretor, (D) the end of RLOF for the accretor, and (E) the end of RLOF for the donor. Unless otherwise indicated, at every stage we shall be discussing the VCA case. The full evolution for the VCA (NVCA) accretion case is shown in Figure 3.2 (Figure 3.3), with subplots showing the behaviour around the contact phase.

3.3.1 | Pre-interaction (stages A–B)

The pre-interaction phase of both stars is identical in both cases, as the key change enabling this work only begins when RLOF occurs. In both cases, it evolves as a fully detached binary for 4.51 Myr, with the donor almost completing its main sequence evolution.

3.3.2 | Pre-contact (stages B–C)

When the donor fills its Roche lobe, it begins to efficiently transfer material to its companion. The increased drag causes the orbit to become hardened; shrinking by approximately 23 per cent. The accretor begins to accrete material, bleeding angular momentum into the system and causing the accretor to spin up. As the

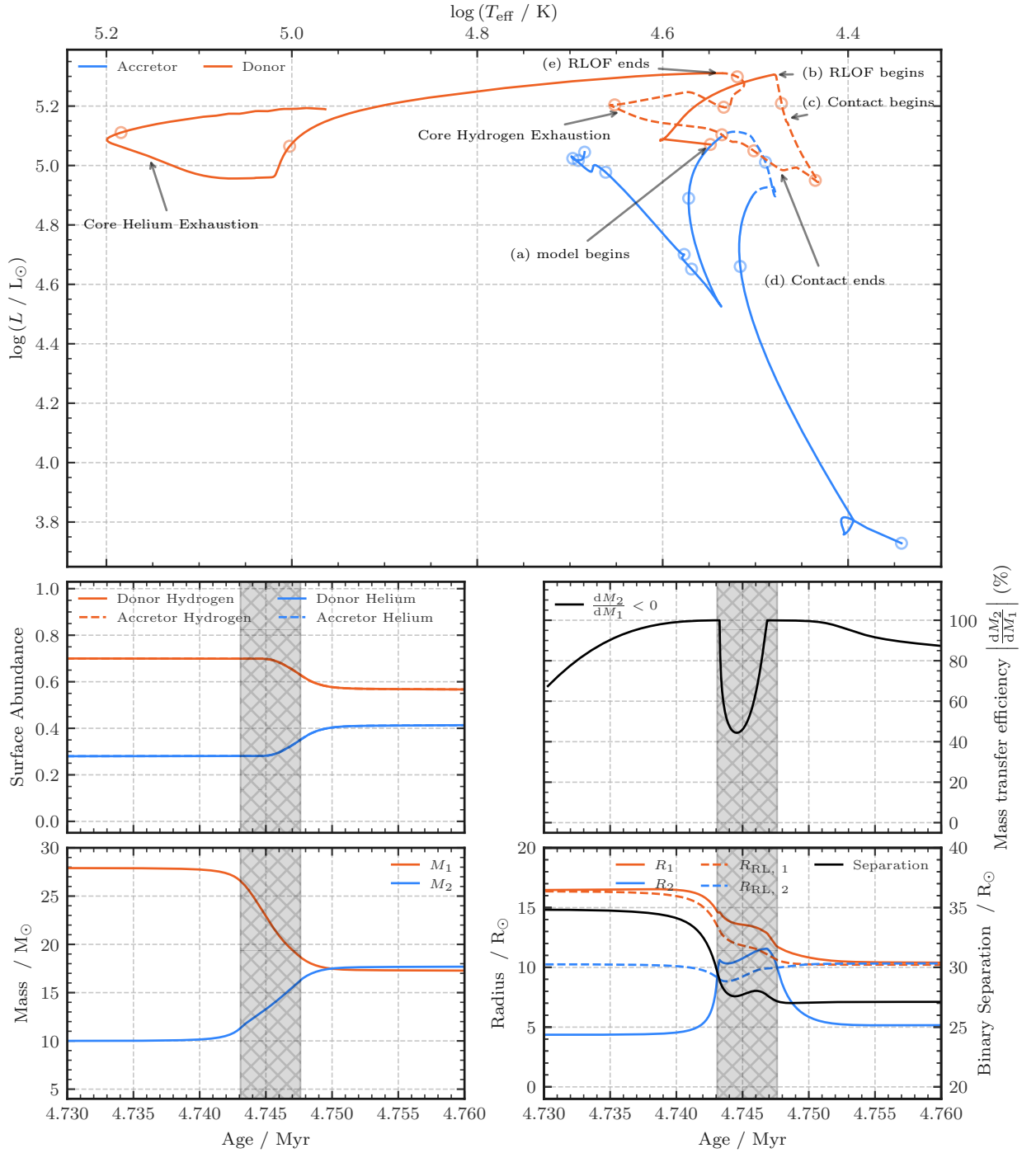


Figure 3.2: The main evolution of a $30 + 10 M_{\odot}$ binary with VCA enabled. Clockwise from top: the HRD with stages in Figure 3.7 and Table 3.2, the mass transfer efficiency, the separation / radius, the mass, and the surface abundances. In the lower four plots, the gray region shows the contact phase, and the plots are cropped to show the ~ 13 kyr on each side of the contact phase. The dashed line on each HRD track shows when the relevant star is overfilling $\mathcal{L}_1$, and contact is when both stars are dashed.

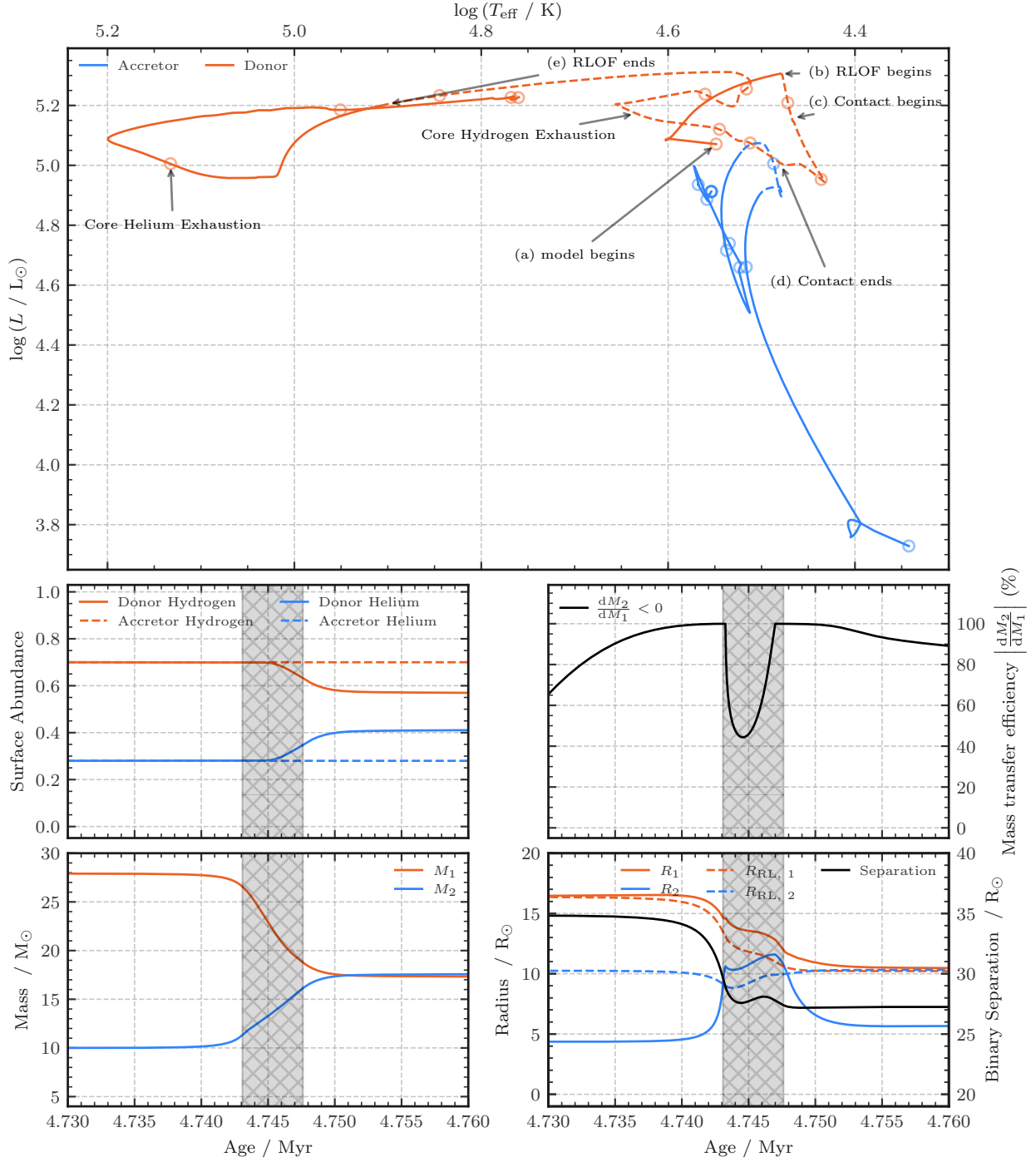


Figure 3.3: As for Figure 3.2, but for the NVCA model.

contact phase nears, the accretor approaches 40 per cent of the critical rotation rate, as shown in Figure 3.4. It reaches approximately 20 per cent when the contact phase begins.

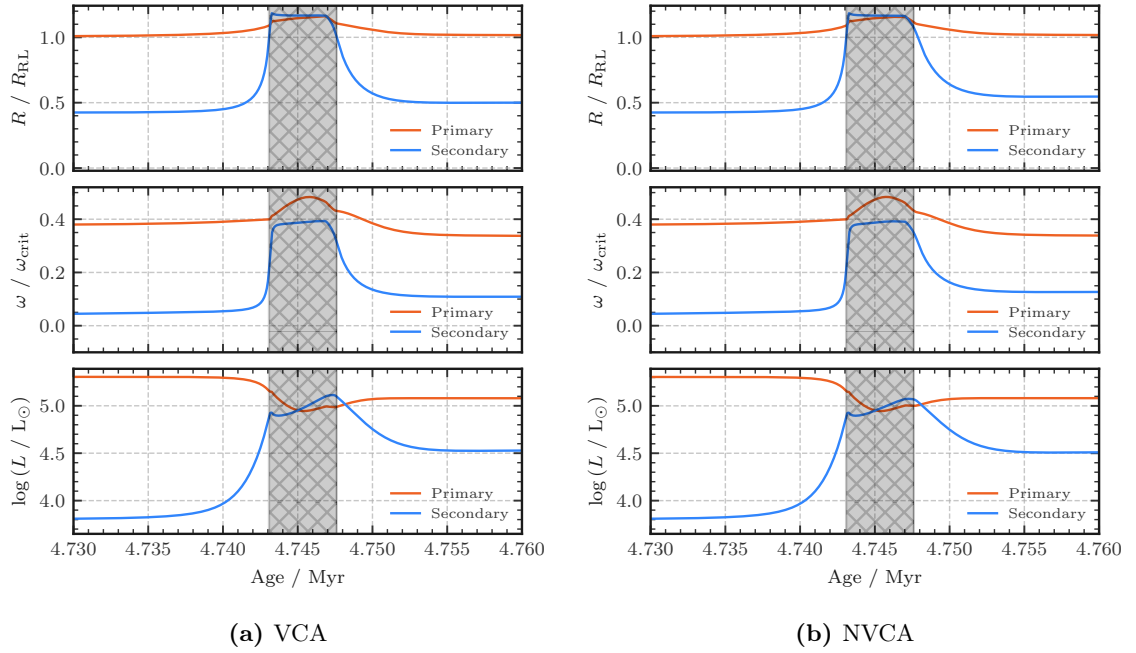


Figure 3.4: The evolution of the radius, rotation rate, and luminosity for the VCA and NVCA models, shown around the contact phase.

3.3.3 | Contact phase (stages C–D)

When the primary star exceeds its Roche lobe during its main sequence, it begins to efficiently transfer material onto the secondary. This mass transfer shrinks the orbital separation of the binary by about 23 per cent, and causes the secondary to swell and fill its own Roche lobe, forming a contact binary. The evolution in the lead up to, during, and immediately after the contact phase is shown in Figure 3.2. Notably, when the system becomes a contact binary, the separation shrinks dramatically owing to conservation of angular momentum. This results in a final period of ~ 2.8 d.

Because of the short initial period of the binary, this stellar model is an example of a system that experiences a contact phase. This is due to the large amount of mass transferred in the immediate lead-up to this phase, causing the accretor to react by increasing in radius. The process of generating and sustaining the contact phase can be broken down into the following stages, summarised in Figure 3.4:

1. In the immediate lead-up to the contact phase, the accretor begins to accrete material, causing it to spin up. The rotation rate also increases due to transfer of angular momentum and the accretor approaches 40 per cent of its critical rotation rate.
2. This accretion breaks thermal equilibrium in the accretor and causes it to expand (Kippenhahn & Meyer-Hofmeister, 1977; Neo et al., 1977; Lau et al., 2024)
3. The donor dims slightly as the accretor brightens by 63.4 per cent.

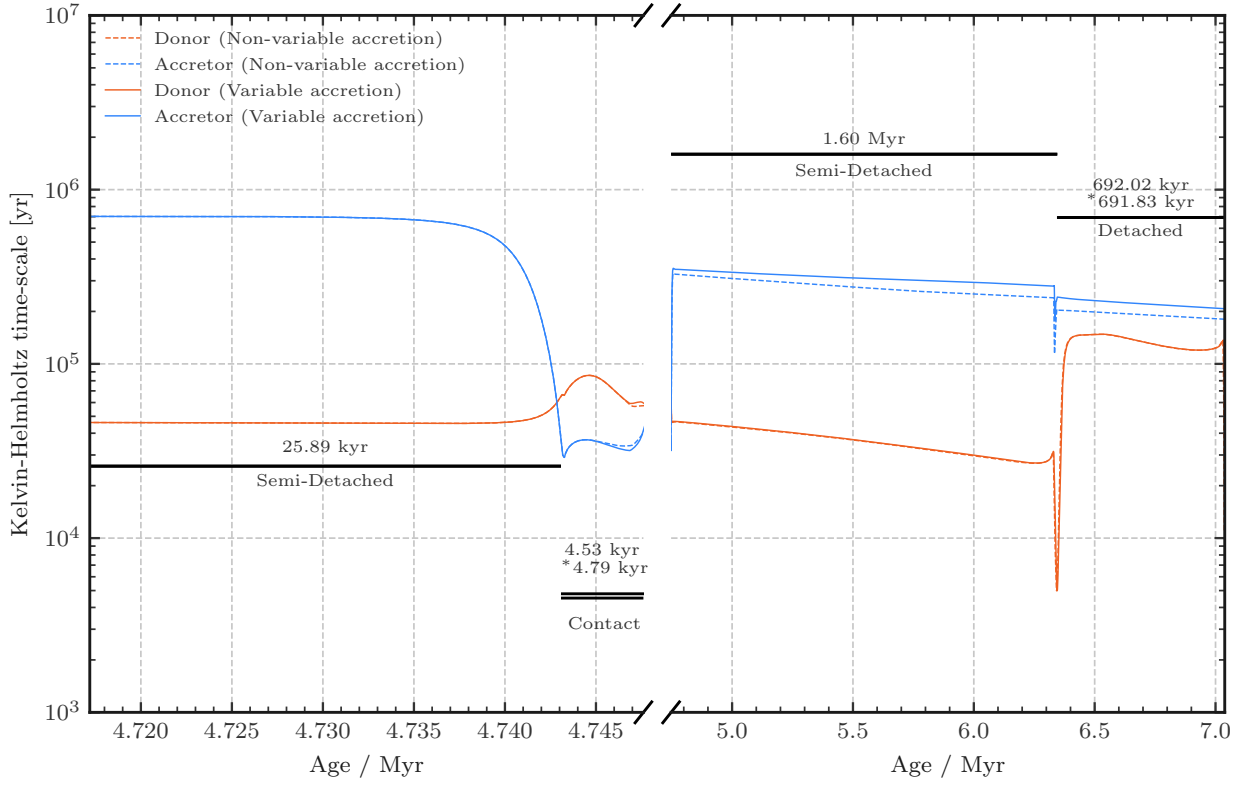


Figure 3.5: The Kelvin-Helmholtz time-scales of both stars in the VCA and NVCA regime. VCA prolongs the final detached phase and shrinks the contact phase. The solid black lines show the length τ of each phase – thermal equilibrium is broken if $\tau_{\text{KH}} > \tau$, i.e., in the first semidetached phase and the contact phase. In the contact and detached phases, the asterisk represents the duration of the NVCA phase.

4. The donor then begins to increase in luminosity, but its radius returns to approximately the size it was prior to the onset of RLOF, which is below the expected radius for a star of that luminosity.
5. The disruption of thermal equilibrium in the accretor only lasts for some time until the accretor's thermal time-scale increases so it can equilibrium. This means that the contact phase lasts for 5 kyr.

During the contact phase, the accretion of material onto the accretor is less efficient. The mass gain is limited from the the balance between its own mass loss due to filling its Roche lobe and the mass gain from the donor star. The mass transfer efficiency during this phase drop. We can see the instantaneous mass transfer efficiency fluctuates during the contact phase in Figure 3.2. The efficiency drops to below 50 per cent during the contact phase.

Eventually, the accretor shrinks within its Roche lobe and the contact phase terminates. The reason for this is that the accretor's mass has increased to such a point that its thermal timescale is now equivalent to the mass transfer rate from the donor star as shown in Figure 3.5. The accretor is thus able to achieve thermodynamic equilibrium and return to its more compact state. This shrinking is not due to the heavier helium-rich material that has been accreted, although as we note below, post-contact the accretor with VCA is smaller in radius due to this effect, as shown in Figure 3.6 and Table 3.2.

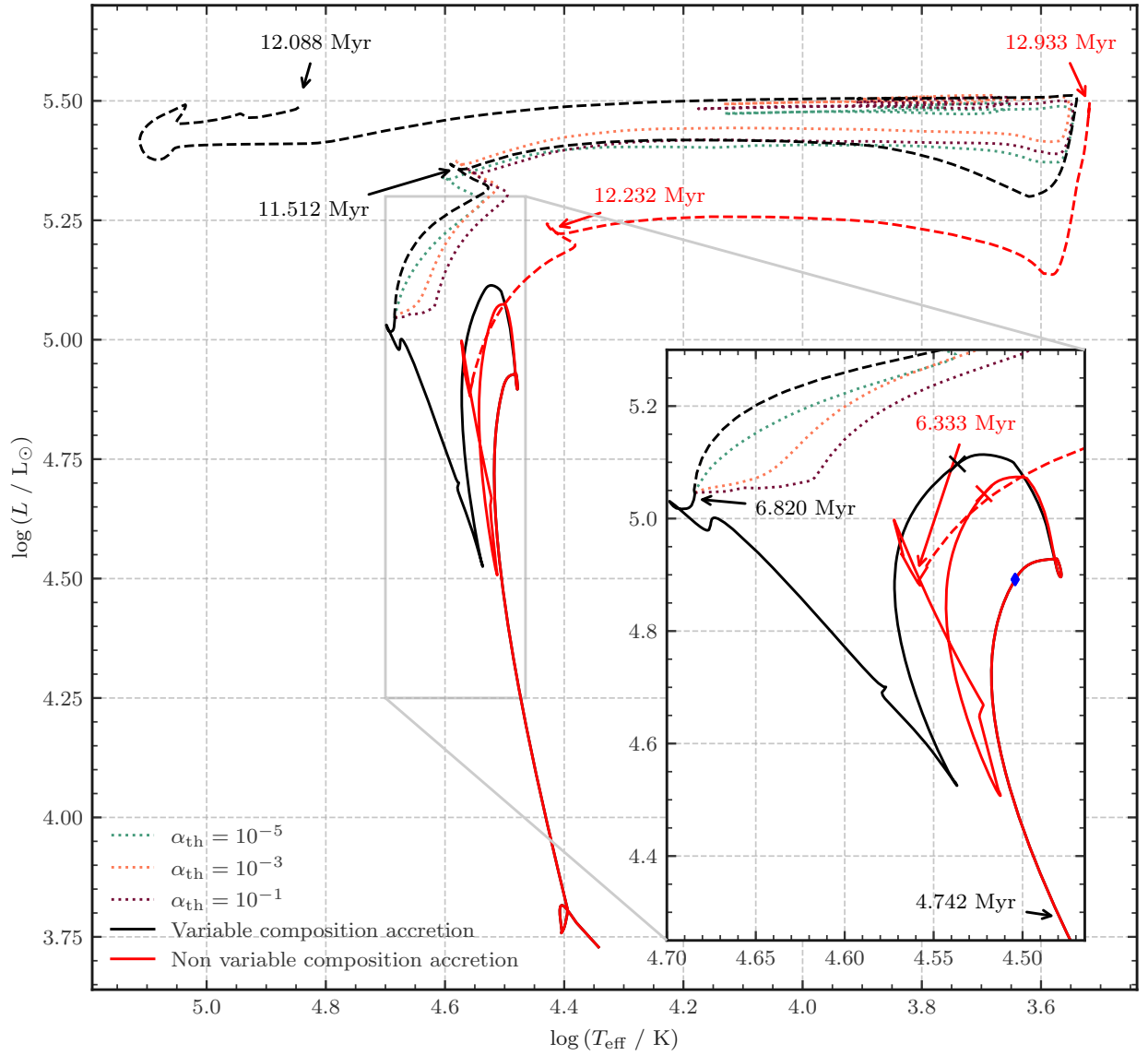


Figure 3.6: An HRD showing the evolution of our model both with (black lines) and without (red lines) the VCA prescription. In the inset, the diamond shows the start of the contact phase (identical for both models), and the coloured crosses show the end of the contact phase for their respective models. The dotted lines show models with both thermohaline mixing and VCA enabled. Models with thermohaline mixing enabled were computed with 199 meshpoints, whereas ones without were computed with 999 meshpoints. The thermohaline mixing models undergo breathing pulses prior to model termination.

3.3.4 | Post-contact (stages D–E)

There is a clear and consistent difference between the post-contact models. As shown in the inset of Figure 3.6, the models diverge shortly before the termination of the contact phase. At this point, however, the process has become irreversible. The VCA model has become 0.01 dex, or ~ 1.16 kK hotter than the NVCA model. This in turn leads the VCA accretor to be hotter and eventually more luminous than its NVCA counterpart, a theme that is reflected once the post-interaction behaviour is to be studied.

We thus examine the reasons for this bifurcation. In Figure 3.7 we show the internal composition of the two stars in each binary at each timestep. Noticeably for stage (D), the accretion of helium-rich material has introduced a layer of helium *atop* a layer of hydrogen, in the VCA case. This hydrogen is the original photosphere of the accretor, and the material has not yet mixed in at this stage, due to the continued injection of helium from the donor star.

Eventually, the extended envelope of the donor star cannot be supported, as it has lost a substantial amount of material. This leads to the collapse of the envelope, and the donor becomes a compact helium star. Over the contact phase, the average mass transfer efficiency has been 62.62 per cent, with the donor star losing $7.81 M_{\odot}$ of material and the accretor star gaining $4.89 M_{\odot}$ of material. The balance is assumed to be lost to the ISM. The instantaneous efficiency, shown in Figure 3.2 and Figure 3.3, peaks at 100 per cent immediately prior to the start and end of the contact phase, and post-contact stays that way for ~ 5 kyr after contact terminates as the system settles into a stable state with roughly constant radii, masses, and separation.

3.3.5 | Post-interaction and future history (stage E–onwards)

After the accretor detaches from its Roche lobe, its internal structure is quite different. In both the VCA and NVCA models, the donor has a hydrogen-exhausted core with a hydrogen-poor envelope. However, in the VCA model, the profile of the accretor is more complex. As shown in Figure 3.7 (stage D), the accretor has a helium-rich core ($Y_{\text{core}} \sim 0.35$) from the result of ongoing core hydrogen burning, a hydrogen-rich lower-envelope region above the core, with an outer-envelope composition profile of increasing helium abundance towards the surface. The formation of helium stars must require some amount of helium-rich mass transfer to a companion, thus, such accretor structures must be common among the companions of observed helium dwarfs (cf. Section 3.6.3).

The evolution of these composition gradients is determined by thermohaline mixing and rotationally induced mixing, as discussed by [Renzo & Götberg \(2021\)](#). Firstly, thermohaline mixing allows the helium-rich material to mix into the stellar interior, although the time-scale of this process is uncertain. We discuss the impact of this below. Second, if the star is rapidly rotating, rotationally induced mixing (which is not included in these models) may induce further mixing of material to smooth out the composition gradient (e.g. [Eggenberger et al., 2010](#)). As noted in [Renzo & Götberg \(2021\)](#), these mixing processes create a large level of complexity in the structure and evolution of the secondary, with the speed with which the composition

gradient can be removed. However, there is still an impact on the evolution of the companion given that the average mean molecular weight of the star increases because of the extra helium that has been transferred.

3.4 | Atypical abundance profiles

As previously discussed, the accretor star has an atypical mean molecular weight profile. The discrepancy can be most seen in Figure 3.7. If thermohaline mixing is included, it would at this stage act to attempt to homogenise the star, with the heavier helium material effectively sinking to the core of the star. The effect of this will vary depending on the efficiency of thermohaline mixing, which is not conclusively known. To that end, the model of Eggleton (1972); Kippenhahn et al. (1980); Stancliffe et al. (2007) is employed, where the free parameter α_{th} controlling the strength of thermohaline mixing is allowed to vary. $\alpha_{\text{th}} = 0$ corresponds to no thermohaline mixing, whereas $\alpha_{\text{th}} = 1$ represents strong (virtually instantaneous) thermohaline mixing.

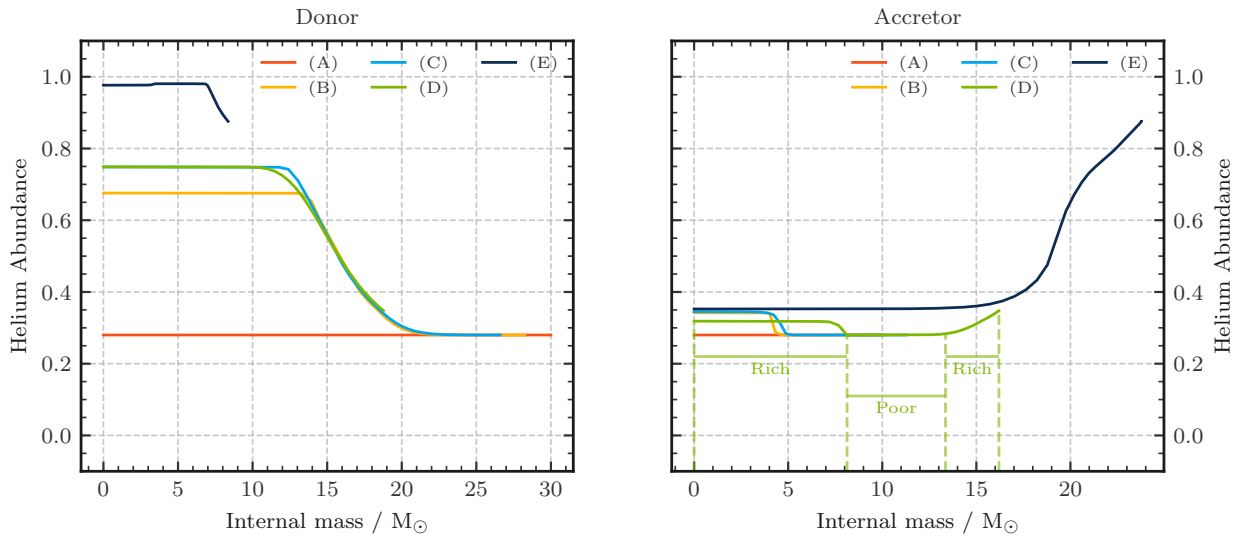


Figure 3.7: The abundance of helium at the five critical stages of the binary evolution. The green lines under the accretor’s plot show the regions where the accretor is helium-rich and helium-poor.

We show in Figure 3.9 the effect that thermohaline mixing has on the accretor. Varying the thermohaline mixing coefficients allows for the shape of the HRD track to change noticeably, with the locations of the hook changing, as well as the trajectory as the model crosses the Hertzsprung gap. However, and critically, even the most efficient thermohaline mixing included – $\alpha_{\text{th}} = 0.1$ – was unable to return the model to its location on the HRD, even though it homogenised the model only within a few timesteps.

3.5 | Comparison between VCA and NVCA

In Figure 3.6 we display a comparative HRD between the cases of VCA and NVCA. In each case an $30+10 M_{\odot}$ binary was evolved with the only difference between the two being whether material was allowed to have VCA or NVCA.

The first and most striking conclusion that we can draw from this result is that the companion is, in the

		$t = 0$ (A)	Star 1 RLOF		Star 2 RLOF		Star 1 RLOF		Star 2 RLOF	
			Begin	End	Begin	End	Begin	End	Begin	End
			(B)	(E)	(C)	(D)	(B)	(E)	(C)	(D)
Donor	$M / M_{\odot}$	30.00	27.93	7.56	26.91	17.43	27.93	7.57	26.91	17.27
	$R / R_{\odot}$	8.82	16.32	10.20	13.88	9.76	16.32	10.15	13.88	9.77
	$\log L / L_{\odot}$	5.07	5.30	5.25	5.16	4.94	5.30	5.25	5.16	4.98
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.09	0.70	0.57
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.89	0.28	0.42
Accretor	$M / M_{\odot}$	10.00	9.99	20.97	11.00	14.57	9.99	20.77	11.00	14.52
	$R / R_{\odot}$	4.87	4.36	4.02	9.23	8.98	4.36	6.53	9.23	9.02
	$\log L / L_{\odot}$	3.73	3.81	4.89	4.90	5.06	3.81	4.73	4.90	4.95
	X_{surf}	0.70	0.70	0.09	0.70	0.57	0.70	0.70	0.70	0.70
	Y_{surf}	0.28	0.28	0.89	0.28	0.41	0.28	0.28	0.28	0.28
Binary	$P_{\text{bin}} / \text{d}$	3.98	3.85	4.41	3.14	2.52	3.85	4.38	3.14	2.54
	$a / R_{\odot}$	36.14	34.74	34.56	30.28	24.71	34.74	34.33	30.28	24.78
	Age / kyr	0.00	4707.16	6450.23	4712.75	4716.42	4707.16	6448.11	4712.75	4716.70

(a) VCA
(b) NVCA

Table 3.2: The key evolutionary parameters of both stars and the binary configuration at five critical points: the initial parameters, and the beginning and ends of both epochs of RLOF. All values are rounded to 2 decimal places. The letter in parentheses represents which stage of Figure 3.7 the abundances are given in.

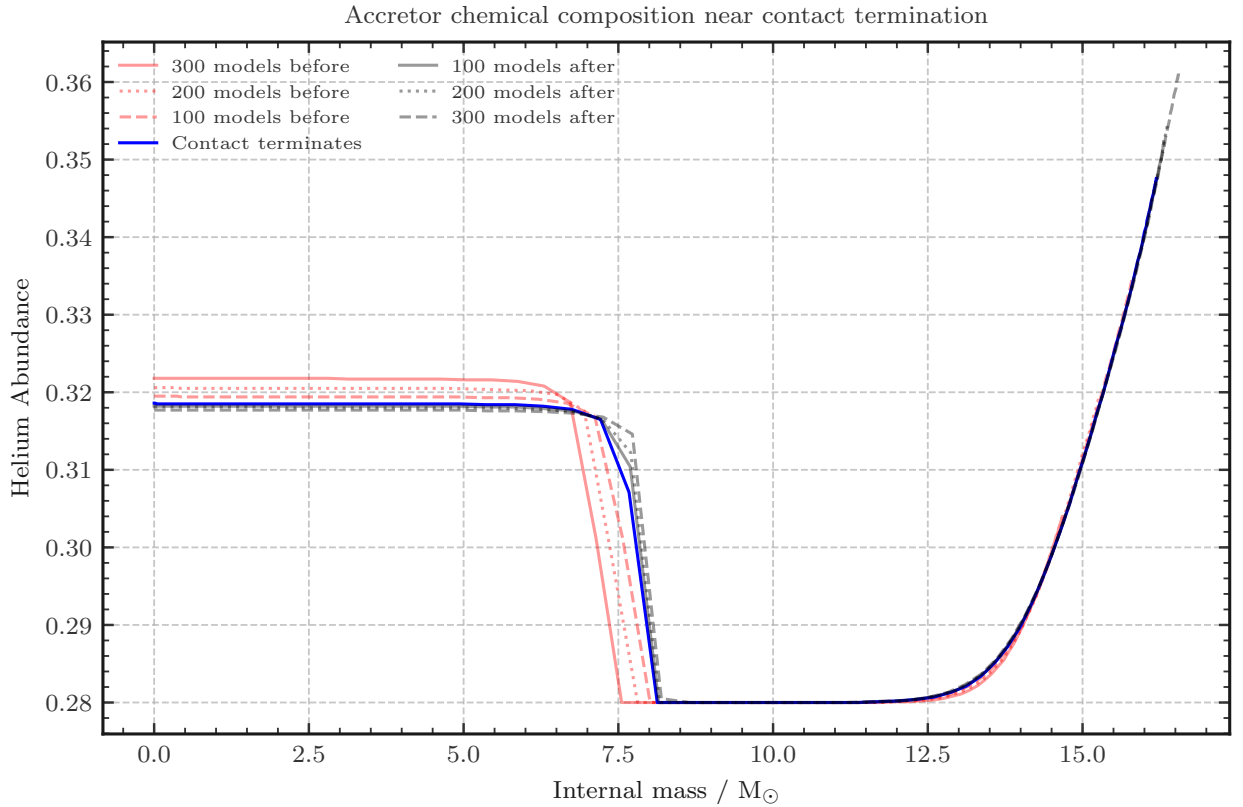


Figure 3.8: The helium abundance in the accretor near when the model detaches. The star contracts as mass transfer ends, resulting in a $\sim 9R_{\odot}$ accretor 300 models after collapse, compared to $\sim 11.5R_{\odot}$ 300 models before (a difference of ~ 1.6 kyr).

VCA case, hotter and more luminous than the NVCA case. As previously mentioned, even near-instantaneous thermohaline mixing to homogenise the star was insufficient to return the accretor to a position on the HRD typified by NVCA models.

Indeed, none of our single-star models are able to replicate the NVCA model. However, a $30M_{\odot}$ model has a similar luminosity to the accretor but its surface temperature is 0.15 dex (~ 9.6 kK) higher. This is due to the lower opacity of the helium-rich surface of the accretor. Similarly, we see that during the main sequence the accretor with VCA is also approximately 0.15 dex (~ 11.3 kK) hotter in surface temperature than the NVCA model.

Within the VCA model, the formation of an inverted mean molecular weight gradient means that thermohaline mixing must be considered. In Figure 3.6 we can see that higher values of α_{th} lead to a more rapid decrease in surface temperature by approximately 0.05 to 0.08 dex. The most efficient thermohaline mixing that we include removes the composition gradient in a few time-steps (see Figure 3.10), but it remains hotter and more luminous throughout the main sequence compared to the NVCA model. This is because while the composition gradient is removed the star has a higher helium abundance. Figure 3.6 shows that when thermohaline mixing is included we see this temperature difference decrease to approximately 0.1 dex (~ 1.54 kK). Thermohaline mixing has a bigger difference on the accretor tracks earlier in the main sequence evolution than at the end of the main sequence, where the tracks are more similar in their evolutionary path.

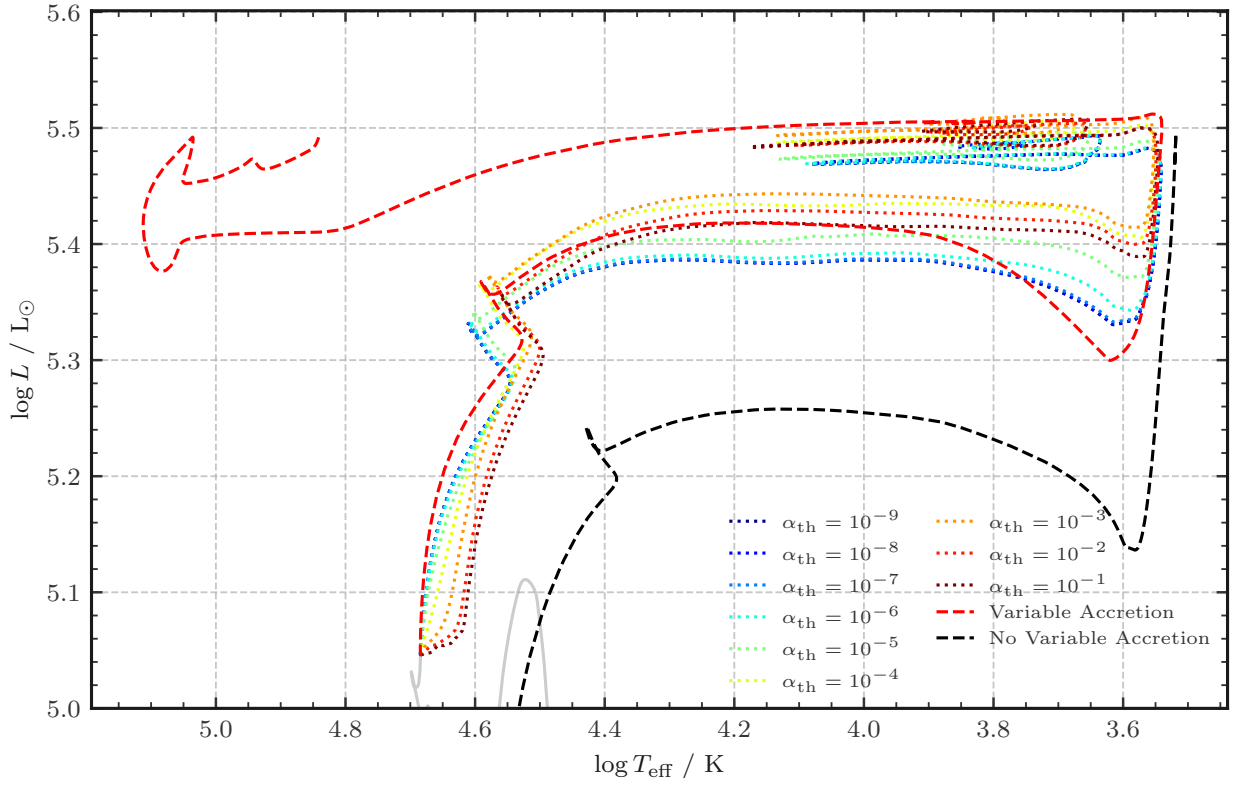


Figure 3.9: An HRD of the post-mass transfer secondary model evolved as a single star with different thermohaline mixing coefficients. The diagram has been truncated to have the y -axis begin at $\log L = 5$, with the input model shown in faint grey. The “variable accretion” and “non-variable accretion” models were evolved with thermohaline mixing entirely disabled and with 199 meshpoints. The thermohaline mixing models were evolved with 999 meshpoints, and all experience breathing pulses prior to model termination.

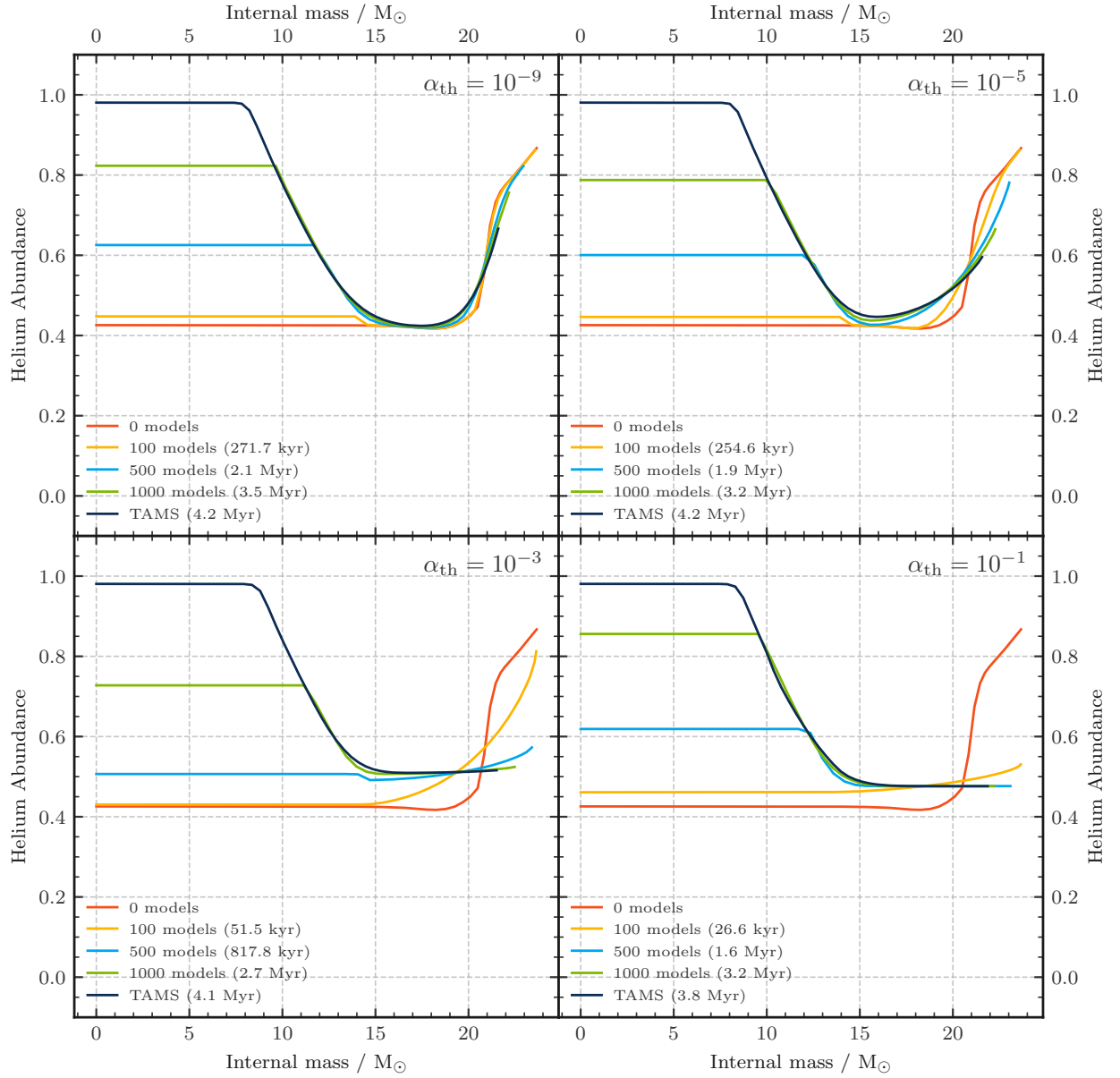


Figure 3.10: The helium abundance profile for each model of thermohaline mixing, as a function of the internal mass of the star. The legend key shows the model number and age corresponding to each line, measured from the beginning of the single-star evolution procedure described in Section 3.3.5. Only the VCA models are shown in this plot. The NCVA models have a flat composition profile at the surface.

3.6 | Implications

An important question to consider is what the VCA model would be observed as. Some of the parameters of the star are typical of single main sequence stars but with higher luminosities, helium-rich abundances, and higher surface temperatures they may be able to be identified from observations. The effective temperature of our VCA models is in the range of O5 to O3 stellar types. The hottest examples might therefore look similar to the WN/O3 stars identified by [Massey et al. \(2023\)](#), although it has previously been suggested that such stars were in fact the helium donors from binary interaction ([Götberg et al., 2018](#); [Smith et al., 2018](#)). The idea that the WN/O3 stars could in fact be the accretor stars does link well with their apparent single-star nature ([Massey et al., 2023](#)) and their relative isolation compared to other massive stars ([Smith et al., 2018](#)), if their donor stars have already exploded in a core-collapse supernova. Alternatively, it is also possible that the accretors may be observed as luminous-blue variable (LBV) stars. [Smith et al. \(2012\)](#) suggested that LBV stars may arise from accretors in binaries.

Given that the evolution of the accretor star remains different from that of a single star, the impact of VCA may provide at least part of the answer to some outstanding questions in stellar evolution. The first question is the known mass discrepancy for some stars between their spectroscopic and evolutionary masses (e.g. [Herrero et al., 1992](#); [Massey et al., 2009](#); [Bouret et al., 2013](#); [Massey et al., 2013](#); [McEvoy et al., 2015](#); [Markova et al., 2018](#); [Ramachandran et al., 2018](#); [Bestenlehner et al., 2020](#)). Comparing the luminosities of our accretor models with single-star models in Figure 3.11, we can see that a star with helium-rich accretion has a luminosity similar to a star that is 20 per cent more massive. Such a difference may explain why for some stars' mass estimates can differ significantly.

The second implication of our VCA model is that the higher effective temperatures of the models lead to more of a star's luminosity being radiated at wavelengths below 912 Å which can help drive ionisation of the surrounding media. This has important implications for understanding the nebula emission from the HII regions and reionization (e.g., [Stanway et al., 2016](#); [Eldridge & Stanway, 2022](#); [Ghodla et al., 2023](#)).

Third, work has shown that mass transfer can lead to accretors with less bound envelopes ([Laplace et al., 2021](#); [Renzo et al., 2023](#)), resulting in systems for which the energy required to explode the star in a supernova is decreased compared to an equivalent single-star model. As such, as in this case (where the mass transfer is sufficiently efficient that the final core collapse could give rise to unusual supernovae, i.e., an event that does not fit into one of the common supernovae types, IIP, IIL, IIB, Ib or Ic. The recent comprehensive work by [Schneider et al. \(2025\)](#) shows how the accretors can have very different supernovae compared to single stars. We note that it is accepted that stellar mergers also give rise to unusual supernovae (e.g. [Menon et al., 2019](#)) and future study is required to identify signatures in supernovae to differentiate between an accretor progenitor and the result of a stellar merger.

3.6.1 | Spectroscopic masses

Animals can talk.

— THIRD OF THE SIX IMPOSSIBLE THINGS

Given that the evolution of the accretor star remains different from that of a single star, the impact of VCA may provide at least part of the answer to some outstanding questions in stellar evolution. The first question is the known mass discrepancy for some stars between the mass predicted by models (the “evolutionary mass”) and the mass inferred from spectroscopic studies of stars (the “spectroscopic mass”). Summaries are found in works such as [Herrero et al. \(1992\)](#); [Massey et al. \(2009\)](#); [Bouret et al. \(2013\)](#); [Massey et al. \(2013\)](#); [McEvoy et al. \(2015\)](#); [Markova et al. \(2018\)](#); [Ramachandran et al. \(2018\)](#); [Bestenlehner et al. \(2020\)](#). The reason has been left as either deficiencies in the physics of model atmospheres ([Herrero et al., 1992](#)) or non-included or not well-understood physics ([Bestenlehner et al., 2020](#)).

This phenomenon is not unique to one code or methodology. To name but four, it has been observed in:

1. BONNSAI (work in [Bestenlehner et al. 2020](#), BONNSAI in [Schneider et al. 2014](#)),
2. the work of [Maeder \(1990\)](#) ([Herrero et al., 1992](#)),
3. the Geneva tracks (work by [Markova et al. 2018](#), Geneva code by [Ekström et al. 2012](#)), and
4. the PoWR code (work by [Ramachandran et al. 2018](#), PoWR code in [Gräfener et al. 2002](#); [Hamann & Gräfener 2003](#); [Sander et al. 2015](#)).

If luminosities are systematically incorrect, as discussed above, then there would be a corresponding systematic incorrect calculation of the mass; that is, a traditional mass-luminosity relationship may not hold. We show in Figure 3.11 the relationship between the evolutionary mass and luminosity for both VCA and NVCA cases, as evolved in subsection 3.3.5. We also display this for integer mass single-star models. Unsurprisingly, the NVCA case evolves like a $24 M_{\odot}$ model – the initial mass of the model when this single-star evolution began was $23.71 M_{\odot}$, and the abundance profile typifies a single star; since the mass transfer was hydrogen-dominated, the profile is not inverted and thus resembles a single star.

On the other hand, the VCA case would (if observed) initially appear to be more massive than a $24 M_{\odot}$ star (despite being $23.66 M_{\odot}$ initially), eventually appearing to be approximately $21 M_{\odot}$ – i.e., $\sim 3 M_{\odot}$ less massive than it actually would be. Thermohaline mixing suppresses this change somewhat, but even with stronger thermohaline mixing still ultimately results in an accretor that is systematically different to a NVCA model.

Consequently, any masses inferred from spectroscopy or luminosity measurements may be an underestimate of the true (evolutionary) mass. This would affect, for example, the inferred masses of compact objects and thus may have implications for our understanding of the progenitors of gravitational waves.

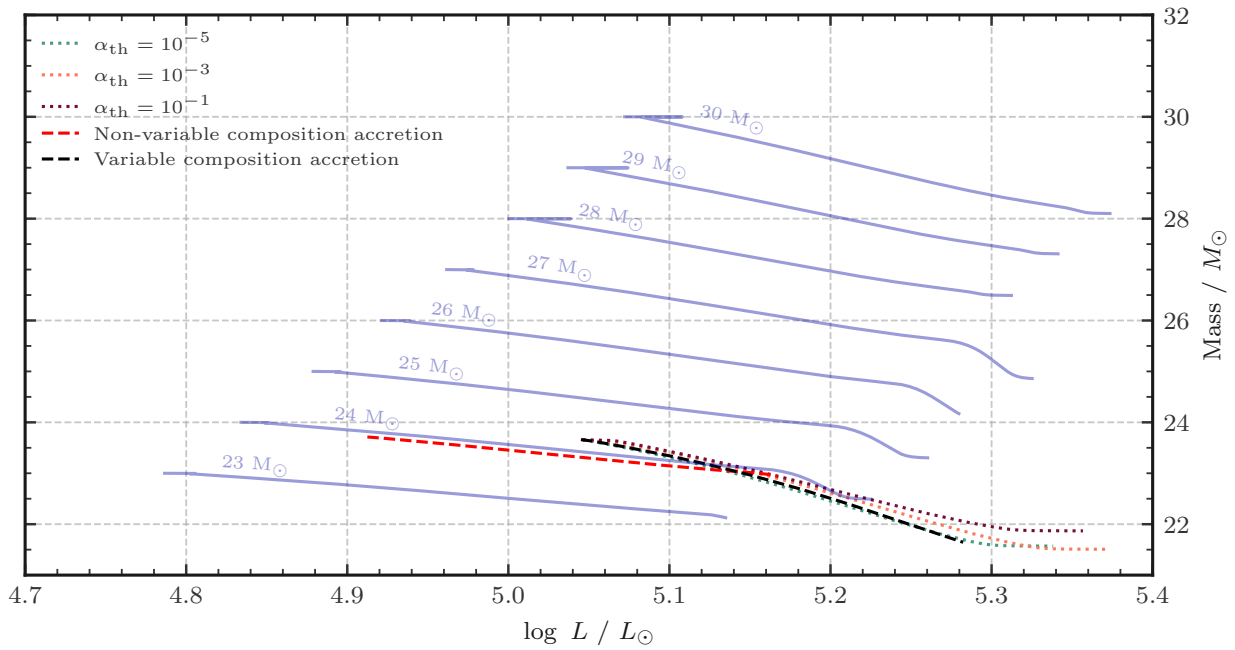


Figure 3.11: The (evolutionary) mass-luminosity relationship in the presence and absence of VCA. Also shown are the curves for reference models of between $M_{\text{ZAMS}} = 21 M_{\odot}$ and $26 M_{\odot}$, which were evolved as a single star.

3.6.2 | Supernovae

Previous work has shown that stellar rejuvenation leads to accretors with less bound envelopes (Renzo et al., 2023), resulting in systems for which the energy required to collapse the core in a supernova is decreased compared to an equivalent single-star model. As such, rejuvenation such as in this case (where the mass transfer is sufficiently efficient that the accretor becomes rejuvenated) could give rise to unusual SNe. SN 1987A, the most widely studied supernova, is often conjectured to be the result of a binary merger (e.g., Podsiadlowski, 2017) due to its unusual explosion as a blue supergiant instead of a red supergiant, as predicted by theory. A rejuvenated star would appear younger, providing a potential avenue of formation for unusual stellar transients.

The surface temperature of the accretor has important implications for the observations of SN progenitors. The donor in our presented model is massive enough to explode as a Type Ib/c SN. A number of observations exist of the pre-SN site of many Type Ib/c SNe (Eldridge et al., 2013; Eldridge & Maund, 2016). One issue is that in many of the observations not only is the progenitor unseen, so too is a massive main sequence star companion. We note that the higher surface temperature of the accretor makes it more difficult to detect in the optical bands which normally exist for this work. The different structure also makes it more difficult to predict the impact of the SN on the companion (Hirai et al., 2018). Thus another reason to study the accretors in more detail is to understand the progenitor detections of Type Ib/c SNe when so few companions have been detected in post-SN observations (Eldridge & Maund, 2016).

3.6.3 | The prevalence of helium-rich mass transfer

Although we have presented only one example of helium-rich mass transfer in this work, it is not unreasonable to assume that this process may be common. To estimate how often such evolution may occur, we have used the binary star population of models from the Binary Population and Spectral Synthesis (BPASS) v2.2 code suite (Eldridge et al., 2017; Stanway & Eldridge, 2018). BPASS only models one of the stars in a binary in detail at a time, but enable us to understand how many stars may experience helium-rich mass transfer.

We used an initial metallicity of $Z = 0.020$ for the models, including every binary where the initial mass of the donor is greater than $7 M_{\odot}$ and the accretor accretes more than 0.1 per cent of its initial mass. For an instantaneous burst of a $10^6 M_{\odot}$ with an upper mass limit of $300 M_{\odot}$, an initial mass function from Kroupa (2001) and initial binary parameter distribution from Moe & Stefano (2017), we find that 3543 ± 68 binary stars undergo this phase of evolution, of which 34 ± 1 per cent accrete more than 5 per cent of their initial mass. Of these 3543 ± 68 models, 23 ± 1 per cent accrete material, which reaches a maximum helium abundance of greater than 0.8. Thus, the evolution of such accretors is common and is of vital importance to understand.

3.7 | Chapter Summary

We have exhibited a case study of a close $30 + 10 M_{\odot}$ binary, in which mass transfer becomes dominated by helium. By contrasting this to a binary with hydrogen-dominated mass transfer, we have shown that this results in an atypical composition profile that produces unexpected accretors. These accretors have systematically different luminosity compared to their hydrogen-dominated alternatives, translating to a different (incorrect) inferred mass.

In summary, the key conclusions of this chapter are as follows:

1. If a close binary system has a helium star within it, it can only have been formed by binary interactions. The accretor is likely to have accreted some helium-rich material, and as shown by our VCA model, the accretor is hotter and more luminous than if the transferred material was not helium-rich. In addition, the eventual fate supernova time of the model is altered from a hydrogen-rich type II supernova to a hydrogen-free type Ib/c supernova.
2. Accretor stars with a helium-rich surface layer on top of a hydrogen-rich lower envelope and helium-rich convective core can be very different to a single star of the same mass. In the case presented, a non-thermal-equilibrium $24 M_{\odot}$ accretor star was 0.2 dex more luminous and 12.5 kK hotter than a single star of the same mass. This means that in a binary system post-mass transfer, normal mass-luminosity relationships may not hold for accretor stars, and stellar masses may be overestimated from luminosity.
3. When thermohaline mixing is included in these models, the surface composition gradient is reduced. This leads to the accretor becoming cooler by 0.1 dex but it remains overluminous for its mass having the luminosity of a $30 M_{\odot}$ single star despite being only $24 M_{\odot}$. The strength of the thermohaline

mixing determines how quickly the accretor stars achieve a lower surface temperature and impacts the lifetime on the main sequence.

4. The mass transfer efficiency in this binary interaction evolves with time through the mass transfer phase. A contact phase of evolution limits the mass transfer efficiency because the secondary cannot accrete more material whilst filling its Roche Lobe. But later, once the contact phase ends, the accretor becomes massive enough for its thermal time-scale to come close to the mass transfer time-scale, mass transfer efficiency increases.
5. We used BPASS v2.2 to perform a population synthesis and calculate the expected number of systems that experience helium-rich mass transfer. At Solar metallicity, 23 per cent of all systems in which mass transfer occurs, accrete helium-rich material with a maximum helium mass fraction of greater than 0.8. Suggesting helium-rich mass transfer is a common and important process to account for in models of binary evolution.

VCA has a clear influence on the future evolution of an accretor in a binary star system. What remains to be seen is the influence on a population writ large; the effect of VCA should be included in binary population synthesis projects to understand the broader implications of VCA. We have suggested that these could be to explain some of the discrepancy between spectroscopic and evolutionary masses of massive stars, lead to greater ionising output for a stellar population and possibly explain some exotics stellar objects such as LBVs. However, the biggest uncertainty is how efficient thermohaline mixing is in such stars. However, observations such as those demonstrated by [Miszuda et al. \(2021\)](#) of stellar oscillations in post-interaction binary stars will be key to understanding this physical process.



CHAPTER 4

Post-Supernova Kinematics: The Natal Kick



And the memories were lost long ago

But at least you have beautiful ghosts.

— *Taylor Swift* —



THE TERMINUS OF A MASSIVE STAR’S life is a core-collapse supernova, when the hydrostatic equilibrium between the pressure of the star and its self-gravity is broken, and the stellar core collapses in on itself. Once the star reaches nuclear densities, neutron degeneracy is able to stabilise the remnant, causing a shockwave to propagate outwards from the core. The energy released is $\sim 10^{53}$ erg, of which $\sim 10^{51}$ erg is transferred to the star (Eldridge & Tout, 2018), resulting in the ejection of an amount of mass we denote M_{ej} and leaving a remnant of mass M_{rem} behind. If the ejection is asymmetric, the collapse endows the nascent remnant with a velocity v_{k} whose overall direction is antialigned to the more massive ejecta. This motion is termed a natal supernova kick.

Given an observation of the proper velocity of a compact remnant, it is in principle possible to trace its motion back and ascertain its kick velocity. However, this is not necessarily a straightforward endeavour. If the proper velocity is high and the remnant old, it has travelled quite a distance from its explosion site to be located at the point the observation has been taken. In addition, the star has potentially had its proper velocity modified by gravitational attraction from either other runaway stars or its host galaxy’s potential. One would require the exact location in 3D space as well as the 3D velocity and remnant age in order to ascertain the motion, but from there the inverse problem becomes straightforward.

4.1 | Overview

As a consequence of conservation of momentum, it can be shown that if the ejecta is assumed to eject along a preferred axis, the velocity of the kick can be written as

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta \left(\frac{M_{\text{NS}}}{M_{\text{rem}}} \right), \quad (4.1)$$

for free parameters α and β (both measured in km s^{-1}). Here, M_{NS} is the mass of a NS; if the kick is applied to a NS, then $M_{\text{NS}} = M_{\text{rem}}$ and the coefficient reduces to unity. If the kick is applied to a BH, then we can take $M_{\text{NS}} = 1.4 M_{\odot}$, which serves to convert the distribution from one in velocity-space to one in momentum-space; see Bray & Eldridge (2016, 2018); Richards et al. (2023) for further detail.

This parameterization introduces, effectively, four free parameters in $\alpha, \beta, \theta, \phi$ (the latter two being azimuthal and equatorial angles of the ejecta mass). We assume that the preferential axis is sampled uniformly across the 4π steradians of the star, and that our only two free parameters are α and β . It is pertinent to note two things due to this assumption: firstly, we are not assuming that the ejection occurs along one axis and that the kick is aligned along a different, random, axis. Rather, we are assuming that the asymmetric ejection occurs along a randomly sampled axis and that a positive-velocity kick is antialigned to that. The second note, explored further in Section 4.1.2, is that there is considerable uncertainty around the nature of the kick and the precise mechanism that causes it to eject along a given axis. This is the reason we sample the kick direction uniformly, though we will simulate sufficiently many systems (on the order of $\mathcal{O}(10^{12})$) that this will be marginalised over.

The existence of two free parameters, α and β , mean that a choice on the values must be made. The

“traditional” way to choose these parameters, as outlined in [Bray & Eldridge \(2016, 2018\)](#), is to refine them based on isolated pulsar velocities. This methodology, however, has significant drawbacks. Notably, the reliance on one observation can engender biases in the result at the expense of no longer predicting accurate results for other observables. For instance, in the limit of ultra-stripped supernovae (USSNe) where the ejecta-remnant mass ratio becomes small, the fiducial kick in [Bray & Eldridge \(2018\)](#) would produce kick velocities that are strongly negative (indicating a strong kick aligned with the direction of motion of the ejecta), contrasted with the weak kicks we expect from these supernovae (e.g., [Willcox et al., 2021](#)). It becomes evident that a new series of empirical tests are needed, whereupon more constraints than simply pulsar velocities are tested. In this Chapter, we outline four such tests that form the basis of the work of [Richards et al. \(2023\)](#). These tests – using observed LVK Collaboration merger rates, period-eccentricity distributions of Binary Neutron Stars (BNSs), isolated pulsar velocities, and production of USSNe – allow us to narrow the parameter space significantly, with the promise of future refinement with more observations (such as through LVK Collaboration O4).

4.1.1 | Parameter space

We perform our computation on three populations, termed FINEGRID, WIDEGRID, and HOBBS, whose properties are outlined in Table 4.1. These populations simulate 1,000 kicks per supernova in each stellar evolution model. The aim of the three populations is twofold. Firstly, by generating FINEGRID and WIDEGRID, we are able to look both at fine detail in the region at which the most likely value is expected to lie and also the large-scale behaviour over wider parameter ranges.

Population Name	α_{low} [km s ⁻¹]	α_{high} [km s ⁻¹]	β_{low} [km s ⁻¹]	β_{high} [km s ⁻¹]	Step Size [km s ⁻¹]
FINEGRID	0	400	-200	200	10
WIDEGRID	0	700	-350	350	50
HOBBS	-	-	-	-	-

Table 4.1: The populations under study. HOBBS is a fiducial kick prescription as given by [Hobbs et al. \(2005\)](#), which is a Maxwellian distribution with distribution parameter $\sigma = 265 \text{ km s}^{-1}$, based on an empirical fit to the velocities of 233 pulsars. All three prescriptions simulated 1,000 kicks for each system, with the resultant kick being the mean velocity sampled, and the direction of the kick being sampled uniformly across the 4π steradians of the orbit.

Secondly, by using [Hobbs et al. \(2005\)](#), we are able to create a population governed by a kick that has been well-tested in literature (see, for example, [Bouffanais et al., 2021](#); [Broekgaarden et al., 2021](#); [Fragione, 2021](#); [Igoshev et al., 2021](#); [Mandel & Broekgaarden, 2022](#)) which serves as a fiducial comparison. From a Bayesian perspective, we are essentially choosing the HOBBS distribution as our prior for the express purpose of a Bayesian test in Section 4.2.2.

The choice of extremal values for α and β are somewhat arbitrary. Figure 4.2 shows a wider parameter space with $\alpha \in [0, 1000]$, $\beta \in [-500, 500]$. Regions in the upper right (bottom left) corner of the plot have quite high (low) mean kick velocities. Although both grids encroach on these regions somewhat, there is

a tradeoff between exploring the grid space fully and using computational resources efficiently. Our mean kick velocities should not be too extreme; the mean for a [Hobbs et al. \(2005\)](#) kick is $\sigma = 265 \text{ km s}^{-1}$, for a [Verbunt et al. \(2017\)](#) is $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$, and for the observed [Willcox et al. \(2021\)](#) kick $\sigma_1 \approx 30 \text{ km s}^{-1}$, $\sigma_2 \approx 230 \text{ km s}^{-1}$. These¹ are shown on the grid as black lines.

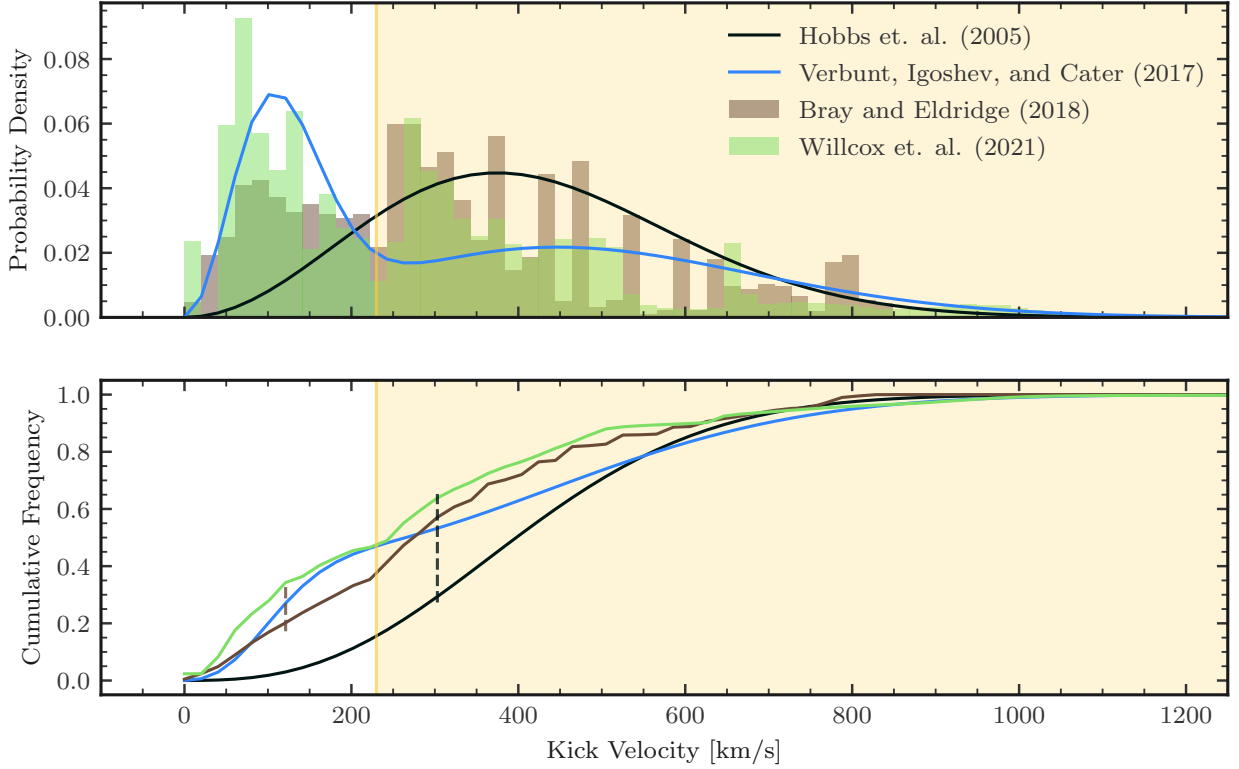


Figure 4.1: The Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of the distributions using kicks from [Hobbs et al. \(2005\)](#); [Verbunt et al. \(2017\)](#); [Bray & Eldridge \(2018\)](#); [Willcox et al. \(2021\)](#). [Hobbs et al. \(2005\)](#) has a closed-form expression, whereas the others are computed empirically, hence the Bray and Willcox CDFs are less smooth than that of Hobbs. The two dashed lines indicate where and how the Kolmogorov-Smirnov (KS) statistic is measured for the associated distribution. The yellow region indicates the region for which $v_{\text{kick}} \geq v_{\text{TGRV}}$, the transverse Galactic rotation velocity of the Sun, $\sim 230 \text{ km s}^{-1}$.

Whilst the exact parameters can be shifted somewhat and maintain the statistical integrity given these constraints, there are a few additional heuristic constraints we place on the regions to select. First, we wish to capture low-velocity USSN kicks (as will be discussed in Section 4.2.4), necessitating a region near $\alpha = \beta = 0$ (where the kick would be identically zero for all ejecta-remnant mass ratios). Secondly, the fiducial work of [Bray & Eldridge \(2018\)](#) found $\alpha = 100, \beta = -170$. Although a goal was to (ideally) remove $\beta < 0$ as a possible value (for USSN again), to impose that as a prior when the fiducial value sat within that range would have gone contrary to typical Bayesian practices. Thirdly, we expected but could not guarantee that the magnitude of each parameter would shrink with further tests, given the large uncertainty bands that [Bray & Eldridge \(2018\)](#) found (hence broadening our search area beyond $|\alpha| = 100, |\beta| = 170$). Finally, we elected to increase the grid search region by 75 per cent for WIDEGRID as this would allow us to just reach $\bar{v}_k = 700 \text{ km s}^{-1}$, which is where the CDF of the [Hobbs et al. \(2005\)](#); [Verbunt et al. \(2017\)](#); [Willcox et al. \(2021\)](#) distributions

¹Due to the grid size and the random sampling, a true contour is difficult to observe due to sampling noise. The contours shown are thus linear fits to the actual contours.

all approximately begin to level out (signifying few velocities above those), as shown in Figure 4.1.

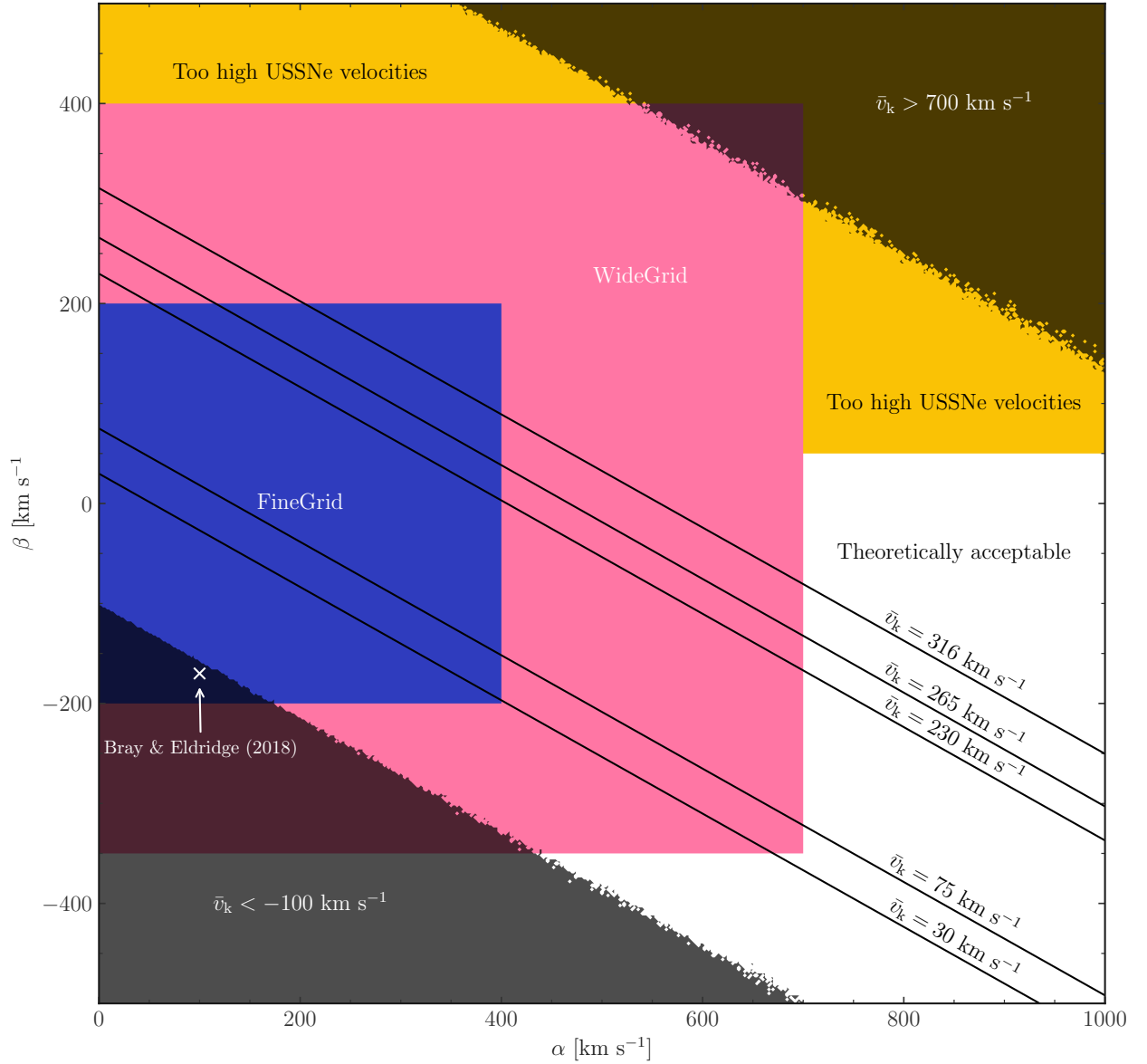


Figure 4.2: A schematic of how FINEGRID and WIDEGRID are related. The mean kick velocity v_k was computed by averaging the kick velocities of 1,000 simulated kicks for each combination of α and β , with $m_{\text{rem}} \in [0.1, 2.5]$ and $m_{\text{ej}} \in [0.1, m_{\text{rem}}]$ sampled uniformly. The white cross represents the point $\alpha = 100, \beta = -170$, which are the fiducial kick parameters from [Bray & Eldridge \(2018\)](#).

4.1.2 | Natal Kick Prescriptions

Cats can disappear.

— FOURTH OF THE SIX IMPOSSIBLE THINGS

There are conflicting accounts as to the cause of the kick (see, for example, [Fryer et al., 1998](#); [Arzoumanian et al., 2002](#); [Bombaci & Popov, 2004](#); [Hobbs et al., 2005](#); [Bray & Eldridge, 2016](#); [Verbunt et al., 2017](#)). There are conflicting accounts as to the form and distribution of the kick too, as discussed in, for example, [Blaauw \(1961\)](#); [Gunn & Ostriker \(1970\)](#); [de Loore et al. \(1975\)](#); [Sutantyo \(1978\)](#); [Hobbs et al. \(2005\)](#); [Baker et al.](#)

(2008); Wongwathanarat et al. (2013); Bray & Eldridge (2018); Giacobbo & Mapelli (2020); Mandel & Müller (2020).

We stress that we are not attempting to prescribe *which* prescription is somehow “best”, but seek an easy-to-configure kick which allows us to configure kicks without changing any of the physics of how our population was formed. This can be done by modifying the kick of Bray & Eldridge (2016), which takes two free parameters which can be changed without imposing constraints on physics. This prescription is hereinafter referred to as the “Bray kick”. Similarly, we require for the purposes of computing log-Likelihoods in Section 4.2.2 a comparison kick, for which we select the kick of Hobbs et al. (2005), a well-known empirically derived kick. This prescription is hereinafter referred to as the “Hobbs kick”.

The Bray kick velocity is prescribed by Eqn(4.1). When applied to a NS, M_{NS} is by definition the remnant mass, so the kick velocity becomes

$$v_k = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta. \quad (4.2)$$

On the other hand, the Hobbs kick velocity is given by the PDF

$$P(v_k) dv_k = \sqrt{\frac{2}{\pi}} \frac{v_k^2}{\sigma^3} \exp \left(-\frac{v_k^2}{2\sigma^2} \right) dv_k, \quad (4.3)$$

where $\sigma = 265 \text{ km s}^{-1}$ is the distribution parameter found by fit in Hobbs et al. (2005) to the velocities of 233 pulsars.

4.1.3 | The TAKAHE orbital evolution code

As part of this work, we have developed the TAKAHE² compact object binary orbit evolution code, which solves the coupled equations of motion Eqn(2.25) and Eqn(2.26). Originally, to facilitate the equations being solved rapidly, we wrote the code as a PYTHON wrapper to a JULIA (Bezanson et al., 2017) integrator. However, the PYJULIA library (Arakaki et al., 2023) proved to be fragile³ and is not in fact supported officially on recent Python versions. To facilitate the potential interoperability between KAITIAKI and TAKAHE (and to remove a prolonged loading time when the JULIA runtime was loaded, given that PYJULIA precompiles our modules), we elected to rewrite the core integrator into PYTHON and to optimise it using traditional PYTHON methods instead.

There are two relevant functions in TAKAHE. The remaining functions are utility functions that allow us to (for instance) convert between lookback time / redshift, and period / separation. These functions are `integrate_eoms()` (Code Sample F.1), which solves the coupled differential equations for $\frac{da}{dt}$ and $\frac{de}{dt}$ (Eqn(2.25) and Eqn(2.26)), and `integrate_timescale()` (Code Sample F.2), which solves Eqn(2.29) for the coalescence time of the binary. With sufficient performance optimisations,⁴ `integrate_timescale()` is able

²The Takahē is an endangered New Zealand bird, which is viewed as taonga to the Ngāi Tahu iwi. There are approximately 500 left, all in conservation areas in the Te Waipounamu / the South Island.

³It requires custom installation procedures (so cannot be included as a dependency in e.g. a PYTHON package), would hang with no discernible output or ability to ascertain why, and could not handle KeyboardInterrupt exceptions in PYTHON.

⁴Primarily, just-in-time compilation with the NUMBA (Lam et al., 2015) library.

to achieve speeds of approximately 18,000 numerical integrations per second on an office PC⁵ on a sample of Neutron Star - Black Hole Binary (NS-BH) binaries in the LISA dataset (Tang, priv. comm.).

Algorithmically, both codes employ a simple Euler integration routine (where $f_{n+1} = f_n + \frac{df_n}{dt} dt$) and the choices work as follows:

1. When `integrate_eoms()` takes a step in time, it evaluates da and de . It then computes the relative changes in both e and a that arise by taking this timestep (through $\delta f = \left| 0.01 \frac{f}{f'} \right|$), and takes a timestep corresponding to the smallest change between a and e , or half of the width of the smallest BPASS time bin.
2. `integrate_timescale()` breaks up Eqn(2.29) into four cases dependent on the initial eccentricity. Let τ be the coalescence time of a circular binary ($\tau = \frac{a_0^4}{4\beta}$), then the cases are:

$$t_c = \begin{cases} \tau & e_0 = 0 \\ \frac{c_0^4}{4\beta} e_0^{\frac{48}{19}} & e_0 < 0.01 \\ \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de & 0.01 \leq e_0 \leq 0.99 \\ \tau^{\frac{768}{425}} (1-e_0^2)^{\frac{7}{2}} & 0.99 < e_0 \end{cases}, \quad (4.4)$$

where all symbols have the same meaning as in Section 2.5. The integral in the intermediate case is evaluated numerically, taking a fixed timestep of $de = \frac{e_0}{1000}$ such that the discretised sum becomes nothing more than

$$t_c (0.01 \leq e_0 \leq 0.99) = \frac{e_0}{1000} \frac{12}{19} \frac{c_0^4}{\beta} \sum_{e=0}^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}}, \quad (4.5)$$

and dependence of the coalescence time t_c on the initial separation a_0 is handled through the c_0 term. Since the step size is an arbitrary choice, a tradeoff can be made between accuracy and runtime. Figure 4.3 shows the effect of changing the step size ($\frac{e_0}{1000}$ in Eqn(4.5)) on convergence to a final result, and Figure 4.4 shows the runtime as a function of step size on an office computer.

4.2 | Observational Constraints

In this section, we outline the four tests we use to refine the acceptable parameters of the Bray kick.

4.2.1 | Merger Rates of BNS systems

The first test is to match the observed merger rates of BNS systems. To do this, we use the LVK Collaboration catalogue of results from Operating Run 3 (O3). The simplest test is to match the merger rate today, which we denote as $\mathcal{R}_0$, to that observed by LVK Collaboration (LIGO-Virgo, 2021; LIGO-Virgo-KAGRA,

⁵A Dell OptiPlex 7480 AIO with a 12-core 3.1 GHz Intel Core i5-10500 CPU, running no other programs except for during the final few iterations.

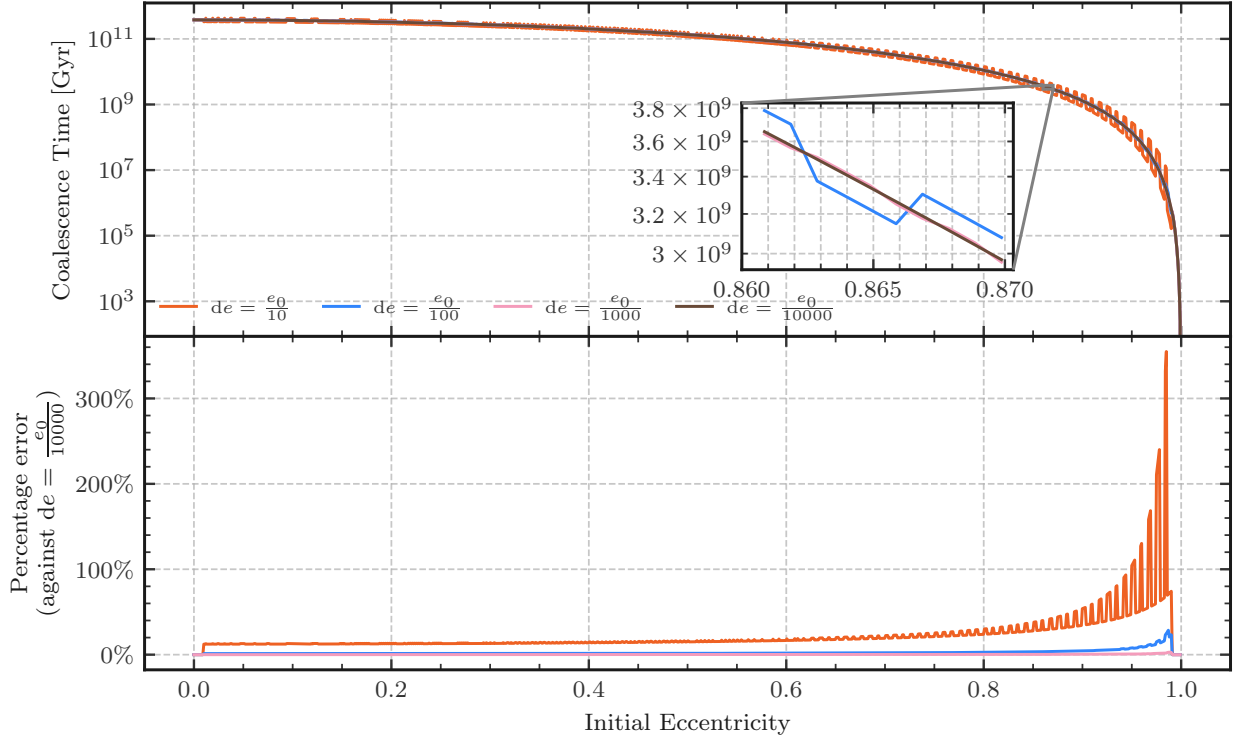


Figure 4.3: The effect of varying de in Eqn(4.5) on the coalescence time. In the inset plot, the $de = \frac{e_0}{10}$ line has been removed for clarity. This binary was assumed to have $m_1 = m_2 = 1.44 M_\odot$ and $a_0 = 1970.5 R_\odot$ (the mean separation of a sample of binaries with $\alpha = 100 \text{ km s}^{-1}$, $\beta = -170 \text{ km s}^{-1}$).

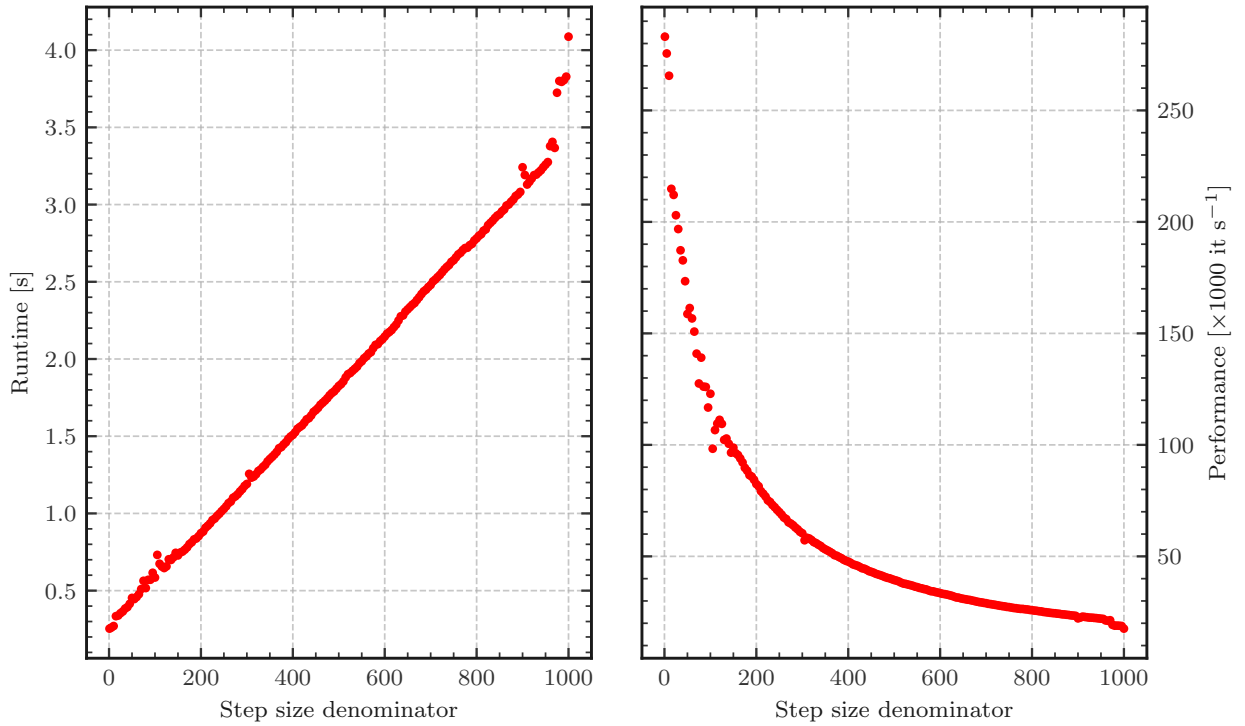


Figure 4.4: The effect of varying de in Eqn(4.5) on the runtime, which is demonstrably $\mathcal{O}(n)$, and the performance of the algorithm. The input data to this algorithm were the LISA binary sample from Tang (priv. comm.). The performance is shown in units of thousands of iterations per second.

2021). We display this in Figure 4.5 alongside the previously reported fiducial values of $\alpha = 100 \text{ km s}^{-1}$ and $\beta = -170 \text{ km s}^{-1}$, the theoretical merger rate calculation performed by Eldridge et al. (2019), and the observed merger rates from LIGO-Virgo (2021) and LIGO-Virgo-KAGRA (2021). We term the contour defined by LIGO-Virgo (2021) (in black) the “LIGO contour”.

Selecting the value to calibrate to is not a straightforward task. Although the LIGO-Virgo (2021) contour is superseded by the LIGO-Virgo-KAGRA (2021) contour, the separation into different inference models means that the merger rates for BNS systems have intractably large error bars covering the entire plane; for instance, their MULTISOURCE (MS) model alone spans $2.11 \leq \log \mathcal{R}_0 \leq 3.23$ which covers FINEGRID entirely and almost entirely covers WIDEGRID. In addition to the lack of constraint, we would be imposing additional parameters into the model insofar as the selection of model – whether, for example, PDB was taken as truth over BGP, MS, or one of the other population models. Whilst an interesting and assuredly insightful endeavour, the aim of this work is instead to develop a simple test that can be used in future on extra populations such as these. A simple criterion which can be extended is valuable, and future work may extend these tests in multiple directions.

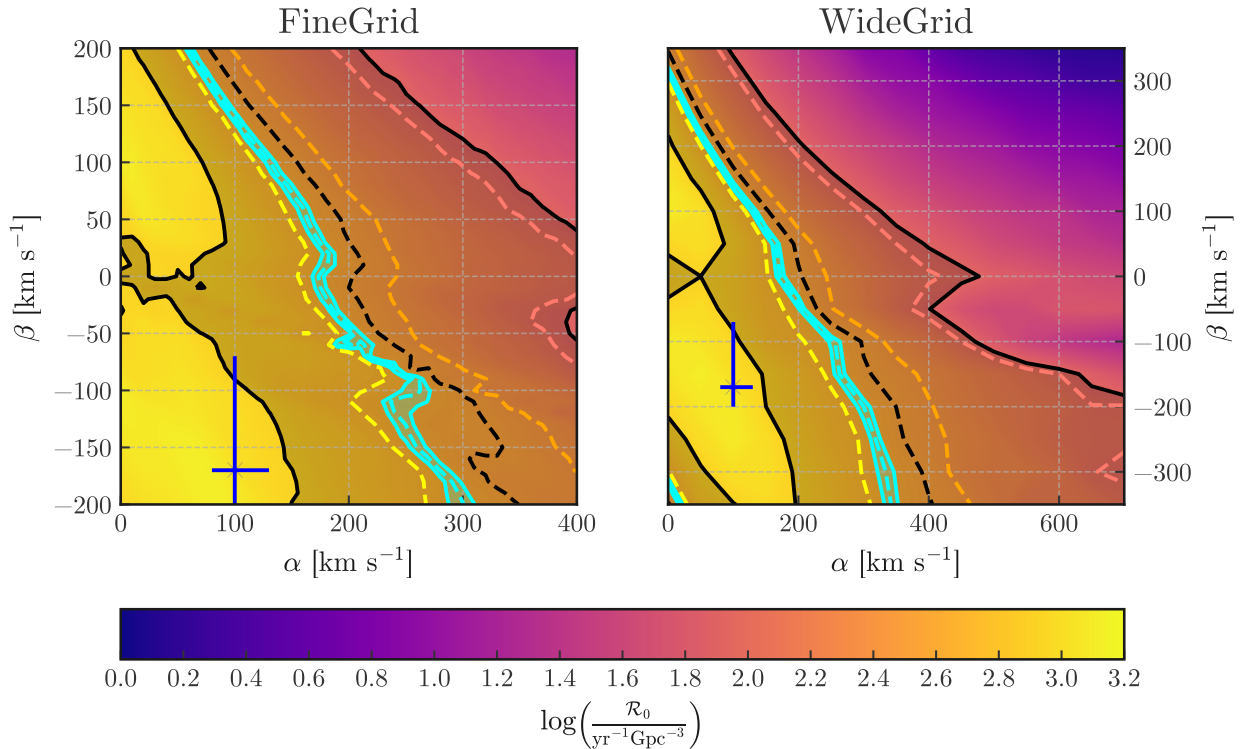


Figure 4.5: The merger rate as a function of kick parameters α and β . The cyan contour is the theoretical merger rate as reported by Eldridge et al. (2019). The black contour is the merger rate observed in LIGO-Virgo (2021). The remaining contours are the merger rates reported in LIGO-Virgo-KAGRA (2021) (pink: BGP, yellow: MS, orange: PDB [ind], see LIGO-Virgo-KAGRA (2021) for an explanation of each model). Shaded regions represent uncertainties, and uncertainties are not reported for the LIGO-Virgo-KAGRA (2021) results as they would cover the entire plane. The blue cross is the fiducial value of α and β from Bray & Eldridge (2018). This value lies off the previous work’s contours as the rejuvenation age calculation has been improved in TUI in preparation for the next major BPASS release.

4.2.2 | Period-Eccentricity Distributions

Since, in the absence of any mass transfer, the motion of two compact binaries is wholly dependent on their initial separation and orbital eccentricity, they trace out paths in a two-dimensional phase space. In general, since binaries circularise over time, and by the emission of gravitational waves their orbits shrink, a binary will start in the high-period, high-eccentricity region of period-eccentricity space, and evolve towards the low-period, low-eccentricity region. As a result, given a population of binary stars, we can design a density plot, where the density is effectively the likelihood of a binary selected at random having the parameters we sample.

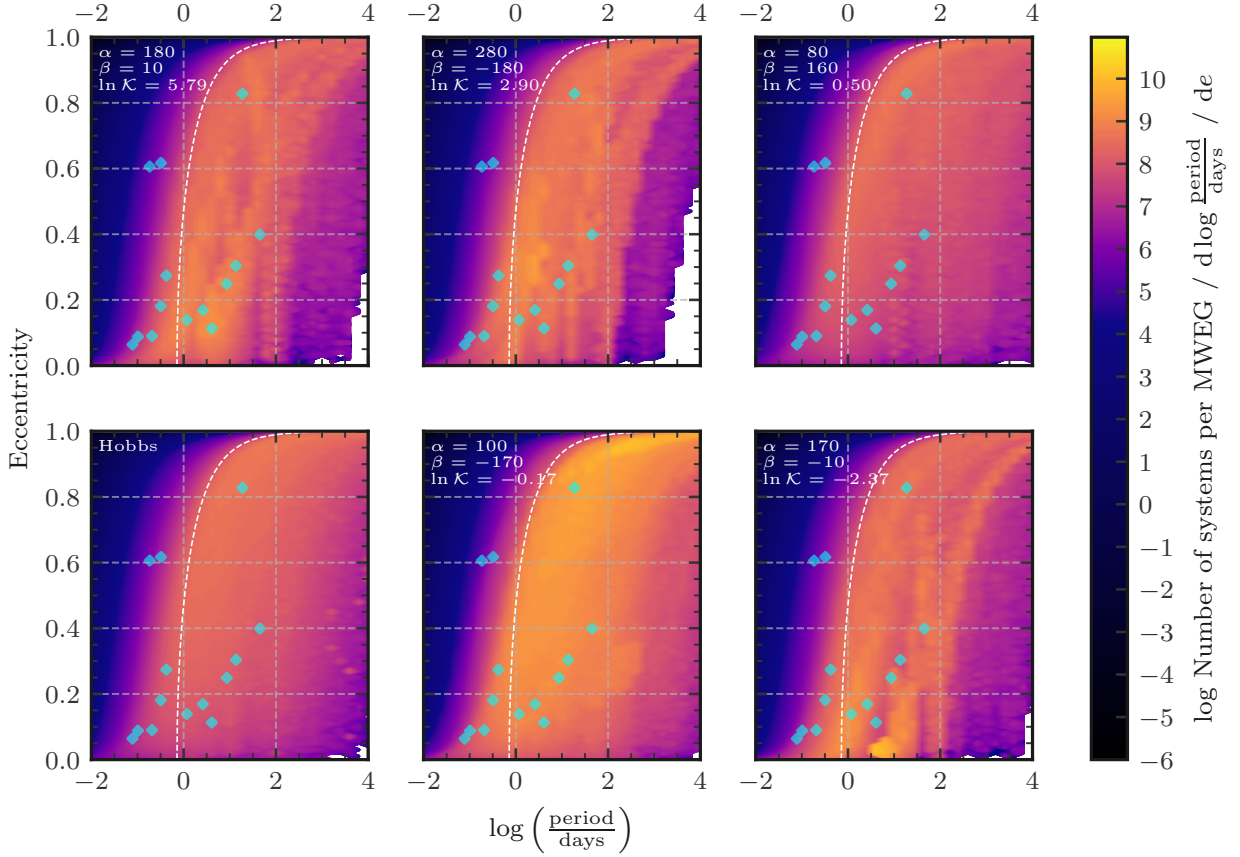


Figure 4.6: The period-eccentricity distributions for six selected [Bray & Eldridge \(2016\)](#) kicks. Shown are the kicks for $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ as the system with the highest $\ln \mathcal{K}$ across the LIGO contour, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ as the lowest, and the balance to illustrate structural differences. White regions show regions where no systems evolve to. The white line represents a constant-time coalescence time of the Hubble time ($\sim 13.8 \text{ Gyr}$), computed using [Mandel \(2021\)](#). The cyan diamonds are the observed systems enumerated in Table 4.2. In white text are the values of (α, β) , and the $\ln \mathcal{K}$ values for each plot.

Computationally, for each binary in our sample, we trace its path through period-eccentricity space, and at every timestep we add $w \times dt$ to the density plot, where dt is the timestep, and w is the weight term output from the TUI code proportional to the IMF weight (where the constant of proportionality is related to the uniformly sampled kick direction). The result of this computation, for the parameter values that lie along the LIGO contour, are shown in Appendix C. We also show the same result in Figure 4.6. Given a catalogue of N (observed) binary stars, we can then compute the likelihood of the model as

Name	$\log(P/\text{days})$	e	Original Reference	Name	$\log(P/\text{days})$	e	Original Reference
J0453+1559	0.609808	0.113	Martinez et al. (2015)	J0737-3039	-0.991400	0.088	Kramer et al. (2006)
B1534+12	-0.375718	0.274	Fonseca et al. (2014)	J1756-2251	-0.494850	0.181	Faulkner et al. (2005)
B1913+16	-0.490797	0.617	Hulse & Taylor (1975)	J1913+1102	-0.686133	0.090	Lazarus et al. (2016)
J1757-1854	-0.735182	0.606	Cameron et al. (2018)	J1518+4904	0.936212	0.249	Janssen et al. (2008)
J1811-1736	1.273672	0.828	Corongiu et al. (2007)	J1829+2456	0.070407	0.139	Champion et al. (2004)
J1930-1852	1.653791	0.399	Swiggum et al. (2015)	J1753-2240	1.134751	0.304	Keith et al. (2009)
J1411+2551	0.417638	0.169	Martinez et al. (2017)	J1946+2052	-1.107905	0.064	Stovall et al. (2018)

Table 4.2: The systems we use in our study. These data were originally compiled in [Vigna-Gómez et al. \(2018\)](#).

$$\ln \mathcal{L} = \sum_{n=0}^N \ln p(\ln P_n, e_n), \quad (4.6)$$

where $p(\ln P_n, e_n)$ is the content of the density plot at the location of the n^{th} observed BNS system. Then, the Bayes factor (defined as the likelihood ratio) can be computed as

$$\mathcal{K} \equiv \frac{\mathcal{L}}{\mathcal{L}_{\text{Hobbs}}} \implies \ln \mathcal{K} = \ln \mathcal{L} - \ln \mathcal{L}_{\text{Hobbs}}, \quad (4.7)$$

where $\mathcal{L}_{\text{Hobbs}}$ is the log-Likelihood of the Hobbs kick. We use a catalogue of fourteen BNS systems originally compiled in [Vigna-Gómez et al. \(2018\)](#), shown in Table 4.2. From this catalogue, we sample the period-eccentricity distribution at that space and take that as the probability of observing that binary, thus allowing us to use the content of the period-eccentricity distribution as the $\ln p$ term in our summand of Eqn(4.6).

By doing this we effectively construct a Bayes-factor map, where $\ln \mathcal{K}$ can be shown evaluated at every combination of α and β . We term this value $\ln \mathcal{K}_{\alpha\beta}$. This is shown in Figure 4.7 for both of our considered grids. In general we see that approximately 60% of FINEGRID has a positive log-Bayes factor (indicating any level of support of the model compared to Hobbs), whereas approximately 40% of WIDEGRID has a positive log-Bayes factor. In a Bayesian analysis, the constraint that $\ln \mathcal{K} \geq 3.4$ indicates strong support for the model contrasted against the reference (i.e., [Hobbs et al., 2005](#)), which contains approximately 9.1% of FINEGRID and 4.4% of WIDEGRID.

4.2.3 | Single-Star Kick Velocities

The traditional methodology for refining a kick prescription is to consider the velocity distribution of isolated pulsars. Thus, given a suitable observational dataset, we would be able to compute the expected kick distribution and compare it, via a KS test, to the observed data set. The KS statistic between distributions $\mathcal{F}$ (the observed distribution) and $\mathcal{G}$ (the compared distribution) is given by

$$\text{KS} = \max |\text{cdf}(\mathcal{F}) - \text{cdf}(\mathcal{G})|, \quad (4.8)$$

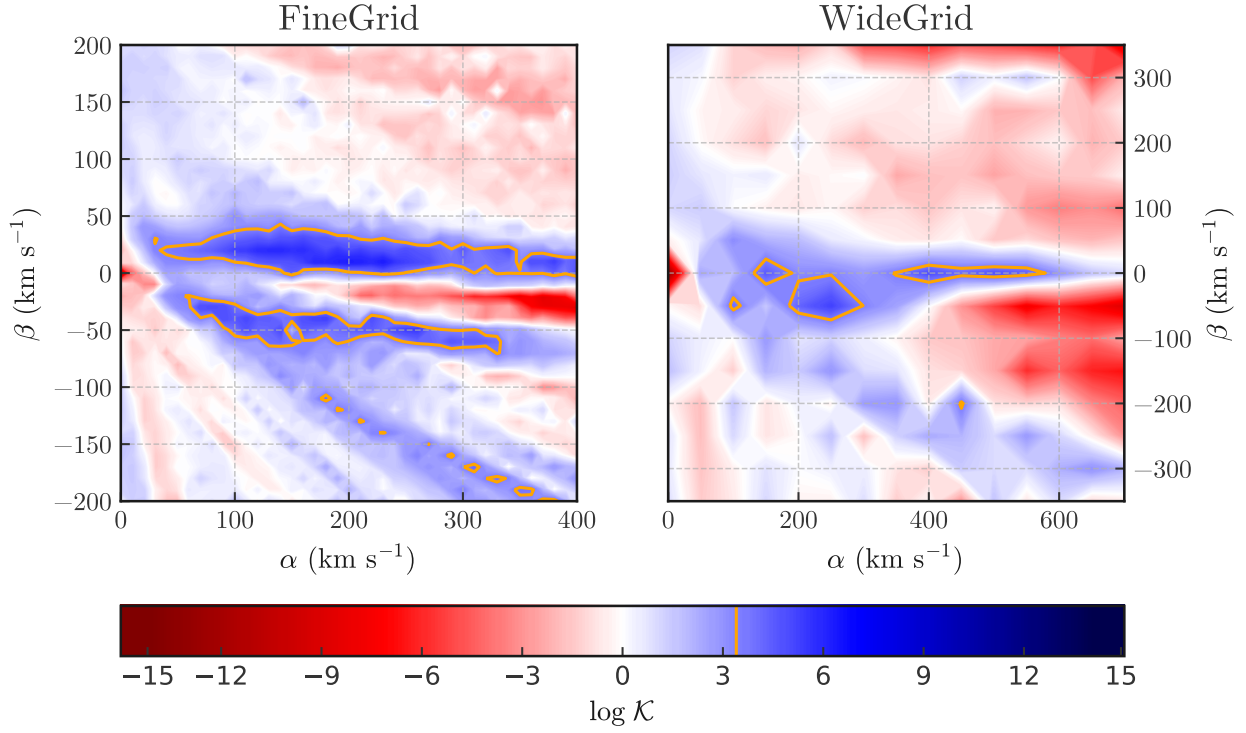


Figure 4.7: The $\ln \mathcal{K}$ values across $\alpha - \beta$ space. The orange line demarcates $\ln \mathcal{K} = 3.4$, which in a Bayesian test represents conclusive evidence in favour of the distribution compared to the [Hobbs et al. \(2005\)](#) kick.

where cdf means the CDF function, defined as the integral of the PDF. Given the CDF is bounded between 0 and 1 inclusive, the KS statistic is also bounded between this region. A KS statistic of 1 means that the compared distribution diverges perfectly from the observed distribution, whereas a KS statistic of 0 means it matches perfectly. This test of divergence was the refinement performed in [Bray & Eldridge \(2016, 2018\)](#), hence we select this (with an independently generated dataset to those works) to be consistent with prior work.

Pulsar velocity dataset

We use the same dataset of 81 pulsars from [Willcox et al. \(2021\)](#), and filter the dataset in the same way that they do. We remove millisecond pulsars, pulsars in globular clusters, and pulsars known to be in binaries, in order to homogenise the dataset. The PDF and CDF of [Willcox et al. \(2021\)](#), as well as those generated from [Hobbs et al. \(2005\)](#) and [Bray & Eldridge \(2018\)](#), is displayed in Figure 4.1.

This homogenised dataset gives the transverse velocities of each pulsar alongside the right ascension / declination, the proper motion in the right ascension / declination directions, and parallax. We assume that the transverse velocity is approximately the original kick velocity. As mentioned earlier in this chapter, old remnants with high velocities may have moved substantially – a 20 Myr remnant with a transverse velocity of 500 km s^{-1} will have moved $\sim 10 \text{ kpc}$, absent external forces ([Willcox et al., 2021](#)). However, this is likely only to be an issue for pulsars whose transverse velocity substantially exceeds the transverse Galactic rotational velocity of the Sun $v_{\text{TGRV}} \approx 230 \text{ km s}^{-1}$ ([Bhattacharjee et al., 2014](#)), shown in yellow on Figure 4.1. Although the dataset of [Willcox et al. \(2021\)](#) has pulsars with velocities higher than this value – peaking at $\sim 850 \text{ km s}^{-1}$

– this is not a grave concern for this work, for two reasons. Firstly, the kick velocities output from TUI are taken immediately after the supernova. Secondly, the analysis Willcox et al. (2021) performs is focused on low-velocity pulsars, and this test limits the kick velocity of an USSNe to 30 km s^{-1} , well below v_{TGRV} .

The Hobbs et al. (2005) kick has a closed-form expression for the PDF $P(v_k)$, given by Eqn(4.3). Consequently, it has a smooth curve in Figure 4.1, whereas the Bray & Eldridge (2018) and Willcox et al. (2021) PDFs are presented as histograms as they are computed.

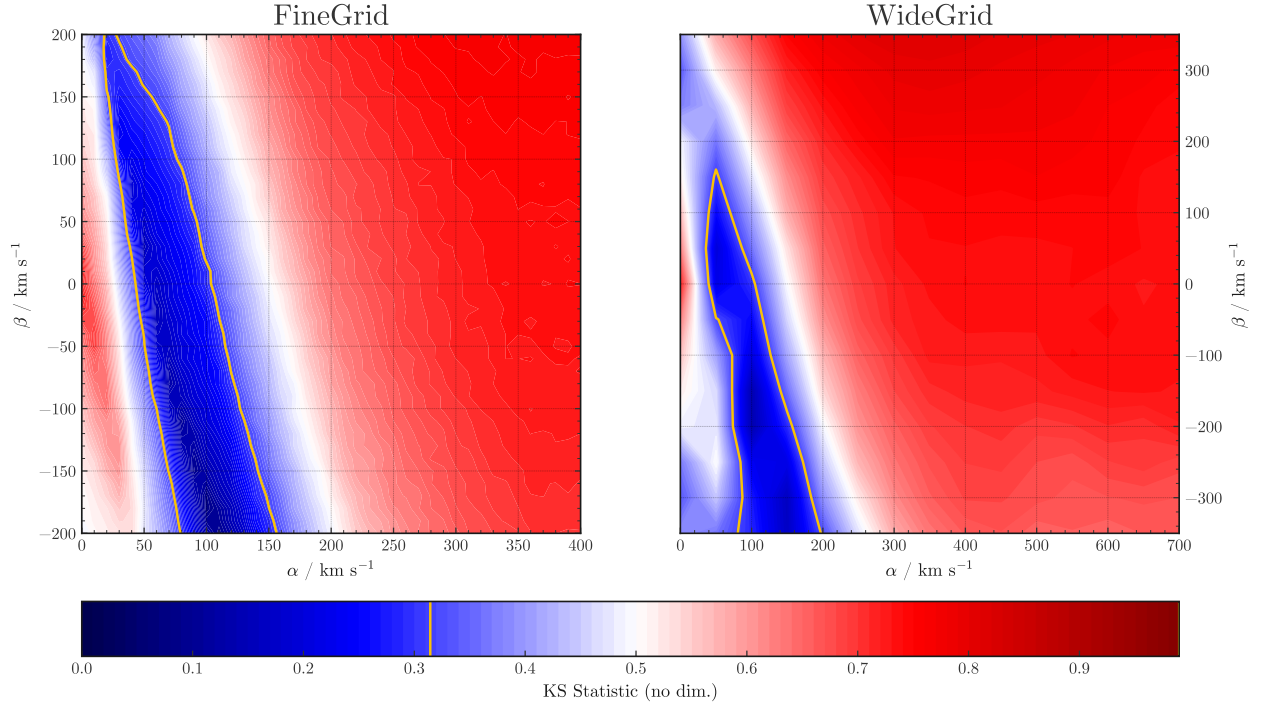


Figure 4.8: The KS statistic for FINEGRID and WIDEGRID. The orange line denotes the KS statistic for the Hobbs kick, which has a value of 0.34.

Comparison

Figure 4.8 shows the KS statistic as a function of α and β across our two parameter spaces. We also show the KS statistic of the Hobbs kick, which has a value of 0.34. We find in general a valley of models with good agreement to Willcox et al. (2021), and find that the global minimum KS statistic is at $\alpha = 100, \beta = -180$ – agreeing with the prior work of Bray & Eldridge (2018) (who find $\alpha = 100, \beta = -170$).

We also show in Figure 4.9 a selection of four PDFs, alongside their associated CDF and the CDFs for Willcox et al. (2021) and the Hobbs kick. We note that the numerical values shown in Figure 4.9 differs somewhat from those shown in Richards et al. (2023). This was due to a minor bug in the code incorrectly labelling the KS statistics for the figure. The actual result of the paper does not change with the correction of the bug, however the values reported in Figure 5 of Richards et al. (2023) do change.

The four values we present on the left side of Figure 4.9 correspond to $\alpha = 100 \text{ km s}^{-1}, \beta = -170 \text{ km s}^{-1}$ for the fiducial Bray kick, $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ to represent the system with the highest $\ln \mathcal{K}$ value from Section 4.2.2, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ the lowest, and $\alpha = 40 \text{ km s}^{-1}, \beta = 0 \text{ km s}^{-1}$ the system with low α and zero β , corresponding to a kick that would *a priori* generate logical velocities in the USSNe

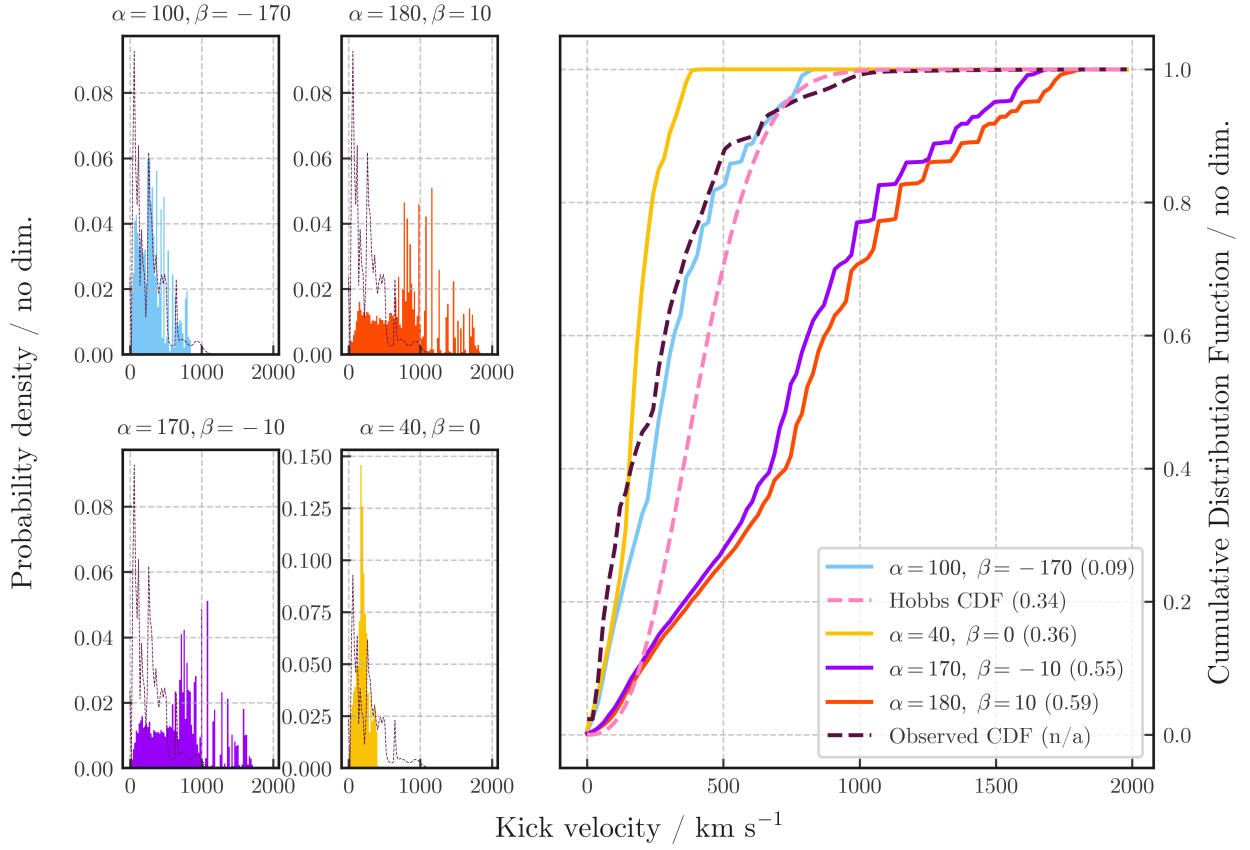


Figure 4.9: The PDFs and CDFs of four systems, alongside the Hobbs CDF and Willcox et al. (2021) CDF. The dashed line on the PDF plots is the PDF of the observed distribution from Willcox et al. (2021). On the right hand plot, the number in parentheses represents the KS statistic for the distribution, measured relative to Willcox et al. (2021), which by definition has no KS statistic.

limit.

4.2.4 | Production of Ultra-Stripped Supernovae

As discussed earlier, the fiducial values of the Bray kick from [Bray & Eldridge \(2018\)](#) admit some problematic extremal cases. If $m_{\text{ej}} \rightarrow 0$, as is the case in a USSN, the kick velocity would become negative. For instance, an “average” USSN with $m_{\text{ej}} = 0.15 M_{\odot}$ and $m_{\text{rem}} = 1.45 M_{\odot}$ would have a kick velocity of $\sim -159 \text{ km s}^{-1}$. The concern here is twofold. Firstly, a negative kick velocity would mean that the velocity vector of the kick would be antialigned with the direction of the more massive remnant; this is a violation of conservation of momentum. Secondly, a kick with high magnitude means that, irrespective of the alignment of the kick, the kick itself is strong. As discussed in [Willcox et al. \(2021\)](#), USSNe are associated with weak kicks, peaking at $v_k \sim 30 \text{ km s}^{-1}$. This means that the fiducial Bray kick is unable to correctly account for USSNe.

To account for these supernova kicks, we introduce a new constraint onto our parameters. Following [Tauris et al. \(2015\)](#), we define a USSN as follows:

1. The ejecta mass is in the range $0 \leq m_{\text{ej}} \leq 0.3 M_{\odot}$, admitting the USSN SN2019dge from [Yao et al. \(2020\)](#),
2. The remnant mass is in the range $1.1 \leq m_{\text{rem}} \leq 1.8 M_{\odot}$,
3. The envelope mass $m_{\text{env}} \leq 0.2 M_{\odot}$.

For every combination α, β , we record whether it is capable of producing one USSN. The constraint is that this heuristic is met. This constraint is extremely conservative, insofar as we do not constrain on the prevalence of USSNe in our sample, but rather by whether or not they are possible at all. The reason for the conservatism here is that the rates of Type Ic USSNe are expected to be very low. In the BPASS dataset, they comprise 1.8 per cent of all CCSNe (Stevance, priv. comm.). The paucity of data alongside the paucity of observational evidence suggests a low rate, but since the prevalence is unknown, we place the conservative constraint of at least one USSN to be agnostic. This effectively widens our error bars such that when the USSNe rate is refined further, it will shrink the parameter space.

4.2.5 | Refinement of the parameter space

We are now in a position to impose constraints on our parameter space based on the analysis provided above. We apply the following four constraints:

1. **DNS Merger Rate** (Section 4.2.1): Does the merger rate produced by these parameters sit along the LIGO contour?
2. **DNS Period-Eccentricity Distributions** (Section 4.2.2): Is $\ln \mathcal{K}_{\alpha\beta} \geq 3.4$ compared to the [Hobbs et al. \(2005\)](#) kick (representing conclusive evidence in favour of the distribution)?

3. **Isolated Pulsar Velocities** (Section 4.2.3): Is the KS statistic produced by this choice of parameters, in comparison to the [Willcox et al. \(2021\)](#) kick, less than 0.5 (indicating it agrees with observational data more than it disagrees)?
4. **Logical USSN velocities** (Section 4.2.4): Is this choice of parameters able to produce at least one USSN?

These can be overlaid onto our parameter space, which is shown in Figure 4.10. The center of mass of our fit region is located at $\alpha = 117^{+36}_{-51} \text{ km s}^{-1}$, $\beta = 13^{+8}_{-14} \text{ km s}^{-1}$, however this value is computed on a finer grid than the parameter space and is thus likely to be an overfit to the data. We thus report the final recommended model value to the nearest half-bin-width of the parameter space, broadening error bars where necessary, and find $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}$, $\beta = 15^{+10}_{-15} \text{ km s}^{-1}$ as the recommended model value. Error bars are always broadened here; traditional rounding would result in the upper bound on α (being 153 km s^{-1}) becoming 150 km s^{-1} , which artificially removes potentially permissible regions of the parameter space. Broadening the bars removes this at the cost of potentially admitting values that do not belong within the “acceptable” zone. Further constraints, however, will shrink the region, so this is an acceptable tradeoff.

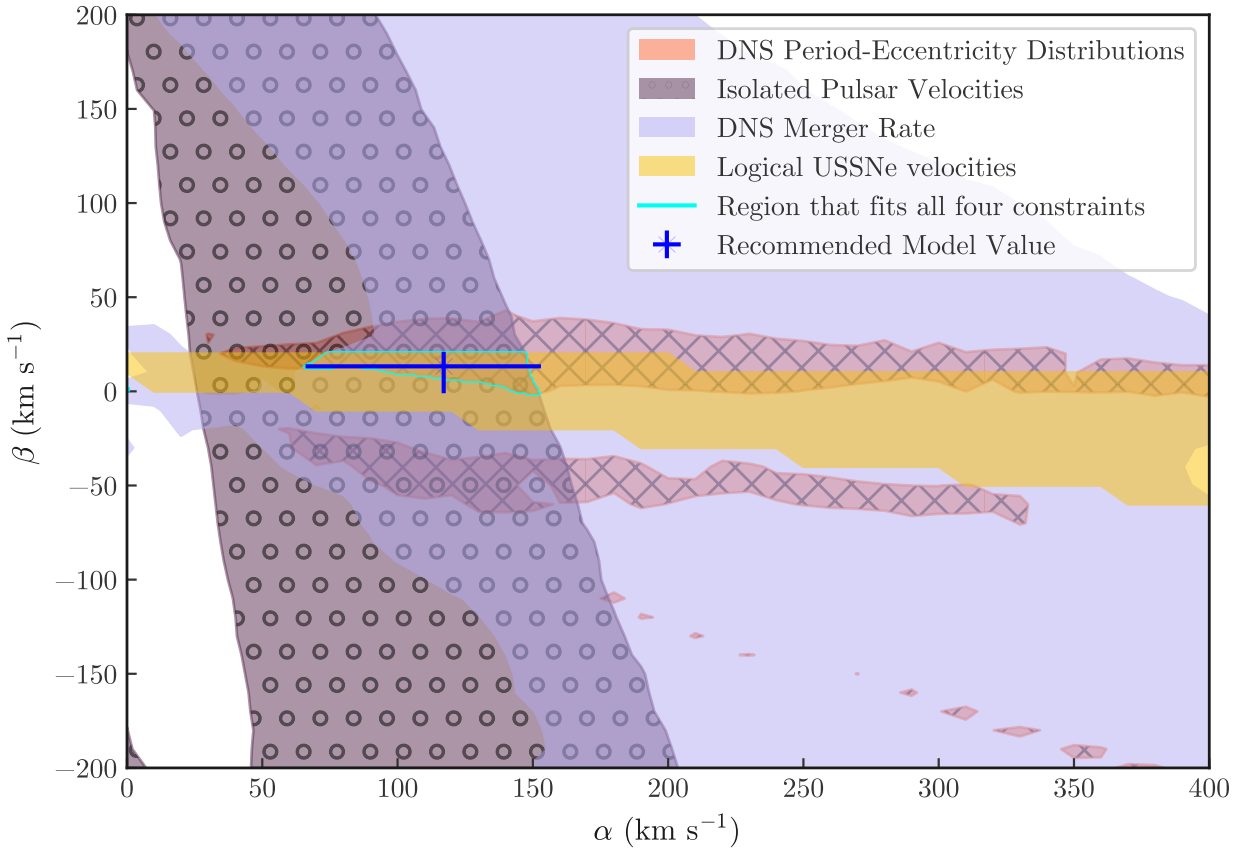


Figure 4.10: The four constraints we apply overlaid on our FINEGRID parameter space. The cyan region represents the region which fits all four parameters, and the blue cross represents the “center of mass” of this region.

In Figure 4.11, we present the two major inferred properties of our recommended model value. First, we present the CDFs of a population governed by a Bray kick with the recommended parameters, contrasted

with the associated Hobbs kick and observed Willcox et al. (2021) data. We also show the period-eccentricity distribution of the population, with the observed systems from Table 4.2 overlaid atop the distribution.

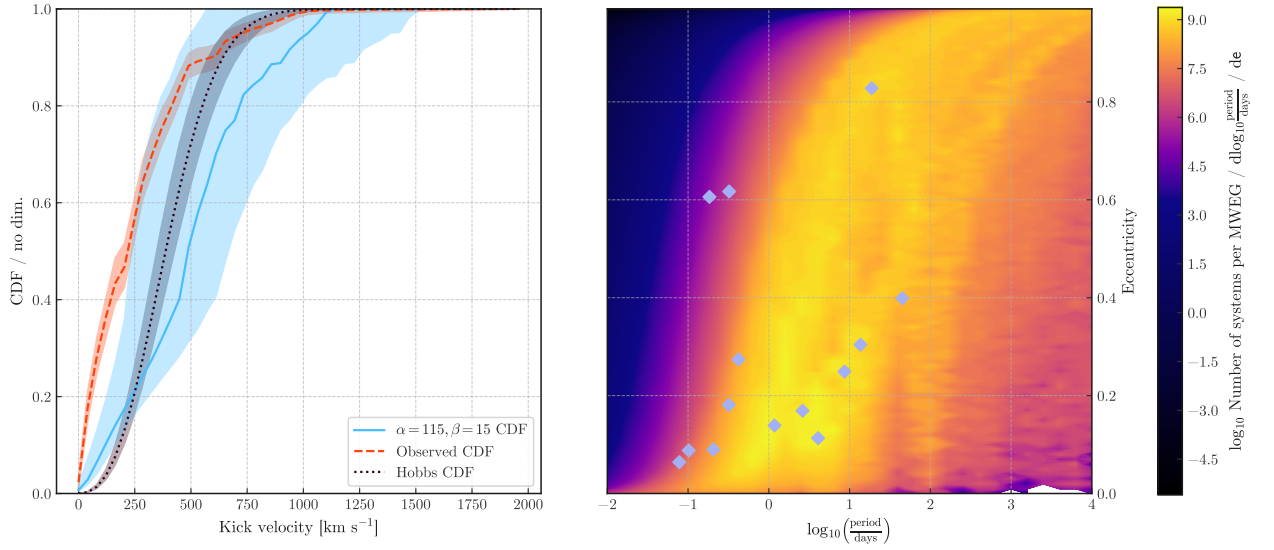


Figure 4.11: The CDF for a population of isolated pulsars (left) and the period-eccentricity distribution (right) for $\alpha = 115 \text{ km s}^{-1}$, $\beta = 15 \text{ km s}^{-1}$. The blue region on the CDF plot shows the uncertainty region for the reported value. The red region shows the uncertainty band on the Willcox et al. (2021) CDF, which was computed by sampling the PDF 10,000 times, taking 2,000 samples each time, and plotting the extent of the simulated distribution. The black band is the uncertainty on the Hobbs distribution, taking $\sigma_{\text{max}} = 291 \text{ km s}^{-1}$ and $\sigma_{\text{min}} = 239 \text{ km s}^{-1}$ from Verbunt et al. (2017).

We are also able to extrapolate this to comment on the expected peak of the event-rate distribution. Given that the event rate is a convolution between the merger rate of our population and the SFRD we choose (cf. Eqn(2.34)), by affixing the SFRD across our populations, by Eqn(2.30) we can vary $\mathcal{R}(t)$ by varying α and β . Since the SFRD is of a Gaussian form (with a peak at $z \sim 1.863$), we can also extract the peak of the distribution assuming it is governed by a particular α - β combination. The general results across the parameter space are shown in Figure 4.12. The best model choice parameters we find fall within the valley of $z = 0$, indicating that the peak of the event rate distribution is the present day.

We note that there is a single peak at $\alpha = 330 \text{ km s}^{-1}$, $\beta = -70 \text{ km s}^{-1}$ which appears to be an outlier when compared to its neighbours. We display in Figure 4.13 the local behaviour of the 9 cells around this region. One can see that this beak is bimodal with a second peak at $z = 0$, which is only 0.098 per cent lower than the reported peak, clearly resulting in a simple outlier.

Unfortunately, there is a paucity of observational evidence to confirm whether or not the peak is truly the present day. Given that a redshift of $z = 2$ corresponds to a lookback time of $t_{\text{LB}} = 10 \text{ Gyr}$, we lack evidence to corroborate this prediction. The next generation of telescopes, including JWST, may yet be able to constrain this result further. In the interim, indirect evidence may be able to be applied via the r-process enrichment history of the Milky Way. Van de Voort et al. (2022), for example, finds that greater velocity natal kicks would be correlated with fewer r-process enhanced stars in the Milky Way, as higher kicks would unbind more binaries, meaning that r-process elements are no longer able to accrete onto star-forming regions. Kobayashi et al. (2023) indicates that current population synthesis models⁶ are unable to reproduce the abundance

⁶Explicitly considered in the paper are COMPAS (Team COMPAS: Riley et al., 2022), StarTrack (Belczynski et al., 2020),

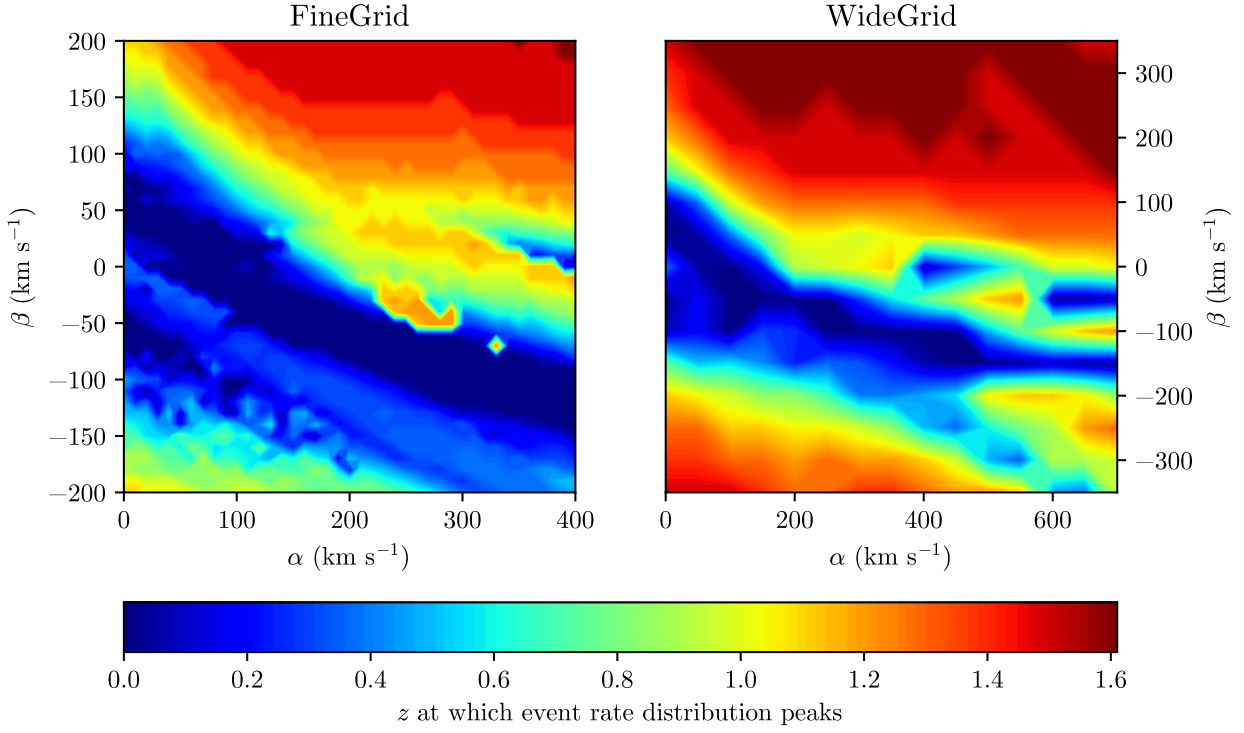


Figure 4.12: The peak of the event rate $\mathcal{R}(t)$ as a function of α and β . Varying these two parameters generally brings the peak closer to the present day ($z = 0$). Absent a merger rate (or, more formally, assuming the merger rate is instantaneous and given by a Dirac-delta function), the SFRD peaks at $z \sim 1.863$. The region surrounding the peak at $\alpha = 330, \beta = -70$ is shown in Figure 4.13.

distributions of certain high-mass isotopes, including the abundances of Oxygen, Iron, Europium, Gold, and Uranium.

4.3 | Discussion

There are limitations to this analysis. The first is that we did not preferentially weight our binaries. [Tauris et al. \(2017\)](#) for example places particular weight on PSR J1930-1852, which is a particularly wide binary. Per [Swiggum et al. \(2015\)](#) (who reported it), “its unique spin and orbital parameters challenge models that describe BNS formation”, which implies that any credible model of BNS formation should replicate its orbital parameters. We did find in Figure C.1 that some parameters had non-zero probability of replicating this pulsar, which indicates that the Bray kick is able to reproduce it.

Secondly, we have assumed Newtonian (specifically, Keplerian) orbits. Formally, the derivations presented earlier (cf. Section 2.4.1) assumed general relativity for the existence of the gravitational wave equation,⁷

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\sqrt{32\pi G}T_{\mu\nu}, \quad (4.9)$$

but the injected physics was Newtonian (see Section A.3 for a derivation). Strictly speaking, relativistic corrections apply to these kinematic equations too, though these only become dominant when the two stars are

Brussels ([Mennekens & Vanbeveren, 2014, 2016](#)), ComBinE ([Kruckow et al., 2018](#)), and BPASS ([Eldridge et al., 2017](#)).

⁷Here, $\bar{h}_{\mu\nu}$ is the trace inverse of a weak perturbation to locally Minkowski spacetime with positive-definite trace, and $\square$ is the d’Alembertian operator.

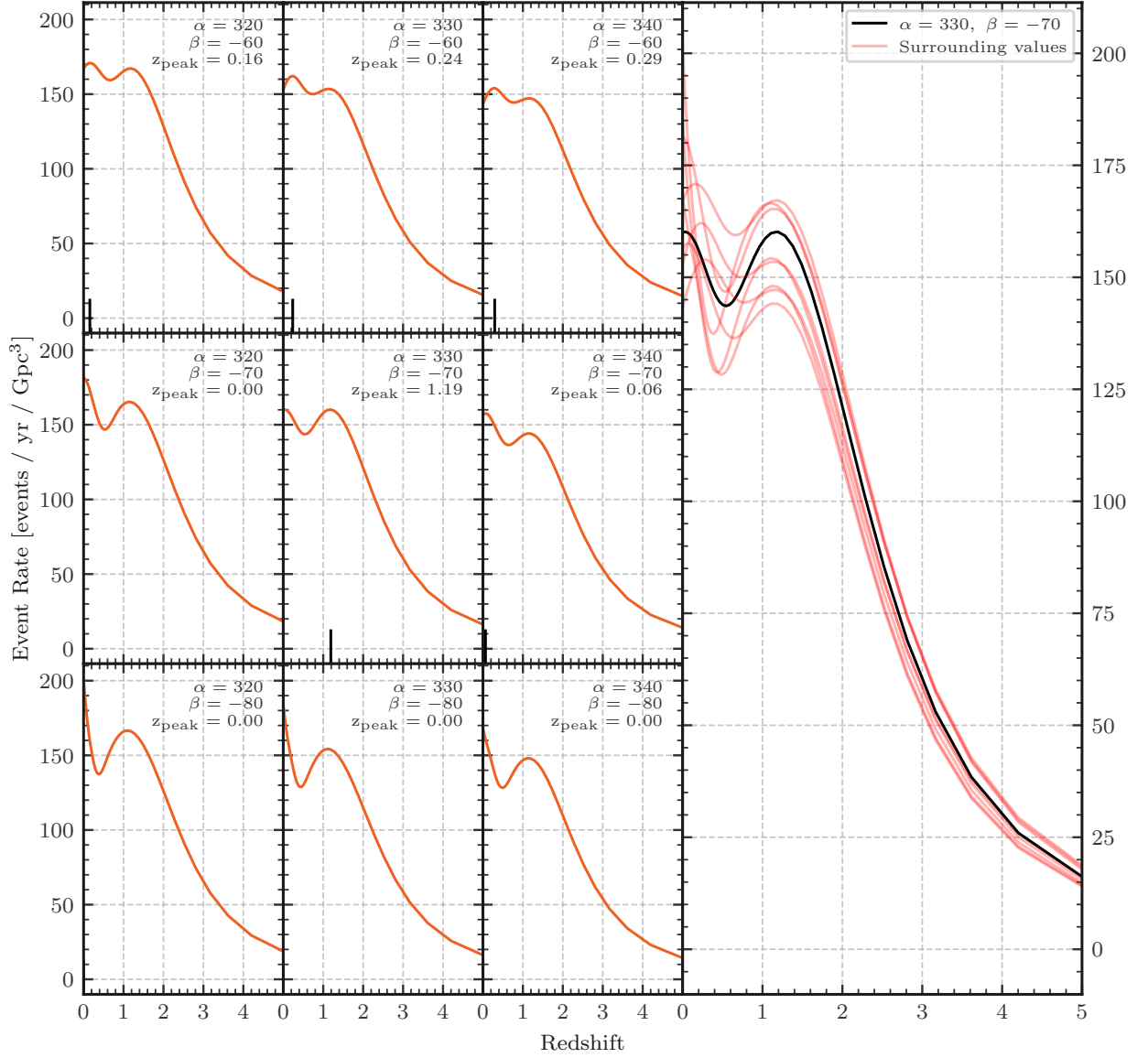


Figure 4.13: The event rate $\mathcal{R}(z)$ for a region surrounding $\alpha = 330, \beta = -70$ on FINEGRID. This region corresponds to the “outlier” in Figure 4.12. Shown on the left are the outlier bin in the center and its neighbours in both α -space and β -space. The black line on each plot denotes the location in redshift of the event rate distribution peak. The right plot shows the outlier in black and its neighbours in faded red behind.

closer than the innermost stable circular orbit (ISCO), given by $r_{\text{isco}} = \frac{6GM}{c^2}$, where matter can be exchanged. To account for this, one would need to apply corrections to the two-body equation of motion, termed the post-Newtonian approximation. The general equation becomes

$$\frac{d\vec{v}}{dt} = \frac{G(m_1 + m_2)}{r^2} \left(-\hat{n} + \sum_{k=2}^N \frac{1}{c^k} \vec{A}_{\frac{k}{2}\text{PN}} \right), \quad (4.10)$$

where N is the highest order term considered, and $\vec{A}_{\frac{k}{2}\text{PN}}$ is termed the “ $\frac{k}{2}$ -th Post-Newtonian” coefficient. Testing with the RIRORIRO code from [van Zeist et al. \(2021\)](#) indicated that including terms up to $N = 6$ (3PN) increases the merger time by fractions of a year, and thus does not significantly alter the merger rate density we observe. Typically 3.5PN ($N = 7$) is used as an upper bound ([Itoh & Futamase, 2003](#)), though up to 10PN ($N = 20$) has been used before ([Blanchet, 2014](#), and references therein).

Relativistic corrections also typically manifest in using more sophisticated spacetime metrics. Eqn(4.9) assumes a metric which is locally Minkowski, that is

$$g_{\mu\nu} = \eta_{\mu\nu} + \sqrt{32\pi G} h_{\mu\nu}, \quad (4.11)$$

where $\eta_{\mu\nu}$ is the positive-definite Minkowski metric. This “weak-field” approximation breaks down when a compact object enters within its companion’s ISCO. Given gravitational waves result in strains on the order of 10^{-19} ([van Zeist et al., 2023](#)), representing observed fractional shifts on Earth smaller than the diameter of a proton, ([Moore et al., 2014](#)), the weak field regime suits, and again modelling the relativistic corrections to the post-ISCO regime only increases the merger rate by fractions of years.

To contextualise this case study, other authors such as [Mapelli & Giacobbo \(2018\)](#); [Wiktorowicz et al. \(2019\)](#); [Giacobbo & Mapelli \(2020\)](#); [Santoliquido et al. \(2021\)](#); [Kapil et al. \(2023\)](#) have explored the parameter space of BNS kicks, though none have performed the analysis in the way we have. These works typically vary the progenitor population properties, in such a way that the formed binaries can in principle vary strongly. However, in our population synthesis procedure, we are able to effectively affix the physics between different models such that *only* the kick velocities are expected to vary between individual populations. Since the Bray kick has two parameters which can be altered, we are able to modify the kick velocity without changing the physics, thus allowing us to isolate and study the kick mechanism alone.

We found a selection of 13 different pairs of parameters that fit our four constraints. For values of $\alpha \geq 100 \text{ km s}^{-1}$, we found that low magnitude, but non-zero, values of β replicated the BNS merger rate from [LIGO-Virgo-KAGRA \(2021\)](#) and replicating the observed BNS systems compiled in [Vigna-Gómez et al. \(2018\)](#). A true ‘best’ set of parameters likely lie within this subsample of our parameter space.

Whilst the current observed BNS merger rate is limited, and has wide uncertainty bands, further observations will allow the distribution of the merger rate over redshift to become more refined. Since the peak of the event rate is dependent on the DTD, this distribution is also sensitive to the natal kick and thus can be used as a further constraint on the Bray kick parameters. In Figure 4.12 we display the location in redshift of this peak for our FINEGRID and WIDEGRID model sets. We find that the peak redshift across both grids

corresponds to $z < 1.5$, with a very clear and defined valley of the present day. Across the 13 pairs of parameters that fulfil all of our constraints, we also find that the peak corresponds to $z \leq 0.26$, and the best model $\alpha = 115 \text{ km s}^{-1}, \beta = 15 \text{ km s}^{-1}$ peaks in the present-day, at $z = 0$.

We have also highlighted the caveat that $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}, \beta = 15^{+10}_{-15} \text{ km s}^{-1}$ should not be interpreted as ‘the correct choice of parameters’ but rather ‘the mean of the set of parameters which replicates the four observables we test against’. It is also pertinent to note that these values are calibrated against the BPASS version 2.2.1 models (and should not change significantly when used with future stellar models from the same stellar evolution code). Using these values with other BPS codes should be approached with caution, and readers using other codes should calibrate their choice of parameters against their BPS codes in the same way that we have.

As part of the calibration and selection of binary parameters, we simulated 1,000 kicks per binary. Testing showed a slight, but not unexpected, dependence on number of simulated kicks. We found that adjusting the number of kicks simulated by a factor of ten resulted in a shift of $\log_e \mathcal{K}$ by approximately one.

Authors such as [Fryer et al. \(1998\)](#); [Arzoumanian et al. \(2002\)](#); [Bombaci & Popov \(2004\)](#) suggest that the kick distribution should ostensibly contain two peaks. In testing, we found no combination of α and β for the Bray kick produced an obviously bimodal single-star velocity distribution, though we do observe interesting structure in the resultant PDFs. Figure 4.11 clearly shows several changes of gradient in the CDF, which can be interpreted as several ranges of kick behaviour. This behaviour arises because the Bray kick is linked to the ejecta mass distribution. Thus the behaviour of low/high kick populations could be due to the results of stellar evolution causing low/high ejecta masses for CCSNe that form neutron stars.

The reason we did not include a bimodal Hobbs kick in this study is twofold. Firstly, the unimodal Hobbs kick has been extensively discussed in literature (see, among others, [Nakar, 2007](#); [Belczynski et al., 2008](#); [Lorimer, 2008](#); [Dominik et al., 2012](#); [Bray & Eldridge, 2016](#); [Eldridge et al., 2017](#); [Bray & Eldridge, 2018](#)) and thus was a good starting point for our investigation. Secondly, as a point of consideration, we computed the log-Likelihood of reproducing our BNS catalog against a weighted model from [Verbunt et al. \(2017\)](#), taking $w = 0.42, \sigma_1 = 75 \text{ km s}^{-1}, \sigma_2 = 316 \text{ km s}^{-1}$. We found that $\log_e \mathcal{K} = 0.8$. Thus, if this value were to be used as the reference model for the study, it would shift our log-Likelihoods by approximately one order of magnitude and would only admit an extra handful of systems.

In the future, an additional prior can be used on our dataset. The natural interpretation of the Bray kick expression is that β is the ‘limiting value’ that the kick velocity should approach as the ejecta mass gets smaller (as in the case of USSNe and ECSNe). In events such as the direct collapse of a star to a black hole in a failed SNe, where the ejecta mass is zero, β would identically be the kick velocity. Thus, in order for conservation of momentum to be respected for low-ejecta mass kicks, we expect $|\beta| > 0$ – i.e., we expect the magnitude of β to be low, but most likely non-zero. This prior was not imposed on our parameter space – the $\log_e \mathcal{K}$ and USSNe constraints independently restricted β to be > 0 . It is pertinent to note, as well, that even in the limit as the ejecta mass goes to zero, a value of $\beta < 0$ would imply that the velocity vector is anti-aligned with the chosen direction of the kick. $\beta < 0$ also allows for a zero kick velocity when the ejecta

mass is non-zero. At this stage, this scenario cannot be ruled out, thus we cannot yet include this constraint as a prior.

In addition, we found that the mean pair of parameters – the center of the cyan region in Figure 4.10 – corresponded to $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$. Although this work does not purport to claim a single ‘best’ pair of parameters, these ones can be interpreted as the pair that optimises each of the individual tests that we have performed. We note that this value lies off-grid, and hence in Figure 4.11 we generated a bespoke data file for this set of two parameters. The optimal form of the kick, for the BPASS models, is therefore

$$v_{\text{kick}} = 115_{-55}^{+40} \left(\frac{m_{\text{ejecta}}}{m_{\text{remnant}}} \right) + 15_{-15}^{+10} \text{ km s}^{-1}. \quad (4.12)$$

In sum, the key conclusions we draw are:

1. Providing multiple constraints allows for the parameter space of possible kick configurations to be drastically reduced.
2. Admissibility of kick velocities of $\lesssim 30 \text{ km s}^{-1}$ for USSNe is one of the tightest constraints we can provide.
3. The peak of the merger rate distribution is likely to be the present-day, and this is a further constraint we can apply to the dataset when more information is available.
4. The center of mass of the permitted region of the Bray parameter space is $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$, although caution should be exercised before using this value without calibration.

The analysis presented here is an analysis pipeline to refine the properties of kick models against four different observables. In principle, it can be applied to any kick model to validate the model. Our statistical analysis provides a simple way to validate kick models: it can be used to rule out either magnitudes of kick velocities or entire physical kick prescriptions.

4.4 | Chapter Summary

We have presented a formalism for evaluating and quantifying the goodness-of-fit of the natal kick mechanism against four independent tests. The tests we have presented – constraining the merger rate of BNS systems, the period-eccentricity distribution of Galactic BNS systems, the kick velocities of single-star pulsars, and the production of ultra-stripped supernovae – greatly constrain the parameter space to be less than 1 per cent of the possible parameters. Furthermore, the data suggest that the peak of the merger rate distribution is the present day, with $z \simeq 1.5$ as the upper bound of the entire parameter space, and $z = 0$ for our recommended value.



CHAPTER 5

Detailed Models of Overcontact Binary Stars



I can remember... sitting on a rooftop. Looking up and wondering what the stars were. I assumed I'd never know.

— *Hoid* —





AS PREVIOUSLY DISCUSSED, STARS undergoing mass transfer will generally experience RLOF during their lifetimes. In a binary where both stars are undergoing mass transfer, both stars may undergo RLOF simultaneously, resulting in a contact binary system in which both stars physically touch. Contact binaries are not uncommon – they form one in every approximately 100 low-mass main-sequence stars (Rucinski, 1998) and 39 in every 100 binary systems (Green et al., 2023).

Modelling their evolution generally offers understanding into difficult-to-probe stellar phenomena and evolutionary phases, such as the common envelope phase of binary evolution, as well as providing vital insights into the immediate pre-merger environment of a binary system. They are characterised by extremely short periods – typically with $P_{\text{orb}} \lesssim 2$ d, but potentially as low as $P_{\text{orb}} = 85$ min in theory (El-Badry et al., 2022) or ~ 167 min in practice (Koen, 2022). The lower bound of the period distribution is an element of some debate and ongoing research; Qian et al. (2020) finds it to be $0.15 \text{ d} \sim 200$ min, and Zhang et al. (2023) finds it to be at $0.12 \text{ d} \sim 170$ min. The contact binary phase of evolution can give rise to pre-supernova mergers, chemically abundant mass-transfer drastically altering the interior structure of the accretor (Chapter 3), and other modified evolutionary behaviour of the binary due to interaction effects stronger than that of wider-period binaries.

As a result of this close interaction, contact binaries may also be a formation channel for stellar exotica such as TŻOs, first proposed by Thorne & Żytkow (1977), where the capture of a neutron star by a supergiant companion in a close binary causes the formation of a TŻO (Taam et al., 1978). It is pertinent to note that these are not the *sole* formation channels for such exotica – for instance, a TŻO could be formed via a primary supernova kicking a nascent neutron star into its companion (Leonard et al., 1994; Podsiadlowski et al., 1995) or via collisions in clusters (Podsiadlowski et al., 1995) – but are instead potential formation channels.

Accurately modelling close-contact binaries, then, is of crucial importance to population synthesis models. Recent studies such as Menon et al. (2021b); Abdul-Masih et al. (2022); Sen et al. (2022) have explored some of the close-contact binaries in the Magellanic Clouds. Menon et al. (2021b) for example finds that for both Magellanic Clouds the majority of contact binaries are expected to have total masses of under $\sim 40 M_{\odot}$ and periods of less than ~ 1 d, and Abdul-Masih et al. (2022) finds that the sample of close-contact binaries that they analyse are likely to form with relatively longer periods than most systems.

Despite the importance of modelling close-contact binaries, only ~ 25 massive close-contact binaries are known to exist (Abdul-Masih, 2023). There are more known so-called W Ursae Majoris variables, with 700 compiled by Latković et al. (2021). These stars, typified by their namesake W Ursa Majoris, are low-mass binaries, typically with mass ratios of ~ 0.3 , and similar surface temperatures, though this study focuses on massive binaries. The paucity of observations is a reflection of the difficulty of distinguishing these systems from a more massive single star, especially since the two stars would share a common temperature (Kuiper, 1941; Latković et al., 2021) and thus similar observational properties. Spectroscopic observations also do not constrain physical parameters of contact binaries as well as they do for regular binaries (or single stars); for instance Gazeas et al. (2021) discuss the O’Connell effect and surface spots of the stars as being sources that modulate the luminosity of the binaries, but especially of the secondary.

Binary name	M_1	M_2	P	Z	$\log T_1$	$\log L_1$	$\log T_2$	Ref.
LSS 3074	14.8 ± 2.2	17.2 ± 4.2	2.183 ± 1	0.020^a	4.6 ± 0.2	5.15 ± 0.2	4.53 ± 0.5	A
MY Cam	37.7 ± 3.2	31.6 ± 4.2	$1.175^{+6}_{-1.175}$	0.020^a	4.62 ± 0.2	5.2 ± 0.5	4.59 ± 0.5	B
SMC 108086	16.9 ± 1.2	14.3 ± 5.1	0.883 ± 0.88	0.004^b	4.53 ± 0.1	4.608 ± 0.4	4.48 ± 0.5	C
TU Mus	16.7 ± 1.6	10.4 ± 1.4	1.387 ± 1	0.020^c	4.57 ± 0.2	5.02 ± 0.4	4.54 ± 0.5	D
V382 Cyg ^d	26.1 ± 3.2	19.0 ± 4.2	1.886 ± 1	0.020^e	4.57 ± 0.2	5.47 ± 0.4	4.58 ± 0.5	E
VFTS 352	28.6 ± 3.6	28.9 ± 1.8	$1.124^{+2}_{-1.124}$	0.010^f	4.63 ± 0.45	5.19 ± 0.5	4.61 ± 0.5	F

Table 5.1: Search parameters for the six close-contact binaries compiled by [Abdul-Masih et al. \(2022\)](#), as used in our fitting to the BPASS v2.2.1 grid. The uncertainties presented are not the physical uncertainties on their parameters, but instead represent the size of the range that we search within, e.g., $P = 1.175^{+6}_{-1.175}$ indicates that we match on all periods between 0 and 7.165 d inclusive. Masses and luminosities are measured relative to Solar, periods are measured in days, and temperatures measured in Kelvin. **Notes:** ^aSource paper assumes solar metallicity with no further qualification ^bUsing the mean metallicity of the SMC ^cBased on [Andersen & Grønbech \(1975\)](#) claiming 0.021 and [Terrell et al. \(2003\)](#) confirming a comparison with synthetic spectra “does not support a carbon content, and therefore metallicity, far from the solar value” ^dA mass ratio inversion occurred for this binary (see [Martins et al., 2017](#)) and hence we use the definition of “primary” from within the paper when reporting in this table. ^eUsing the mean metallicity of the Milky Way ^fAs [Crowther et al. \(2018\)](#) reports “half-solar” for the Tarantula Nebula, the host nebula of VFTS 352. **References:** A: [Raucq et al. \(2017\)](#), B: [Lorenzo et al. \(2014\)](#), C: [Hilditch et al. \(2005\)](#), D: [Penny et al. \(2008\)](#), E: [Martins et al. \(2017\)](#), F: [Almeida et al. \(2015\)](#)

5.1 | Observations of contact binaries

There’s a place called Wonderland.

— FIFTH OF THE SIX IMPOSSIBLE THINGS

Observations of contact binaries date back to the 1940s ([Kuiper, 1941](#)), and in the past eighty years the catalogue has grown to somewhere between $\sim 42\,000$ ([Petrosky et al., 2021](#)) and $\sim 86\,000$ ([Qian et al., 2020](#)) contact binaries, representing between 26.7 and 54.8 per cent of all observed binaries (those from the Washington Double Star catalogue; [Mason et al. 2001](#)). The binaries we study in this chapter are summarised in Table 5.1, and were originally compiled by [Abdul-Masih et al. \(2022\)](#). Their sample focused on measuring the period stability of contact binaries and is thus a subset of the known sample. They rejected a handful of other non O+O overcontact binaries, for reasons that include short-period von Zeipel-Kozai-Lidov oscillations, incomplete photometric / radial velocity fits, non-MS binaries, unknown separations, and irregular light curves.

Ages for the six binaries have not been published, though a soft restriction can be placed on their evolutionary status; since [Abdul-Masih et al. \(2022\)](#) compile specifically massive MS contact binaries, their interactions must take place whilst the progenitor stars are on their respective main sequences.

5.1.1 | Progenitor history

We now discuss each of the overcontact systems in detail, providing observational context to each system. For each binary, we list a selection of key measurements, including the fillout factor for each star. A fillout

factor is a measurement of the degree to which a system overfills its Roche lobes, with $1 < f < 2$ denoting an overcontact binary, and a high fillout factor representing a more swelled and in-contact system. This definition of the fillout factor is a property of the *system*, not the *constituent stars*, and originates from [Mochnicki & Doughty \(1972\)](#).

LSS 3074

The binarity of LSS 3074 ([Raucq et al., 2017](#)) was first observed in 1985 by [Morrell & Niemela \(1990\)](#) with an observed period of 2.19 d, with its HeI and HeII spectra causing the authors to conclude that it was an O4f+ and O6-7 binary, with inclination-weighted masses of $M_1 \sin^3 i = 8 M_\odot$ and $M_2 \sin^3 i = 9 M_\odot$. With later measurements of the inclinations to be $i \approx 24^\circ$, these masses were refined to $M_1 = 17.2 M_\odot$ and $M_2 = 14.8 M_\odot$. It is located behind the Coalsack Nebula, at a distance of 2256 pc, with a B-band magnitude of 12.81 mag ([Stephenson & Sanduleak, 1971](#)). It has a fillout factor $f = 1.05$.

MY Camelopardalis

MY Camelopardalis (MY Cam; [Lorenzo et al., 2014](#)) is a binary in the cluster Alicante 1, and its binarity was established in 2003 by [Negueruela & Marco \(2003\)](#). It is likely to be a very massive merger progenitor, though is barely in contact with $f = 1.01$ ([Abdul-Masih et al., 2022](#)).

SMC 108086

OGLE SMC-SC10 108086 (SMC 108086; [Hilditch et al., 2005](#)) is a O9 + O9.5/B0 binary. It is a system deeply in contact, with $f = 1.70$ ([Hilditch et al., 2005](#); [Abdul-Masih et al., 2022](#)) – the deepest known – and has a short period of $P = 0.88$ d. It is located at a distance of 56.5 kpc, in the Small Magellanic Cloud (SMC).

TU Muscae

TU Muscae (TU Mus; [Penny et al., 2008](#)) is an O8V + B0V binary discovered by [Oosterhoff \(1928, 1930\)](#). Its binarity was confirmed as late as 1971, by [Knipe \(1971\)](#). The system was profiled by [Andersen & Grønbech \(1975\)](#), who concluded that, based on comparison to theoretical models by [Stothers \(1972, 1973\)](#), the system was of solar composition ($Z = 0.021 \approx Z_\odot$) and on the main sequence with $X_H = 0.739$. It has a fillout factor of $f = 1.12$.

V382 Cygni

V382 Cygni (V382 Cyg; [Martins et al., 2017](#)) is an O6.5 + O6 binary whose components are oblate spheroids (deduced from the β -Lyrae light curves; [Pearce 1952](#)). It was originally discovered by [Morgenroth \(1935\)](#); [Petrov \(1946\)](#). It has a fillout factor of $f = 1.10$.

VFTS 352

VFTS 352 was detected in the VLT-FLAMES survey of the Tarantula Nebula (Evans et al., 2011) and is the most recent detection in our sample. The Tarantula Nebula / 30 Doradus is a large H(II) region in the Large Magellanic Cloud (LMC). The progenitor of VFTS 352 is likely an equal mass, short-period binary with no strong helium or nitrogen surface abundances (Abdul-Masih et al., 2019), which may be due to biases in the CNO abundances, or similar, in that atmospheric study. It has a fillout factor of $f = 1.28$.

5.1.2 | Data matching with BPASS v2.2

To constrain the nature and prior evolution of these contact binaries we attempt to match stellar models from the BPASS v2.2 models to these binaries. We perform our data matching using the HOKI code (Stevance et al., 2020). We look for matches in masses, periods, temperature / luminosity of the primary, and temperature of the secondary. The luminosity of the secondary in each case is less constrained, and in some cases unpublished. The search parameters are shown in Table 5.1. We query the BPASS v2.2.1 dataset and record whether or not they enter the (widened) parameter values. Parameters are widened as some of the periods are tighter than the BPASS bins and thus can be missed in the search.

We note three key caveats in our search criteria.

1. We have performed our matching based on parameters which are difficult to constrain for an eclipsing binary. The better choice would be to match primarily on the masses / radii, which are known more precisely for eclipsing binaries (as the measurement dependence is on the times of first and second contact, and the relative velocity of the stars, which can be determined through radial velocity measurements). The rationale is that the BPASS v2.2 models use the Hurley et al. (2000) approximations, which become highly approximate during the mass transfer phase – resulting in the mass, temperature, and luminosity of the companion becoming more approximate during mass transfer. Consequently, we must preference matching on the donor parameters over the accretor parameters, though we include the accretor mass and effective temperature to allow for it to have some influence on the output models. This also means that we must be content with the wider error bars on the search parameters this necessitates. Matching on six parameters instead of four also means that our results are more likely to be outliers; they may either be overfits or not fit at all, with both a result of necessarily matching on more parameters.
2. As a consequence of the first caveat, the search error bars are widened, in some cases substantially. The known parameters by Abdul-Masih et al. (2022) are so tightly constrained that the parameter grid spacing in BPASS becomes significant, and thus the search bars must be widened to account for this. This is also made manifest by the approximations in BPASS v2. BPASS uses the approximations of Hurley et al. (2000) for the secondary radius, temperature, and luminosity. Thus, we widen our parameters for the secondary to compensate for this approximation.
3. As alluded to previously, luminosities are not uniformly reported for the companions across the grid.

In particular, the companion luminosity for VFTS 352 is unreported, and thus we approximate it using the Stefan-Boltzmann law,

$$\frac{L}{L_{\odot}} = \left(\frac{R}{R_{\odot}}\right)^2 \left(\frac{T}{T_{\odot}}\right)^4, \quad (5.1)$$

where $T_{\odot} = 5778$ K. This induces two assumptions: that the companion is a perfect blackbody, and that it is spherical. The first assumption is typically assumed regardless, and the secondary is a necessary condition of our 1D stellar evolution approach. Although the star is, physically, distorted with a fillout factor of $f = 1.28$, the key assumption underpinning 1D stellar evolution is spherical symmetry.

The matching values HOKI finds are shown in Table 5.2. The primary models are also shown on Figure 5.1. We also report in the table the inferred ages at match, which lie between $\mu_{\log(\text{age})} = 6.13$ (for MY Cam) and $\mu_{\log(\text{age})} = 7.14$ (for TU Mus).

To determine the likely future history of this sample, we must use the TUI code to match pre- and post-supernova models. To that end, in Table 5.3, we enumerate every resultant match to the observed parameters from Table 5.1. For each match, we display the masses, radii, and effective temperatures of each star in the binary. We also track whether the stars are – at the time of match – filling their Roche lobes.

Our modelling shows that all of these contact binaries survive their contact phase, eventually exiting to continue on as semi-detached and detached binaries. Importantly, none overflow their $\mathcal{L}_2$ points – which is commonly taken as a precursor signal of a merger event in other codes such as POSYDON (Fragos et al., 2023) – and none of them merge. As they evolve, as shown in Table E.1, the primary typically ends up as a helium star or WR star, and the secondaries become more massive, typically with a post-contact mass ratio inversion occurring.

The full future history is completed by examining the remnants both after the primary and secondary supernova. To account for the uncertainties present in the natal kick prescription (Chapter 4), we consider three kick prescriptions: the Hobbs et al. (2005) and Richards et al. (2023) kicks from Chapter 4, and the bimodal Verbunt et al. (2017) kick. The Hobbs et al. (2005) and Richards et al. (2023) kicks were described in Chapter 4, and the Verbunt et al. (2017) kick motivates its choice with a second peak at lower kick velocities to account for phenomena such as electron-capture supernovae and USSNe. It samples the kick velocity from a modified Maxwellian distribution, a la Hobbs et al. (2005), though the distribution is given by

$$P(v_k) dv_k = \sqrt{\frac{2}{\pi}} v_k^2 \left[\frac{w}{\sigma_1^3} \exp\left(-\frac{v_k^2}{2\sigma_1^2}\right) + \frac{1-w}{\sigma_1^3} \exp\left(-\frac{v_k^2}{2\sigma_1^2}\right) \right], \quad (5.2)$$

where $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$ are the distribution parameters and $w = 0.42$ is the weight of the lower peak. For each binary, we compute the post-supernova kick parameters M_1, M_2, a_0 and e_0 in the same way as in Chapter 4, and – after the secondary supernova – use these to calculate the coalescence time in Gyr.

For each system, we record the type of binary left after the secondary supernova – NS-BH, Black Hole – White Dwarf Binary (BH-WD), Neutron Star - White Dwarf Binary (NS-WD), BNS, Binary Black Hole

Binary	BPASS v2.2.1 matches	$M_{1,ZAMS}$	$M_{2,ZAMS}$	q	P_{ZAMS}	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$
LSS 3074	sneplot-z020-30-0.3-0.6	30.0	9.0	0.3	3.98	6.54	0.64
	sneplot-z020-25-0.4-0.6	25.0	10.0	0.4	3.98	6.80	0.12
	sneplot-z020-25-0.5-0.6	25.0	12.5	0.5	3.98	6.80	0.13
	sneplot-z020-24-0.4-0.6	24.0	9.6	0.4	3.98	6.83	0.09
	sneplot-z020-24-0.5-0.6	24.0	12.0	0.5	3.98	6.83	0.09
MY Cam	sneplot-z020-50-0.5-0.6	50.0	25.0	0.5	3.98	6.35	0.67
	sneplot-z020-50-0.4-0.8	50.0	20.0	0.4	6.31	6.31	0.72
	sneplot-z020-50-0.5-0.8	50.0	25.0	0.5	6.31	6.21	0.81
	sneplot-z020-50-0.6-0.8	50.0	30.0	0.6	6.31	6.13	0.88
	sneplot-z020-40-0.7-0.8	40.0	28.0	0.7	6.31	6.43	0.66
SMC 108086	sneplot-z004-24-0.5-0.4	24.0	12.0	0.5	2.51	6.47	0.77
	sneplot-z004-23-0.4-0.4	23.0	9.2	0.4	2.51	6.63	0.63
	sneplot-z004-23-0.5-0.4	23.0	11.5	0.5	2.51	6.62	0.65
	sneplot-z004-22-0.4-0.4	22.0	8.8	0.4	2.51	6.70	0.57
	sneplot-z004-22-0.5-0.4	22.0	11.0	0.5	2.51	6.70	0.57
	sneplot-z004-21-0.4-0.4	21.0	8.4	0.4	2.51	6.77	0.50
	sneplot-z004-21-0.5-0.4	21.0	10.5	0.5	2.51	6.76	0.51
	sneplot-z004-20-0.4-0.4	20.0	8.0	0.4	2.51	6.79	0.49
	sneplot-z004-20-0.5-0.4	20.0	10.0	0.5	2.51	6.78	0.49
	sneplot-z004-19-0.4-0.4	19.0	7.6	0.4	2.51	6.80	0.49
TU Mus	sneplot-z020-25-0.3-0.4	25.0	7.5	0.3	2.51	6.69	0.56
	sneplot-z020-24-0.3-0.4	24.0	7.2	0.3	2.51	6.71	0.56
	sneplot-z020-23-0.3-0.4	23.0	6.9	0.3	2.51	6.77	0.52
	sneplot-z020-16-0.6-0.4	16.0	9.6	0.6	2.51	7.14	0.05
V382 Cyg	sneplot-z020-50-0.2-0.6	50.0	10.0	0.2	3.98	6.44	0.56
	sneplot-z020-40-0.3-0.6	40.0	12.0	0.3	3.98	6.52	0.54
	sneplot-z020-40-0.4-0.6	40.0	16.0	0.4	3.98	6.51	0.55
	sneplot-z020-35-0.4-0.6	35.0	14.0	0.4	3.98	6.55	0.56
	sneplot-z020-30-0.4-0.6	30.0	12.0	0.4	3.98	6.60	0.57
	sneplot-z020-30-0.5-0.6	30.0	15.0	0.5	3.98	6.59	0.57
VFTS 352	sneplot-z010-40-0.5-0.6	40.0	20.0	0.5	3.98	6.55	0.53

Table 5.2: Matches from the HOKI (Stevance et al., 2020) software. Masses are measured in solar masses, periods in days, and ages in years. Ages were computed as the mean of the ages in log-space where a match occurs, and similar for the standard deviations.

ZAMS parameters												Match Parameters			
System Name	$M_1 / M_\odot$	$M_2 / M_\odot$	$\log P / \text{d}$	$M_1 / M_\odot$	$M_2 / M_\odot$	$\log R_1 / R_\odot$	$\log R_2 / R_\odot$	$\log T_1 / \text{K}$	$\log T_1 / \text{K}$	Star 1 RLOF	Star 2 RLOF				
LSS 3074	55.00	15.00	0.50	36.02	17.92	1.03	0.81	4.54	4.58	✓	✓				
LSS 3074	50.00	15.00	0.90	26.98	22.14	1.06	0.81	4.53	4.61	✓	✓				
TU Mus	30.00	5.00	0.70	15.46	8.58	0.91	0.80	4.55	4.52	✓	✓				
TU Mus	55.00	10.00	0.90	26.25	17.99	0.96	0.66	4.55	4.56	✓	✓				
V382 Cyg	45.00	15.00	0.50	27.98	19.29	0.98	0.90	4.54	4.61	✓	✓				
V382 Cyg	50.00	15.00	0.50	31.74	18.66	1.00	0.86	4.54	4.61	✓	✓				
V382 Cyg	50.00	20.00	0.50	36.95	23.71	1.07	0.90	4.55	4.58	✓	✓				
V382 Cyg	45.00	25.00	0.50	28.55	29.34	0.97	0.93	4.54	4.56	✓	✓				
V382 Cyg	35.00	10.00	0.60	20.64	14.32	0.99	0.92	4.54	4.56	✓	✓				
V382 Cyg	40.00	10.00	0.60	23.91	13.98	0.99	0.88	4.57	4.57	✓	✓				
V382 Cyg	55.00	10.00	0.60	30.19	13.38	0.97	0.80	4.56	4.61	✓	✓				
V382 Cyg	40.00	15.00	0.60	25.57	20.74	1.03	0.96	4.54	4.60	✓	✓				
V382 Cyg	45.00	15.00	0.60	30.64	19.51	1.07	0.94	4.54	4.58	✓	✓				
V382 Cyg	45.00	20.00	0.60	27.08	26.03	0.99	0.88	4.54	4.61	✓	✓				
V382 Cyg	45.00	15.00	0.70	26.15	20.82	1.01	0.83	4.54	4.61	✓	✓				
V382 Cyg	50.00	20.00	0.70	27.80	29.75	0.98	0.85	4.58	4.62	✓	✓				
LSS 3074	30.00	15.00	0.60	17.23	22.90	1.01	0.84	4.53	4.59	✓	✗				
TU Mus	30.00	10.00	0.50	16.44	14.94	0.90	0.73	4.56	4.56	✓	✗				
TU Mus	30.00	10.00	0.60	15.39	16.34	0.93	0.67	4.56	4.54	✓	✗				
TU Mus	30.00	5.00	0.80	14.81	9.36	0.96	0.71	4.55	4.55	✓	✗				
V382 Cyg	35.00	15.00	0.50	19.62	21.78	0.92	0.82	4.57	4.61	✓	✗				
V382 Cyg	35.00	20.00	0.50	20.82	28.25	0.96	0.90	4.56	4.61	✓	✗				
V382 Cyg	40.00	20.00	0.50	16.71	28.23	0.91	0.89	4.59	4.61	✓	✗				
V382 Cyg	35.00	20.00	0.60	20.84	30.86	1.02	0.89	4.56	4.61	✓	✗				
V382 Cyg	35.00	10.00	0.70	20.12	15.43	1.03	0.91	4.54	4.57	✓	✗				

Table 5.3: Matches from the TU1 code. All parameters are rounded to two decimal places, and are given as at the first moment of matching. Only binaries that have at least one component filling their Roche lobe are shown in the list, with the balance listed in Table D.1.

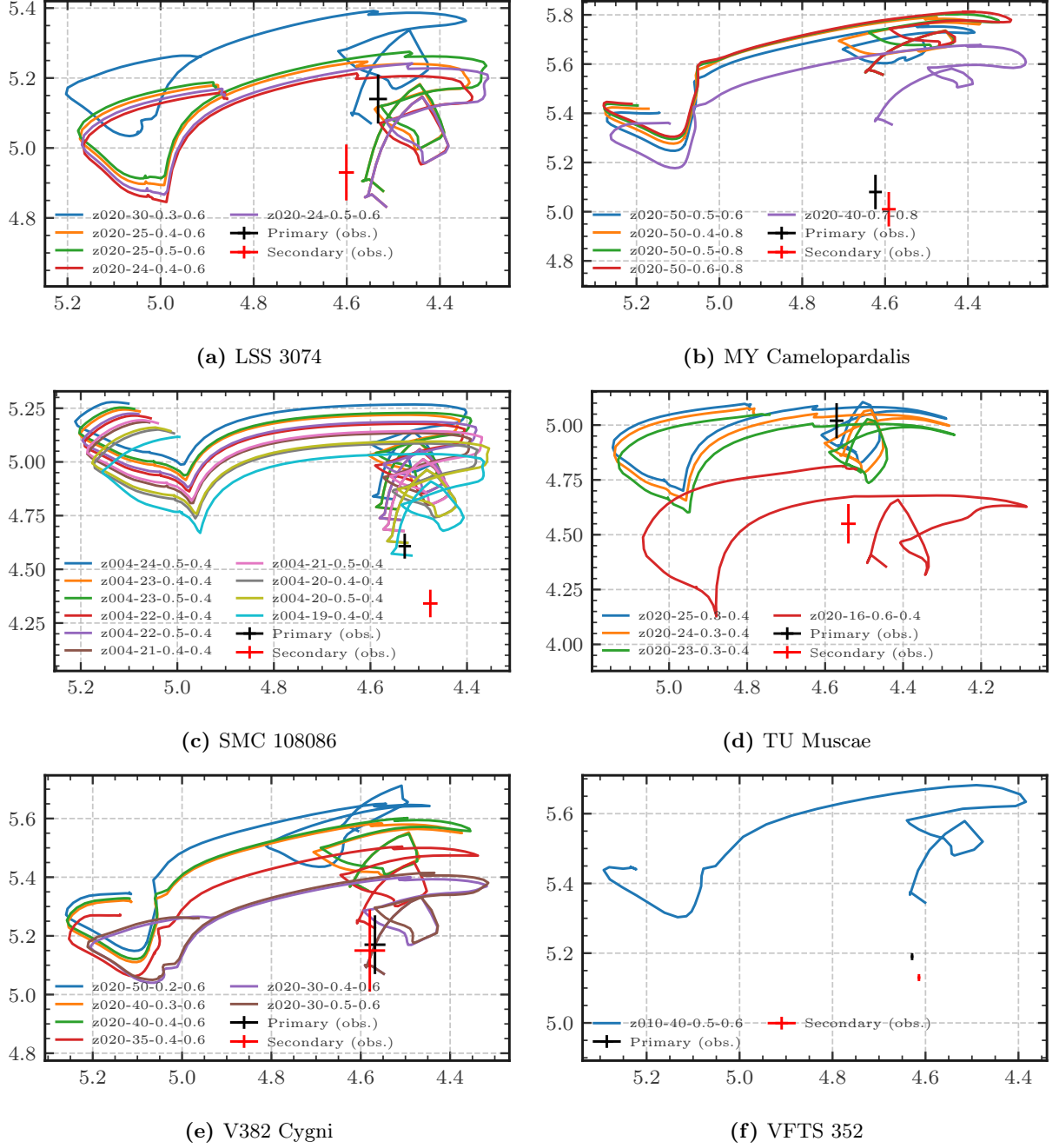
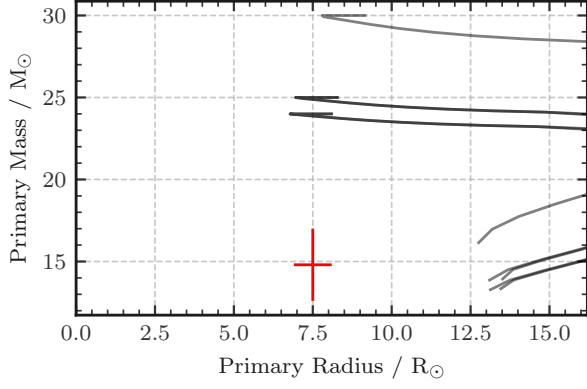
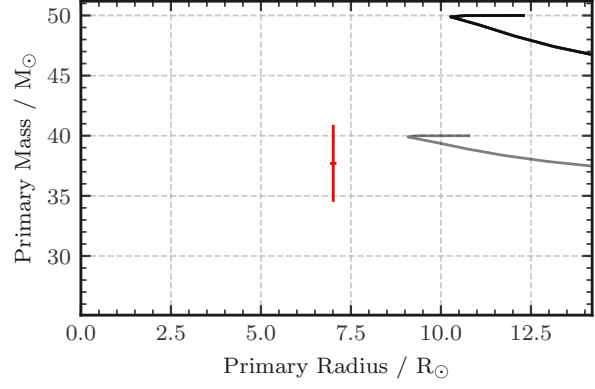


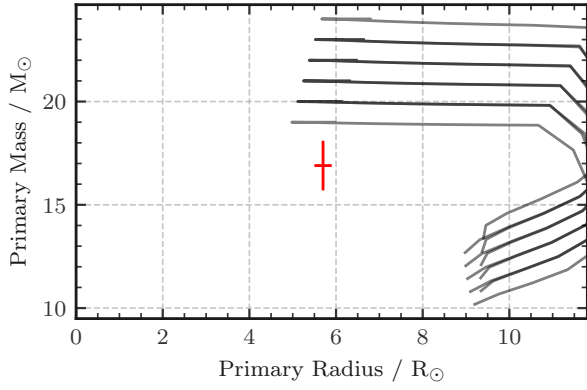
Figure 5.1: BPASS v2.2.1 primary models for the matches obtained by HOKI for the parameters in Table 5.1. The legend key is the BPASS stellar model file name (omitting the “sneplot-” prefix common to every entry), and is broken down into its parts in Table 5.2. The overplotted observed systems do not match every system, and the search error bars were widened somewhat as a result (see Table 5.1 for details).



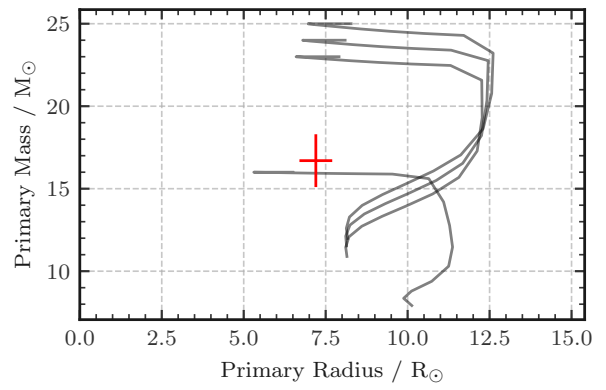
(a) LSS 3074



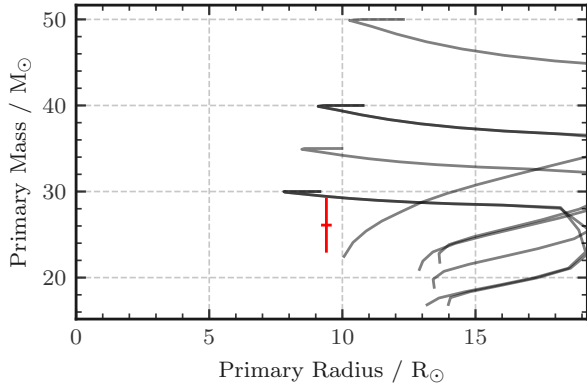
(b) MY Camelopardalis



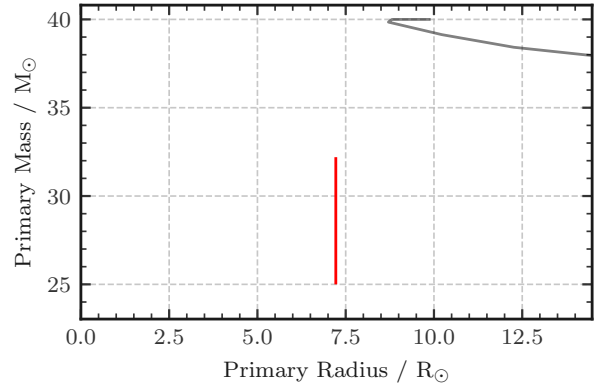
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure 5.2: BPASS v2 model tracks for our six close-contact binaries. Models shown are restricted to only during their core hydrogen burning phase, which for this plot we assume to be when the helium core mass $M_{\text{He}} < 10^{-5} M_{\odot}$. The x -axis is limited to double the maximum value of the observed radius.

(BBH). If the mean eccentricity is less than zero or strictly greater than one, we record it as “Unbound”, and present the percentage of systems in which the binary unbinds. If the binary remains bound, we present the coalescence time, which is allowed to be arbitrarily large (with the largest being 1.75 Pyr). In practice, arbitrarily large coalescence times are likely unphysical, with Pyr coalescence times approaching the end of the stelliferous era. Nevertheless, we include them as predictions made by the integration of the equations of [Peters \(1964\)](#).

We assume that a white dwarf is left if the CO core mass of the star is less than $1.38 M_{\odot}$, the total mass of the star is less than $1.5 M_{\odot}$, or if core carbon burning does not occur. Otherwise, we assume that remnants with $M_{\text{rem}} \leq 2.5 M_{\odot}$ leave behind a NS, and otherwise a BH is left.

Across the three kick models, in approximately 55.2 per cent of the binaries, the binary remains bound after the secondary supernova. Most of the unbound models (44.4 per cent) originate from the [Verbunt et al. \(2017\)](#) kick, and the [Richards et al. \(2023\)](#) kick from Chapter 4 contributes 17.8 per cent of the unbound models. The bound models typically have initial post-secondary-supernova periods of tens of years, with a notable outlier in MY Camelopardalis, which has two systems with separations $> 100 R_{\odot}$, creating large periods.

Binary	$\bar{a}_0 / R_{\odot}$	$\bar{e}_0[no.dim]$	$\bar{P}_0 / \text{yr}$
LSS 3074	6.77 ± 0.10	0.55 ± 0.09	1.433 ± 0.034
MY Cam	70.17 ± 0.10	0.68 ± 0.07	56.23 ± 0.05
SMC 108086	8.05 ± 0.16	0.67 ± 0.12	1.72 ± 0.05
TU Mus	7.69 ± 0.20	0.59 ± 0.12	2.04 ± 0.05
V382 Cyg	17.38 ± 0.08	0.66 ± 0.06	5.376 ± 0.034
VFTS 352	29.62 ± 0	0.9 ± 0	7.31 ± 0

Table 5.4: The mean post-secondary-supernova parameters for the bound remnants of our binary models. Uncertainties are the 1σ uncertainties from taking the mean across all parameters, and the uncertainty on VFTS 352 is due to there only being one bound remnant in the calculation. Individual kick prescriptions are not distinguished for this calculation.

A wide variety of remnants are possible; these are summarised below:

1. LSS 3074

- NS-NS (9 occurrences)
- NS-WD (3 occurrences)

2. MY Cam

- NS-NS (5 occurrences)
- BH-NS (4 occurrences)
- NS-WD (3 occurrences)
- NS-BH (1 occurrence)

3. SMC 108086

- NS-NS (15 occurrences)
- NS-WD (2 occurrences)

4. TU Mus

- NS-NS (7 occurrences)
- NS-WD (2 occurrences)

5. V382 Cyg

- NS-NS (7 occurrences)
- NS-WD (1 occurrence)

6. VFTS 352

- BH-BH (1 occurrence)

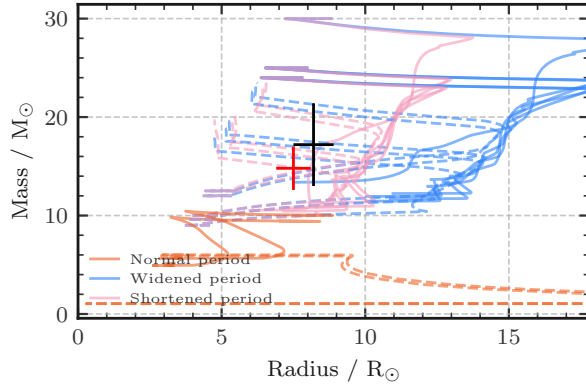
We present the full table of results in Appendix E. We can see a number of extremely low-mass WDs, the lightest of which are $0.28 M_{\odot}$, barely above the current lowest-mass WD ($0.17 M_{\odot}$; [Kilic et al., 2007](#)). It is possible that these may be numerical artefacts, and that the supernova has either destroyed the star leaving effectively no remnant, or that the stars merged during the supernova event. This would be compounded by the stars appearing to be unbound, with high frequency. The TUI code does not record whether a merger occurs during the supernova, so this is not possible to test given the current architecture of the data.

5.2 | Creation of bespoke binary star evolution models

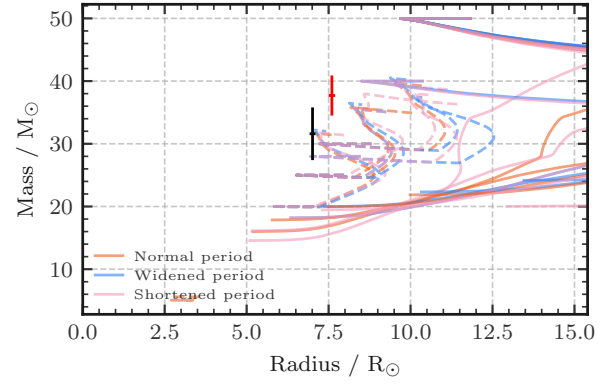
Having now established the ZAMS parameters for our binaries, we can take these as input to our STARS code to determine their future history. For each matched model, we use the initial masses and periods as inputs to the STARS evolution code. We use the AOTEAROA STARS binary star evolution program as described in Chapters 2 and 3.

While we use the periods from the BPASS model fits we also expand our search to two extra periods per pair of initial masses. The first is the fiducial period, as reported from the HOKI code. The second is a set of binaries with identical parameters except for periods 0.1 dex higher, and the third is binaries with periods 0.1 dex lower. Our aim, in evaluating the expanded period grid, is to determine the sensitivity of the grid to this parameter choice, given that the major aim of this chapter is to investigate, effectively, short-period binaries.

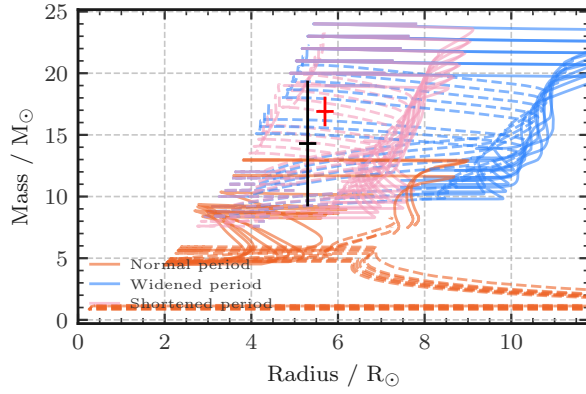
The results of our forward evolution are shown in Appendix D. We also show summary plots for the six close-contact binaries in Figure 5.3, where we plot the primary mass against the primary radius. As the six contact binaries are all eclipsing binaries, the masses and radii are the most well-constrained parameters.



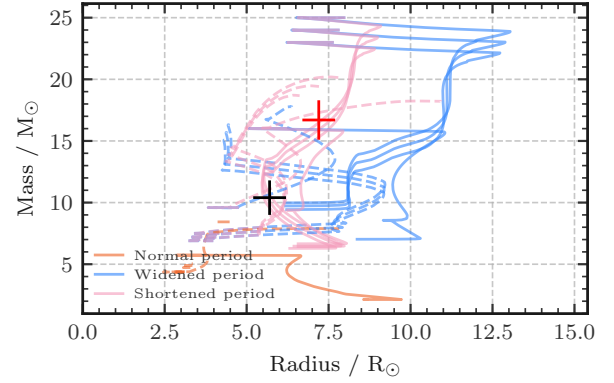
(a) LSS 3074



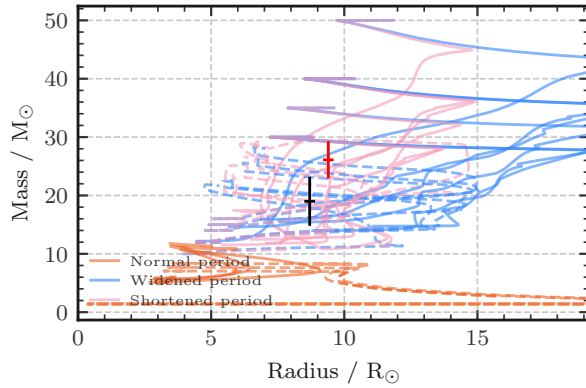
(b) MY Camelopardalis



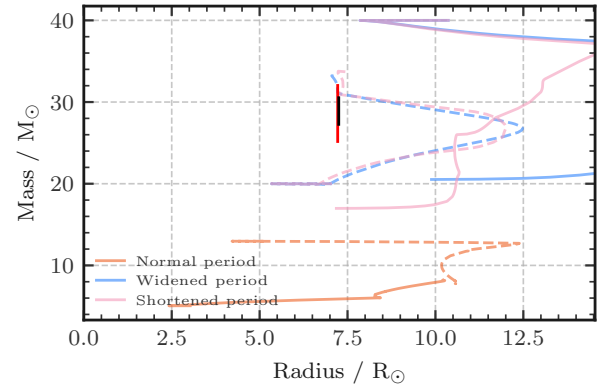
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure 5.3: Detailed model tracks for our six close-contact binaries. Solid (dashed) lines represent the donor (accretor), which has its observed parameters shown in red (black). Models shown are restricted to only during the core hydrogen burning phase of the donor. The x -axis is limited to double the maximum value of the observed radius of either star.

As a result, however, there is a dearth of matching models. LSS 3074, TU Muscae, and V382 Cygni are the only three systems with any matches to the observed masses and periods, which are enumerated in Table 5.5. We also take each of the matching models and display them in Figure 5.4. There are 14 matching models in total.

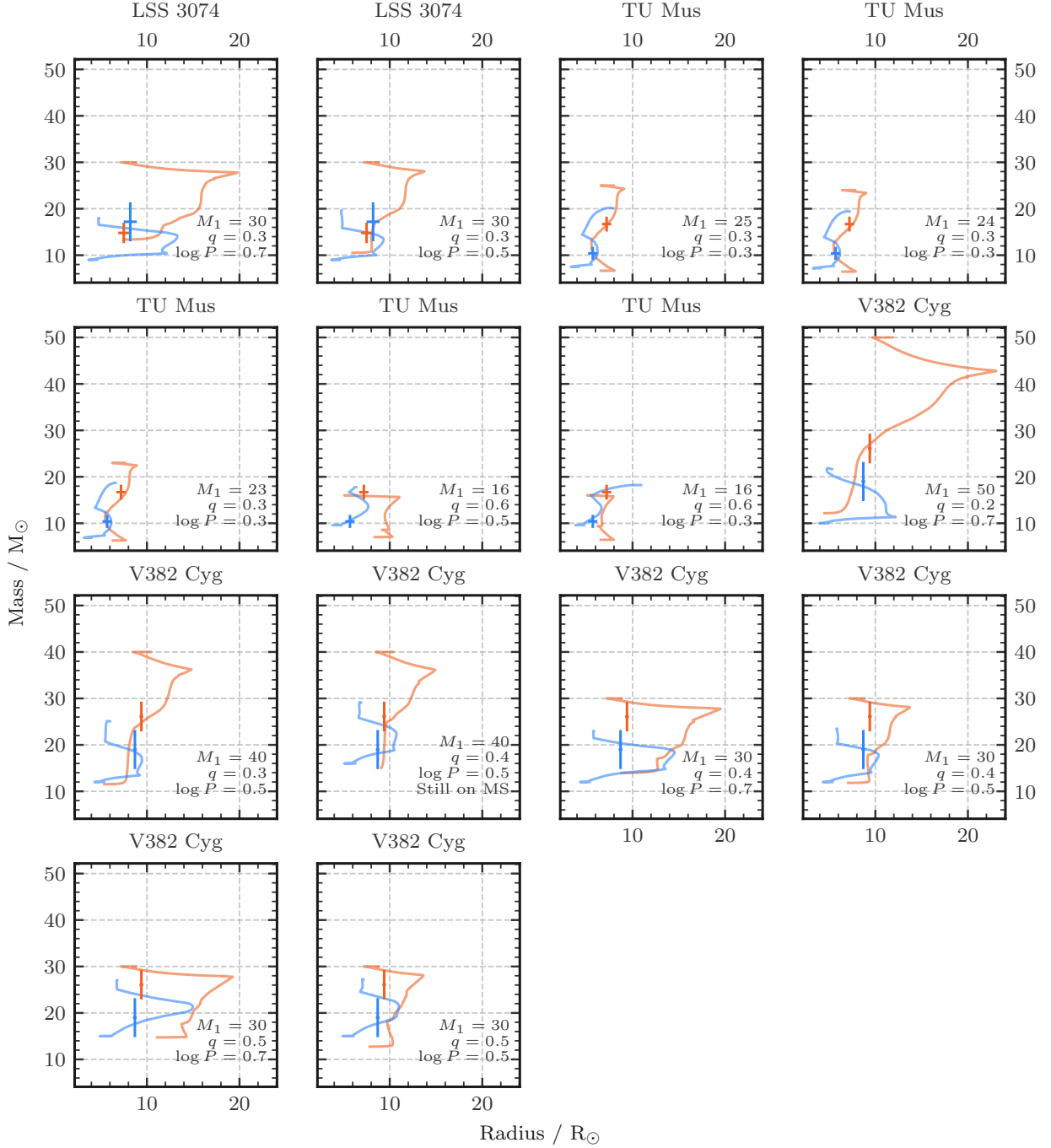


Figure 5.4: The mass-radius plots for our donor (orange) and accretor (blue) models that match their observations. The tracks are restricted to whilst the donor is still undergoing core hydrogen fusion, with the exception of one model which terminated on the main sequence and is marked on its plot.

Binary	ZAMS parameters				Ages			Match details			
	Z	$M_1 / M_\odot$	q	$\log P / d$	Begin / Myr	End / Myr	Length / kyr	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$
LSS 3074	z020	30	0.3	0.7	6.1087	6.1141	5.43	13.37 $\pm$ 0.01	7.40 $\pm$ 0.33	17.96 $\pm$ 0.00	4.82 $\pm$ 0.00
LSS 3074	z020	30	0.3	0.5	5.4656	6.1309	665.33	13.73 $\pm$ 0.63	8.03 $\pm$ 0.03	18.39 $\pm$ 0.27	4.82 $\pm$ 0.03
TU Mus	z020	25	0.3	0.3	3.2888	3.2934	4.63	16.50 $\pm$ 0.59	7.20 $\pm$ 0.29	11.55 $\pm$ 0.54	6.17 $\pm$ 0.06
TU Mus	z020	24	0.3	0.3	3.5528	3.5574	4.61	16.01 $\pm$ 0.57	7.24 $\pm$ 0.27	10.92 $\pm$ 0.50	6.15 $\pm$ 0.04
TU Mus	z020	23	0.3	0.3	3.8383	3.8415	3.28	15.77 $\pm$ 0.41	7.42 $\pm$ 0.17	10.07 $\pm$ 0.32	6.13 $\pm$ 0.01
TU Mus	z020	16	0.6	0.5	5.3727	7.1860	1813.26	15.91 $\pm$ 0.01	7.18 $\pm$ 0.29	9.58 $\pm$ 0.00	4.36 $\pm$ 0.04
TU Mus*	z020	16	0.6	0.3	5.1722	6.9546	1782.43	15.90 $\pm$ 0.01	7.18 $\pm$ 0.29	9.58 $\pm$ 0.00	4.37 $\pm$ 0.04
TU Mus*	z020	16	0.6	0.3	7.3024	7.3026	0.27	15.12 $\pm$ 0.01	7.69 $\pm$ 0.01	10.33 $\pm$ 0.01	5.02 $\pm$ 0.01
V382 Cyg	z020	50	0.2	0.7	3.0350	3.0399	4.96	27.18 $\pm$ 0.19	9.43 $\pm$ 0.11	21.16 $\pm$ 0.17	4.79 $\pm$ 0.04
V382 Cyg	z020	40	0.3	0.5	2.9967	2.9982	1.45	24.70 $\pm$ 0.19	9.40 $\pm$ 0.11	20.05 $\pm$ 0.19	6.40 $\pm$ 0.30
V382 Cyg	z020	40	0.4	0.5	3.2253	3.6514	426.16	23.61 $\pm$ 0.41	9.54 $\pm$ 0.03	26.69 $\pm$ 0.21	6.72 $\pm$ 0.02
V382 Cyg	z020	30	0.4	0.7	2.3847	2.6565	271.78	29.24 $\pm$ 0.03	9.43 $\pm$ 0.10	12.00 $\pm$ 0.00	4.70 $\pm$ 0.01
V382 Cyg	z020	30	0.4	0.5	2.2749	2.5998	324.97	29.17 $\pm$ 0.04	9.40 $\pm$ 0.12	12.00 $\pm$ 0.00	4.70 $\pm$ 0.01
V382 Cyg	z020	30	0.5	0.7	2.3662	2.6543	288.04	29.24 $\pm$ 0.03	9.42 $\pm$ 0.10	14.98 $\pm$ 0.00	5.48 $\pm$ 0.02
V382 Cyg	z020	30	0.5	0.5	2.2768	2.6024	325.57	29.17 $\pm$ 0.04	9.40 $\pm$ 0.12	14.98 $\pm$ 0.00	5.49 $\pm$ 0.02

Table 5.5: The matches in primary mass / primary radius to our overcontact models. Shown here are the ZAMS parameters. An asterisk next to the star name indicates that the binary undergoes two periods where it matches the donor mass / radius, and each match is shown. Quoted uncertainties are the 1σ confidence levels, and all quantities are (for brevity) always rounded to 2dp.

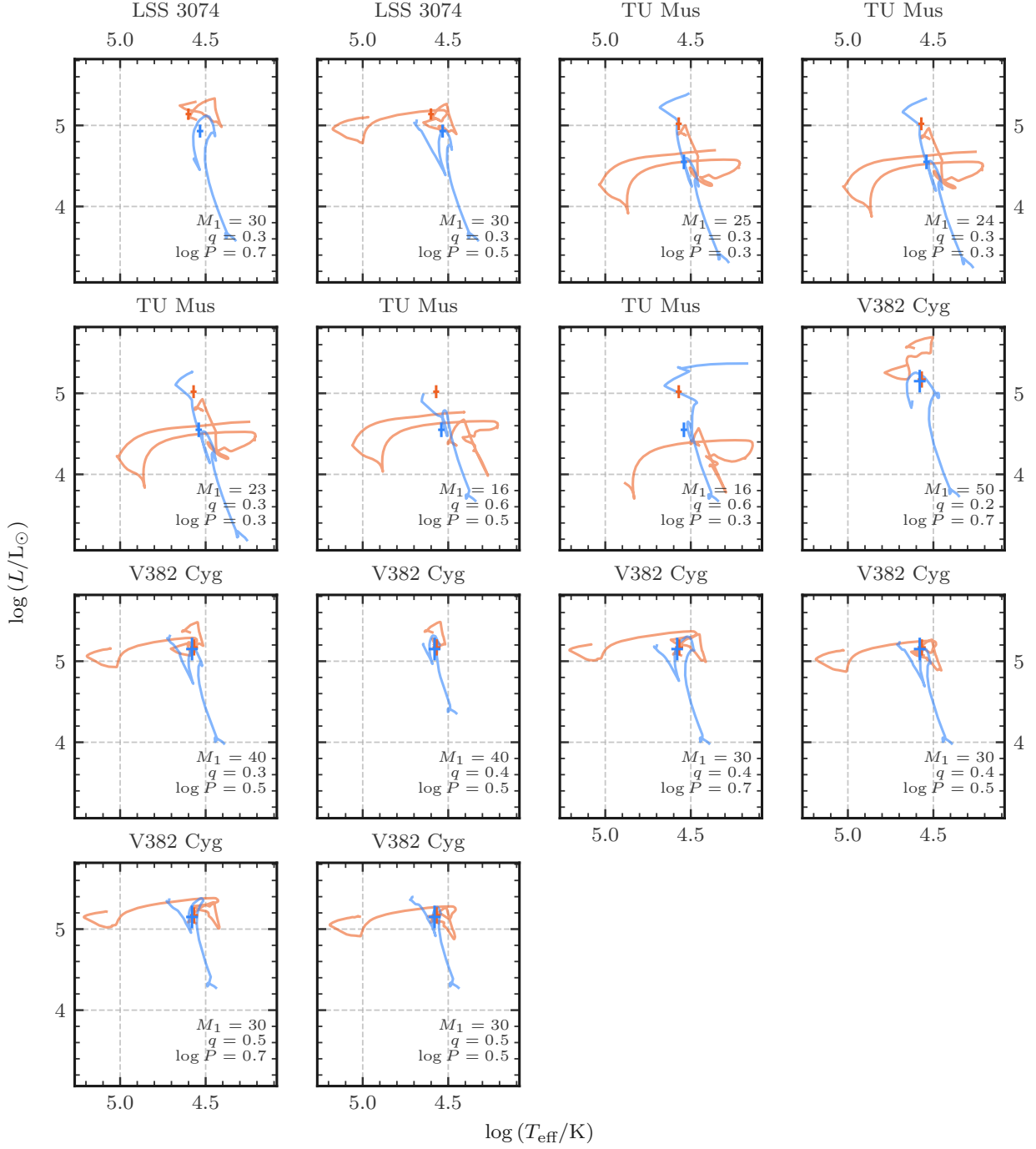
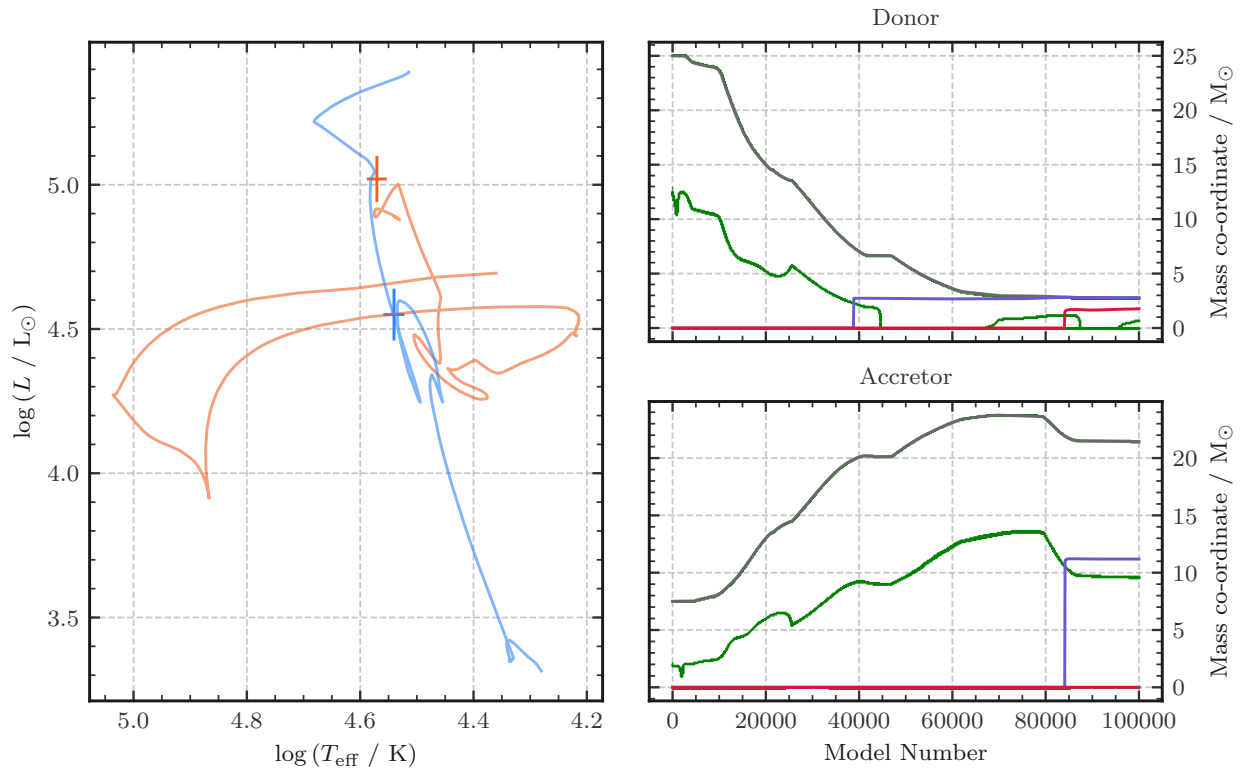
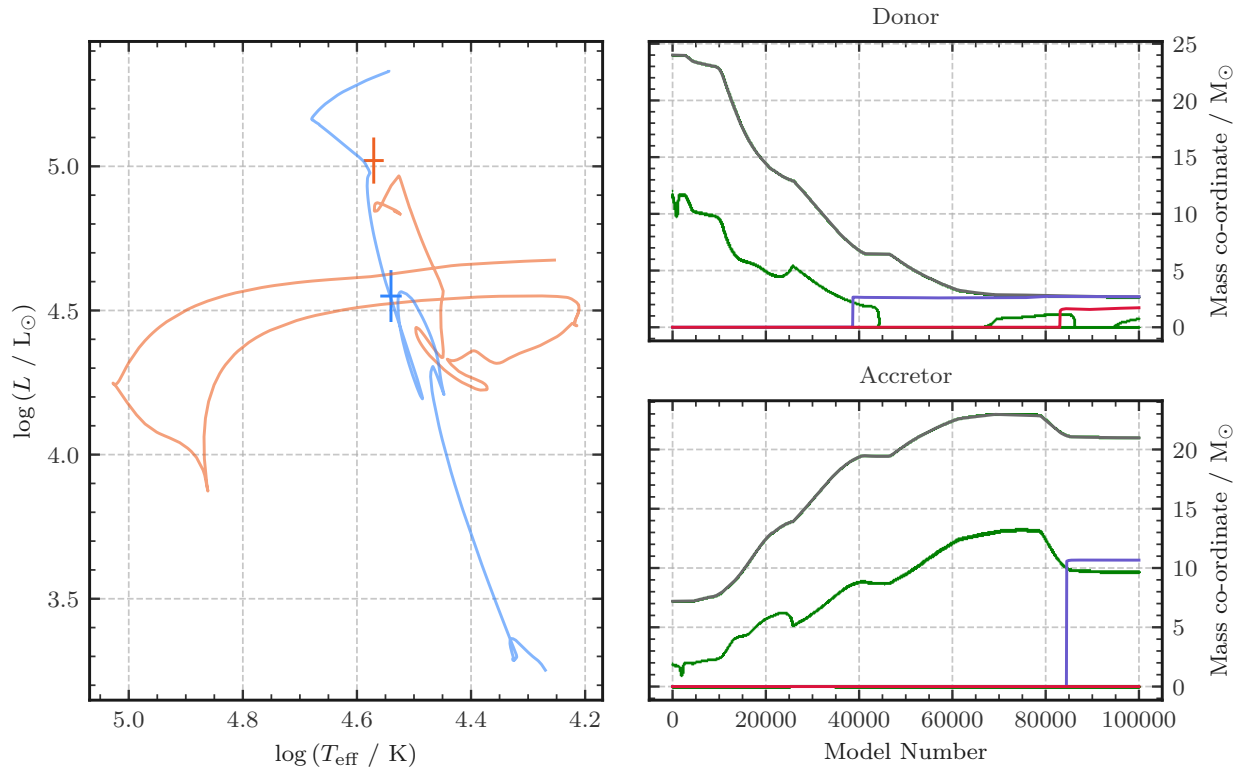


Figure 5.5: HRDs for the matching overcontact binary models. Also shown are the crosses representing the locations of the observed systems as reported in their original papers. The secondary luminosities in particular are not well constrained, and thus their locations must be taken with caution.



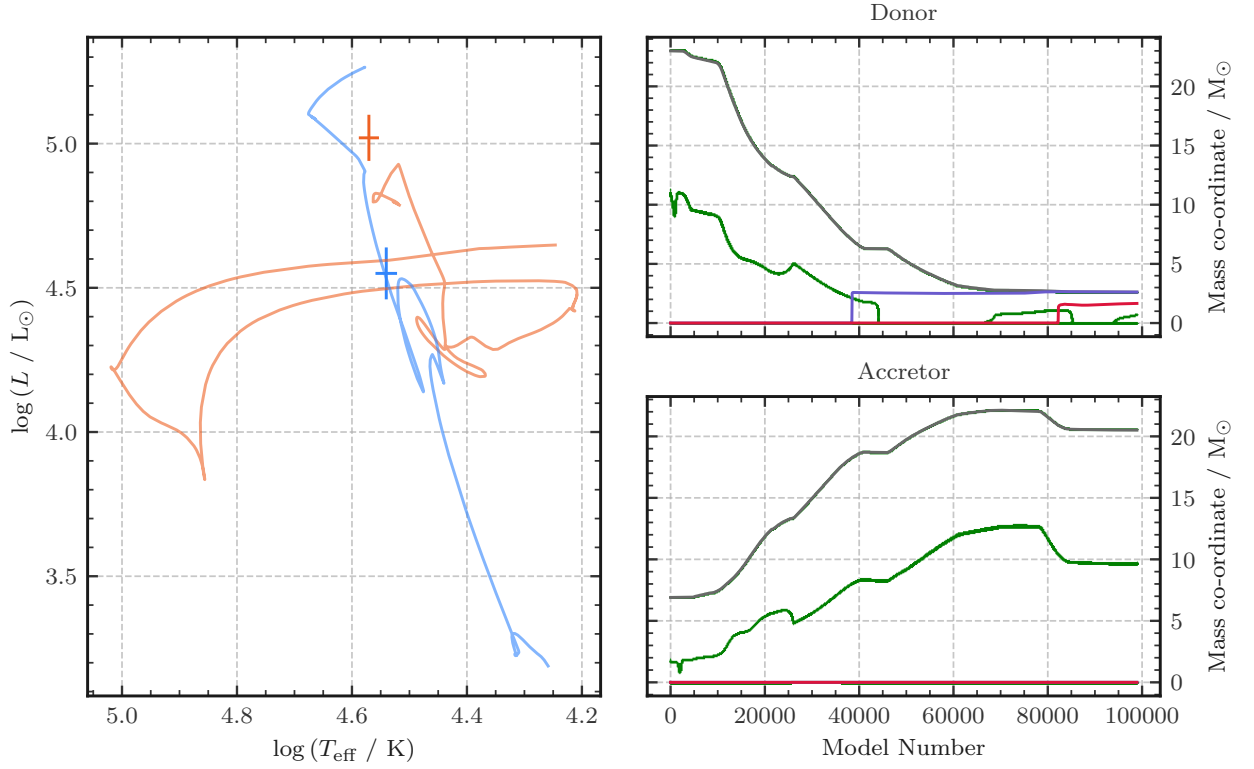
(a) The first match for TU Muscae ($M_1 = 25 M_{\odot}$, $M_2 = 7.5 M_{\odot}$, $\log(P / \text{d}) = 0.3$).



(b) The second match for TU Muscae ($M_1 = 24 M_{\odot}$, $M_2 = 7.2 M_{\odot}$, $\log(P / \text{d}) = 0.3$).

Binary	ZAMS parameters			Observed details					Match details				Matches	
	$M_1 / M_\odot$	q	$\log P / d$	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$	$M_1 / M_\odot$	$R_1 / R_\odot$	$M_2 / M_\odot$	$R_2 / R_\odot$	*1	*2	
LSS 3074	30	0.3	0.7	14.8 ± 2.2	7.5 ± 0.6	17.2 ± 4.2	8.2 ± 0.7	13.37 ± 0.01	7.40 ± 0.33	17.96 ± 0.00	4.82 ± 0.00	✓	✕ _R	
LSS 3074	30	0.3	0.5	14.8 ± 2.2	7.5 ± 0.6	17.2 ± 4.2	8.2 ± 0.7	13.73 ± 0.63	8.03 ± 0.03	18.39 ± 0.27	4.82 ± 0.03	✓	✕ _R	
TU Mus	25	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	16.50 ± 0.59	7.20 ± 0.29	11.55 ± 0.54	6.17 ± 0.06	✓	✓	
TU Mus	24	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	16.01 ± 0.57	7.24 ± 0.27	10.92 ± 0.50	6.15 ± 0.04	✓	✓	
TU Mus	23	0.3	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.77 ± 0.41	7.42 ± 0.17	10.07 ± 0.32	6.13 ± 0.01	✓	✓	
TU Mus	16	0.6	0.5	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.91 ± 0.01	7.18 ± 0.29	9.58 ± 0.00	4.36 ± 0.04	✓	✕ _R	
TU Mus*	16	0.6	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.90 ± 0.01	7.18 ± 0.29	9.58 ± 0.00	4.37 ± 0.04	✓	✕ _R	
TU Mus*	16	0.6	0.3	16.7 ± 1.6	7.2 ± 0.5	10.4 ± 1.4	5.7 ± 0.5	15.12 ± 0.01	7.69 ± 0.01	10.33 ± 0.01	5.02 ± 0.01	✓	✕ _R	
V382 Cyg	50	0.2	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	27.18 ± 0.19	9.43 ± 0.11	21.16 ± 0.17	4.79 ± 0.04	✓	✕ _R	
V382 Cyg	40	0.3	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	24.70 ± 0.19	9.40 ± 0.11	20.05 ± 0.19	6.40 ± 0.30	✓	✕ _R	
V382 Cyg	40	0.4	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	23.61 ± 0.41	9.54 ± 0.03	26.69 ± 0.21	6.72 ± 0.02	✓	✕ _{MR}	
V382 Cyg	30	0.4	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.24 ± 0.03	9.43 ± 0.10	12.00 ± 0.00	4.70 ± 0.01	✓	✕ _{MR}	
V382 Cyg	30	0.4	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.17 ± 0.04	9.40 ± 0.12	12.00 ± 0.00	4.70 ± 0.01	✓	✕ _{MR}	
V382 Cyg	30	0.5	0.7	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.24 ± 0.03	9.42 ± 0.10	14.98 ± 0.00	5.48 ± 0.02	✓	✕ _R	
V382 Cyg	30	0.5	0.5	26.1 ± 3.2	9.4 ± 0.2	19.0 ± 4.2	8.7 ± 0.2	29.17 ± 0.04	9.40 ± 0.12	14.98 ± 0.00	5.49 ± 0.02	✓	✕ _R	

Table 5.6: The observed vs. matched parameters for each of the 14 overcontact matches we find. An asterisk next to the star name indicates that the binary undergoes two periods where it matches the donor mass / radius, and each match is shown. In the “matches” column, ✓ represents that both the masses and the radii overlap for a given model, and ✗ represents that at least one of the parameters does not match. In these instances, the parameter(s) which do not match are shown as subscripts next to the cross.



(c) The third match for TU Muscae ($M_1 = 23 M_\odot$, $M_2 = 6.9 M_\odot$, $\log(P/d) = 0.3$).

Figure 5.6: The three fits for TU Muscae we recover. Shown here are the HRDs and the Kippenhahn diagrams for both the donor and the accretor. In the Kippenhahn diagrams, the blue line is the helium core, and the red is the CO core.

5.3 | Matches to observed close contact models

The matches that we find are shown in Table 5.6 next to their search parameters. Three of the binaries have matches on mass and radius for both stars simultaneously. The balance generally do not match on the companion radius, with three systems also failing to match on the companion mass as well. We consider a mass or radius to have overlapped to its observed value if the error bars at all overlap, and caveat that the reported error bars on the observed parameters are the search uncertainties, as per Table 5.1.

The three systems which match on both the primary and secondary mass and radius all model TU Muscae. We show in Figure 5.6 the HRDs and Kippenhahn diagrams of these three matches. The relative positions of the observations on the HRD suggest that a mass ratio inversion occurred for this system, and thus our labellings of the primary / secondary are incorrect.

The three matches for TU Muscae differ only by their initial masses. The primary masses are $25 M_\odot$ (Figure 5.6a), $24 M_\odot$ (Figure 5.6b), and $23 M_\odot$ (Figure 5.6c). All three have initial periods of $10^{0.3} \text{ d} \sim 2 \text{ d}$, and all three have mass ratios of $1/3$. Of these three matches, the closest match is the $25 M_\odot$ progenitor in Figure 5.6a.

What then remains is to evaluate the goodness-of-fit of the detailed model code results. The three results are summarised in Table 5.7. The three models all fulfil their matching criteria at similar ages to their v2 counterparts. The differences can be ascribed to two key factors. Firstly, the BPASS v2 ages were computed

as matches on the HR diagram to primarily the donor, as BPASS v2 models the secondary as a point mass during the pre-supernova evolution of the donor. This is in contrast to the STARS matches, which matched masses and radii of *both* stars. Secondly, the ZAMS periods were 0.1 dex wider than the ones used in the computation of the STARS model, and originate in our lower-period grid as described at the beginning of Section 5.2.

Binary	ZAMS Parameters			BPASS v2		STARS code	
	$M_1 / M_\odot$	q	$\log P / \text{d}$	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$	$\mu_{\log(\text{age})}$	$\sigma_{\log(\text{age})}$
TU Mus	25	0.3	0.3	6.69	0.56	6.5174	0.0002
TU Mus	24	0.3	0.3	6.71	0.56	6.5509	0.0002
TU Mus	23	0.3	0.3	6.77	0.52	6.5843	0.0001

Table 5.7: The match results with associated mean ages at match. The BPASS v2 periods were computed with 0.1 dex periods higher, hence a slight mismatch in the ages is not unexpected. Ages from BPASS v2 were first presented in Table 5.2, and ages from STARS are computed in the same way, although on masses and radii compared to temperatures and luminosities.

5.3.1 | The length of the contact phase

We also briefly investigate the mean length of the contact phase. For each binary in our sample, we determine the age at which it fills both Roche lobes, and the age at which it detaches. If the binary undergoes two contact phases, as is the case in one model of TU Muscae (see Table 5.5), we record this information for each phase. If the binary is still in a contact phase at the end of the model sequence, we record the final age as the termination age. We find that as the initial period increases, the mean contact time of the binary decreases, as expected. We find that the mean contact time in the widest period grid is $\mu_{\text{wide}} = 3.819 \text{ kyr}$, $\sigma_{\text{wide}} = 1.723 \text{ kyr}$, of the normal period grid $\mu_{\text{normal}} = 5.871 \text{ kyr}$, $\sigma_{\text{normal}} = 2.773 \text{ kyr}$, and of the shorter period grid $\mu_{\text{wide}} = 20.897 \text{ kyr}$, $\sigma_{\text{wide}} = 73.755 \text{ kyr}$. We show in Figure 5.7 the dependence of the mean contact length on the initial parameters. For (comparatively) low initial primary masses $M_1 \lesssim 24 M_\odot$, there is no dependence. Comparatively, we can see that the mean contact length generally decreases with initial period and mass ratio.

5.4 | Discussion

The most obvious limitation is that we were able to only produce three matching systems on one of the six close-contact binaries. Figure 5.3 shows that, especially for SMC 108086 and MY Camelopardalis, the parameter space exploration was insufficient and that, were the space widened, more matches may be able to be found.

In addition, the systems here were assumed to be initially nonrotating. Adjusting the initial rotation rate of either or both stars may allow us to obtain a better fit, however this immediately increases the number of models one would need to consider, as there is no immediate prior on what the initial rotation rate of the star(s) would need to be.

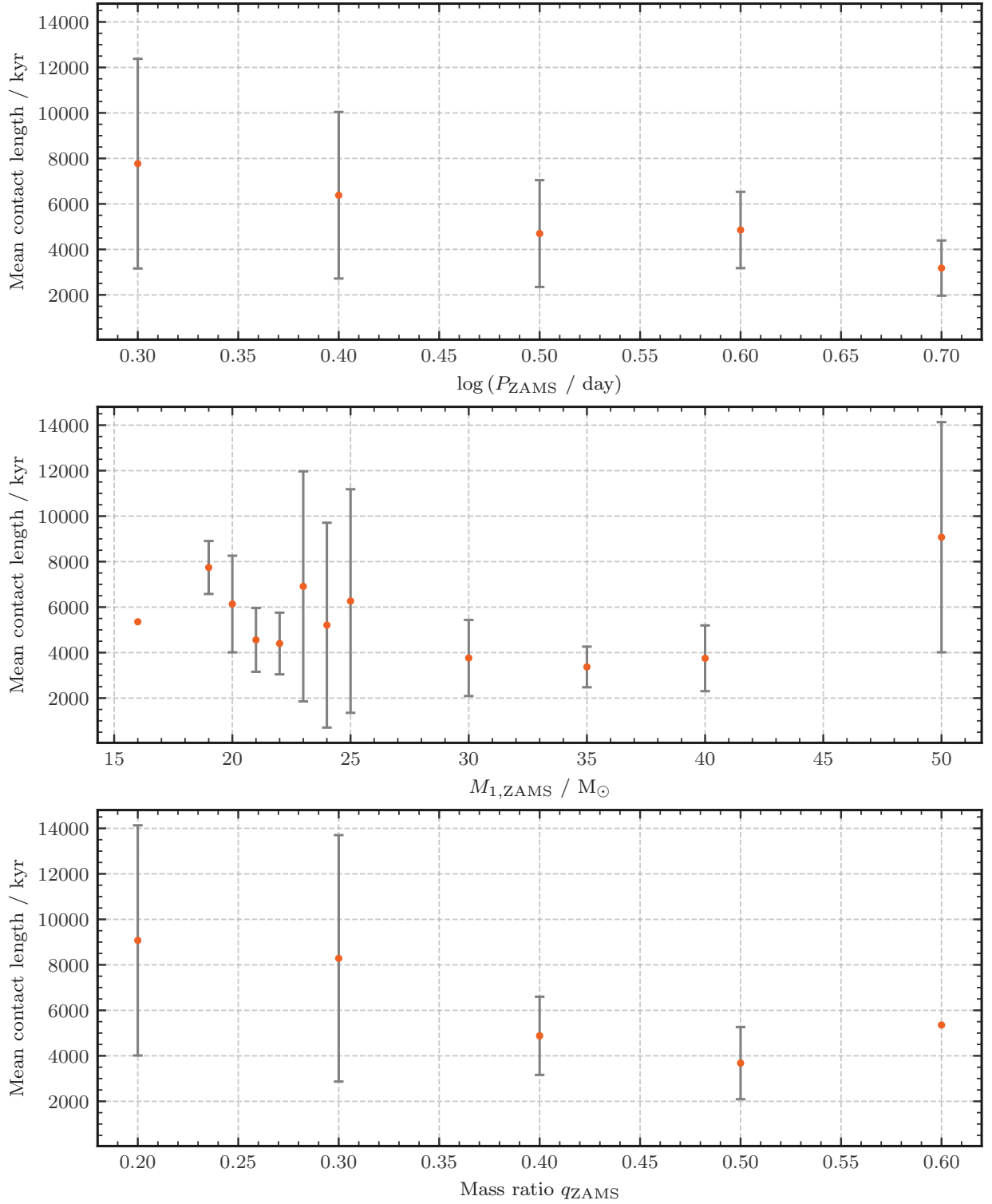


Figure 5.7: The dependence of the mean contact length on the initial parameters $\log P$, M_1 , and q . Error bars correspond to the standard deviations (i.e., 1σ uncertainties) of the measurements. The data points with no error bar only contain one model. One outlier, corresponding to $M_1 = 50 M_{\odot}$, $q = 0.2$, $\log(P / \text{d}) = 0.5$, was manually removed, as it had a contact length of ~ 400 Myr, and rendered the rest of the plot unreadable.

The initial matches for systems, as listed in Table 5.1, were performed as matches to positions on an HRD, which presents difficulty. First, as contact binaries, the secondary luminosity is difficult to constrain, and the assumption of the Stefan-Boltzmann law (Eqn(5.1)) is unlikely to hold proper. This resulted in intentionally wide search bars, to encapsulate this uncertainty, but also resulted in binaries that are unlikely to actually match the search parameters.

This study also relies on the BPASS v2 models to evaluate the ZAMS parameters of our models. BPASS uses the approximations by [Hurley et al. \(2000\)](#) to approximate the evolution of the secondary star, contributing further to the aforementioned difficulty in constraining the search parameters for the accretor.

In order to obviate these limitations, a wide grid in both primary and secondary mass could be computed. By doing so, one removes the dependence on the BPASS secondary grid and inherent uncertainties arising from [Hurley et al. \(2000\)](#), at the computational expense of having to search a much larger range of models for matches.

5.5 | Chapter Summary

We started by compiling the six contact binaries in the sample of [Abdul-Masih et al. \(2022\)](#). We matched these binaries to their ZAMS parameters using the BPASS v2.2 and HOKI codes, finding between one and ten viable sets of initial parameters to each model. We also found their mean ages to be between 1.34 Myr and 13.8 Myr, depending on the model. We also used the BPASS v2.2 models to predict the eventual outcome of these contact binaries, and found that a range of possible compact object binaries could occur, with initial periods on the order of tens of years.

By using the BPASS fits as a starting point for the initial parameters of the contact binaries, we created bespoke binary evolution models for the contact binaries with the STARS code, we found that 14 model tracks matched, for LSS 3074, TU Muscae, and V382 Cygni. We found that these matches were short compared to the lifetime of the binary, generally existing for ~ 200 kyr – ~ 600 kyr. Of the 14 matches, we found that only three matched the observed masses and radii of both components in the binary, all for TU Muscae on the shortest grid. The models for LSS 3074 and V382 Cygni, as well as three of the models for TU Muscae were unable to match the radius of the secondary, and three of the models for V382 Cygni were unable to match the secondary mass as well. We conjectured that this may be an artifact of the parameter space explored, and that a wider grid would enable us to find closer matches.



CHAPTER 6

Conclusion



The glow of summer has faded, now... and the moonlight jellies carry on towards
the great unknown.

— *Stardew Valley* —



THIS THESIS HAS PRIMARILY concerned itself with improving an established stellar modelling code for future population synthesis projects. In doing so, we discussed several key aspects of binary evolution, which shall be recapped here. To wit, in Chapter 1 we introduced three major themes guiding this research:

- I. Can we study contact binary systems, and refine our treatment of RLOF to overcome numerical issues inherent in these models?
- II. Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?

In this chapter, we examine these themes in detail and review the implications of them in terms of the results we have obtained.

6.1 | Summary

*Oh! I have slipped the surly bonds of Earth
And danced the skies on laughter-silvered wings;
...
And, while with silent lifting mind I've trod
The high untrespassed sanctity of space,
Put out my hand, and touched the face of God.
— "HIGH FLIGHT", BY JOHN MAGEE JR.*

In Chapter 3, we introduced the notion of helium-rich mass transfer, that being when the mass transfer within a binary continues for a high rate for a long time, thus stripping material from the exposed core of a donor. We found that this produced small differences in the contact phase which were magnified in the post-contact phase. We found that, on net, this produced hotter and at times more luminous accretors in binaries, and concluded that these are potential formation channels for stars such as WN/O3 stars, and may in part explain the discrepancy between spectroscopic and evolutionary masses that persist throughout various modelling efforts.

In Chapter 4, we examined the form and structure of the natal supernova kick. We introduced the [Bray & Eldridge \(2016\)](#) kick and, by direct comparison with the [Hobbs et al. \(2005\)](#) kick, developed a testing pipeline with multiple orthogonal constraints. In turn, we were able to constrain the Bray kick based on four fundamental phenomena: the detected LVK Collaboration merger rate (Section 4.2.1), the observed period-eccentricity distribution of Galactic BNSs (Section 4.2.2), the observed isolated pulsar velocity distribution (Section 4.2.3), and the existence of USSNe (Section 4.2.4). We found that the region of the parameter space that was able to replicate all four phenomena was 1 per cent of the search region, and concluded that the optimal form of the kick was

$$v_{\text{kick}} = 115_{-55}^{+40} \left(\frac{m_{\text{ejecta}}}{m_{\text{remnant}}} \right) + 15_{-15}^{+10} \text{km s}^{-1}. \quad (6.1)$$

In Chapter 5, we began to search for BPASS matches to six observed close-contact binaries as compiled in [Abdul-Masih et al. \(2022\)](#). We found 31 matches in the BPASS v2.2 dataset. Using these 31 matches as a starting grid, and allowing for ± 0.1 dex in the period grid, we found 15 matching models in total, for LSS 3074, TU Muscae, and V382 Cygni. Of these, three of the models of TU Muscae were found to match the observed masses and radii, with the others all matching on the donors and most frequently failing to match on the radii of the accretors. This was found to not be of large concern, as the companion radius is the most difficult to constrain parameter of the four for an eclipsing binary.

6.2 | Limitations

Though in reviewing the incidents of my administration I am unconscious of intentional error, I am nevertheless too sensible of my defects not to think it probable that I may have committed many errors.

— GEORGE WASHINGTON

There are three major limitations to our study: that we have assumed circular orbits, that we have focused predominantly on O-type stars, and that we have had to make sometimes seemingly arbitrary parameter choices in order for the code to perform. These are detailed in the following subsections.

6.2.1 | Circular Orbits

A common thread throughout the detailed model tracks in this work is the assumption of a circular orbit. Binary stars are not obliged to be circular, and indeed [Moe & Stefano \(2017\)](#) suggest quite the opposite; that the eccentricity distribution varies with the stellar type. Generalising our results to arbitrary eccentricity is, however, a complex undertaking for potentially little payoff. [Eldridge et al. \(2017\)](#) argues, and [Hurley et al. \(2002\)](#) shows, that the result of a binary computation is the same for an elliptical orbit and a circular orbit, if the circle has a radius equal to the semi-latus rectum of the ellipse. This is illustrated geometrically in Figure 6.1.

6.2.2 | Our parameter choices

Most of our parameter choices have in some way been physically motivated; for instance the choice of k in Section 2.4.3, corresponding to the efficiency of RLOF, was chosen via a combination of work by [Hurley et al. \(2002\)](#) and [Claeys et al. \(2014\)](#). Through testing, it has become apparent that the [Kolb & Ritter \(1990\)](#) treatment of mass transfer may be more appropriate; an implementation of this as a mass loss scheme would

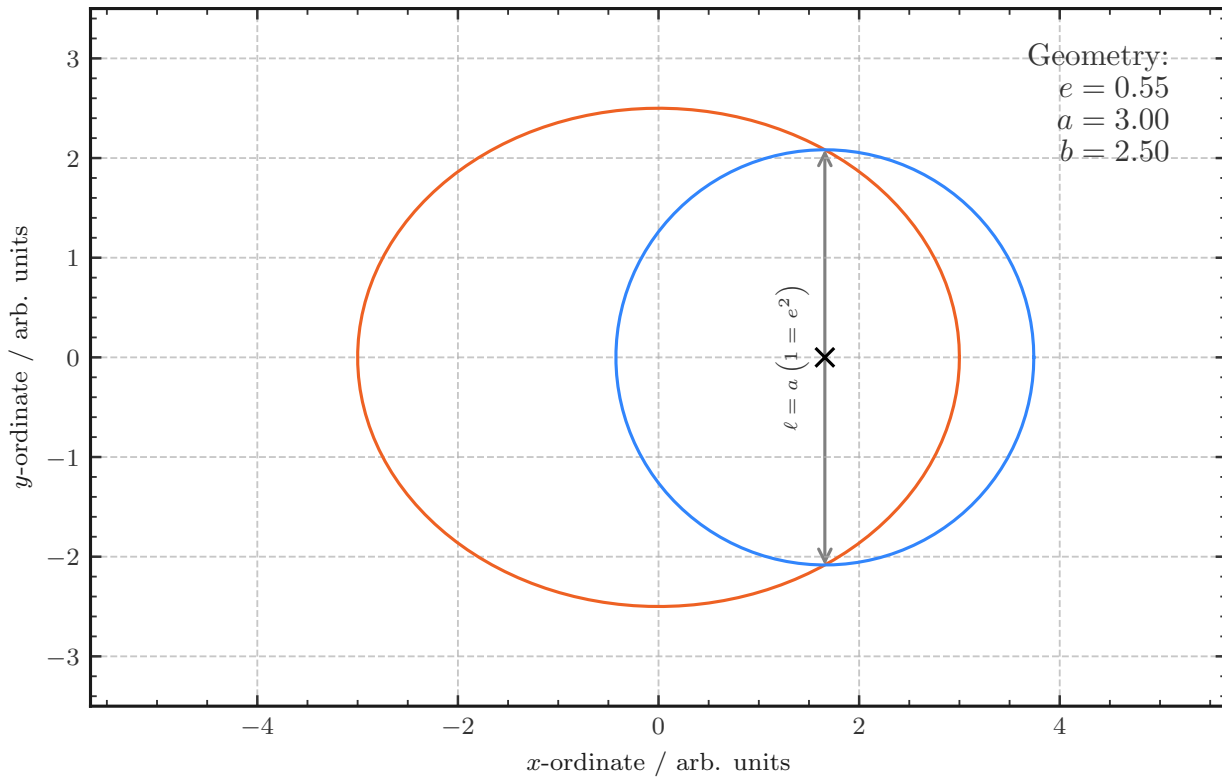


Figure 6.1: The geometry of an ellipse and the circle with a radius equal to the semi-latus rectum. The ellipse was chosen arbitrarily to have $a = 3$, $b = 2.5$. The black cross is at the focus of the ellipse, which is also the center of the circle.

be a potential avenue for future exploration of the constraints of mass transfer.

Stellar modelling is a nonlinear task, and the clearest example of this in this thesis is the thermohaline mixing coefficients α_{th} in Chapter 3, where the interior composition profile was changed quite substantially depending on the strength of the thermohaline mixing. Most of these somewhat arbitrary parameter choices are a reflection of uncertainty in the physics; thermohaline mixing in particular is a difficult process to constrain.

6.3 | Suggested Future Work

*A still more glorious dawn awaits
 – not a sunrise, but a galaxy rise.*

— CARL SAGAN

I can slay the Jabberwocky.

— LAST OF THE SIX IMPOSSIBLE THINGS

There are multiple avenues open to future research questions with the STARS code, and in this section we

outline but a few.

Chapter 4 examined the form and structure of the natal supernova kick, with a particular emphasis on [Bray & Eldridge \(2016\)](#) type kicks. Looking forward, LVK Collaboration O4 will soon release, offering constraints on the observed BNS merger rate density. In addition, the same pipeline can be performed on other kick prescriptions that offer free parameters, such as [Verbunt et al. \(2017\)](#), which would allow for their refinement.

Chapter 5 explored a rather limited parameter space. The question thus remains, *to what extent do the conclusions drawn hold true on a population of binary stars*, and to much the same extent, *do the population prevalences derived from the BPASS v2 models match those from a grid of these models?* Exploration of the parameter space is firmly a big data project. For a reasonably trivial grid of $m_1 \in [5, 100] \text{ M}_\odot$ in steps of $5 \text{ M}_\odot$, $q \in [0, 0.9]$ in steps of 0.1, and $\log P \in [0.5, 3]$ in steps of 0.5 dex, one arrives at 1200 models. Models typically take, conservatively, between 20 min and 60 min to run on the cluster, which has up to 72 cores available for use. Table 6.1 shows some estimated runtimes for different grid fidelities given this setup. An investigation into the population properties would likely need to use model B as a coarse grid and model D as a finer grid, though the span of periods is wider here than is strictly necessary.

Model Name	Grid Specification				Runtime (DD:HH:MM:SS)		
	Δm_1	Δq	$\Delta \log P$	N_{models}	20 min	40 min	60 min
A	5	0.1	0.5	1200	00:05:33:20	00:11:06:40	00:16:40:00
B	5	0.1	0.1	5200	01:00:04:26	02:00:08:53	03:00:13:20
C	1	0.1	0.5	5760	01:02:40:00	02:05:20:00	03:08:00:00
D	1	0.1	0.1	24960	04:19:33:20	09:15:06:40	14:10:40:00

Table 6.1: The runtimes for a grid of STARS models on a 72-core cluster, assuming each model takes 20 min/40 min/60 min respectively. This calculation assumes all cores begin operating simultaneously, and that runs are homogeneous; that any two runs have the same run time. In this grid, $m_1 \in [5, 100] \text{ M}_\odot$, $q \in [0, 0.9]$, and $\log P \in [0.5, 3]$.

Looking towards the eastern horizon, the BPASS project and other similar population synthesis codes will soon be releasing new versions. The STARS code seems uniquely positioned, especially compared to alternatives such as MESA, to offer new and exciting insights into the field of binary stellar evolution.

And that is something you might not have known.

— TOM SCOTT

Appendices



APPENDIX A

Additional Derivations



Algebra's like sheet music. The important thing isn't can you read music, it's can you *hear* it. Can you hear the music, Robert?

— *Niels Bohr, in Oppenheimer* —



In this section, we present additional derivations for expressions used in Chapter 2.

A.1 | Roche lobe formula

We follow the derivation given in [Rose \(1998\)](#) to find an expression for the Roche potential. First we note that in a rotating reference frame in Cartesian coordinates, we have

$$\overbrace{\frac{GM_1M_2}{a^2}\hat{\mathbf{x}}}^{\text{gravitational force}} = \underbrace{-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1)}_{\text{centrifugal force}}, \quad (\text{A.1})$$

where $\hat{\mathbf{x}}$ is the line between the center of mass of both stars, a the separation, $\vec{\omega}$ the angular velocity, and $\vec{r}_1$ the position vector of the primary star. The x-ordinates of the centres of mass are given by

$$x_1 = \frac{M_2}{M_1 + M_2}a \quad (\text{A.2})$$

$$x_2 = \frac{-M_1}{M_1 + M_2}a \quad (\text{A.3})$$

Since we are considering a rotating coordinate system we have that the centrifugal force is given by

$$-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1) = M_1\omega^2 x_1, \quad (\text{A.4})$$

and hence

$$\begin{aligned} \frac{GM_1M_2}{a^2}\hat{\mathbf{x}} &= M_1\omega^2 x_1 \\ \omega^2 &= \frac{GM_1M_2}{M_1x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2}{x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2(M_1 + M_2)}{M_2 \cdot a \cdot a^2}\hat{\mathbf{x}} \\ &= \frac{G(M_1 + M_2)}{a^3}\hat{\mathbf{x}}. \end{aligned} \quad (\text{A.5})$$

The centrifugal potential of a test mass m at (x, y) is hence

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2) = -\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}, \quad (\text{A.6})$$

and thus its energy is

$$E = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + V(x, y), \quad (\text{A.7})$$

where

$$V(x, y) = - \overbrace{\frac{GmM_1}{\sqrt{(x-x_1)^2 + y^2}}}^{\text{G.P.E. due to primary}} - \overbrace{\frac{GmM_2}{\sqrt{(x-x_2)^2 + y^2}}}^{\text{G.P.E. due to secondary}} - \overbrace{\frac{Gm(M_1 + M_2)(x^2 + y^2)}{2a^3}}^{\text{centrifugal potential}}. \quad (\text{A.8})$$

We introduce the dimensionless coordinates measured in units of the separation of the binary (or reduced masses)

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{a}, \quad \xi_1 = \frac{M_2}{M_1 + M_2}, \quad \xi_2 = \frac{M_1}{M_1 + M_2} \quad (\text{A.9})$$

Our potential now becomes

$$V(\xi, \eta) = -\frac{Gm(M_1 + M_2)}{a} \times \left[\frac{\xi_2}{\sqrt{(\xi - \xi_1)^2 + \eta^2}} + \frac{\xi_1}{\sqrt{(\xi - \xi_2)^2 + \eta^2}} + \frac{1}{2}(\xi^2 + \eta^2) \right]. \quad (\text{A.10})$$

Taking the partial derivative along the ξ -axis (the rescaled x -axis), we get

$$\frac{\partial V}{\partial \xi} = -\frac{Gm(M_1 + M_2)}{a} \times \left[-\frac{\xi_2(\xi - \xi_1)}{\left((\xi - \xi_1)^2 + \eta^2\right)^{\frac{3}{2}}} - \frac{\xi_1(\xi - \xi_2)}{\left((\xi - \xi_2)^2 + \eta^2\right)^{\frac{3}{2}}} + \xi \right] \quad (\text{A.11})$$

Hence the extrema of the potential V are at the points ξ where (taking $\eta = 0$ on the ξ -axis)

The formula for the Roche lobes of binary stars.

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}, \quad (\text{A.12})$$

which was to be shown.

A.2 | Kepler's Three Laws

This section is adapted from [Richards \(2021\)](#). Kepler's three laws such that they apply to binary stars are:

1. All binary stars orbit the barycentre in an ellipse with the barycentre as a focus.
2. A line segment joining the two stars sweeps out equal areas during equal intervals of time.
3. The square of the period of each star is proportional to the cube of the semimajor axis (i.e., $T^2 \propto a^3$).

The derivations are presented below.

A.2.1 | The First Law

Given the orbital setup in Figure A.1, the distance vector $\vec{r}$ between two stars in a binary is

$$\vec{r} = \vec{r}_2 - \vec{r}_1, \quad (\text{A.13})$$

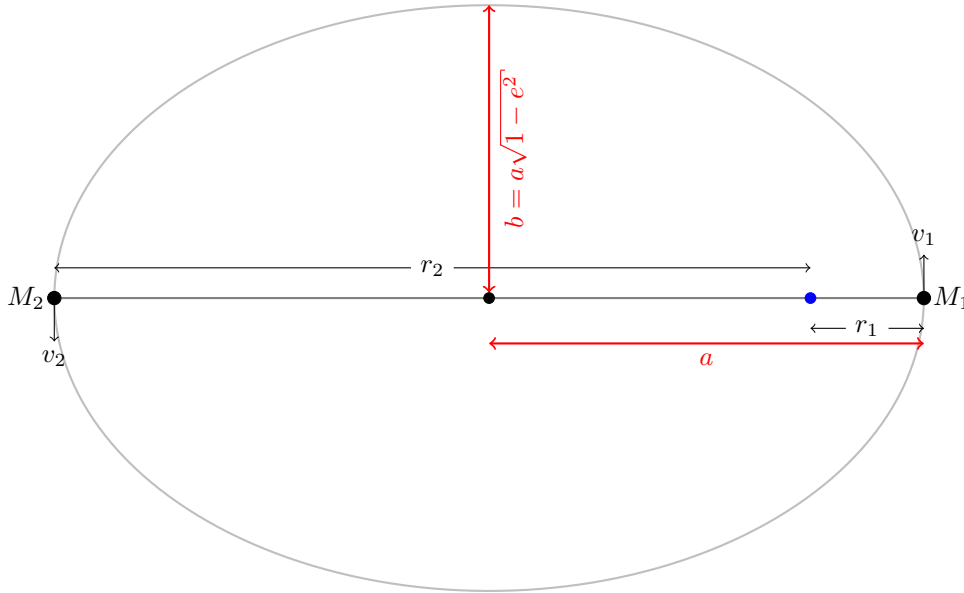


Figure A.1: Figure 2.8 again; A schematic of a simple binary star system with eccentricity e , semimajor axis a , and semiminor axis b . The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance r_1 to the barycentre by Kepler's first and Newton's second laws. The elliptical orbital trajectory is shown in grey. Figure adapted from [Richards \(2021\)](#).

with the reduced mass

$$\mu = \frac{M_1 M_2}{M_1 + M_2}, \quad (\text{A.14})$$

and total mass

$$M = M_1 + M_2. \quad (\text{A.15})$$

Since the angular momentum vector is given by

$$\vec{L} = \vec{r} \times \vec{p}, \quad (\text{A.16})$$

its derivative is

$$\frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} = 0, \quad (\text{A.17})$$

where the equality follows as $\vec{v} \equiv \frac{d\vec{r}}{dt}$ is parallel to $\vec{p}$, and $\vec{r}$ is parallel to $\frac{d\vec{p}}{dt} \equiv \vec{F}$. This is equivalent to a one-body problem where a mass $\mu = \frac{M_1 M_2}{M_1 + M_2}$ orbits a central mass of mass $M_1 + M_2$. The angular momentum is

$$\vec{L} = \mu \vec{r} \times \vec{v} \quad (\text{A.18})$$

$$\begin{aligned} &= \mu |\vec{r}| \hat{\mathbf{r}} \times \frac{d}{dt} (|\vec{r}| \hat{\mathbf{r}}) \\ &= \mu |\vec{r}| \hat{\mathbf{r}} \times \left(\frac{dr}{dt} \hat{\mathbf{r}} + r \frac{d\hat{\mathbf{r}}}{dt} \right) \\ &= -\frac{dr}{dt} \hat{\mathbf{r}} \times \mu |\vec{r}| \hat{\mathbf{r}} - r \frac{d\hat{\mathbf{r}}}{dt} \times \mu |\vec{r}| \hat{\mathbf{r}} \\ &= -\frac{dr}{dt} \mu |\vec{r}| (\hat{\mathbf{r}} \times \hat{\mathbf{r}}) - |\vec{r}|^2 \mu \left(\frac{d\hat{\mathbf{r}}}{dt} \times \hat{\mathbf{r}} \right) \\ &= -|\vec{r}|^2 \mu \left(\frac{d\hat{\mathbf{r}}}{dt} \times \hat{\mathbf{r}} \right) \\ &= |\vec{r}|^2 \mu \left(\hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\ &= \mu |\vec{r}|^2 \hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt}. \end{aligned} \quad (\text{A.19})$$

We seek an expression for $|\vec{r}|$ to determine the trajectory of the stars. We write

$$\begin{aligned} \vec{a} \times \vec{L} &= -\frac{GM}{|\vec{r}|^2} \hat{\mathbf{r}} \times \left(\mu |\vec{r}|^2 \hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\ &= GM\mu \hat{\mathbf{r}} \times \left(\hat{\mathbf{r}} \times \frac{d\hat{\mathbf{r}}}{dt} \right) \\ &= -GM\mu \left[\left(\hat{\mathbf{r}} \cdot \frac{d\hat{\mathbf{r}}}{dt} \right) \hat{\mathbf{r}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}) \frac{d\hat{\mathbf{r}}}{dt} \right] \\ &= GM\mu \frac{d\hat{\mathbf{r}}}{dt}. \end{aligned}$$

Now consider

$$\begin{aligned} \frac{d}{dt} (\vec{v} \times \vec{L}) &= \frac{d\vec{v}}{dt} \times \vec{L} + \vec{v} \times \frac{d\vec{L}}{dt} \\ &= \vec{a} \times \vec{L} \\ &= \frac{d}{dt} (GM\mu \hat{\mathbf{r}}), \end{aligned} \quad (\text{A.20})$$

where $\frac{d\vec{L}}{dt} = 0$ by Eqn(A.17). Hence

$$\vec{v} \times \vec{L} = GM\mu \hat{\mathbf{r}} + \vec{D}, \quad (\text{A.21})$$

where $\vec{D}$ is a constant of integration that exists in the plane of the orbit. We can then introduce the vector $\vec{r}$ into this equation:

$$\begin{aligned}\vec{r} \cdot (\vec{v} \times \vec{L}) &= GM\mu |\vec{r}| \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} + \vec{r} \cdot \vec{D} \\ (\vec{r} \times \vec{v}) \cdot L &= GM\mu r + rD \cos \theta.\end{aligned}\tag{A.22}$$

Now by Eqn(A.19), this is simply

$$\frac{|\vec{L}|^2}{\mu} = GM\mu |\vec{r}| \left(1 + \frac{|\vec{D}| \cos \theta}{GM\mu} \right).\tag{A.23}$$

We define the eccentricity of the ellipse as

$$e = \frac{|\vec{r}|}{GM\mu}\tag{A.24}$$

and hence we have

Kepler's First Law

$$|\vec{r}| = \frac{|\vec{L}|^2 / \mu^2}{GM(1 + e \cos \theta)},\tag{A.25}$$

which is the equation defining an ellipse, provided that¹ $e < 1$ – thus proving the First Law. Since the general equation for an ellipse with semimajor axis a is

$$|\vec{r}| = \frac{a(1 - e^2)}{1 + e \cos \theta} \quad \text{for } 0 \leq e < 1,\tag{A.26}$$

it is immediately apparent that we can write

$$|\vec{L}| = \mu \sqrt{GMa(1 - e^2)},\tag{A.27}$$

and since $|\vec{L}|$ is constant by Eqn(A.17), presuming total mass M remains constant we have that μ is constant as well.

A.2.2 | The Second Law

Consider the area of an elliptical orbit (see the shaded area in Figure A.2). We have

$$dA = dr(r d\theta) = r dr d\theta,\tag{A.28}$$

and integrating over r gives

$$dA = \frac{1}{2} r^2 d\theta,\tag{A.29}$$

¹Or equivalently that $|\vec{D}| < GM\mu$, which we are free to select as required as the vector $\vec{D}$ is arbitrary.

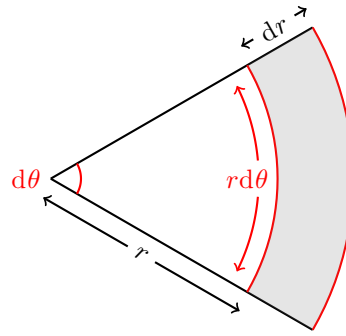


Figure A.2: An infinitesimal area segment of an orbital trajectory. Given the definitions in the figure, we have the area of the shaded segment is $dA = dr(rd\theta) = r dr d\theta$.

thus

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}. \quad (\text{A.30})$$

We can write the velocity of the star in the plane of the ellipse in terms of its radial component $\vec{v}_r$ and its angular component $\vec{v}_\theta$,

$$\vec{v} = \vec{v}_r + \vec{v}_\theta = \frac{dr}{dt} \hat{r} + |\vec{r}| \frac{d\theta}{dt} \quad (\text{A.31})$$

Importantly we have

$$v_\theta = r \frac{d\theta}{dt}, \quad (\text{A.32})$$

such that

$$\frac{dA}{dt} = \frac{1}{2} r v_\theta. \quad (\text{A.33})$$

Since r and v_θ are orthogonal, we have

$$r v_\theta = \underbrace{|\vec{r} \times \vec{\theta}|}_{\text{by Eqn(A.18)}} = \left| \frac{\vec{L}}{\mu} \right| = \frac{L}{\mu}, \quad (\text{A.34})$$

hence

Kepler's Second Law

$$\frac{dA}{dt} = \frac{1}{2} \frac{L}{\mu}. \quad (\text{A.35})$$

By Eqn(A.17), L is constant (and by Eqn(A.27), μ is constant as well), hence we have that $\frac{dA}{dt}$ is constant over the orbit, and Kepler's second law is established.

A.2.3 | The Third Law

Let the period of the binary star system be given by T . The area of the ellipse swept out by the binary over its orbit is

$$A = \int_0^T \frac{dA}{dt} dt = \frac{1}{2} \frac{L}{\mu} \int_0^T dt = \frac{1}{2} \frac{L}{\mu} T, \quad (\text{A.36})$$

and since the orbit is elliptical, this is equal to πab , and thus

$$\pi ab = \frac{1}{2} \frac{L}{\mu} T, \quad (\text{A.37})$$

or, equivalently,

$$T^2 = \frac{4\pi^2 a^2 b^2 \mu^2}{L^2}. \quad (\text{A.38})$$

Further, from the geometry of an ellipse we have

$$b^2 = a^2 (1 - e^2). \quad (\text{A.39})$$

Using this and Eqn(A.27) in tandem, we have

Kepler's Third Law

$$T^2 = \frac{4\pi^2}{GM} a^3, \quad (\text{A.40})$$

which proves Kepler's third law.

A.3 | Equations of Motion of decaying binaries

The following derivation is adapted from the derivation presented in [Richards \(2021\)](#). In this section we use a positive-definite metric signature ($\text{Tr } \boldsymbol{\eta} = +2$), and the Einstein summation convention where repeated indices are summed shall apply. Further, Greek indices span from 0 to 3, and Latin indices span from 1 to 3.

We have two goals. The first is to derive expressions for the coupled equations $\frac{da}{dt}$ and $\frac{de}{dt}$ describing the decay of a binary due to gravitational radiation. The second is to determine an expression for $T_c(a_0, e_0)$, the coalescence time, which is the time-scale for this to occur.

A.3.1 | $\frac{da}{dt}$ and $\frac{de}{dt}$

In the weak-field limit, the spacetime metric is presumed to be a perturbation of a locally Minkowski metric, that is, it is given by

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}, \quad (\text{A.41})$$

with $\boldsymbol{\eta}$ the Minkowski metric, $|h| \ll 1$, and $\kappa^2 = 32\pi G$. This metric induces (see, among others, [Einstein, 1916](#); [Peters, 1964](#)) the wave equation

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\kappa T_{\mu\nu}, \quad (\text{A.42})$$

where $\mathbf{T}$ is the stress-energy tensor. Given some plane wave solution to this wave equation, of the form

$$h_{\mu\nu} = A e_{\mu\nu} \cos(\vec{\mathbf{k}} \cdot \vec{\mathbf{x}} - \omega t), \quad (\text{A.43})$$

where $\mathbf{e}$ is a unit polarization tensor, we have the energy density

$$U = \frac{1}{2}c^2\omega^2 A^2. \quad (\text{A.44})$$

The two possible non-zero spacelike and transverse polarizations for a wave travelling (without loss of generality) in the z -direction are

$$e_1 = \frac{1}{\sqrt{2}}(\hat{x}\hat{x} - \hat{y}\hat{y}), \quad (\text{A.45})$$

and

$$e_2 = \frac{1}{\sqrt{2}}(\hat{x}\hat{y} - \hat{y}\hat{x}), \quad (\text{A.46})$$

where we denote the “plus” polarisation as e_1 and the “cross” polarisation as e_2 . The power P radiated into a solid angle Ω is given by

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left(\frac{\partial^3 Q_{ij}}{\partial t^3} e_{ij} \right)^2, \quad (\text{A.47})$$

where the components of the mass tensor $\mathbf{Q}$ are given by

$$Q_{ij} = m^\alpha x_{\alpha i} x_{\alpha j}. \quad (\text{A.48})$$

The total power is obtained reasonably straightforwardly:

$$\frac{dP}{d\Omega} = \frac{G}{8\pi c^5} \left[\left(\frac{d^3 Q_{ij}}{dt^3} \right)^2 - 2n_i \frac{d^3 Q_{ij}}{dt^3} n_k \frac{d^3 Q_{kj}}{dt^3} - \frac{1}{2} \left(\frac{d^3 Q_{ii}}{dt^3} \right)^2 + \frac{1}{2} \left(n_i n_j \frac{d^3 Q_{ij}}{dt^3} \right)^2 + \frac{d^3 Q_{ii}}{dt^3} n_j n_k \frac{d^3 Q_{jk}}{dt^3} \right], \quad (\text{A.49})$$

with $\hat{n}$ a unit vector in the direction of radiation. The total power is then found by integrating this equation over all directions, viz.

$$P = \frac{G}{5c^5} \left(\frac{d^3 Q_{ij}}{dt^3} \frac{d^3 Q_{ij}}{dt^3} - \frac{1}{3} \frac{d^3 Q_{ii}}{dt^3} \frac{d^3 Q_{jj}}{dt^3} \right). \quad (\text{A.50})$$

Let us now inject the physics. Suppose the primary star has mass M_1 and the secondary mass M_2 . In the xy -plane, with the origin at the center of mass, we can parameterise the positions of the two stars as

$$\vec{x}_1 = \begin{bmatrix} d_1 \cos \varphi \\ d_1 \sin \varphi \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} -d_2 \cos \varphi \\ -d_2 \sin \varphi \end{bmatrix}, \quad (\text{A.51})$$

with

$$d_1 = \frac{m}{m+M}d \quad \text{and} \quad d_2 = \frac{M}{m+M}d, \quad (\text{A.52})$$

and φ our angle parameter. The components of the tensor $\mathbf{Q}$ are

$$Q_{xx} = \mu d^2 \cos^2 \varphi, \quad (\text{A.53})$$

$$Q_{yy} = \mu d^2 \sin^2 \varphi, \quad (\text{A.54})$$

$$Q_{xy} = Q_{yx} = \mu d^2 \sin \varphi \cos \varphi, \quad (\text{A.55})$$

with the rest vanishing, and with μ as the reduced mass. Now, as binary stars obey Kepler's laws, Kepler's orbit equation (Eqn(A.25)) holds:

$$d = \frac{a(1-e^2)}{1+e \cos \dot{\varphi}}, \quad (\text{A.56})$$

whilst the angular velocity is given by

$$\dot{\varphi} = \frac{\sqrt{G(M_1 + M_2)(1-e^2)a}}{d^2}. \quad (\text{A.57})$$

We therefore have

$$\frac{d^3 Q_{xx}}{dt^3} = \gamma(1+e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + 3e \sin \dot{\varphi} \cos^2 \dot{\varphi}), \quad (\text{A.58})$$

$$\frac{d^3 Q_{yy}}{dt^3} = -\gamma(1+e \cos \dot{\varphi})^2 (2 \sin(2\dot{\varphi}) + e \sin \dot{\varphi} (1 + 3 \cos^2 \dot{\varphi})), \quad (\text{A.59})$$

$$\frac{d^3 Q_{xy}}{dt^3} = \frac{d^3 Q_{yx}}{dt^3} = -\gamma(1+e \cos \dot{\varphi})^2 (2 \cos(2\dot{\varphi}) - e \cos \dot{\varphi} (1 - 3 \cos^2 \dot{\varphi})), \quad (\text{A.60})$$

where the quantity γ is defined as

$$\gamma^2 \equiv \frac{4G^3 M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1-e^2)^5}. \quad (\text{A.61})$$

The total power is therefore

$$P = \frac{8}{15} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a(1 - e^2)^5} (1 + e \cos \varphi)^4 (12(1 + e \cos \varphi)^2 + e^2 \sin^2 \varphi). \quad (\text{A.62})$$

Averaging this over one orbit yields the average power, which we write now as $\langle \frac{dE}{dt} \rangle$,

$$\left\langle \frac{dE}{dt} \right\rangle = -\frac{32}{5} \frac{G^4}{c^5} \frac{M_1^2 M_2^2 (M_1 + M_2)}{a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right). \quad (\text{A.63})$$

The average angular momentum is

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{32}{5} \frac{G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right). \quad (\text{A.64})$$

We now write the gravitational energy from the two-body problem,

$$E = -\frac{GM_1 M_2}{2a} \quad (\text{A.65})$$

$$\Rightarrow \frac{dE}{dt} = \frac{GM_1 M_2}{2} \frac{1}{a^2} \frac{da}{dt}. \quad (\text{A.66})$$

Define the constant β as

$$\beta \equiv \frac{64}{5} \frac{G^3 M_1 M_2 (M_1 + M_2)}{c^5}. \quad (\text{A.67})$$

Now insert Eqn(A.66) into Eqn(A.63) to obtain

$$\frac{da}{dt} = -\frac{2a^2}{GM_1 M_2} \frac{32}{5} \frac{G^4 M_1^2 M_2^2 (M_1 + M_2)}{c^5 a^5 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (\text{A.68})$$

$$= -\frac{\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right), \quad (\text{A.69})$$

which is our first equation. The derivation for $\frac{de}{dt}$ is more involved. The angular momentum from the two-body problem is

$$L^2 = \frac{GM_1^2 M_2^2 a (1 - e^2)}{M_1 + M_2} \quad (\text{A.70})$$

Implicit differentiation reveals that

$$2L \frac{dL}{dt} = \frac{GM_1^2 M_2^2}{M_1 + M_2} \frac{d}{dt} (a (1 - e^2)) \quad (\text{A.71})$$

$$= \frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right]. \quad (\text{A.72})$$

Multiplying each side of Eqn(A.64) by $2L$ and substituting this in, we obtain

$$\begin{aligned}
\frac{GM_1^2 M_2^2}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -L \frac{64 G^{\frac{7}{2}} M_1^2 M_2^2 \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right) \\
\frac{1}{M_1 + M_2} \left[\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} \right] &= -\sqrt{\frac{GM_1^2 M_2^2 a (1 - e^2)}{M_1 + M_2}} \frac{64 G^{\frac{5}{2}} \sqrt{M_1 + M_2}}{c^5 a^{\frac{7}{2}} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right) \\
\frac{da}{dt} (1 - e^2) - 2ae \frac{de}{dt} &= -\frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) \\
2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) + \frac{da}{dt} (1 - e^2) \\
2ae \frac{de}{dt} &= \frac{\beta}{a^3 (1 - e^2)^{\frac{3}{2}}} \left(1 + \frac{7}{8} e^2 \right) - \frac{\beta}{a^3 (1 - e^2)^{\frac{5}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \\
\frac{de}{dt} &= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \frac{(1 - e^2) \left(1 + \frac{7}{8} e^2 \right) - \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right)}{2e} \\
&= \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left[-\frac{19}{12} e^3 - \frac{121}{192} e \right] \\
&= -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right), \tag{A.73}
\end{aligned}$$

which is our second equation. Hence we arrive at the coupled differential equations describing the decay of a binary star system:

Peters (1964) equations for the inspiral of two compact objects

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \tag{A.74a}$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right) \tag{A.74b}$$

A.3.2 | The coalescence time

The coalescence time is by definition the time it takes for the binary system to merge after the secondary supernova. We can step towards this in a similar way to [Peters \(1964\)](#). First, dividing Eqn(A.74a) by Eqn(A.74b) gives

$$\frac{da}{de} = \frac{12}{19} \frac{a}{e} \frac{1 + \frac{73}{24} e^2 + \frac{37}{96} e^4}{(1 - e^2) \left(1 + \frac{121}{304} e^2 \right)} \tag{A.75}$$

We can integrate this however we choose to obtain $a(e)$, namely

$$a(e) = \frac{c_0 e^{\frac{12}{19}}}{(1 - e^2)} \left(1 + \frac{121}{304} e^2 \right)^{\frac{870}{2299}} \tag{A.76}$$

with

$$c_0 \equiv a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304} e_0^2 \right)^{-\frac{870}{2299}} \quad (\text{A.77})$$

embodying the initial parameters. Now, substituting Eqn(A.76) into Eqn(A.74b), we see

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right) \quad (\text{A.78})$$

$$= -\frac{19}{12} \frac{\beta}{\left(\frac{c_0 e^{\frac{12}{19}}}{(1 - e^2)} \left(1 + \frac{121}{304} e^2 \right)^{\frac{870}{2299}} \right)^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right) \quad (\text{A.79})$$

$$= -\frac{19}{12} \frac{\beta}{c_0^4} \frac{(1 - e^2)^{\frac{3}{2}} e}{e^{\frac{48}{19}}} \frac{\left(1 + \frac{121}{304} e^2 \right)}{\left(1 + \frac{121}{304} e^2 \right)^{\frac{3480}{2299}}} \quad (\text{A.80})$$

$$= -\frac{19}{12} \frac{\beta}{c_0^4} \frac{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}}{\left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}} \quad (\text{A.81})$$

$$(\text{A.82})$$

Now write this as $\frac{dt}{de}$ and integrate to get the coalescence time:

$$\frac{dt}{de} = -\frac{12}{19} \frac{c_0^4}{\beta} \frac{\left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} \quad (\text{A.83})$$

$$T_c = \int_0^{T_c} dt = -\frac{12}{19} \frac{c_0^4}{\beta} \int_{e_0}^0 \frac{\left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{A.84})$$

$$= \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{\left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}} e^{-\frac{29}{19}}} de \quad (\text{A.85})$$

i.e.,

The coalescence time of inspiralling compact objects

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de \quad (\text{A.86})$$

If $e_0 = 0$ then the orbit is circular, and we immediately see concerning behaviour – the integral collapses to zero, necessitating a different approach.

One of the critical mistakes here is presuming that c_0 in Eqn(A.77) is valid for all physical eccentricities. In fact, in the limit as $e_0 \rightarrow 0$, c_0 diverges. We retreat to before this critical assumption, and examine Eqn(A.74a). In the limiting case of a circular orbit, $e = e_0 = 0$, so the expression becomes

$$\frac{da}{dt} = -\frac{\beta}{a^3}. \quad (\text{A.87})$$

and by separation of variables we obtain

$$a^3 da = -\beta dt, \quad (\text{A.88})$$

of which the solution is

$$\frac{a^4}{4} = -\beta t + c, \quad (\text{A.89})$$

for constant $c \in \mathbb{R}$. The boundary conditions are that when $t = 0, a = a_0$, and when $t = T_c, a = 0$. Applying the first boundary condition, we have $c = a_0^4$. Hence,

$$a(t) = (a_0^4 - 4\beta t)^{\frac{1}{4}}, \quad (\text{A.90})$$

and imposing our second boundary condition we obtain the coalescence time

$$T_c = \frac{a_0^4}{4\beta}. \quad (\text{A.91})$$

But if $e \neq 0$, then the orbit is eccentric and the integral must be solved numerically, as no closed form solution exists.

A.4 | Construction of the wind mass-loss rate

We construct our wind mass loss rates as follows. The total pre-WR mass loss is given by [de Jager et al. \(1988\)](#), namely

$$\log \dot{M}_{\text{de Jager}} = - \sum_{n=0}^6 \sum_{\substack{i=0 \\ j=n-i}}^n a_{ij} \cos(i \arccos x_1) \cos(j \arccos x_2), \quad (\text{A.92})$$

$$\dot{M}_{\text{pre-WR}} = \sqrt{\frac{Z}{Z_{\odot}}} 10^{\log \dot{M}_{\text{de Jager}}}, \quad (\text{A.93})$$

where $x_1 = \frac{\log T - 4.05}{0.75}$, $x_2 = \frac{\log L - 4.6}{2.1}$, and the coefficients a_{ij} are given by the entries in the matrix

$$\mathbf{a}_{ij} = \begin{pmatrix} 6.34916 & -5.04240 & -0.83426 & -1.13925 & -0.12202 \\ 3.41678 & 0.15629 & 2.96244 & 0.33659 & 0.57576 \\ -1.08683 & 0.41952 & -1.37272 & -1.07493 & - \\ 0.13095 & -0.09825 & 0.13025 & - & - \\ 0.22427 & 0.46591 & - & - & - \\ 0.11968 & - & - & - & - \end{pmatrix}. \quad (\text{A.94})$$

This mass loss rate is based upon the rates for 271 stars, the majority of which are Population I. Individual terms in the summation are Chebyshev polynomials of the first kind, restricted over the domain $\log T_{\text{eff}} \in [3.3, 4.8]$, $\log L \in [2.5, 6.7]$. As the free parameters here are temperature and luminosity, the result can be plotted over a HRD. The results of this are shown in Figure A.3. Generally we constrain the inputs to the arccosine function to be between $-1 \leq x \leq 1$, which permits them to be defined outside of the defined area from [de Jager et al. \(1988\)](#); the effect of this is shown in the right-hand panel of Figure A.3. Without this restriction, the mass loss rates are simply undefined there, and are thus set to zero within the code.

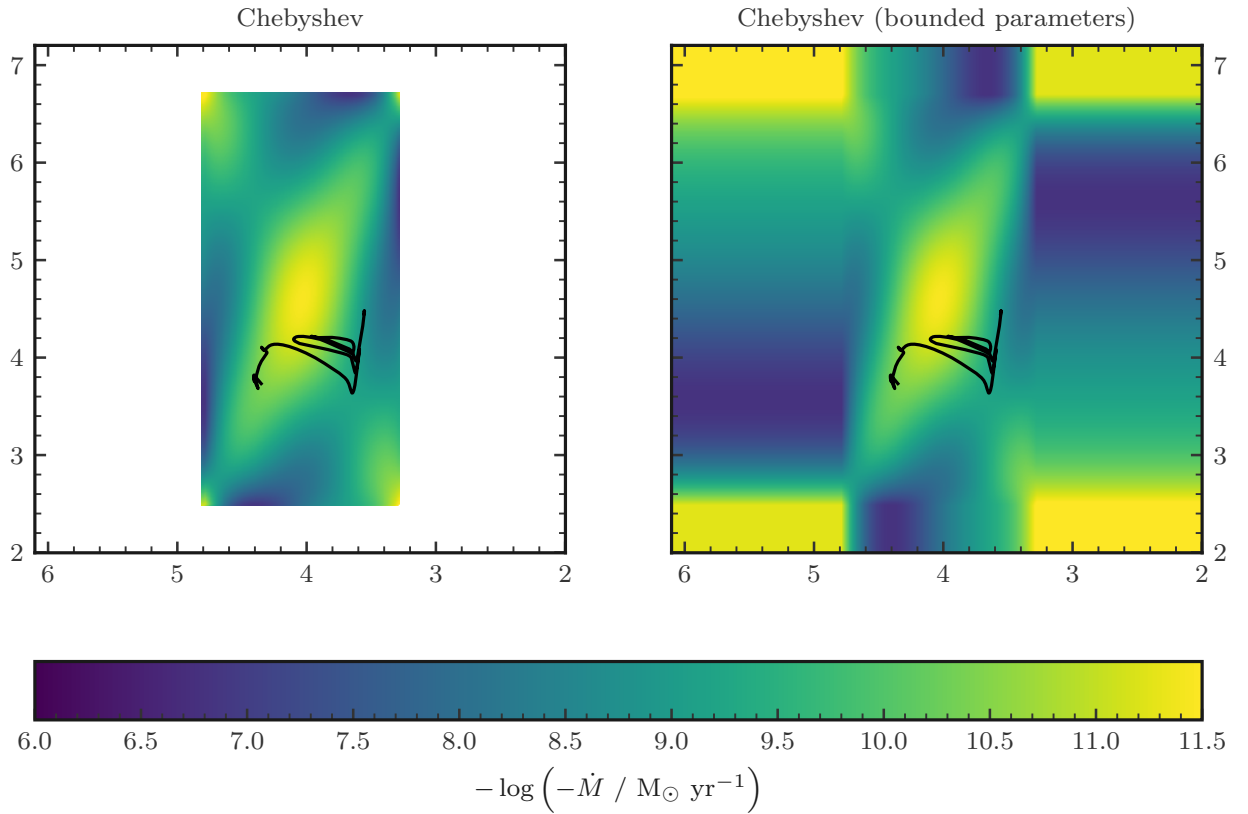


Figure A.3: The Chebyshev polynomial expansion of Eqn(A.92). A $10 M_{\odot}$ model is overplotted in black for reference. The “bounded parameters” plot restricts the arguments to arccos in Eqn(A.92) to be between $-1 \leq x_i \leq 1$.

We interpolate the bi-stability jump around 25 kK (see, for example, [Pauldrach & Puls, 1990](#); [Vink et al., 1999](#)) with the following regime. Define the constants

$$C_1 = \left(\frac{Z}{Z_\odot} \right)^{0.13}, \quad (\text{A.95})$$

$$C_2 = -c_1 + c_2 \log \left(\frac{L}{c_{16}} \right) - c_3 \log \left(\frac{M}{c_{15}} \right) - c_4 \log (1.3C_1) + c_5 \log \left(\frac{T}{c_{14}} \right) - c_6 \left(\log \left[\frac{T}{c_{14}} \right] \right)^2 + c_7 \log \left(\frac{Z}{Z_\odot} \right), \quad (\text{A.96})$$

and

$$C_3 = -c_8 + c_9 \log \left(\frac{L}{c_{16}} \right) - c_{10} \log \left(\frac{M}{c_{15}} \right) - c_{11} \log (0.65C_1) + c_{12} \log \left(\frac{T}{2 \times 10^4} \right) + c_{13} \log \left(\frac{Z}{Z_\odot} \right), \quad (\text{A.97})$$

where the constants c_i are given in Table A.1.

Constant	Value	Constant	Value	Constant	Value	Constant	Value
c_1	6.697	c_5	0.933	c_9	2.210	c_{13}	0.85
c_2	2.194	c_6	10.92	c_{10}	1.339	c_{14}	$4 \times 10^4 \text{ K}$
c_3	1.313	c_7	0.85	c_{11}	1.601	c_{15}	$30 \text{ M}_\odot$
c_4	1.226	c_8	6.688	c_{12}	1.07	c_{16}	$10^5 \text{ L}_\odot$

Table A.1: The mass loss rate constants for interpolation around the bi-stability jump (Eqns (A.95–97)). This interpolation is shown pictorially in Figure A.4.

Our mass loss rate becomes:

$$\dot{M}_{\text{BS}} = \begin{cases} \frac{(T-T_1)10^{C_3} + (T_2-T)\dot{M}_{\text{pre-WR}}}{2.5 \times 10^3} & T_1 < T < T_2 \\ 10^{C_2} & T_2 \leq T \leq T_3 \\ \frac{(T-T_3)10^{C_2} + (T_4-T)10^{C_3}}{5 \times 10^3} & T_3 < T < T_4 \\ \frac{(6T_1-T)10^{C_2} + (T-5T_1)\dot{M}_{\text{pre-WR}}}{10^4} & 5T_1 < T < 6T_1 \\ \dot{M}_{\text{pre-WR}} & \text{otherwise} \end{cases} \quad (\text{A.98})$$

where $T_1 = 10 \text{ kK}$, $T_2 = 12.5 \text{ kK}$, $T_3 = 22.5 \text{ kK}$, $T_4 = 27.5 \text{ kK}$. This interpolation is shown in Figure A.4.

We define

$$\mathcal{M} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-13.6 + 1.63 \log \left(\frac{L}{L_\odot} \right) + 2.22 \log X}, \quad (\text{A.99})$$

where X is the surface hydrogen abundance of the primary, and then define

$$\dot{M}_{\text{WNL}} = \begin{cases} 5 \left(\mathcal{M} - \dot{M}_{\text{BS}} \right) (0.42 - X) + \dot{M}_{\text{BS}} & 0.42 > X > 0.4 \\ \mathcal{M} & \text{otherwise} \end{cases}. \quad (\text{A.100})$$

Our total mass loss (provided the star has not depleted its surface helium, which will be discussed below) is now

$$\dot{M}_A = \begin{cases} 10\dot{\mu}(\log T - 3.9) + \dot{M}_{BS} & X < 0.42 \text{ and } \log T > 3.9 \\ \dot{M}_{WNL} & X > 0.42 \text{ and } \log T > 4.0, \\ \dot{M}_{BS} & \text{otherwise} \end{cases} \quad (\text{A.101})$$

where $\dot{\mu} = \dot{M}_{WNL} - \dot{M}_{BS}$. Finally, if the surface helium depletes we modify the mass loss rate further, defining

$$\dot{M}_{WC} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-8.3+0.84 \log\left(\frac{Z}{Z_\odot}\right)+2.04 \log X_{\text{He}}+1.04 \log(1-X_{\text{He}})} \quad (\text{A.102})$$

such that our total mass loss rate becomes (provided $X_{\text{H}} < 10^{-3}$ and $\log T > 4$)

$$\dot{M}_B = \begin{cases} 100 \left(\dot{M}_{WC} - \dot{M}_{WNL} \right) (\mathcal{X} - 0.02) + \dot{M}_{WNL} & \mathcal{X} > 0.02 \\ \dot{M}_{WNL} & \mathcal{X} < 0.02 \end{cases}, \quad (\text{A.103})$$

where $\mathcal{X} = \frac{\frac{1}{3}X_{\text{C}} + \frac{1}{4}X_{\text{O}}}{X_{\text{He}}}$. The total wind mass-loss rate is then

$$\dot{M}_{\text{wind}} = \begin{cases} \dot{M}_B & \text{surface helium depleted} \\ \dot{M}_A & \text{otherwise} \end{cases}. \quad (\text{A.104})$$

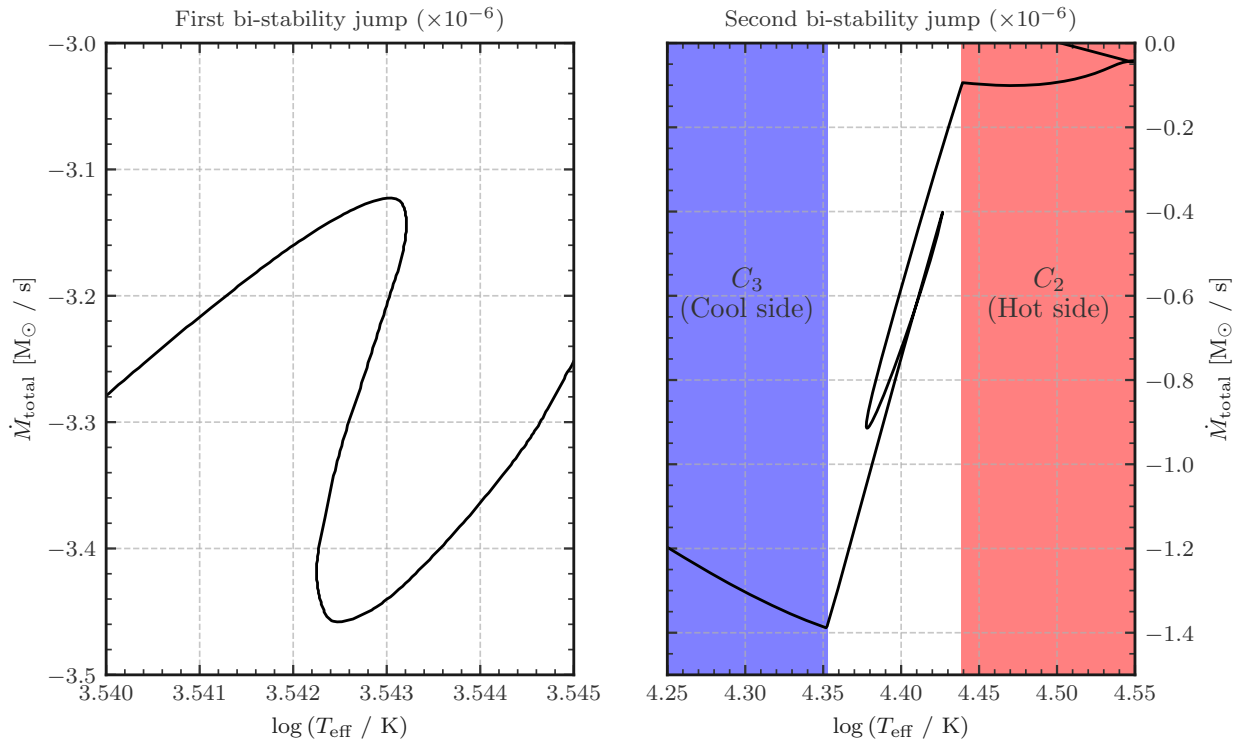


Figure A.4: An illustration of the bi-stability jumps in the mass-loss prescription. This illustration was computed for a $21 \text{ M}_{\odot}$ star. *Left plot:* The lower bi-stability jump, *Right plot:* The upper bi-stability jump. Shown in blue is the cool side of the bi-stability jump, where C_3 (Eqn(A.97)) is the interpolation, and shown in red is the hot side, where C_2 (Eqn(A.96)) is the interpolation.



APPENDIX B

Code Metadata



First learn computer science and all the theory. Next develop a programming style.

Then forget all that and just hack.

— *George Carrette* —



B.1 | Code Versions

Code Name	Version	Reference
HOKI	1.7.1	Stevance et al. (2020)
KAITIAKI	1.0.0	This work
MATPLOTLIB	3.8.2	Hunter (2007)
NUMBA	0.58.1	Lam et al. (2015)
NUMPY	1.26.3	Harris et al. (2020)
PANDAS	2.1.4	McKinney (2010)
PYTHON	3.11.5	-
SCIPY	1.11.4	Virtanen et al. (2020)
TAKAHE	1.0.0	Richards et al. (2023)
TUI	8	Ghodla et al. (2023)*

Table B.1: The code versions we use in this thesis. *TUI was described in earlier papers, but [Ghodla et al. \(2023\)](#) was the first to name it.

B.2 | Parameter Configuration

In this appendix we list the default parameters chosen for our binary evolution runs, as detailed in subsection 2.6.1.

The possible mass loss prescriptions (IML1/IML2) are:

0. No mass loss
1. [Reimers \(1975\)](#)
2. [Blöcker \(1995\)](#)
3. [Vassiliadis & Wood \(1993\)](#)
4. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with interpolation errors described in [Richards et al. \(2024\)](#)
5. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) (described in [Eldridge et al., 2017](#))
6. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with slowly turning on WR rates described in [Richards et al. \(2024\)](#)
7. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with no WR rates, described in [Richards et al. \(2024\)](#)
8. Unimplemented

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
NM2	199	Meshpoints	ICL	1		IT1	40	
ION	2		IT2	20		IAM	1	
JIN	15		IOP	5		OUT	15	
INUC	0		NCH	3		IBC	0	
JP	1		ICN	0		ITH	1	
IML1	5		IX	1		IML2	5	Mass loss (*2)
IY	1		ISGTH	0		IZ	1	
IMO	0		IMODE	2		IDIFF	0	
NWRT1	100		EPS	10^{-6}		NWRT2	1	
DEL	0.01		NWRT3	2		DHO	10^{-8}	
NWRT4	1		DT3	0		NWRT5	0	
DDD	1.0		NSAVE	10		NMONT	0	
DT1	0.1		DT2	1.2		CT1	3.0	
CT2	3.0		CT3	0.05		CT4	0.01	
CT5	0.6		CT6	0.35		CT7	0.45	
CT8	10^{-4}		CT9	0.1		CT10	0.3	

Table B.2: Our general initial conditions. Unless otherwise described, these are assumed to be enabled for all runs.

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
ZS	0.02		ALPHA	1.925		CH	0.7	
CC	0.173		CN	0.053		CO	0.482	
CNE	0.099		CMG	0.038		CSI	0.08	
CFE	0.072		RCD	10^6		OS	0.12	
RML	0		RMG	0		ECA	0	
XF	0.1		DR	0.001		RMT	1000.0	
RHL	0		AC	0		AK1	1.0	
AK2	10^{-4}		ECT	0		TRB	0	
IRAM	1		IRS1	1		VROT1	0	
IRS2	1		VROT2	10.0		FMAC	1.0	
FAM	0		IVMC	0		TRC1	700.0	
IVMS	0		TRC2	0		MWTS	0	
IAGB	0		ISGFAC	0		FACSGMIN	0.01	
SGTHFAC	0		ISTART	0		HKH	1.0	
GFF	10.0	Removed	NNMOD	0		CEPR	0	CEE prescription
ALPHACE	1.0		-	-	-	-	-	-

Table B.2: Continued

9. ZAMS evolution to a specific ZAMS mass, custom routine by Jan Eldridge



APPENDIX C

Period-Eccentricity distributions along the LVK Collaboration O3 contour



Statistics is the grammar of science.

— *Karl Pearson* —



In Figure C.1 we display the period-eccentricity distributions along the LVK Collaboration O3 contour. We also display a white dashed line corresponding to a merger time contour of the Hubble time. Most plots have a white region in the lower right corner of the plot, which corresponds to no systems passing through those bins. We use the definition of a Milky-Way Equivalent Galaxy (MWEG) from [Kasen et al. \(2017\)](#), namely that there is one MWEG per $(4.4 \text{ Mpc})^3$.

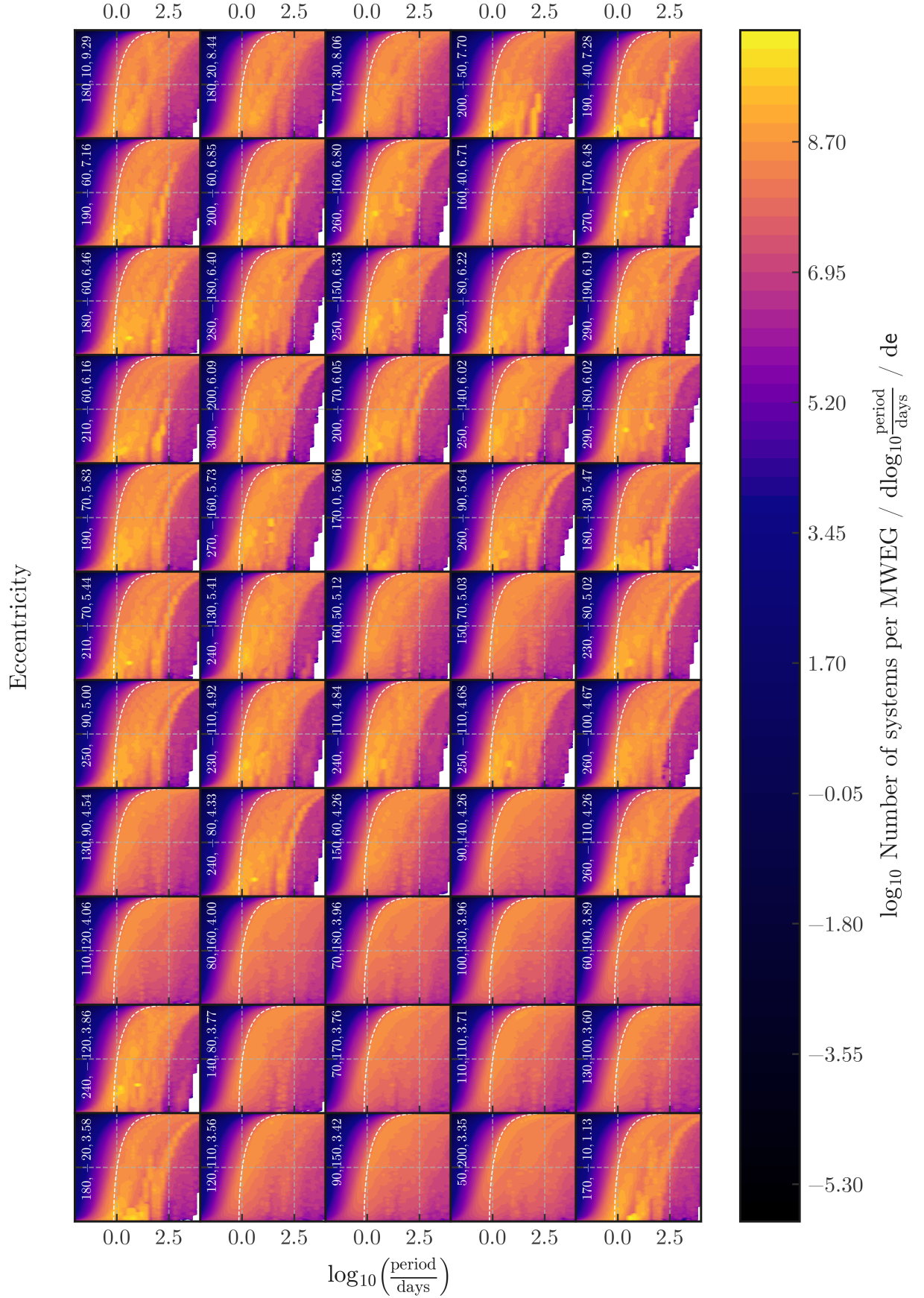


Figure C.1: The period-eccentricity distributions of populations governed by Bray kicks across the LIGO O3 contour. The combination of α and β used in the kick prescription is given in white along the left side of each plot. White regions denote no systems passing through those bins.



APPENDIX D

Bespoke Overcontact Binary Models



All models are wrong, but some are useful.

— *George E. P. Box* —

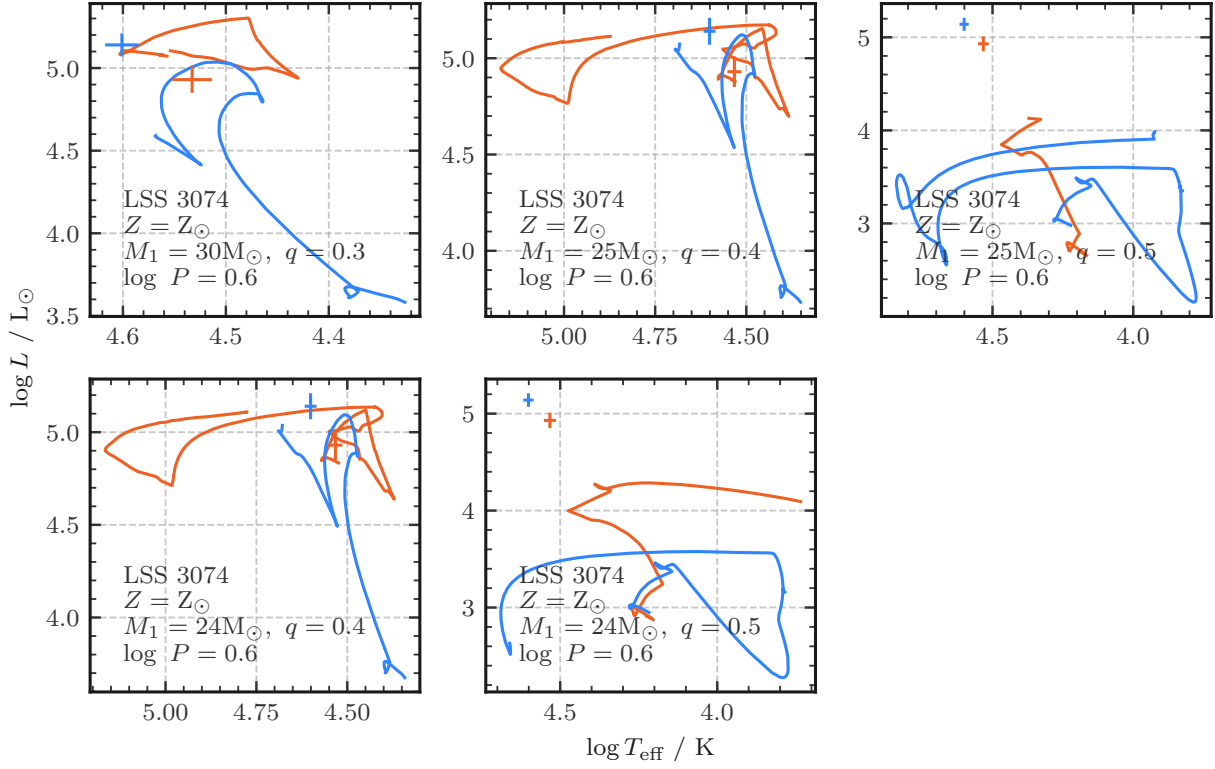


In Table D.1, we list the non-RLOF filling matches to observed contact binaries, contrasted with those that do undergo RLOF as enumerated in Table 5.3. In Figures D.1–D.3, we present the results of our bespoke overcontact binaries as established in Section 5.2. Shown are three parameter configurations, corresponding to the fiducial (normal period) grid in Figure D.1, and our ± 0.1 dex period grid in Figure D.2 (wider periods) and Figure D.3 (shorter periods).

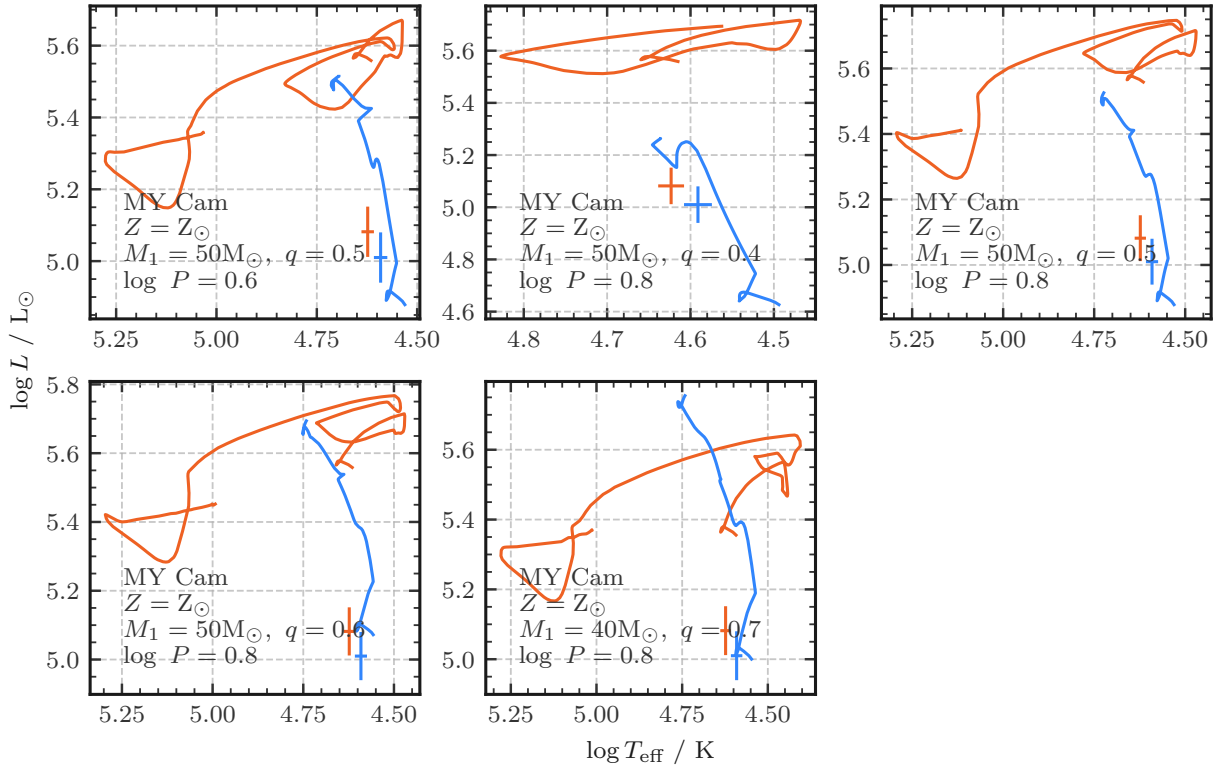
We present the results of our bespoke overcontact binaries as established in Section 5.2. Shown are three parameter configurations, corresponding to the fiducial (normal period) grid in Figure D.1, and our ± 0.1 dex period grid Figure D.2 (wider periods) and Figure D.3 (shorter periods).

System Name	ZAMS parameters				Match Parameters					
	M_1 [$M_\odot$]	M_2 [$M_\odot$]	$\log P$ [d]	M_1 [$M_\odot$]	M_2 [$M_\odot$]	$\log R_1$ [$R_\odot$]	$\log R_2$ [$R_\odot$]	$\log T_1$ [K]	$\log T_1$ [K]	Star 1 RLOF Star 2 RLOF
SMC 108086	20.00	15.00	0.50	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	0.60	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	0.70	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	0.80	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	0.90	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	1.00	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	1.10	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
SMC 108086	20.00	15.00	1.20	20.00	15.00	0.77	0.70	4.53	4.48	✗ ✗
V382 Cyg	35.00	30.00	0.50	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	0.60	29.08	24.58	0.99	0.92	4.57	4.55	✗ ✗
V382 Cyg	35.00	30.00	0.60	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	0.70	29.11	24.60	0.99	0.92	4.57	4.55	✗ ✗
V382 Cyg	35.00	30.00	0.70	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	0.80	29.04	24.59	1.01	0.93	4.56	4.55	✗ ✗
V382 Cyg	35.00	30.00	0.80	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	0.90	29.07	24.60	1.01	0.93	4.56	4.55	✗ ✗
V382 Cyg	35.00	30.00	0.90	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	1.00	29.08	24.60	1.01	0.93	4.56	4.55	✗ ✗
V382 Cyg	35.00	30.00	1.00	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	1.10	29.09	24.60	1.01	0.93	4.56	4.55	✗ ✗
V382 Cyg	35.00	30.00	1.10	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗
V382 Cyg	30.00	25.00	1.20	29.09	24.60	1.01	0.93	4.56	4.55	✗ ✗
V382 Cyg	35.00	30.00	1.20	35.00	30.00	0.98	0.95	4.58	4.56	✗ ✗

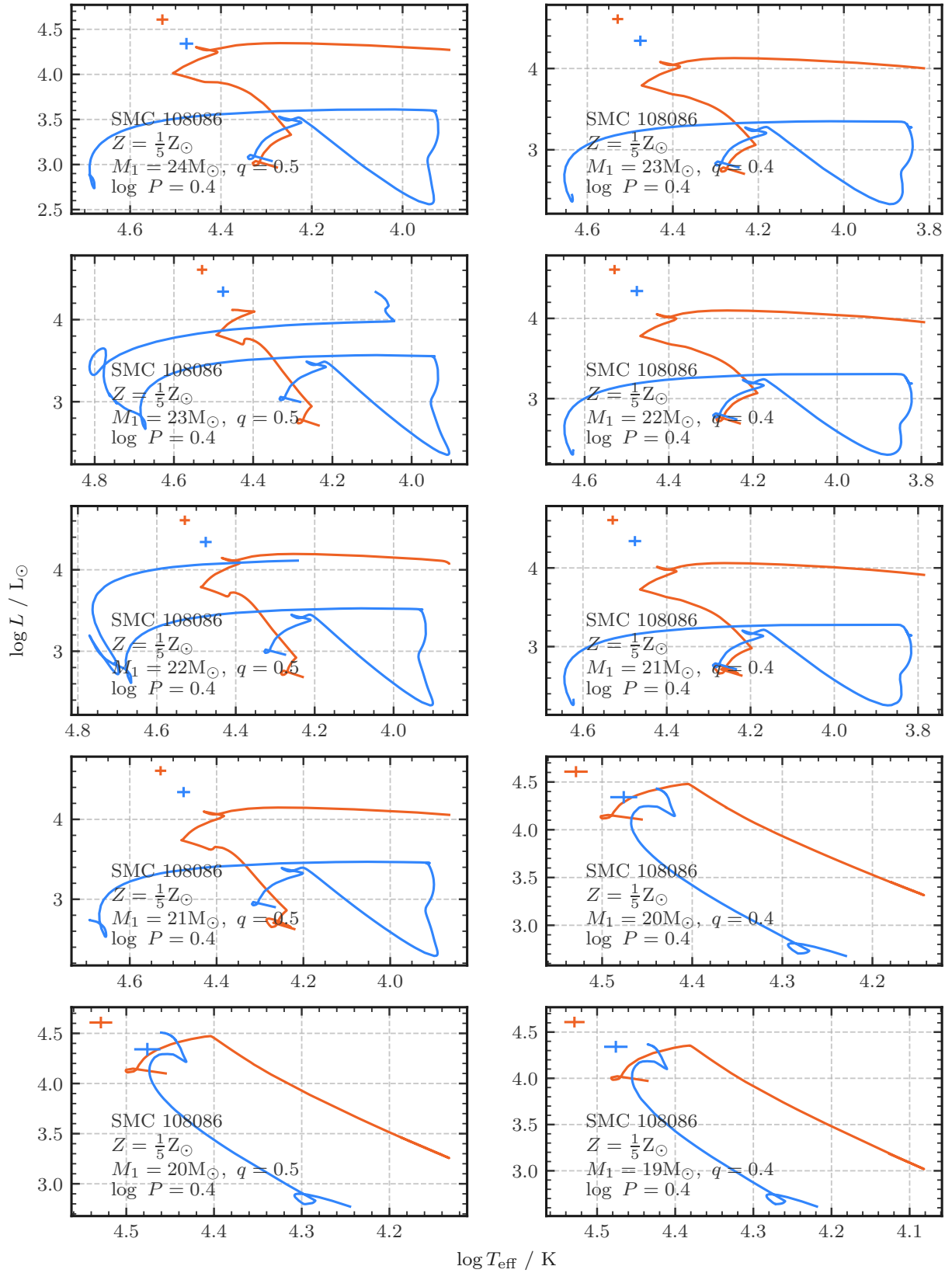
Table D.1: Matches from the TUI code. All parameters are rounded to two decimal places, and are given as at the first moment of matching. Only systems which do not fill their Roche lobes are shown, with the ones filling at least one component shown in Table 5.3.



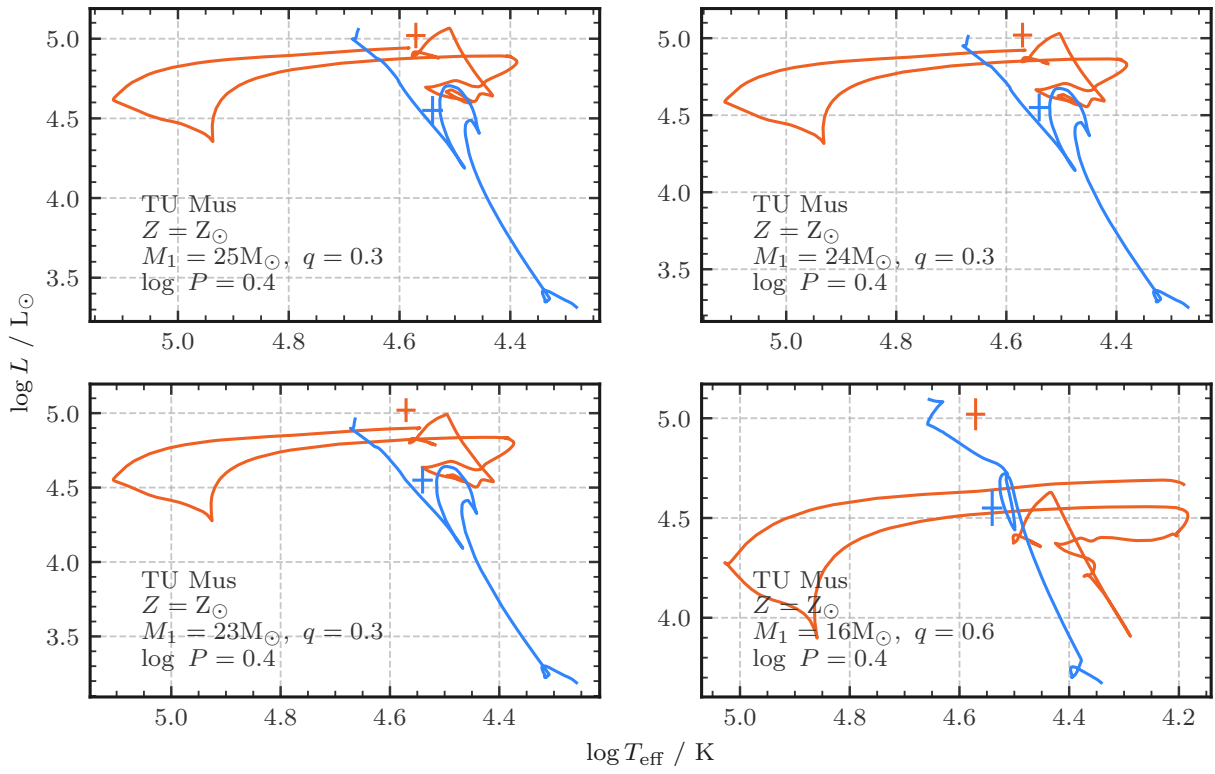
(a) LSS 3074



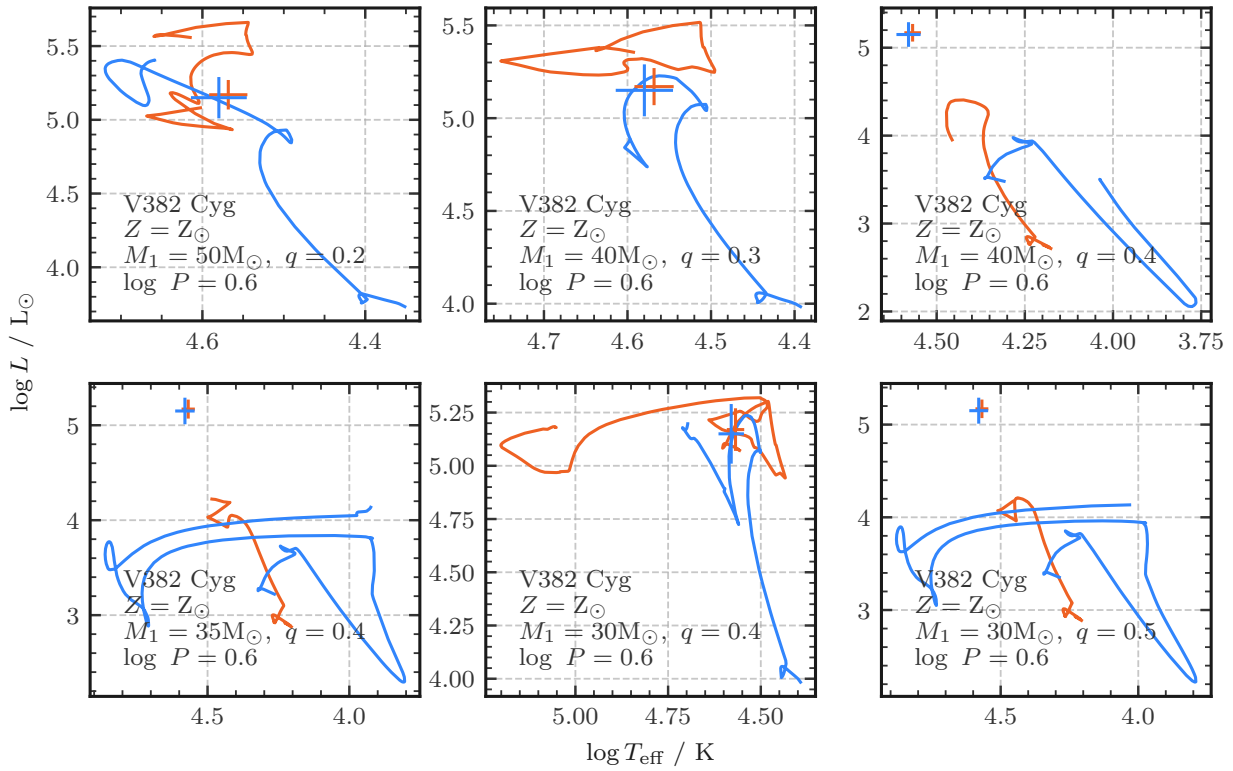
(b) MY Camelopardalis



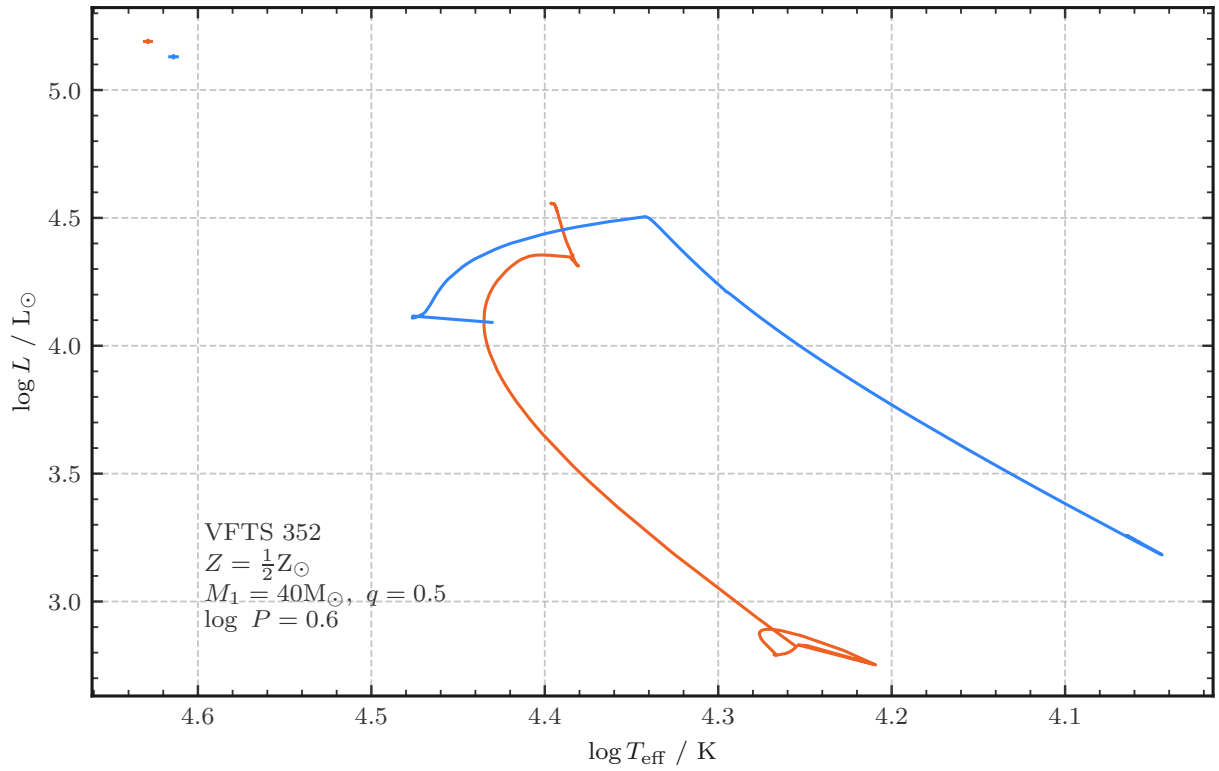
(c) SMC 108086



(d) TU Muscae

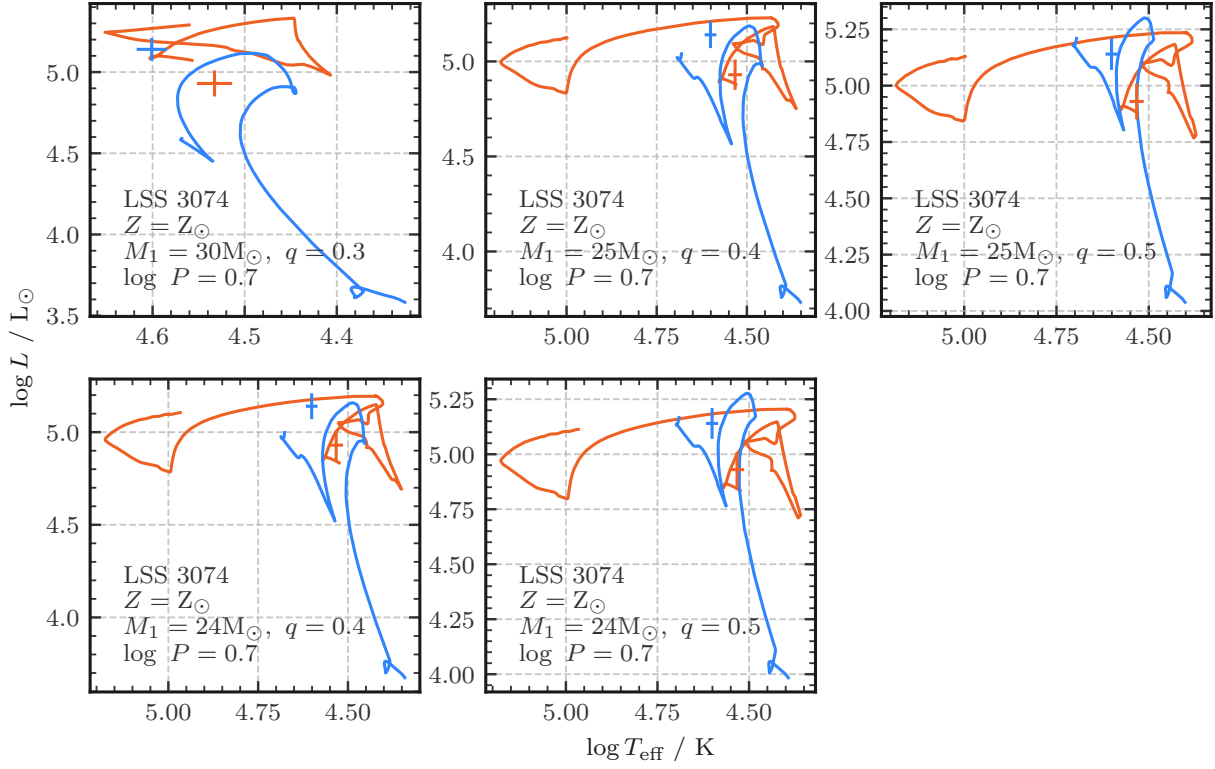


(e) V382 Cygni

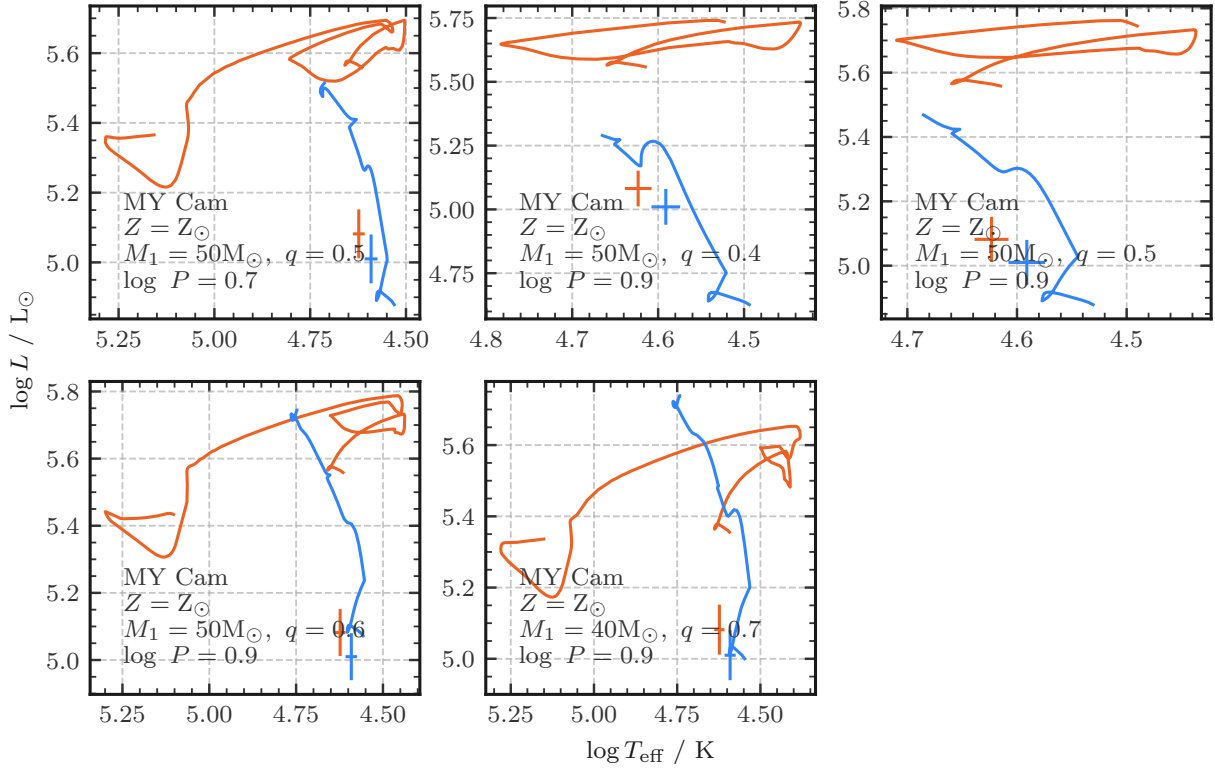


(f) VFTS 352

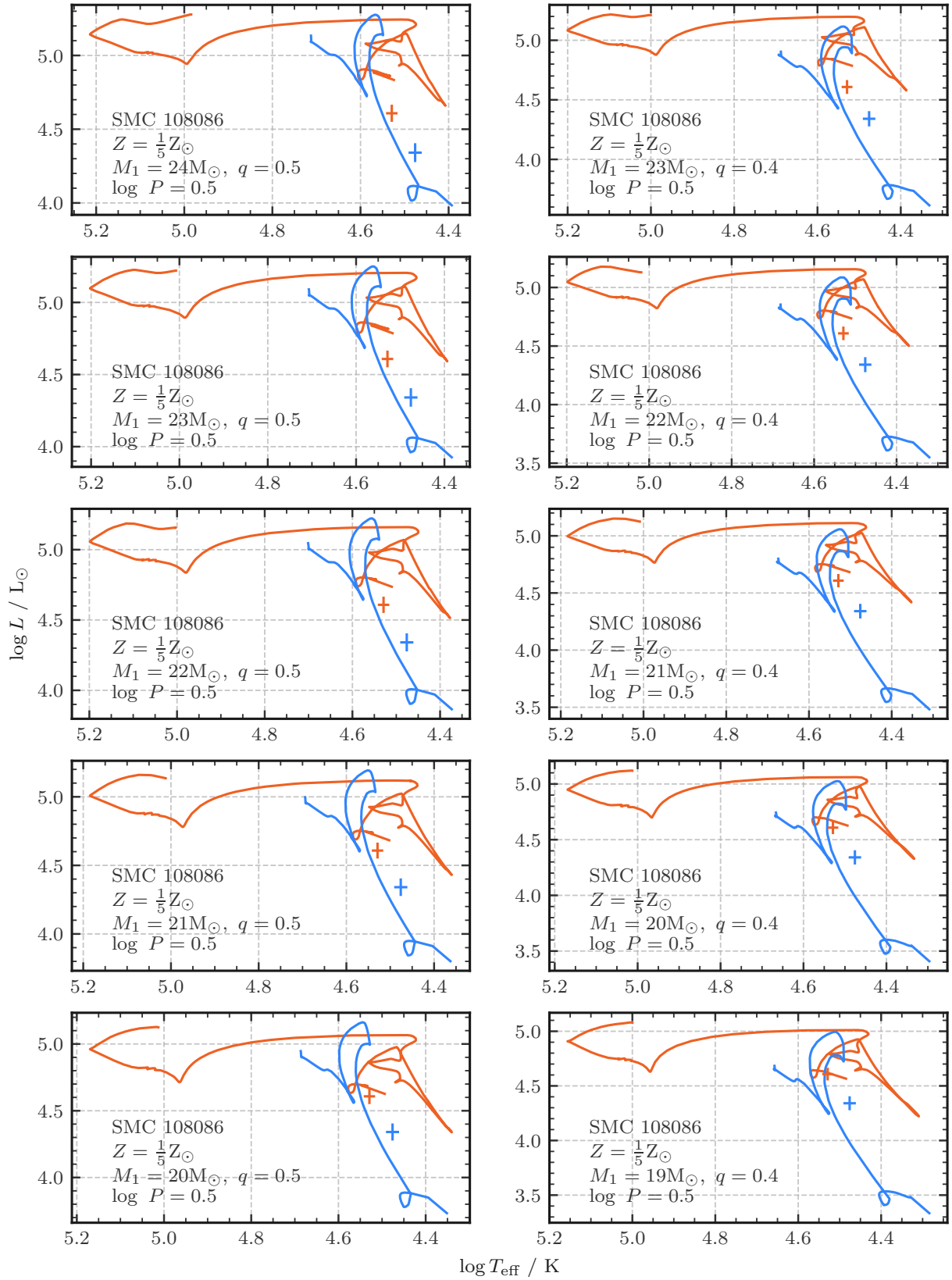
Figure D.1: The results of our bespoke binary evolution for **normal** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.



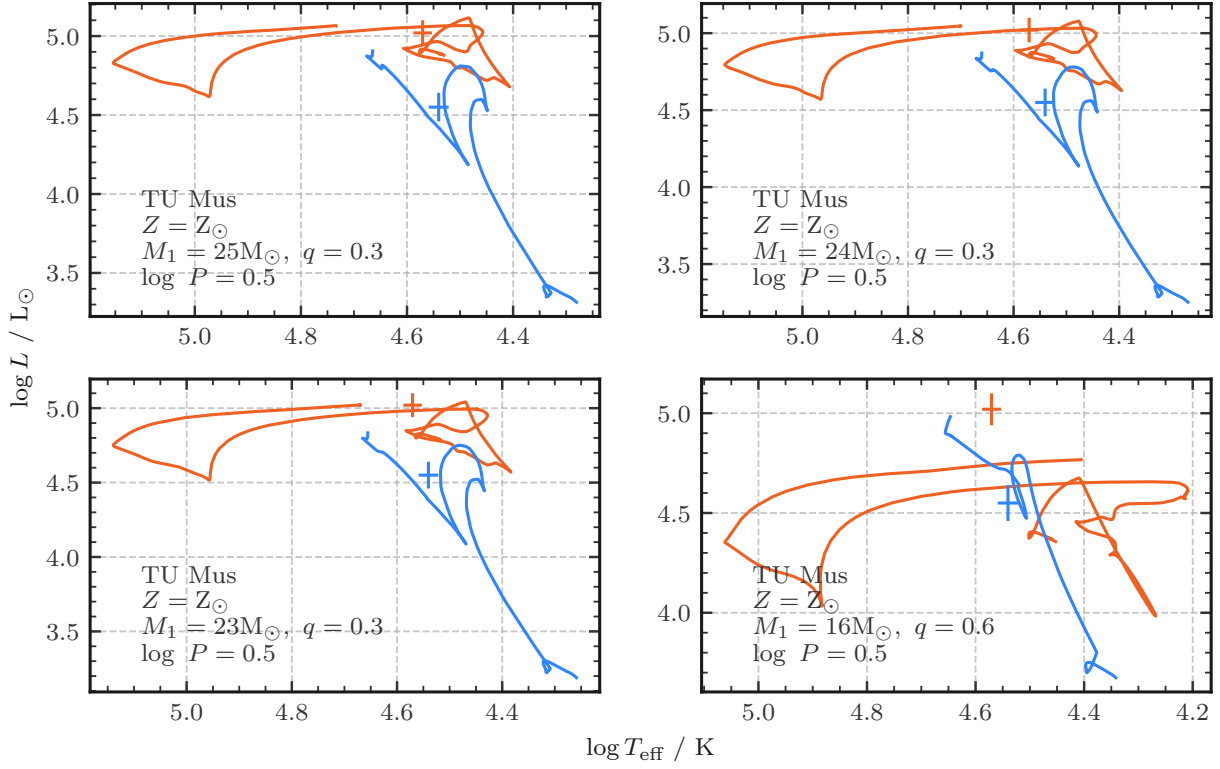
(a) LSS 3074



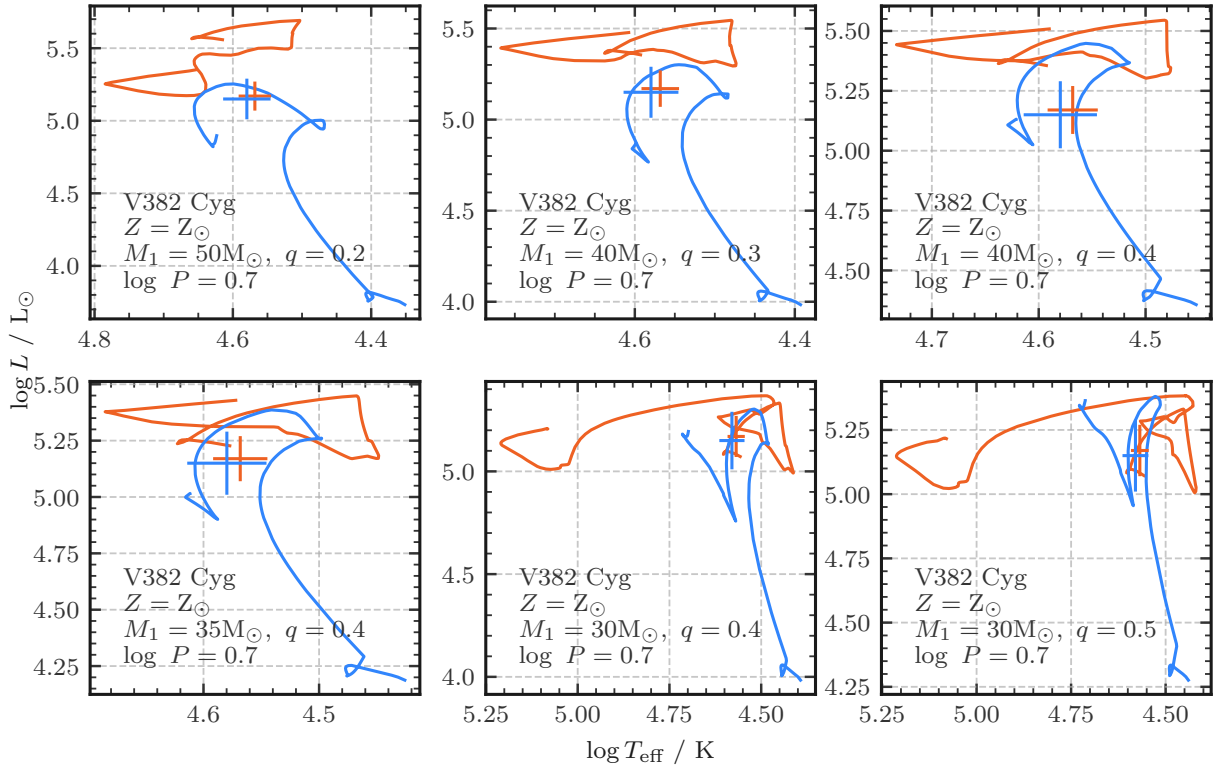
(b) MY Camelopardalis



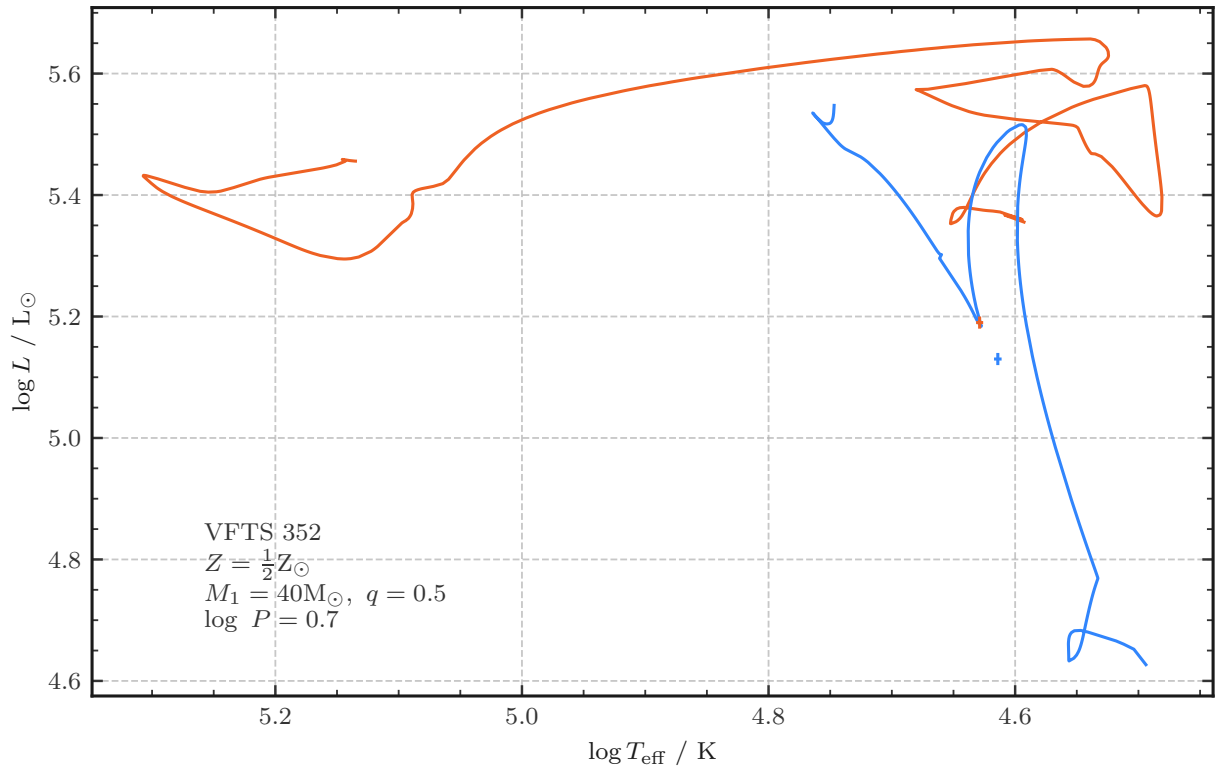
(c) SMC 108086



(d) TU Muscae

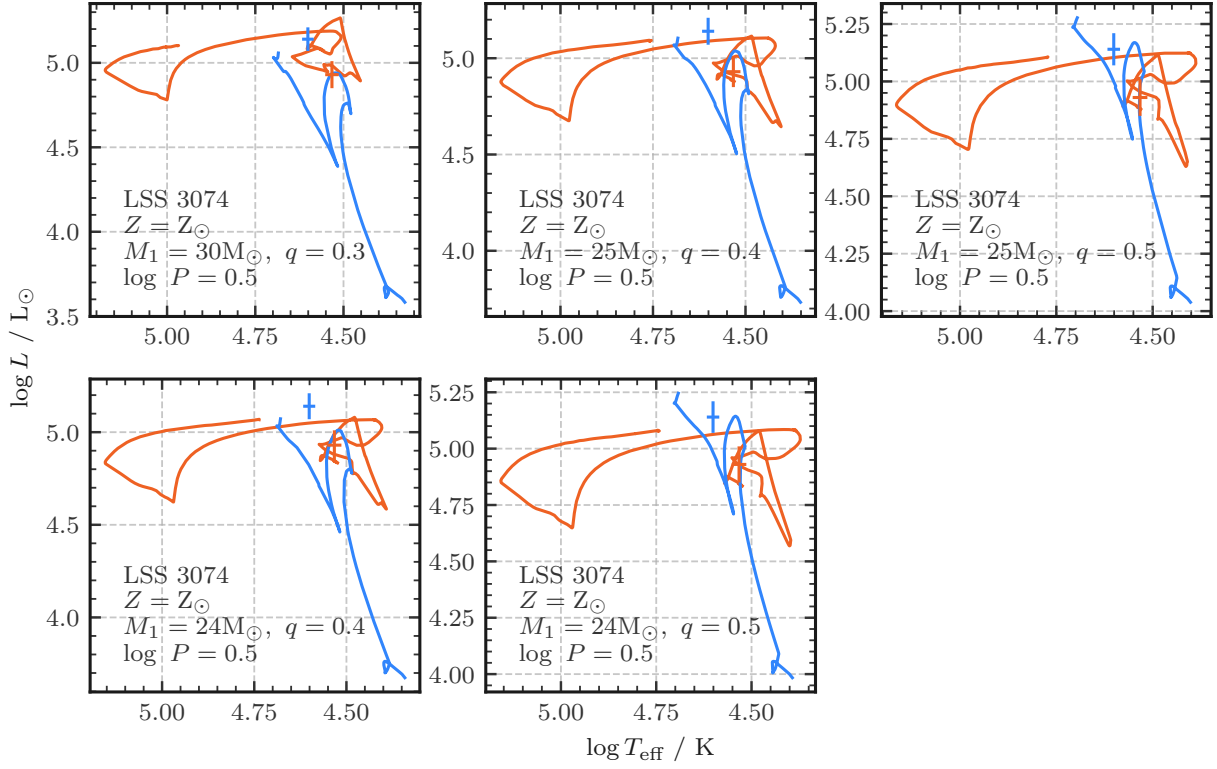


(e) V382 Cygni

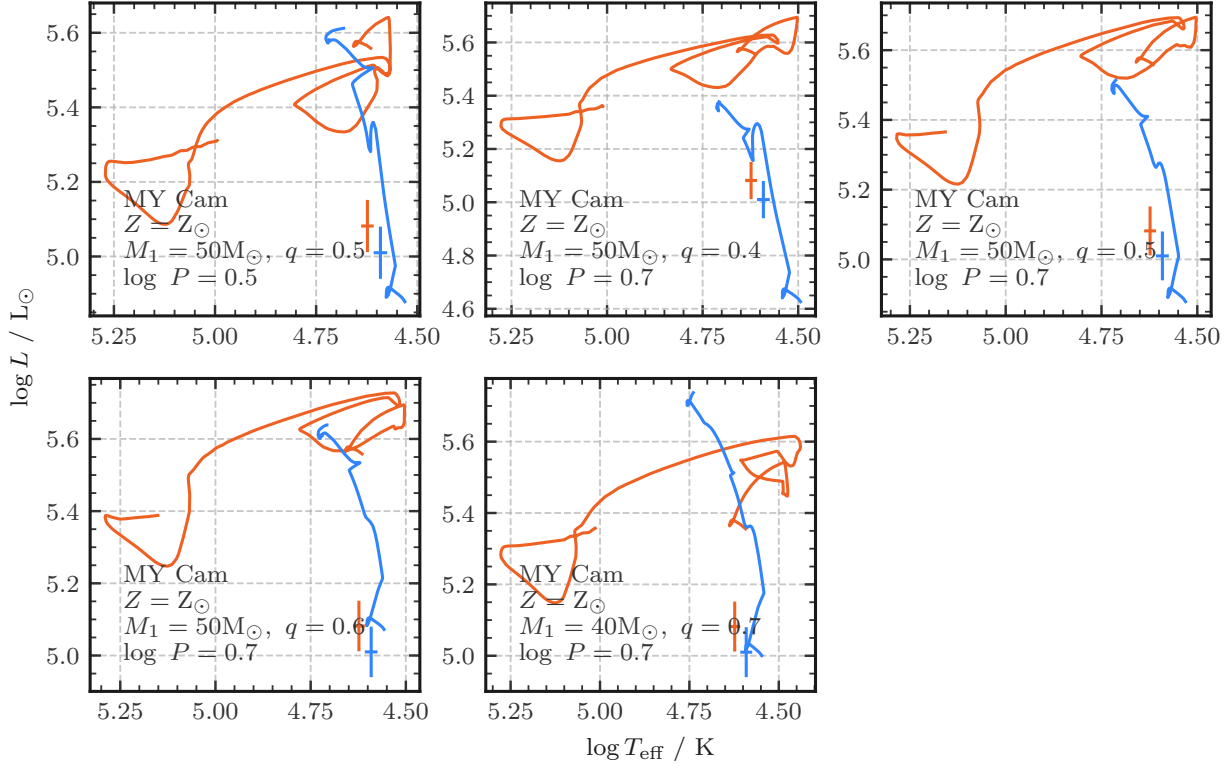


(f) VFTS 352

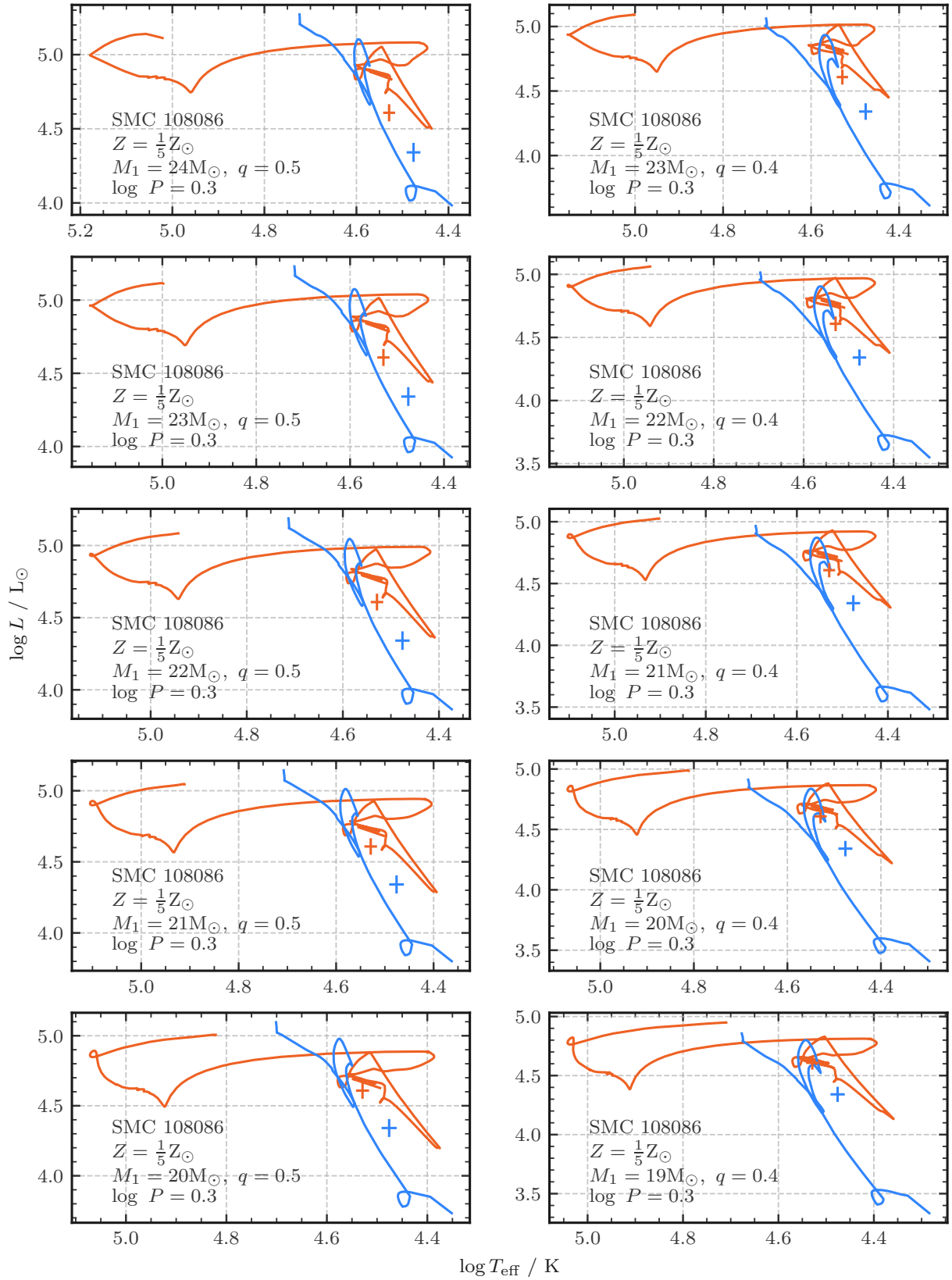
Figure D.2: The results of our bespoke binary evolution for **wider** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.



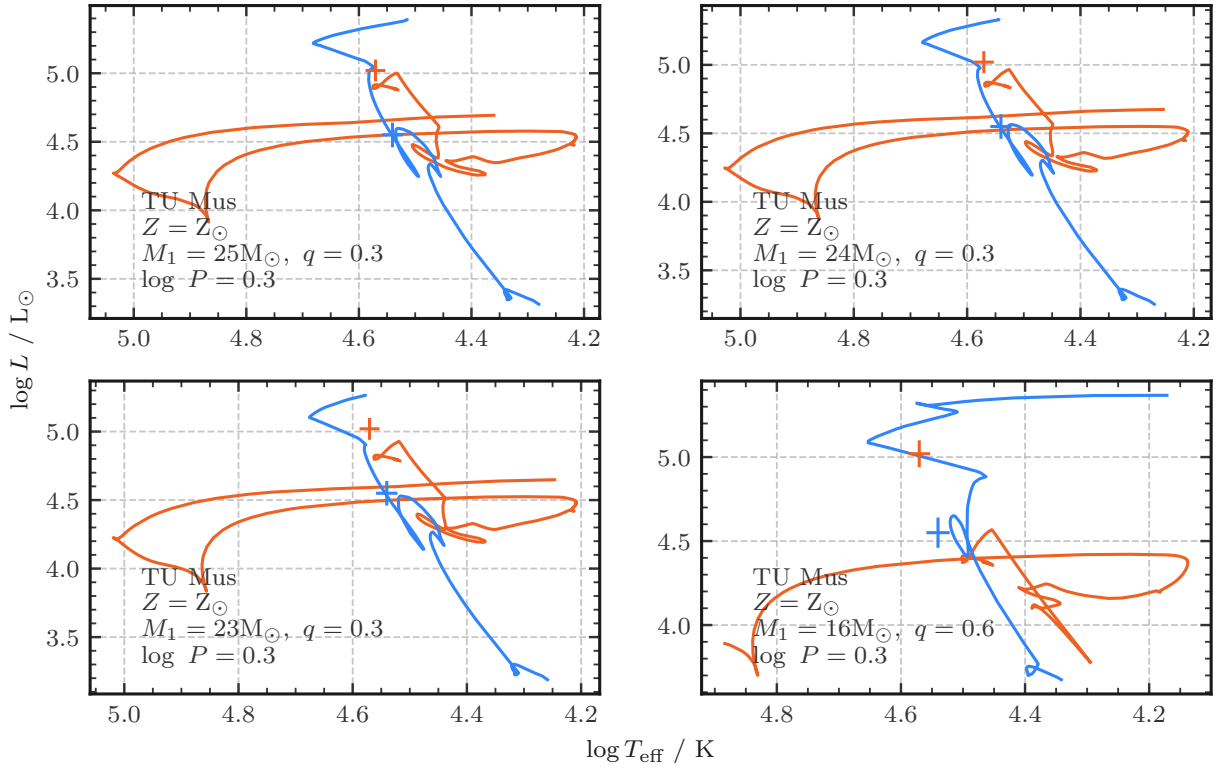
(a) LSS 3074



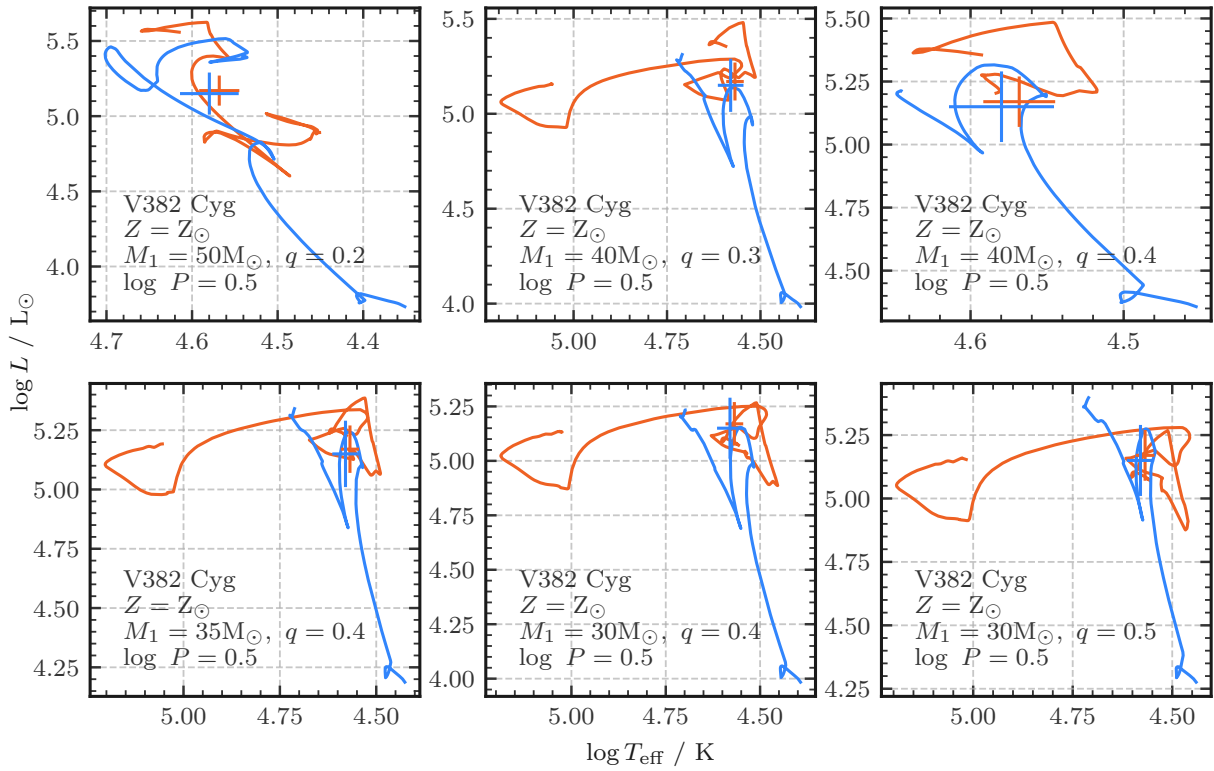
(b) MY Camelopardalis



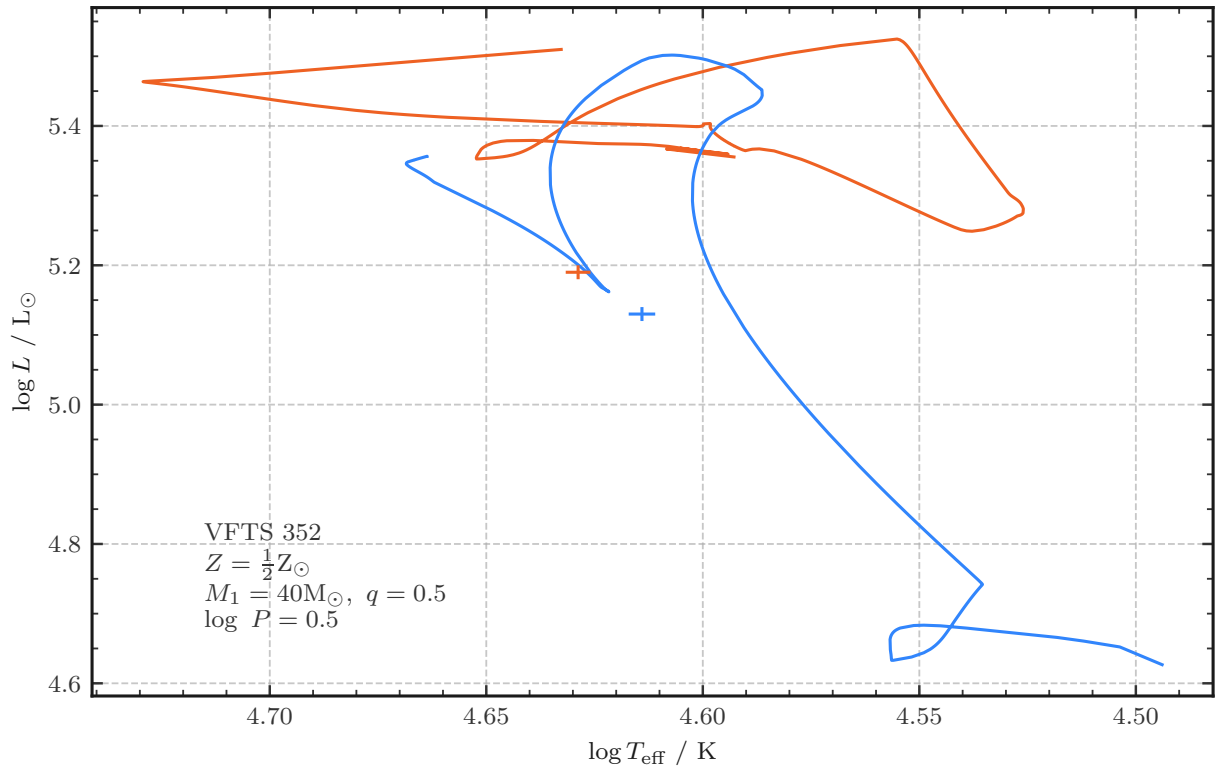
(c) SMC 108086



(d) TU Muscae



(e) V382 Cygni



(f) VFTS 352

Figure D.3: The results of our bespoke binary evolution for **shorter** periods. In these plots, orange lines represent the donors, and blue lines represent the accretors.



APPENDIX E

Overcontact Binary Remnants



I only tell stories, Your Grace. They may be truths, they may be fictions. All I know
is that the stories themselves exist and that I must tell them.

— *Hoid* —



In Table E.1, we list the binaries found as the matches to the BPASS v2 models in Section 5.1.2.

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
LSS 3074	z020	30	0.3	0.6	17	1.4	0.4	NS-WD	1.57	0.42	0.78 (0.05)	0.53 (0.16)	14.27	Hobbs et al. (2005)
LSS 3074	z020	30	0.3	0.6	17	1.4	1.2	NS-NS	1.40	1.53	14.59 (0.74)	0.87 (0.90)	314.01	Verbunt et al. (2017)
LSS 3074	z020	30	0.3	0.6	17	1.4	0.8	NS-NS	1.40	1.51	6.55 (0.00)	0.69 (0.00)	473.61	Richards et al. (2023)
LSS 3074	z020	25	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	Unbound (38.2%)			Hobbs et al. (2005)
LSS 3074	z020	25	0.4	0.6	18	1.4	0.2	NS-WD	1.83	0.76	1.62 (0.33)	0.40 (0.31)	91.40	Verbunt et al. (2017)
LSS 3074	z020	25	0.4	0.6	18	1.4	0.6	NS-NS	1.71	1.45	4.33 (0.00)	0.03 (0.00)	1382.52	Richards et al. (2023)
LSS 3074	z020	25	0.5	0.6	25	1.4	0.6	NS-NS	1.62	1.51	4.11 (0.03)	0.87 (0.08)	9.97	Hobbs et al. (2005)
LSS 3074	z020	25	0.5	0.6	25	1.4	0.2	NS-WD	1.79	1.07	Unbound (54.0%)			Verbunt et al. (2017)
LSS 3074	z020	25	0.5	0.6	25	1.4	1.0	NS-NS	1.47	1.46	Unbound (100.0%)			Richards et al. (2023)
LSS 3074	z020	24	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	13.07 (0.53)	0.88 (0.11)	181.22	Hobbs et al. (2005)
LSS 3074	z020	24	0.4	0.6	18	1.4	1.0	NS-NS	1.40	1.45	10.82 (0.62)	0.65 (0.44)	2632.05	Verbunt et al. (2017)
LSS 3074	z020	24	0.4	0.6	18	1.4	0.6	NS-NS	1.71	1.45	3.99 (0.00)	0.22 (0.00)	931.16	Richards et al. (2023)
LSS 3074	z020	24	0.5	0.6	24	1.4	0.6	NS-NS	1.58	1.47	5.52 (0.19)	0.49 (0.13)	1056.95	Hobbs et al. (2005)
LSS 3074	z020	24	0.5	0.6	24	1.4	0.2	NS-WD	1.79	1.04	1.46 (0.30)	0.39 (0.27)	54.78	Verbunt et al. (2017)
LSS 3074	z020	24	0.5	0.6	24	1.4	1.0	NS-NS	1.40	1.45	14.44 (0.00)	0.61 (0.00)	7860.26	Richards et al. (2023)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
MY Cam	z020	70	0.3	1.0	35	3.16	1.2	BH-NS	3.17	1.50	43.72 (0.78)	0.98 (0.66)	6.58	Hobbs et al. (2005)
MY Cam	z020	70	0.3	1.0	35	3.16	0.2	BH-WD	4.26	0.87	Unbound (50.2%)			Verbunt et al. (2017)
MY Cam	z020	70	0.3	1.0	35	3.16	1.4	BH-NS	3.16	1.56	Unbound (100.0%)			Richards et al. (2023)
MY Cam	z020	60	0.3	0.8	30	1.4	0.8	NS-NS	1.54	1.53	22.05 (0.73)	0.86 (0.18)	1127.07	Hobbs et al. (2005)
MY Cam	z020	60	0.3	0.8	30	1.4	1.6	NS-NS	1.40	1.51	Unbound (28.2%)			Verbunt et al. (2017)
MY Cam	z020	60	0.3	0.8	30	1.4	1.2	NS-NS	1.40	1.46	30.10 (0.00)	0.69 (0.00)	30142.89	Richards et al. (2023)
MY Cam	z020	60	0.4	0.8	40	3.16	1.0	BH-NS	3.42	1.52	27.08 (0.47)	0.92 (0.22)	167.28	Hobbs et al. (2005)
MY Cam	z020	60	0.4	0.8	40	3.16	0.2	BH-WD	4.21	0.90	Unbound (48.2%)			Verbunt et al. (2017)
MY Cam	z020	60	0.4	0.8	40	3.16	1.2	BH-NS	3.26	1.47	454.07 (0.00)	0.97 (0.00)	10326.72	Richards et al. (2023)
MY Cam	z020	60	0.3	1.0	30	3.16	1.6	BH-NS	3.17	1.51	Unbound (82.7%)			Hobbs et al. (2005)
MY Cam	z020	60	0.3	1.0	30	3.16	1.8	BH-NS	3.17	1.52	Unbound (39.9%)			Verbunt et al. (2017)
MY Cam	z020	60	0.3	1.0	30	3.16	1.4	BH-NS	3.17	1.50	68.63 (0.00)	0.52 (0.00)	434468.74	Richards et al. (2023)
MY Cam	z020	50	0.5	0.6	40	1.4	1.4	NS-NS	1.40	1.48	Unbound (73.7%)			Hobbs et al. (2005)
MY Cam	z020	50	0.5	0.6	40	1.4	0.2	NS-WD	1.73	1.26	0.94 (0.03)	0.53 (0.30)	8.31	Verbunt et al. (2017)

Binary Name		Z		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C			
MY Cam	z020	50	0.5	0.6	40	1.4	1.0	NS-NS	1.54	1.56		Unbound (100.0%)		Richards et al. (2023)		
MY Cam	z020	50	0.4	0.8	35	1.4	0.8	NS-NS	1.59	1.51	15.94 (0.34)	0.91 (0.06)	121.62	Hobbs et al. (2005)		
MY Cam	z020	50	0.4	0.8	35	1.4	1.6	NS-NS	1.40	1.46		Unbound (40.2%)		Verbunt et al. (2017)		
MY Cam	z020	50	0.4	0.8	35	1.4	1.2	NS-NS	1.40	1.56	70.36 (0.00)	0.33 (0.00)	1745452.95	Richards et al. (2023)		
MY Cam	z020	50	0.5	0.8	40	1.4	0.8	NS-NS	1.62	1.53		Unbound (50.0%)		Hobbs et al. (2005)		
MY Cam	z020	50	0.5	0.8	40	1.4	0.2	NS-WD	1.73	1.26	1.72 (0.51)	0.48 (0.32)	51.11	Verbunt et al. (2017)		
MY Cam	z020	50	0.5	0.8	40	1.4	1.2	NS-NS	1.47	1.56	60.65 (0.00)	0.62 (0.00)	306467.37	Richards et al. (2023)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.0	NS-NS	1.65	2.29		Unbound (76.0%)		Hobbs et al. (2005)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.8	NS-BH	1.40	3.15		Unbound (60.4%)		Verbunt et al. (2017)		
MY Cam	z020	50	0.6	0.8	50	1.4	1.4	NS-BH	1.47	2.98	115.83 (0.00)	0.59 (0.00)	1275227.75	Richards et al. (2023)		
MY Cam	z020	40	0.7	0.8	50	1.4	1.0	NS-NS	1.65	2.29		Unbound (51.4%)		Hobbs et al. (2005)		
MY Cam	z020	40	0.7	0.8	50	1.4	0.2	NS-WD	1.76	1.98	1.08 (0.03)	0.45 (0.20)	11.60	Verbunt et al. (2017)		
MY Cam	z020	40	0.7	0.8	50	1.4	1.4	NS-BH	1.47	2.98		Unbound (100.0%)		Richards et al. (2023)		
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91		Unbound (100.0%)		Hobbs et al. (2005)		

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	24	0.5	0.4	-	24.0	-	NS-BH	1.00	2.91	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	23	0.4	0.4	18	1.4	0.4	NS-NS	1.92	1.46	7.88 (1.00)	0.58 (0.35)	1411.28	Hobbs et al. (2005)
SMC 108086	z004	23	0.4	0.4	18	1.4	1.0	NS-NS	1.61	1.44	10.63 (0.57)	0.67 (0.28)	1931.17	Verbunt et al. (2017)
SMC 108086	z004	23	0.4	0.4	18	1.4	0.6	NS-NS	1.53	1.47	6.95 (0.00)	0.43 (0.00)	2571.06	Richards et al. (2023)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	23	0.5	0.4	-	23.0	-	NS-BH	1.00	2.77	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	22	0.4	0.4	17	1.4	0.4	NS-NS	2.17	1.44	5.83 (1.30)	0.78 (0.48)	96.60	Hobbs et al. (2005)
SMC 108086	z004	22	0.4	0.4	17	1.4	1.0	NS-NS	1.62	1.44	13.55 (0.91)	0.93 (1.02)	31.29	Verbunt et al. (2017)
SMC 108086	z004	22	0.4	0.4	17	1.4	0.6	NS-NS	1.54	1.45	4.79 (0.00)	0.96 (0.00)	0.42	Richards et al. (2023)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	22	0.5	0.4	-	22.0	-	NS-BH	1.00	2.56	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	21	0.4	0.4	16	1.4	0.4	NS-NS	1.61	1.47	6.98 (0.56)	0.33 (0.30)	3368.40	Hobbs et al. (2005)

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
SMC 108086	z004	21	0.4	0.4	16	1.4	1.0	NS-NS	1.70	1.46	11.09 (0.93)	0.84 (0.91)	256.61	Verbunt et al. (2017)
SMC 108086	z004	21	0.4	0.4	16	1.4	0.6	NS-NS	1.64	1.46	3.75 (0.00)	0.58 (0.00)	226.10	Richards et al. (2023)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Hobbs et al. (2005)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Verbunt et al. (2017)
SMC 108086*	z004	21	0.5	0.4	-	21.0	-	NS-NS	1.00	2.40	Unbound (100.0%)			Richards et al. (2023)
SMC 108086	z004	20	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	20.76 (0.90)	0.75 (0.73)	5803.72	Hobbs et al. (2005)
SMC 108086	z004	20	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	Unbound (50.4%)			Verbunt et al. (2017)
SMC 108086	z004	20	0.4	0.4	15	1.4	0.6	NS-NS	1.65	1.45	5.64 (0.00)	0.64 (0.00)	442.55	Richards et al. (2023)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.4	NS-NS	1.89	1.47	4.42 (0.18)	0.63 (0.10)	221.87	Hobbs et al. (2005)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.2	NS-WD	1.81	0.58	1.63 (0.33)	0.39 (0.38)	121.66	Verbunt et al. (2017)
SMC 108086	z004	20	0.5	0.4	20	1.4	0.8	NS-NS	1.59	1.48	5.19 (0.00)	0.94 (0.00)	1.88	Richards et al. (2023)
SMC 108086	z004	19	0.4	0.4	15	1.4	1.0	NS-NS	1.53	1.46	17.93 (0.88)	0.66 (0.71)	8651.22	Hobbs et al. (2005)
SMC 108086	z004	19	0.4	0.4	15	1.4	0.2	NS-WD	1.85	0.34	2.85 (0.51)	0.96 (0.62)	0.36	Verbunt et al. (2017)
SMC 108086	z004	19	0.4	0.4	15	1.4	0.6	NS-NS	1.65	1.45	7.03 (0.00)	0.33 (0.00)	3429.63	Richards et al. (2023)
TU Mus	z020	25	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	4.32 (1.46)	0.97 (0.44)	0.12	Hobbs et al. (2005)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
TU Mus	z020	25	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	0.87 (0.05)	0.41 (0.24)	15.25	Verbunt et al. (2017)
TU Mus	z020	25	0.3	0.4	16	1.4	0.6	NS-NS	1.72	1.44	4.60 (0.00)	0.16 (0.00)	1484.94	Richards et al. (2023)
TU Mus	z020	24	0.3	0.4	16	1.4	0.2	NS-WD	1.60	0.95	Unbound (89.9%)			Hobbs et al. (2005)
TU Mus	z020	24	0.3	0.4	16	1.4	0.8	NS-NS	1.40	1.44	10.93 (0.93)	0.90 (0.86)	60.41	Verbunt et al. (2017)
TU Mus	z020	24	0.3	0.4	16	1.4	0.6	NS-NS	1.72	1.44	2.73 (0.00)	0.63 (0.00)	66.09	Richards et al. (2023)
TU Mus	z020	23	0.3	0.4	15	1.4	0.8	NS-NS	1.41	1.48	8.44 (0.52)	0.45 (0.40)	4301.09	Hobbs et al. (2005)
TU Mus	z020	23	0.3	0.4	15	1.4	0.2	NS-WD	1.59	0.93	Unbound (43.6%)			Verbunt et al. (2017)
TU Mus	z020	23	0.3	0.4	15	1.4	0.6	NS-NS	1.81	1.45	3.45 (0.00)	0.12 (0.00)	685.72	Richards et al. (2023)
TU Mus	z020	16	0.6	0.4	21	1.4	0.6	NS-NS	1.64	1.50	2.76 (0.01)	0.88 (0.11)	2.73	Hobbs et al. (2005)
TU Mus	z020	16	0.6	0.4	21	1.4	1.6	NS-NS	1.40	1.50	Unbound (19.5%)			Verbunt et al. (2017)
TU Mus	z020	16	0.6	0.4	21	1.4	1.2	NS-NS	1.40	1.48	31.09 (0.00)	0.76 (0.00)	15597.63	Richards et al. (2023)
V382 Cyg	z020	50	0.2	0.6	20	1.4	0.8	NS-NS	1.40	1.44	Unbound (45.2%)			Hobbs et al. (2005)
V382 Cyg	z020	50	0.2	0.6	20	1.4	1.4	NS-NS	1.40	1.47	Unbound (51.9%)			Verbunt et al. (2017)
V382 Cyg	z020	50	0.2	0.6	20	1.4	1.0	NS-NS	1.40	1.55	12.17 (0.00)	0.77 (0.00)	1098.05	Richards et al. (2023)
V382 Cyg	z020	40	0.3	0.6	23	1.4	1.2	NS-NS	1.40	1.47	19.61 (0.60)	0.70 (0.35)	8667.25	Hobbs et al. (2005)

Binary Name	Z	Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
		M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
V382 Cyg	z020	40	0.3	0.6	23	1.4	0.2	NS-WD	1.80	0.99		Unbound (40.4%)		Verbunt et al. (2017)
V382 Cyg	z020	40	0.3	0.6	23	1.4	0.8	NS-NS	1.41	1.45	26.07 (0.00)	0.58 (0.00)	45411.40	Richards et al. (2023)
V382 Cyg	z020	40	0.4	0.6	30	1.4	0.6	NS-WD	1.66	0.28		Unbound (97.3%)		Hobbs et al. (2005)
V382 Cyg	z020	40	0.4	0.6	30	1.4	1.4	NS-NS	1.40	1.51		Unbound (49.9%)		Verbunt et al. (2017)
V382 Cyg	z020	40	0.4	0.6	30	1.4	1.0	NS-NS	1.40	1.45	18.06 (0.00)	0.75 (0.00)	4347.70	Richards et al. (2023)
V382 Cyg	z020	35	0.4	0.6	25	1.4	1.2	NS-NS	1.46	1.45		Unbound (32.5%)		Hobbs et al. (2005)
V382 Cyg	z020	35	0.4	0.6	25	1.4	0.2	NS-WD	1.79	1.07		Unbound (44.4%)		Verbunt et al. (2017)
V382 Cyg	z020	35	0.4	0.6	25	1.4	0.8	NS-NS	1.47	1.49	13.49 (0.00)	0.67 (0.00)	3944.11	Richards et al. (2023)
V382 Cyg	z020	30	0.4	0.6	23	1.4	1.2	NS-NS	1.40	1.47		Unbound (38.9%)		Hobbs et al. (2005)
V382 Cyg	z020	30	0.4	0.6	23	1.4	0.2	NS-WD	1.80	0.99	1.56 (0.24)	0.30 (0.35)	86.59	Verbunt et al. (2017)
V382 Cyg	z020	30	0.4	0.6	23	1.4	0.8	NS-NS	1.41	1.45	19.92 (0.00)	0.50 (0.00)	34461.76	Richards et al. (2023)
V382 Cyg	z020	30	0.5	0.6	30	1.4	0.6	NS-WD	1.66	0.28		Unbound (98.6%)		Hobbs et al. (2005)

Binary		Primary Model			Secondary Model			Remnants			Kick Outcome			Kick Model
Name	Z	M_1	q	$\log P$	M_2	M_{rem}	$\log P$	Type	M_1	M_2	a_0	e_0	T_C	
V382 Cyg	z020	30	0.5	0.6	30	1.4	1.4	NS-NS	1.40	1.51	Unbound (48.3%)			Verbunt et al. (2017)
V382 Cyg	z020	30	0.5	0.6	30	1.4	1.0	NS-NS	1.40	1.45	28.16 (0.00)	0.99 (0.00)	0.88	Richards et al. (2023)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.2	BH-BH	3.33	3.52	Unbound (73.3%)			Hobbs et al. (2005)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.4	BH-BH	3.32	3.92	Unbound (55.4%)			Verbunt et al. (2017)
VFTS 352	z010	40	0.5	0.6	40	3.16	1.0	BH-BH	3.35	3.18	29.62 (0.00)	0.90 (0.00)	240.49	Richards et al. (2023)

Table E.1: The remnants generated by each model from the BPASS release. Post-secondary supernova properties are generated using the mentioned kick, simulating 1,000 kicks and taking the mean post-secondary parameters. Masses are measured in $M_\odot$, separations in $R_\odot$, and coalescence times in Gyr. Asterisks next to binary names denote candidates for a star that undergoes quasi-chemically homogenous evolution in the secondary. In this table, a white dwarf is left if the CO core mass of the star is less than $1.38 M_\odot$, the total mass of the star is less than $1.5 M_\odot$, or if core carbon burning does not occur. Otherwise if all three of those conditions are met, the mass boundary demarcating black holes from neutron stars is placed at $2.5 M_\odot$. Quoted errors on a_0 and e_0 , shown in parentheses underneath the nominal value, are the 68 per cent confidence intervals, and systems flagged as “Unbound” have the percentage of simulated kick events in which they are unbound displayed in parentheses, though for these systems the mean eccentricity falls outside the range of $[0, 1)$.



APPENDIX F

Code Samples



Any fool can write code that a computer can understand. Good programmers write code that humans can understand.

— *Martin Fowler* —



Code Sample F.1: The TAKAHE function `integrate_eoms()`

```

1 def integrate_eoms(a0, e0, p):
2     """
3     General purpose integrator for Nyadzani & Razzaque eqns for a & e.
4
5     Params:
6         a0 - The initial semimajor axis, measured in solar radii
7         e0 - The initial eccentricity, dimensionless
8         p - A vector of parameters:
9             p[1] = m1 (units: Solar Mass)
10            p[2] = m2 (units: Solar Mass)
11            p[3] = unused (will be removed in next version)
12            p[4] = unused (will be removed in next version)
13            p[5] = Lifetime (evolution + rejuvenation)
14
15     Returns:
16         A - An array of the semimajor axes of the binary system over time. (Solar
17            ↪ Mass)
18         E - An array of the eccentricities of the binary system over time. (no
19            ↪ dim.)
20     """
21
22     Solar_Mass = 1.989e30      # kg
23     Solar_Radius = 696340000.0 # m
24     G = 6.67e-11              # m^3 kg^-1 s^-2
25     c = 299792458.0           # m/s
26
27     A = np.array([])
28     E = np.array([])
29     H = np.array([])
30
31     m1, m2, _, _, evotime = p[0], p[1], p[2], p[3], p[4]
32
33     # number of seconds in a year
34     seconds_per_year = 60 * 60 * 24 * 365.25
35
36     #####
37     #      Unit Check      #
38     #####
39     a = a0 * Solar_Radius # Meters
40     e = e0                 # Dimensionless
41     m1 = m1 * Solar_Mass  # Kilogram
42     m2 = m2 * Solar_Mass  # Kilogram
43     #####
44     #      End Unit Check      #
45     #####
46
47     # Beta has units m^4 / s
48     beta = ((64/5) * G**3 * m1 * m2 * (m1 + m2) / (c**5))
49
50     A = np.append(A, a)
51     E = np.append(E, e)
52     H = np.append(H, 0.0)
53
54     total_time = 0
55
56     # Integrate until past the end of the universe, or a 10km orbit
57     while total_time/seconds_per_year + evotime < 1e11 and a > 1e4:
58         # an euler integrator: work out da/dt then times it by dt
59         # to get da, which then we can work out as a = a + da/dt * dt.
60         initial_da = (- beta / ((a**3) * (1 - e**2)**(7/2)))

```

```

59     da = initial_da * (1 + (73/24) * e**2 + (37/96) * e**4)
60
61     intial_de = (((-19/12) * beta) / (a**4*(1-e**2)**(5/2)))
62     de = intial_de * (e + (121/304) * e**3)
63     # Units: s^-1
64
65     timeA = abs(1e-2 * a/da)
66
67     if e > 1e-10:
68         timeE = abs(1e-2 * e/de)
69     else:
70         de = 0
71         e = 1e-10
72         timeE = timeA * 10
73
74     # maximum timestep is half of the width of the smallest BPASS time bin
75     conv_frac = 0.23076752*0.5*seconds_per_year
76
77     dt2 = (evotime + total_time/seconds_per_year)*conv_frac
78
79     # Take a timestep that results in the smallest change: either a
80     # change in E, a change in A, 1/2 the smallest BPASS bin.
81     dt = min(timeE, timeA, dt2)
82
83     a = a + dt * da
84     e = e + dt * de
85
86     A = np.append(A, a)
87     E = np.append(E, e)
88     H = np.append(H, dt)
89
90     total_time = total_time + dt
91
92     # Determine why we stopped, so the parent function can tell what went
93     # "wrong".
94     stop_reason = "flag_not_set"
95
96     if total_time/seconds_per_year + evotime >= 1e11:
97         stop_reason = "out_of_time"
98
99     if a <= 1e4:
100         stop_reason = "merged"
101
102     # Solar Radii, Dimensionless
103     return A / Solar_Radius, E, H, stop_reason

```

Code Sample F.2: The TAKAHE function integrate_timescale()

```

1 def integrate_timescale(m1, m2, p0, e0):
2     """
3     General purpose integrator for eqn(5.14) of Peters, 1964 [1].
4
5     [1] https://ui.adsabs.harvard.edu/abs/1964PhRv..136.1224P/abstract
6
7     Params:
8     m1 -- The mass of the first star (Solar Mass)
9     m2 -- The mass of the second star (Solar Mass)
10    p0 -- The initial period (days)
11    e0 -- The initial eccentricity
12

```

```

13 Returns:
14 tC -- The coalescence time (seconds)
15 """
16 Solar_Mass = 1.989e30      # kg
17 Solar_Radius = 696340000.0 # m
18 G = 6.67e-11              # m3 kg-1 s-2
19 c = 299792458.0           # m/s
20
21 m1 = float(m1)
22 m2 = float(m2)
23 p0 = float(p0)
24 e0 = float(e0)
25
26 #####
27 # BEGIN UNIT CHECK #
28 #####
29 m1 = m1 * Solar_Mass      # Kilograms ##
30 m2 = m2 * Solar_Mass      # Kilograms ##
31 p0 = p0 * 24 * 60 * 60    # Seconds ##
32 e0 = e0                    # Dimensionless ##
33 #####
34 # END UNIT CHECK #
35 #####
36
37 # Use Kepler's Third Law to compute the semimajor axis, in meters
38 a0 = (p0**2.0 * (G * (m1+m2))) / (4.0 * np.pi**2.0)**(1.0/3.0) # meters
39
40 # compute the constant c0
41 c0 = a0 * (1.0-e0**2.0) * e0**(-12.0/19.0) * (1.0+(121.0 / 304.0 *
42 ↪ e0**2.0))**(-870.0/2299.0) # meters
43
44 # Compute the constant beta
45 beta = 64.0 / 5.0 * G**3.0 * m1 * m2 * (m1 + m2) / c**5.0
46 # meters4 / s
47
48 # Circular binary coalescence time
49 tC = a0**4.0 / (4.0*beta)
50
51 if e0 != 0:
52     if e0 < 0.01:
53         # Low ecc - see eqn after eqn(5.14) of Peters, 1964
54         tC = c0**4.0 * e0**(48.0/19.0) / (4.0*beta)
55     elif e0 > 0.99:
56         # High ecc - see eqn after eqn after eqn(5.14) of Peters, 1964.
57         tC = tC * ((768.0 / 425.0) * (1.0-e0**2.0)**3.5)
58     else:
59         # Medium ecc - see eqn(5.14) of Peters, 1964
60         e = 0.0
61         de = e0 / 10000.0
62         summand = 0.0
63
64         while e < e0:
65             this_integral = de * e**(29.0/19.0)
66             this_integral *= (1.0 + (121.0/304.0) * e**2.0)**(1181.0/2299.0)
67             this_integral /= (1.0 - e**2.0)**(1.5)
68
69             summand += this_integral
70             e += de
71
72         e = float(e)

```

```
73         tC = (12.0/19.0) * (c0**4.0 / beta) * summand
74
75     return tC
```


Bibliography

- Abbott B. P., et al., 2016, Phys. Rev. Lett., 116, 061102
- Abdul-Masih M., 2023, Observations of high-mass contact binaries in the local Universe, European Astronomical Society Annual Meeting
- Abdul-Masih M., et al., 2019, ApJ, 880, 115
- Abdul-Masih M., Escorza A., Menon A., Mahy L., Marchant P., 2022, A&A, 666, A18
- Agazie G., et al., 2023, ApJ, 951, L8
- Alighieri D., ca. 1317, Paradiso
- Almeida L. A., et al., 2015, ApJ, 812, 102
- Amaro-Seoane P., et al., 2012, Class. Quan. Grav., 29, 124016
- Andersen J., Grønbech B., 1975, A&A, 45, 107
- Angelou G. C., Church R. P., Stancliffe R. J., Lattanzio J. C., Smith G. H., 2011, ApJ, 728, 79
- Antoniadis J., et al., 2023, A&A, 678, A50
- Arakaki T., et al., 2023, JuliaPy/pyjulia: v0.6.2, doi:10.5281/zenodo.10443741, <https://doi.org/10.5281/zenodo.10443741>
- Arcavi I., et al., 2017, Nature, 551, 64
- Arnett W. D., Bahcall J. N., Kirshner R. P., Woosley S. E., 1989, ARA&A, 27, 629
- Arzoumanian Z., Chernoff D. F., Cordes J. M., 2002, ApJ, 568, 289
- Asplund M., Amarsi A. M., Grevesse N., 2021, A&A, 653, A141
- Baade W., Zwicky F., 1934, Contrib. Mt. Wilson Obs., 3, 73
- Bahcall J. N., Pinsonneault M. H., 1992, Rev. Mod. Phys., 64, 885
- Bahcall J. N., Pinsonneault M. H., Wasserburg G. J., 1995, Rev. Mod. Phys., 67, 781

- Baker J. G., Boggs W. D., Centrella J., Kelly B. J., McWilliams S. T., Miller M. C., van Meter J. R., 2008, ApJ, 682, L29
- Ballesteros F. J., 2012, Europhys. Lett., 97, 34008
- Bars I., Terning J., 2010, Extra Dimensions in Space and Time, doi:10.1007/978-0-387-77638-5. , <https://ui.adsabs.harvard.edu/abs/2010edst.book.....B>
- Barstow M. A., Bond H. E., Holberg J. B., Burleigh M. R., Hubeny I., Koester D., 2005, MNRAS, 362, 1134
- Bavera S. S., et al., 2020, A&A, 635, A97
- Belczynski K., Kalogera V., Rasio F. A., Taam R. E., Zezas A., Bulik T., Maccarone T. J., Ivanova N., 2008, ApJS, 174, 223
- Belczynski K., et al., 2020, A&A, 636, A104
- Bestenlehner J. M., et al., 2020, MNRAS, 499, 1918
- Bezanson J., Edelman A., Karpinski S., Shah V. B., 2017, SIAM J. Appl. Math., 59, 65
- Bhattacharjee P., Chaudhury S., Kundu S., 2014, ApJ, 785, 63
- Blaauw A., 1961, Bull. Astron. Inst. Netherlands, 15, 265
- Blanchet L., 2014, Liv. Rev. Rel., 17, 2
- Blöcker T., 1995, A&A, 297, 727
- Boffin H. M. J., Jorissen A., 1988, A&A, 205, 155
- Bombaci I., Popov S. B., 2004, A&A, 424, 627
- Bondi H., Hoyle F., 1944, MNRAS, 104, 273
- Bouffanais Y., Mapelli M., Santoliquido F., Giacobbo N., Di Carlo U. N., Rastello S., Artale M. C., Iorio G., 2021, MNRAS, 507, 5224
- Bouret J. C., Lanz T., Martins F., Marcolino W. L. F., Hillier D. J., Depagne E., Hubeny I., 2013, A&A, 555, A1
- Bray J. C., Eldridge J. J., 2016, MNRAS, 461, 3747
- Bray J. C., Eldridge J. J., 2018, MNRAS, 480, 5657
- Broekgaarden F. S., et al., 2021, MNRAS, 508, 5028
- Buck P. H., 1938, Vikings of the sunrise. New Zealand Electronic Text Centre, Wellington, N.Z
- Burbidge E. M., Burbidge G. R., Fowler W. A., Hoyle F., 1957, Rev. Mod. Phys., 29, 547

- Burdge K. B., et al., 2020, *ApJ*, 905, L7
- Burns J. A., 1978, *Vistas Astron.*, 22, 193
- Böhm-Vitense E., 1958, *Z. Astrophys.*, 46, 108
- Cameron A. D., et al., 2018, *MNRAS*, 475, L57
- Cantiello M., Yoon S. C., Langer N., Livio M., 2007, *A&A*, 465, L29
- Carroll B. W., Ostlie D. A., 2017, *An Introduction to Modern Astrophysics*, 2 edn. Cambridge University Press, doi:10.1017/9781108380980
- Caughlan G. R., Fowler W. A., 1988, *At. Data Nuc. Data Tab.*, 40, 283
- Champion D. J., Lorimer D. R., McLaughlin M. A., Cordes J. M., Arzoumanian Z., Weisberg J. M., Taylor J. H., 2004, *MNRAS*, 350, L61
- Chandrasekhar S., 1935, *MNRAS*, 95, 207
- Charbonnel C., Zahn J.-P., 2007, *A&A*, 467, L15
- Christensen N., 1992, *Phys. Rev. D*, 46, 5250
- Christensen N., 2018, *Rep. Prog. Phys.*, 82, 016903
- Claeys J. S. W., Pols O. R., Izzard R. G., Vink J., Verbunt F. W. M., 2014, *A&A*, 563, A83
- Coppersmith D., Winograd S., 1987, in *Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing. STOC '87*. Association for Computing Machinery, New York, NY, USA, pp 1–6, doi:10.1145/28395.28396, <https://doi.org/10.1145/28395.28396>
- Corbett G., Webb H., 2016, *Vertical Readings in Dante's Comedy: Volume 2*. JSTOR Open Access monographs, Open Book Publishers, <https://books.google.co.nz/books?id=5hmuDQAAQBAJ>
- Corongiu A., Kramer M., Stappers B. W., Lyne A. G., Jessner A., Possenti A., D'Amico N., Löhmer O., 2007, *A&A*, 462, 703
- Cowley C. R., Bidelman W. P., Hubrig S., Mathys G., Bord D. J., 2004, *A&A*, 419, 1087
- Cox J. P., Giuli R. T., 1968, *Principles of stellar structure*. <https://ui.adsabs.harvard.edu/abs/1968pss.book.....C>
- Crowther P. A., Castro N., Evans C., Vink J., Melnick J., Selman F., 2018, Technical report, *Dissecting the Core of the Tarantula Nebula with MUSE*, <https://ui.adsabs.harvard.edu/abs/2018arXiv180100855C>. <https://ui.adsabs.harvard.edu/abs/2018arXiv180100855C>
- Davis P. J., Siess L., Deschamps R., 2013, *A&A*, 556, A4
- Delgado A. J., Thomas H. C., 1981, *A&A*, 96, 142

- Dominik M., Belczynski K., Fryer C., Holz D. E., Berti E., Bulik T., Mandel I., O’Shaughnessy R., 2012, *ApJ*, 759, 52
- Drout M. R., Götberg Y., Ludwig B. A., Groh J. H., de Mink S. E., O’Grady A. J. G., Smith N., 2023, *Science*, 382, 1287
- Dutta D., Klencki J., 2023, On the evolutionary nature of puffed-up stripped star binaries and their occurrence in stellar populations ([arXiv:2312.12658](https://arxiv.org/abs/2312.12658)), doi:10.48550/arXiv.2312.12658
- Eddington A. S., 1916, *MNRAS*, 77, 16
- Eddington A. S., 1920, *Observatory*, 43, 341
- Eddy J. A., Ise R., 1979, A new sun : the solar results from SKYLAB. National Aeronautics and Space Administration, <https://ui.adsabs.harvard.edu/abs/1979nssr.book.....E>
- Eggenberger P., et al., 2010, *A&A*, 519, A116
- Eggleton P. P., 1967, *MNRAS*, 135, 243
- Eggleton P. P., 1971, *MNRAS*, 151, 351
- Eggleton P. P., 1972, *MNRAS*, 156, 361
- Eggleton P. P., 1983, *ApJ*, 268, 368
- Eggleton P. P., Faulkner J., Flannery B. P., 1973, *A&A*, 23, 325
- Eichler D., Livio M., Piran T., Schramm D. N., 1989, *Nature*, 340, 126
- Einstein A., 1905, *Ann. Physik*, 322, 891
- Einstein A., 1916, *Sitzungsber. Kon. Pre. Akad. Wiss.*, pp 688–696
- Einstein A., 1917, *Sitzungsber. Kon. Pre. Akad. Wiss.*, pp 142–152
- Ekström S., et al., 2012, *A&A*, 537, A146
- El-Badry K., Conroy C., Fuller J., Kiman R., van Roestel J., Rodriguez A. C., Burdge K. B., 2022, *MNRAS*, 517, 4916
- Eldridge J. J., Maund J. R., 2016, *MNRAS*, 461, L117
- Eldridge J. J., Stanway E. R., 2022, *ARA&A*, 60, 455
- Eldridge J. J., Tout C. A., 2004a, *MNRAS*, 348, 201
- Eldridge J. J., Tout C. A., 2004b, *MNRAS*, 353, 87
- Eldridge J. J., Tout C. A., 2018, *The Structure And Evolution Of Stars*. World Scientific

- Eldridge J. J., Izzard R. G., Tout C. A., 2008, MNRAS, 384, 1109
- Eldridge J. J., Langer N., Tout C. A., 2011, MNRAS, 414, 3501
- Eldridge J. J., Fraser M., Smartt S. J., Maund J. R., Crockett R. M., 2013, MNRAS, 436, 774
- Eldridge J. J., Stanway E. R., Xiao L., McClelland L. a. S., Taylor G., Ng M., Greis S. M. L., Bray J. C., 2017, Publ. Astron. Soc. Australia, 34, e058
- Eldridge J. J., Stanway E. R., Tang P. N., 2019, MNRAS, 482, 870
- Evans C. J., et al., 2011, A&A, 530, A108
- Fang C., 2015, Highl. Astron., 16, 19
- Faulkner A. J., et al., 2005, ApJ, 618, L119
- Fawzi A., et al., 2022, Nature, 610, 47
- Fonseca E., Stairs I. H., Thorsett S. E., 2014, ApJ, 787, 82
- Fragione G., 2021, ApJ, 923, L2
- Fragos T., et al., 2023, ApJS, 264, 45
- Fraser A. E., Joyce M., Anders E. H., Tayar J., Cantiello M., 2022, ApJ, 941, 164
- Fratanduono D. E., et al., 2020, Phys. Rev. Lett., 124, 015701
- Fryer C., Burrows A., Benz W., 1998, ApJ, 496, 333
- Fuller R. S., Hamacher D. W., 2017, J. Astron. Hist. Herit., 20, 349
- Fuller J., Lu W., 2022, MNRAS, 511, 3951
- Gamow G., 1928, Nature, 122, 805
- Gazeas K., et al., 2021, MNRAS, 501, 2897
- Ghodla S., Eldridge J. J., Stanway E. R., Stevance H. F., 2023, MNRAS, 518, 860
- Giacobbo N., Mapelli M., 2020, ApJ, 891, 141
- Goodwin S. P., Gribbin J., Hendry M. A., 1998, Observatory, 118, 201
- Götberg Y., de Mink S. E., Groh J. H., 2017, A&A, 608, A11
- Green M. J., Maoz D., Mazeh T., Faigler S., Shahaf S., Gomel R., El-Badry K., Rix H.-W., 2023, MNRAS, 522, 29
- Grevesse N., Noels A., 1993, Cosmic abundances of the elements.. <https://ui.adsabs.harvard.edu/abs/1993oe...conf...15G>

- Groh J. H., Meynet G., Ekström S., Georgy C., 2014, *A&A*, 564, A30
- Gräfener G., Koesterke L., Hamann W. R., 2002, *A&A*, 387, 244
- Gunn J. E., Ostriker J. P., 1970, *ApJ*, 160, 979
- Götberg Y., de Mink S. E., Groh J. H., Kupfer T., Crowther P. A., Zapartas E., Renzo M., 2018, *A&A*, 615, A78
- Götberg Y., et al., 2023, *The Astrophysical Journal*, 959, 125
- Hamacher D., 2022, *The First Astronomers: How Indigenous Elders read the stars*. Allen & Unwin, Crows Nest, New South Wales
- Hamacher D. W., Barsa J., Passi S., Tapim A., 2019, Indigenous use of stellar scintillation to predict weather and seasonal change, doi:10.48550/arXiv.1903.01060, <https://ui.adsabs.harvard.edu/abs/2019arXiv190301060H>
- Hamann W. R., Gräfener G., 2003, *A&A*, 410, 993
- Harris C. R., et al., 2020, *Nature*, 585, 357
- Hartmann D. H., 1999, *Proc. Nat. Acad. Sci*, 96, 4752
- Hawkins G. S., 1973, *Beyond Stonehenge*. Hutchinson
- Hayashi C., 1961, *PASJ*, 13, 450
- Heney L. G., Lelevier R., Levee R. D., 1955, *PASP*, 67, 154
- Heney L. G., Wilets L., Böhm K. H., Lelevier R., Levee R. D., 1959, *ApJ*, 129, 628
- Heney L. G., Forbes J. E., Gould N. L., 1964, *ApJ*, 139, 306
- Herrero A., Kudritzki R. P., Vilchez J. M., Kunze D., Butler K., Haser S., 1992, *A&A*, 261, 209
- Hilditch R. W., Howarth I. D., Harries T. J., 2005, *MNRAS*, 357, 304
- Hirai R., Podsiadlowski P., Yamada S., 2018, *ApJ*, 864, 119
- Hobbs G., Lorimer D. R., Lyne A. G., Kramer M., 2005, *MNRAS*, 360, 974
- Hoffleit D., 1964, *Catalogue of Bright Stars*
- Hoyle F., 1946, *MNRAS*, 106, 343
- Hoyle F., 1960, *MNRAS*, 120, 22
- Hulse R. A., Taylor J. H., 1975, *ApJ*, 195, L51
- Hunter J. D., 2007, *Comput. Sci. Eng.*, 9, 90

- Hunter I., et al., 2008, *ApJ*, 676, L29
- Hurley J. R., Pols O. R., Tout C. A., 2000, *MNRAS*, 315, 543
- Hurley J. R., Tout C. A., Pols O. R., 2002, *MNRAS*, 329, 897
- Igoshev A. P., Chruslinska M., Dorozsmai A., Toonen S., 2021, *MNRAS*, 508, 3345
- Itoh Y., Futamase T., 2003, *Phys. Rev. D*, 68, 121501
- Ivezić u., et al., 2019, *ApJ*, 873, 111
- Janssen G. H., Stappers B. W., Kramer M., Nice D. J., Jessner A., Cognard I., Purver M. B., 2008, *A&A*, 490, 753
- Jermyn A. S., et al., 2023, *ApJS*, 265, 15
- Kapil V., Mandel I., Berti E., Müller B., 2023, *MNRAS*, 519, 5893
- Kasen D., Metzger B., Barnes J., Quataert E., Ramirez-Ruiz E., 2017, *Nature*, 551, 80
- Keith M. J., Kramer M., Lyne A. G., Eatough R. P., Stairs I. H., Possenti A., Camilo F., Manchester R. N., 2009, *MNRAS*, 393, 623
- Kilic M., Allende Prieto C., Brown W. R., Koester D., 2007, *ApJ*, 660, 1451
- Kippenhahn R., Meyer-Hofmeister E., 1977, *A&A*, 54, 539
- Kippenhahn R., Ruschenplatt G., Thomas H. C., 1980, *A&A*, 91, 175
- Klencki J., Nelemans G., Istrate A. G., Chruslinska M., 2021, *A&A*, 645, A54
- Knipe G. F. G., 1971, *Rep. Obs. Johannesburg Circ.*, 8, 8
- Kobayashi C., et al., 2023, *ApJ*, 943, L12
- Koen C., 2022, *MNRAS*, 510, 1857
- Kolb U., Ritter H., 1990, *A&A*, 236, 385
- Kosok P., Reiche M., 1949, *Archaeology*, 2, 206
- Kramer M., et al., 2006, *Science*, 314, 97
- Kroupa P., 2001, *MNRAS*, 322, 231
- Kruckow M. U., Tauris T. M., Langer N., Kramer M., Izzard R. G., 2018, *MNRAS*, 481, 1908
- Kuiper G. P., 1941, *ApJ*, 93, 133
- LIGO Scientific Collaboration and Virgo Collaboration 2016, *Phys. Rev. Lett.*, 116, 061102
- LIGO-Virgo 2021, *ApJ*, 913, L7

- LIGO-Virgo-KAGRA 2021, arXiv e-prints, p. arXiv:2111.03634
- Lam S. K., Pitrou A., Seibert S., 2015, in Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC. LLVM '15. Association for Computing Machinery, New York, NY, USA, pp 1–6, doi:10.1145/2833157.2833162, <https://dl.acm.org/doi/10.1145/2833157.2833162>
- Lamers H. J. G. L. M., Snow T. P., Lindholm D. M., 1995, ApJ, 455, 269
- Landri C., Ricker P. M., Renzo M., Rau S., Vigna-Gómez A., 2024, arXiv e-prints, p. arXiv:2407.15932
- Langer N., 2012, ARA&A, 50, 107
- Langer N., Norman C. A., 2006, ApJ, 638, L63
- Laplace E., Justham S., Renzo M., Göteborg Y., Farmer R., Vartanyan D., de Mink S. E., 2021, A&A, 656, A58
- Latković O., Čeki A., Lazarević S., 2021, The Astrophysical Journal Supplement Series, 254, 10
- Lattimer J. M., Prakash M., 2001, ApJ, 550, 426
- Lau R. M., et al., 2022, Nature Astronomy, 6, 1308
- Lau M. Y. M., Hirai R., Mandel I., Tout C. A., 2024, ApJ, 966, L7
- Law-Smith J. A. P., et al., 2020, Successful Common Envelope Ejection and Binary Neutron Star Formation in 3D Hydrodynamics, doi:10.48550/arXiv.2011.06630, <https://ui.adsabs.harvard.edu/abs/2020arXiv201106630L>
- Lazarus P., et al., 2016, ApJ, 831, 150
- Leonard P. J. T., Hills J. G., Dewey R. J., 1994, ApJ, 423, L19
- Leung K. C., 1989, in Batten A. H., ed., Algols. Springer Netherlands, Dordrecht, pp 279–288, doi:10.1007/978-94-009-2413-0_24
- Levesque E. M., Massey P., Żytkow A. N., Morrell N., 2014, MNRAS: Letters, 443, L94
- Li T., Li X.-D., 2013, ApJ, 780, 47
- Linial I., Sari R., 2017, MNRAS, 469, 2441
- Lorenzo J., Negueruela I., Val Baker A. K. F., García M., Simón-Díaz S., Pastor P., Méndez Majuelos M., 2014, A&A, 572, A110
- Lorimer D. R., 2008, Liv. Rev. Rel., 11, 8
- Madau P., Dickinson M., 2014, ARA&A, 52, 415
- Maeder A., 1990, A&AS, 84, 139

- Mandel I., 2021, RNAAS, 5, 223
- Mandel I., Broekgaarden F. S., 2022, Liv. Rev. Rel., 25, 1
- Mandel I., Müller B., 2020, MNRAS, 499, 3214
- Mapelli M., Giacobbo N., 2018, MNRAS, 479, 4391
- Marchant P., Langer N., Podsiadlowski P., Tauris T. M., Moriya T. J., 2016, A&A, 588, A50
- Markova N., Puls J., Langer N., 2018, A&A, 613, A12
- Marov M. Y., 2015, The Fundamentals of Modern Astrophysics, doi:10.1007/978-1-4614-8730-2, <https://ui.adsabs.harvard.edu/abs/2015fma..book.....M>
- Martinez J. G., et al., 2015, ApJ, 812, 143
- Martinez J. G., et al., 2017, ApJ, 851, L29
- Martins F., Mahy L., Hervé A., 2017, A&A, 607, A82
- Mason B. D., Wycoff G. L., Hartkopf W. I., Douglass G. G., Worley C. E., 2001, AJ, 122, 3466
- Massey P., Zangari A. M., Morrell N. I., Puls J., DeGioia-Eastwood K., Bresolin F., Kudritzki R.-P., 2009, ApJ, 692, 618
- Massey P., Neugent K. F., Hillier D. J., Puls J., 2013, ApJ, 768, 6
- Massey P., Neugent K. F., Morrell N. I., 2023, ApJ, 947, 77
- McEvoy C. M., et al., 2015, A&A, 575, A70
- McKee C. F., Ostriker E. C., 2007, ARA&A, 45, 565
- McKinney W., 2010, in Walt S. v. d., Millman J., eds, Proceedings of the 9th Python in Science Conference. pp 56 – 61, doi:10.25080/Majora-92bf1922-00a
- Mennekens N., Vanbeveren D., 2014, A&A, 564, A134
- Mennekens N., Vanbeveren D., 2016, A&A, 589, A64
- Menon A., Utrobin V., Heger A., 2019, MNRAS, 482, 438
- Menon A., et al., 2021a, MNRAS, 507, 5013
- Menon A., et al., 2021b, MNRAS, 507, 5013
- Menon A., et al., 2024, ApJ, 963, L42
- Miszuda A., Szewczuk W., Daszyńska-Daszkiewicz J., 2021, MNRAS, 505, 3206
- Mochnacki S. W., Doughty N. A., 1972, Monthly Notices of the Royal Astronomical Society, 156, 51

- Moe M., Stefano R. D., 2017, *ApJS*, 230, 15
- Moore C. J., Cole R. H., Berry C. P. L., 2014, *Class. Quan. Grav.*, 32, 015014
- Morgenroth O., 1935, *Astron. Nachr*, 255, 425
- Morrell N. I., Niemela V. S., 1990, in Garmany C. D., ed., *Astronomical Society of the Pacific Conference Series Vol. 7, Properties of Hot Luminous Stars*. pp 57–59
- Nakar E., 2007, *Phys. Rep.*, 442, 166
- Nandez J. L. A., Ivanova N., Lombardi Jr. J. C., 2014, *ApJ*, 786, 39
- Negueruela I., Marco A., 2003, *A&A*, 406, 119
- Neo S., Miyaji S., Nomoto K., Sugimoto D., 1977, *PASJ*, 29, 249
- Noon K., de Napoli K., 2022, *First Knowledges Astronomy: Sky Country*. Thames & Hudson
- Nugis T., Lamers H. J. G. L. M., 2000, *A&A*, 360, 227
- Oosterhoff P. T., 1928, *Bull. Astron. Inst. Netherlands*, 4, 183
- Oosterhoff P. T., 1930, *Bull. Astron. Inst. Netherlands*, 5, 156
- Packet W., 1981, *A&A*, 102, 17
- Pauldrach A. W. A., Puls J., 1990, *A&A*, 237, 409
- Paxton B., Bildsten L., Dotter A., Herwig F., Lesaffre P., Timmes F., 2011, *ApJS*, 192, 3
- Paxton B., et al., 2013, *ApJS*, 208, 4
- Paxton B., et al., 2015, *ApJS*, 220, 15
- Paxton B., et al., 2018, *ApJS*, 234, 34
- Paxton B., et al., 2019, *ApJS*, 243, 10
- Payne C. H., 1925, PhD thesis, Harvard, <https://ui.adsabs.harvard.edu/abs/1925PhDT.....1P>
- Pearce J. A., 1952, *PASP*, 64, 219
- Penny L. R., Ouzts C., Gies D. R., 2008, *ApJ*, 681, 554
- Peters P. C., 1964, *Phys. Rev. Lett.*, 136, 1224
- Petrosky E., Hwang H.-C., Zakamska N. L., Chandra V., Hill M. J., 2021, *MNRAS*, 503, 3975
- Petrov A., 1946, *PZ*, 6, 72
- Pian E., et al., 2017, *Nature*, 551, 67

- Podsiadlowski P., 2017, in Alsabti A. W., Murdin P., eds, , Handbook of Supernovae. p. 635, doi:10.1007/978-3-319-21846-5_123
- Podsiadlowski P., Cannon R. C., Rees M. J., 1995, MNRAS, 274, 485
- Poincaré M. H., 1906, Rend. Circ. Mat. Palermo, 21, 129
- Pols O. R., 1994, A&A, 290, 119
- Pols O. R., Tout C. A., Eggleton P. P., Han Z., 1995, MNRAS, 274, 964
- Pols O. R., Tout C. A., Lattanzio J. C., Karakas A. I., 2001, in Podsiadlowski P., Rappaport S., King A. R., D'Antona F., Burderi L., eds, Astronomical Society of the Pacific Conference Series Vol. 229, Evolution of Binary and Multiple Star Systems. p. 31
- Price D. J., et al., 2018, Publ. Astron. Soc. Australia, 35, e031
- Ptolemy C., Proclus ca. 168, Tetrabiblos
- Qian S.-B., Zhu L.-Y., Liu L., Zhang X.-D., Shi X.-D., He J.-J., Zhang J., 2020, Res. Astron. Astrophys., 20, 163
- Qin Y., Fragos T., Meynet G., Andrews J., Sørensen M., Song H. F., 2018, A&A, 616, A28
- Ramachandran V., Hamann W. R., Hainich R., Oskinova L. M., Shenar T., Sander A. A. C., Todt H., Gallagher J. S., 2018, A&A, 615, A40
- Raucq F., Gosset E., Rauw G., Manfroid J., Mahy L., Mennekens N., Vanbeveren D., 2017, A&A, 601, A133
- Reardon D. J., et al., 2023, ApJ, 951, L6
- Redd N. T., 2018, Proc. Nat. Acad. Sci, 115, 12836
- Reimers D., 1975, Bull. Soc. R. Sci. Liege, 8, 369
- Renzo M., Götberg Y., 2021, ApJ, 923, 277
- Renzo M., Zapartas E., Justham S., Breivik K., Lau M., Farmer R., Cantiello M., Metzger B. D., 2023, ApJ, 942, L32
- Reynolds T. M., Fraser M., Gilmore G., 2015, MNRAS, 453, 2885
- Richards S., 2021, Master's thesis, University of Auckland, doi:10.5281/zenodo.5523933, <https://zenodo.org/record/5523933>
- Richards S. M., Eldridge J. J., Briel M. M., Stevance H. F., Willcox R., 2023, MNRAS, 522, 3972
- Richards S., Eldridge J., Ghodla S., Briel M., 2024, The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material, doi:10.48550/arXiv.2411.03000, <https://ui.adsabs.harvard.edu/abs/2024arXiv241103000R>

- Robbins F. E., 1940, *Tetrabiblos*. Harvard University Press Cambridge, MA
- Robertson J. A., Eggleton P. P., 1977, *MNRAS*, 179, 359
- Rose W. K., 1998, *Advanced Stellar Astrophysics*. Cambridge University Press
- Rosseland S., 1924, *MNRAS*, 84, 720
- Rucinski S. M., 1998, *ApJ*, 116, 2998
- Rueda J. A., Ruffini R., Wu Y.-B., Xue S.-S., 2014, *Phys. Rev. C*, 89, 035804
- Salpeter E. E., 1955, *ApJ*, 121, 161
- Sander A., Shenar T., Hainich R., Gímenez-García A., Todt H., Hamann W. R., 2015, *A&A*, 577, A13
- Santoliquido F., Mapelli M., Giacobbo N., Bouffanais Y., Artale M. C., 2021, *MNRAS*, 502, 4877
- Schmitz K., 2021, *JHEP*, 2021, 97
- Schneider F. R. N., Langer N., de Koter A., Brott I., Izzard R. G., Lau H. H. B., 2014, *A&A*, 570, A66
- Schneider F. R. N., Podsiadlowski P., Laplace E., 2024, *A&A*, 686, A45
- Schneider F. R. N., Laplace E., Podsiadlowski P., 2025, arXiv e-prints, p. arXiv:2507.06391
- Schwarzschild M., Härm R., 1965, *ApJ*, 142, 855
- Schwarzschild M., Härm R., 1967, *ApJ*, 150, 961
- Sen K., et al., 2022, *A&A*, 659, A98
- Sen K., et al., 2023, *A&A*, 672, A198
- Shakespeare W., ca. 1603, *The Tragedy of Othello, the Moor of Venice*
- Smartt S. J., et al., 2017, *Nature*, 551, 75
- Smith N., Li W., Filippenko A. V., Chornock R., 2011, *MNRAS*, 412, 1522
- Smith N., Mauerhan J. C., Silverman J. M., Ganeshalingam M., Filippenko A. V., Cenko S. B., Clubb K. I., Kandrashoff M. T., 2012, *MNRAS*, 426, 1905
- Smith N., Götberg Y., de Mink S. E., 2018, *MNRAS*, 475, 772
- Soberman G. E., Phinney E. S., van den Heuvel E. P. J., 1997, *A&A*, 327, 620
- Southworth J., 2015, in Rucinski S. M., Torres G., Zejda M., eds, *Astronomical Society of the Pacific Conference Series Vol. 496, Living Together: Planets, Host Stars and Binaries*. p. 164 (arXiv:1411.1219), doi:10.48550/arXiv.1411.1219
- Stancilffe R. J., 2005, PhD thesis, <https://ui.adsabs.harvard.edu/abs/2005PhDT.....352S>

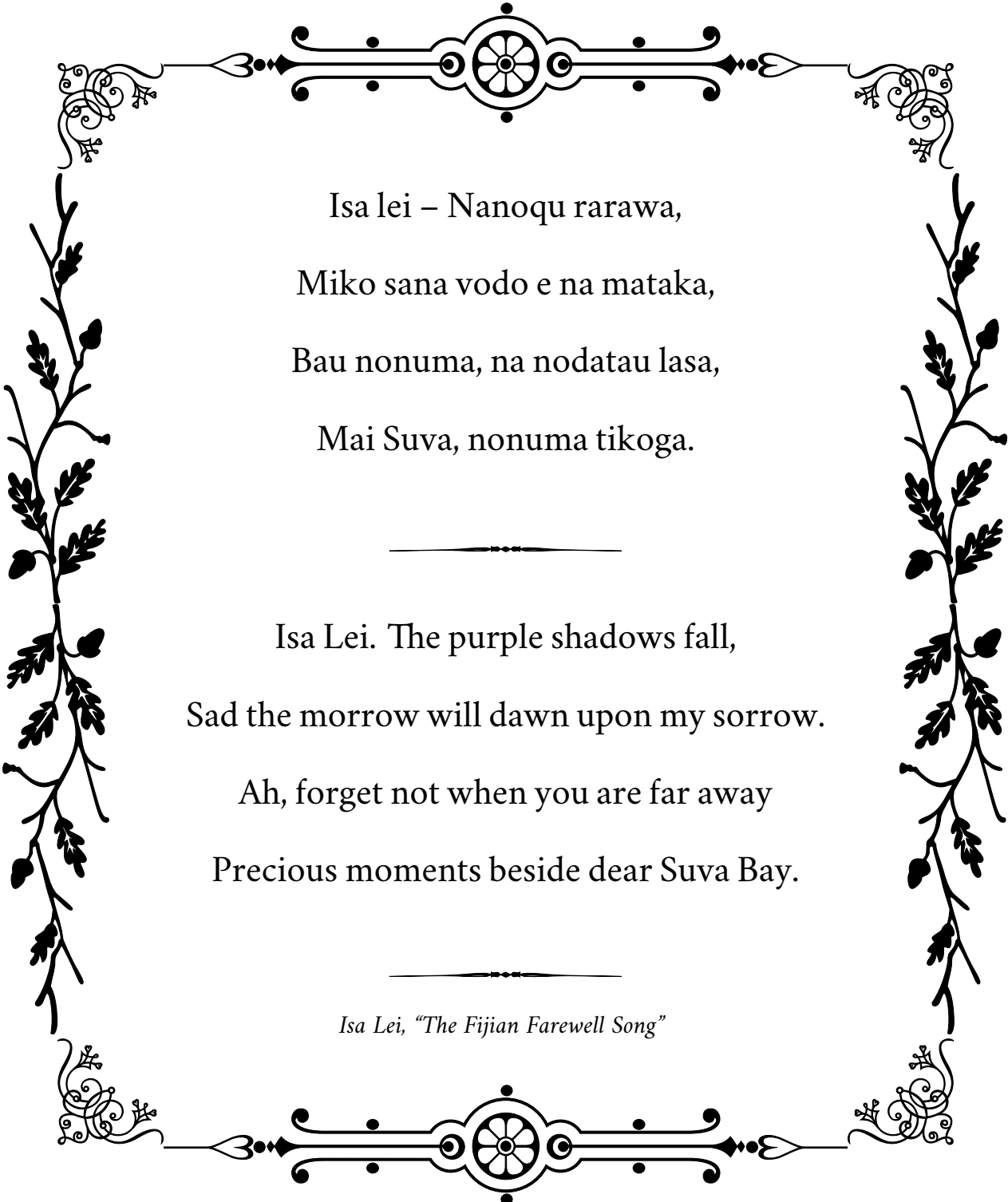
- Stancliffe R. J., 2006, *Mem. Soc. Astron. Italiana*, 77, 792
- Stancliffe R. J., 2016, *IAU Focus Meeting*, 29B, 600
- Stancliffe R. J., Eldridge J. J., 2009, *MNRAS*, 396, 1699
- Stancliffe R. J., Tout C. A., Pols O. R., 2004, *MNRAS*, 352, 984
- Stancliffe R. J., Glebbeek E., Izzard R. G., Pols O. R., 2007, *A&A*, 464, L57
- Stanway E. R., Eldridge J. J., 2018, *MNRAS*, 479, 75
- Stanway E. R., Eldridge J. J., Becker G. D., 2016, *MNRAS*, 456, 485
- Stephenson C. B., Sanduleak N., 1971, *Publ. Warner Swasey Obs.*, 1, 1
- Stephenson F. R., Yau K. K. C., Hunger H., 1985, *Nature*, 314, 587
- Stern M. E., 1960, *Tellus*, 12, 172
- Stevance H., Eldridge J., Stanway E., 2020, *JOSS*, 5, 1987
- Stevance H. F., Eldridge J. J., Stanway E. R., Lyman J., McLeod A. F., Levan A. J., 2023, *Nature Astronomy*, 7, 444
- Stothers R., 1972, *ApJ*, 175, 431
- Stothers R., 1973, *PASP*, 85, 363
- Stovall K., et al., 2018, *ApJ*, 854, L22
- Sutantyo W., 1978, *Ap&SS*, 54, 479
- Swiggum J. K., et al., 2015, *ApJ*, 805, 156
- Taam R. E., Bodenheimer P., Ostriker J. P., 1978, *ApJ*, 222, 269
- Tauris T. M., Takens R. J., 1998, *A&A*, 330, 1047
- Tauris T. M., Langer N., Podsiadlowski P., 2015, *MNRAS*, 451, 2123
- Tauris T. M., et al., 2017, *ApJ*, 846, 170
- Team COMPAS: Riley J., et al., 2022, *ApJS*, 258, 34
- Terrell D., Munari U., Zwitter T., Nelson R., 2003, *ApJ*, 126, 2988
- Thorne K. S., Żytkow A. N., 1977, *ApJ*, 212, 832
- Tylenda R., et al., 2011, *A&A*, 528, A114
- Ulrich R. K., 1972, *ApJ*, 172, 165

- de Donder E., Vanbeveren D., 1998, *A&A*, 333, 557
- de Jager C., Nieuwenhuijzen H., van der Hucht K. A., 1988, *A&AS*, 72, 259
- de Loore C., de Grève J. P., de Cuyper J. P., 1975, *Ap&SS*, 36, 219
- de Mink S. E., Pols O. R., Hilditch R. W., 2007, *A&A*, 467, 1181
- de Mink S. E., Langer N., Izzard R. G., Sana H., de Koter A., 2013, *ApJ*, 764, 166
- van de Voort F., Pakmor R., Bieri R., Grand R. J. J., 2022, *MNRAS*, 512, 5258
- Vanbeveren D., De Loore C., Van Rensbergen W., 1998, *A&ARv*, 9, 63
- Vassiliadis E., Wood P. R., 1993, *ApJ*, 413, 641
- Verbunt F., Igoshev A., Cator E., 2017, *A&A*, 608, A57
- Vigna-Gómez A., et al., 2018, *MNRAS*, 481, 4009
- Vink J. S., de Koter A., Lamers H. J. G. L. M., 1999, *A&A*, 350, 181
- Vink J. S., de Koter A., Lamers H. J. G. L. M., 2001, *A&A*, 369, 574
- Virtanen P., et al., 2020, *Nat. Methods*, 17, 261
- Vitense E., 1953, *Z. Astrophys.*, 32, 135
- Wagg T., Johnston C., Bellinger E. P., Renzo M., Townsend R., Mink S. E. d., 2024, *A&A*, 687, A222
- Walmswell J. J., Tout C. A., Eldridge J. J., 2015, *MNRAS*, 447, 2951
- Wellstein S., Langer N., Braun H., 2001, *A&A*, 369, 939
- Wiktorowicz G., Wyrzykowski Ł., Chruslinska M., Klencki J., Rybicki K. A., Belczynski K., 2019, *ApJ*, 885, 1
- Willcox R., Mandel I., Thrane E., Deller A., Stevenson S., Vigna-Gómez A., 2021, *ApJ*, 920, L37
- Williams V. V., 2012, in *Proceedings of the forty-fourth annual ACM symposium on Theory of computing. STOC '12*. Association for Computing Machinery, New York, NY, USA, pp 887–898, doi:10.1145/2213977.2214056, <https://dl.acm.org/doi/10.1145/2213977.2214056>
- Wongwathanarat A., Janka H.-T., Müller E., 2013, *A&A*, 552, A126
- Xiao L., Stanway E. R., Eldridge J. J., 2018, *MNRAS*, 477, 904
- Xu H., et al., 2023, *Res. Astron. Astrophys.*, 23, 075024
- Yao Y., et al., 2020, *ApJ*, 900, 46
- Zdunik J. L., Fortin M., Haensel P., 2017, *A&A*, 599, A119

Zhang X., Chen X., Fu J., Li Y., 2023, MNRAS, 523, 1394

van Zeist W., Stevance H., Eldridge J., 2021, JOSS, 6, 2968

van Zeist W. G. J., Eldridge J. J., Tang P. N., 2023, MNRAS, 524, 2836



Isa lei – Nanoqu rarawa,
Miko sana vodo e na mataka,
Bau nonuma, na nodatau lasa,
Mai Suva, nonuma tikoga.

Isa Lei. The purple shadows fall,
Sad the morrow will dawn upon my sorrow.
Ah, forget not when you are far away
Precious moments beside dear Suva Bay.

Isa Lei, "The Fijian Farewell Song"